



PARTICIPANT WORKBOOK

Foundations of Clinical Research

From Clinical Question to Publication

TRAINERS

Vương Mỹ Lượng

Bernard Osang'ir

Foundations of Clinical Research

From Clinical Question to Publication

Participant Workbook

TRAINERS

Vương Mỹ Lượng

Bernard Osang'ir



Foundations of Clinical Research

From Clinical Question to Publication

Participant workbook, first edition, 2026

This workbook belongs to: _____

Group: _____ Role in the group: _____

About the data. FLUID-ICU, the course dataset used in Sessions 5–8, is **simulated**: an invented cohort of 620 adults with sepsis in four ICUs, generated by computer for teaching. It describes no real patient, hospital or country. Its results illustrate methods, not clinical findings, and a manuscript written from it must say so and must never be submitted as real research. Papers 1 and 2 are real published articles, cited with their licences in the course folder.

Every file name in this workbook (for example `Templates/Research_Proposal_Template.docx`) refers to the course folder that your trainers share with you. The workbook was typeset with LuaLaTeX in the Neudata house style.

Copyright © Neudata, 2026

All property rights and copyright are reserved. This book contains proprietary information and materials developed by Neudata and may not be copied, reproduced, distributed, published or cited, in whole or in part, without the prior written consent of Neudata. Where explicit written permission has been granted, appropriate attribution must be included, followed by “by courtesy of Neudata”.

Neudata Consulting Ltd · www.neu-data.com · contact@neu-data.com

Contents

Getting started	1
1 ASK — from clinical problem to research question	8
1.1 Before the session — prepare.	8
1.2 Session plan	9
1.3 Key ideas in pictures	9
1.4 Key words.	12
1.5 Exercises	12
1.6 Demo companion.	17
1.7 Proposal builder 1	17
1.8 After the session — homework	18
1.9 Extra practice	19
2 FIND & READ — finding, managing and reading the evidence	21
2.1 Before the session — prepare.	21
2.2 Session plan	22
2.3 Key ideas in pictures	22
2.4 Key words.	25
2.5 Exercises	26
2.6 Demo companion.	31
2.7 Proposal builder 2	31
2.8 After the session — homework	33
2.9 Extra practice	34
3 DESIGN — choosing the right study	36
3.1 Before the session — prepare.	36
3.2 Session plan	37
3.3 Key ideas in pictures	37
3.4 Key words.	41

3.5	Exercises	41
3.6	Demo companion	44
3.7	Proposal builder 3	44
3.8	After the session — homework	45
3.9	Extra practice	45
4	BIAS & EFFECT — can we trust it, how big is it?	48
4.1	Before the session — prepare.	48
4.2	Session plan	49
4.3	Key ideas in pictures	49
4.4	Key words.	53
4.5	Exercises	53
4.6	Demo companion	59
4.7	Proposal builder 4	60
4.8	After the session — homework	60
4.9	Extra practice	61
5	DATA — collect it well, describe it honestly	63
5.1	Before the session — prepare.	63
5.2	Session plan	64
5.3	Key ideas in pictures	64
5.4	Key words.	67
5.5	Exercises	68
5.6	Demo companion	73
5.7	Proposal builder 5	74
5.8	After the session — homework	76
5.9	Extra practice	76
6	ANALYSE — from sample to answer	78
6.1	Before the session — prepare.	79
6.2	Session plan	79
6.3	Key ideas in pictures	79
6.4	Key words.	84
6.5	Exercises	84
6.6	Demo companion	94
6.7	Proposal builder 6	95

6.8	After the session — homework	97
6.9	Extra practice	97
7	PROTECT & WRITE — ethics, integrity, Methods and Results	99
7.1	Before the session — prepare.	100
7.2	Session plan	100
7.3	Key ideas in pictures	100
7.4	Key words.	104
7.5	Exercises	104
7.6	Demo companion	106
7.7	Proposal builder 7 and manuscript workshop 1	106
7.8	After the session — homework	111
7.9	Extra practice	111
8	PUBLISH — finish the paper, get it out, present	113
8.1	Before the session — prepare.	113
8.2	Session plan	114
8.3	Key ideas in pictures	114
8.4	Key words.	118
8.5	Exercises	119
8.6	Demo companion	125
8.7	Manuscript workshop 2 and presentations.	125
8.8	After the session — homework	127
8.9	Extra practice	127
9	Templates and tools	130
9.1	Your toolkit at a glance	130
9.2	Planning the study	131
9.3	Writing and publishing	133
9.4	Appraisal and reporting tools	134
9.5	Calculators and worked examples	139
10	The course dataset — FLUID-ICU	141
10.1	What FLUID-ICU is	142
10.2	Who is in the data: from 623 rows to 527 complete cases	143
10.3	The data dictionary	143

10.4	How the raw data were cleaned	145
10.5	What the data show (simulated): the numbers you will meet	146
10.6	The five group questions	148
10.7	How to read your results pack	149
10.8	The manuscript workflow	150
11	The assignments	152
11.1	Group product 1: the research proposal	152
11.2	The presentation in Session 8.	156
11.3	Group product 2: the manuscript.	157
11.4	The peer-review swap (after the course)	159
11.5	The marking rubrics (summary).	162
A	Answers to the exercises and extra practice	165
A.1	Session 1 · ASK.	165
A.2	Session 2 · FIND & READ	171
A.3	Session 3 · DESIGN	175
A.4	Session 4 · BIAS & EFFECT.	179
A.5	Session 5 · DATA	184
A.6	Session 6 · ANALYSE	188
A.7	Session 7 · PROTECT & WRITE	194
A.8	Session 8 · PUBLISH	198
B	Glossary — English–Vietnamese	202
C	Further reading and free resources	217
D	Planning pages	223
D.1	Proposal planner	223
D.2	Manuscript planner	226

Getting started

Welcome. Every day you make clinical decisions that research could improve. Over **four weekends** this course walks the whole road of research with you: from a question at the bedside, to a sound study design, to analysing data, to writing and publishing the paper. You do not need any research or statistics background. We go slowly, with pictures, examples from the wards and a lot of group work. You will never calculate anything beyond a simple 2×2 table, and you do not need any software: the trainers run the analyses live, and you learn to read them and write them up.

How to use this workbook

This workbook is your companion for the whole course. **Bring it to every session**, write in it, and keep it afterwards: it holds every exercise, every worksheet your group needs, and the templates and checklists you will use when you plan your own studies.

Each of the eight session chapters (Chapters 1–8) has the same parts, in the order you will use them:

Part	What it is	When to use it
Learning objectives	What you should be able to do at the end of the session	Read before the session; tick them off afterwards
Before the session — prepare	A short check list (15–30 minutes)	The evening before
Session plan	The timetable of the 3 hours	During the session
Key ideas in pictures	The main ideas, each with one figure from the slides and a take-away	Before and after the session
Key words	The words of the day in one line each (full English–Vietnamese glossary in Appendix B)	Any time
Exercises	Every exercise of the session, with space to write	In class, and as homework
Demo companion	What you will see in the live demo and what to note	During the demo
Proposal builder / manuscript workshop	Your group's worksheet for the proposal (and, from Session 7, the manuscript)	Group work at the end of the session
After the session — homework	What to finish before the next session	After the session
Extra practice	2–4 extra short exercises to check yourself	At home
My notes	A lined page	Any time

Chapters 9–11 are your **toolkit**: the templates and tools (Chapter 9), the course dataset FLUID-ICU (Chapter 10) and the two group assignments with their rubrics (Chapter 11). The appendices hold the **answers** (Appendix A), the **English–Vietnamese glossary** (Appendix B), **further reading** (Appendix C) and **blank planning pages** for your proposal and manuscript (Appendix D).

The boxes in this workbook

Navy banners mark an exercise from the session; **teal banners** mark extra practice for home. **Grey ruled boxes** are for your handwriting. A box like ☐ is a tick box. A **teal "Take-away" strip** under a figure gives its message in one sentence — if you remember only that sentence, you have the idea.

Try first, then check. The answers to the exercises and the extra practice are in Appendix A. There is often more than one good answer: compare your reasoning, not just your words. The exercises are discussed in the room and are **never marked**.

File names such as `Templates/Research_Proposal_Template.docx` refer to the course folder that the trainers share with you in Session 1.

The course at a glance

Weekend	Session	Theme	Your group builds
1	1 · Saturday	ASK — from clinical problem to research question	Proposal: topic, PICO, question, aims
	2 · Sunday	FIND & READ — searching, managing references, reading a paper in 15 minutes	Proposal: search, 5 references, problem statement, working title
2	3 · Saturday	DESIGN — choosing the right study	Proposal: design, setting, population
	4 · Sunday	BIAS & EFFECT — bias, confounding, RR/OR/NNT, diagnostic tests, forest plots	Proposal: exposure, outcomes, bias plan
3	5 · Saturday	DATA — CRFs, data management, describing data; <i>meet the course dataset</i>	Proposal: variables, CRF, data plan · Manuscript: Table 1 understood
	6 · Sunday	ANALYSE — CIs, p-values, tests, regression, survival, missing data; <i>analyse the course dataset</i>	Proposal: sample size, analysis plan · Manuscript: results interpreted
4	7 · Saturday	PROTECT & WRITE — ethics, registration, integrity; writing Methods and Results	Proposal: ethics, feasibility, publication plan · Manuscript: Methods + Results
	8 · Sunday	PUBLISH — Introduction, Discussion, abstract; journals, submission, peer review; presentations	Manuscript: complete draft · Proposal presented

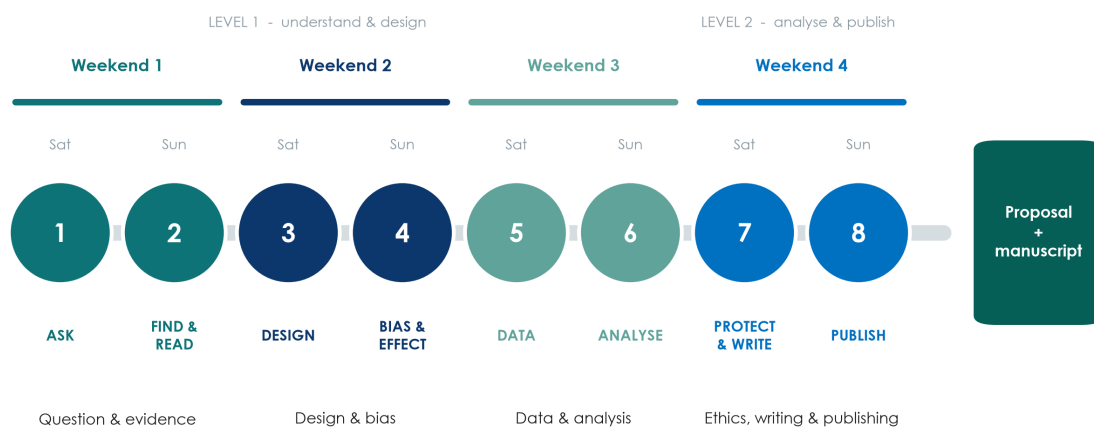


Figure 1: The course road map: four weekends, eight sessions, from asking a question to publishing the answer

Take-away: Weekends 1–2 (Level 1): understand and design a study. Weekends 3–4 (Level 2): analyse a realistic dataset and write it up.

Each session lasts **3 hours with a 15-minute break**. Every session has the same rhythm: a short recap, two teaching blocks with frequent “check your understanding” moments, and time to build your proposal or your paper. Each session also has **extra slides** for anyone who wants to read further at home.

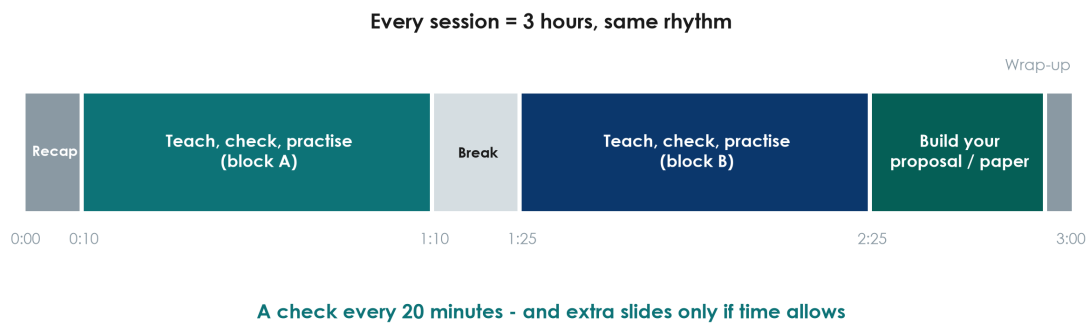


Figure 2: Every session has the same rhythm

One running example

One clinical topic travels with us through the whole course: **fluids in septic shock**.

P — Population	Adults (18+) with septic shock in the ICU
I — Intervention	Restrictive fluid strategy
C — Comparison	Usual care (standard fluids)
O — Outcome	Death from any cause
T — Time	Within 28 days

In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality? In Weekends 1–2 we ask, search and design this question, and look at two real papers that answer it with two different designs:

- **Paper 1** — Boulet N, et al. *Crit Care* 2024;28:429: a randomised controlled **pilot trial** of a restrictive fluid strategy (48 patients analysed).
- **Paper 2** — Hyun D-G, et al. *Ann Intensive Care* 2023;13:105: a **propensity-score-matched cohort** from a sepsis registry.

In Weekends 3–4 the question takes its observational form — *is a positive cumulative fluid balance by day 3 associated with higher 28-day mortality?* — and we analyse **FLUID-ICU**, a realistic **simulated** dataset of 620 ICU patients with sepsis, and write it up as a paper (Chapter 10).

Your group and its two products

You stay in the **same group of about six** for all eight sessions, with a mix of professions. Agree roles in Session 1 (they can rotate between weekends):

Role	What this person does
Facilitator	Keeps the group on task, makes sure every voice is heard, summarises decisions
Scribe	Types into the shared proposal document; keeps the single master version
Timekeeper	Watches the clock in group work; times the rehearsals at 6 minutes
Literature lead	Runs the PubMed search; keeps the references in a reference manager; leads the journal choice
Methods & data lead	Owens design, outcomes, variables, sample size and analysis plan
Presenter	Leads the 6-minute presentation in Session 8

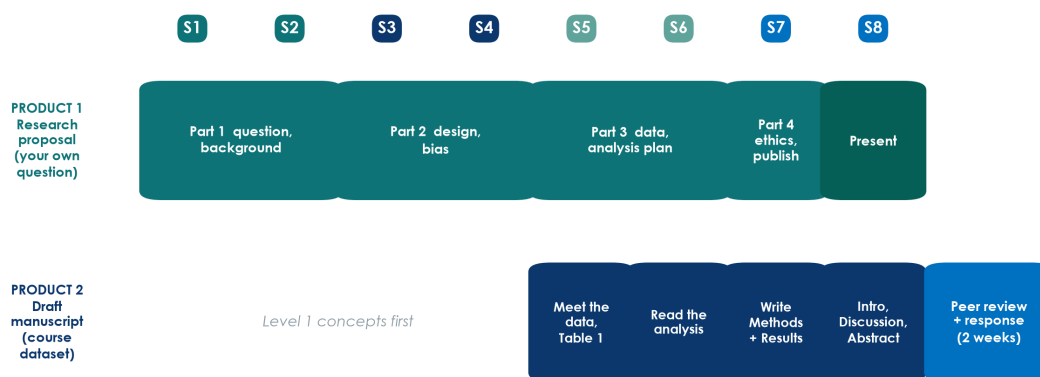


Figure 3: Your group builds two products during the course

Take-away: Product 1 is about your own question (planning a study); product 2 is about someone else's data (reporting a study).

- 1. A research proposal on your own question**, built one part per weekend with the *Proposal builder* worksheets and **presented in Session 8** (6 minutes + 2 minutes of feedback), with a publication plan. See Chapter 11.
- 2. A draft manuscript from the course dataset.** Each group answers a different question on FLUID-ICU using its **results pack**. You draft Methods and Results in Session 7 and the rest in Session 8. After the course, groups **review each other's drafts and reply like real authors** (peer-review swap, 2 weeks).

My group: _____ My role(s): _____ Our FLUID-ICU question: Q _____

Our group members and their roles:

Between sessions

Plan about **30–60 minutes** between sessions. Each chapter ends with a homework list; this is the overview.

After	Do this
Session 1	Think of synonyms for your PICO terms
Session 2	Finish Proposal Part 1 and the literature matrix; read the abstract of Paper 2
Weekend 2	Finish Proposal Part 2
Session 5	Read your group's results pack (Table 1 and figures)
Weekend 3	Finish Proposal Part 3; read Papers 1 and 2; start your 6-slide presentation
Session 7	Finish Proposal Part 4 and your presentation; re-read your Methods + Results
Session 8	Polish the manuscript; peer-review swap and response letter within 2 weeks

What you need: a pen, this workbook, and (optional) a laptop or phone for PubMed and the online quiz. No software to install. If you have done the free Neudata course *Clinical Data Analysis in R*, you may re-run the course analyses with the scripts in Data/R/ (optional).

The quiz. In Session 1 and again at the end of Session 8 you answer the same 24 questions. It is anonymous (participant number only) and measures what the **course** achieved, not you.

What we expect from you

Course expectations

Over the four weekends we expect you to:

1. **attend the sessions** (at least 6 of the 8),
2. **present your group's research proposal** in Session 8, and
3. with your group, **submit the manuscript draft** (by 23:59 on the day of Session 8).

The marks on the proposal and the manuscript are **formative**: they are there to help you improve. The peer review and the response letter in the two weeks after the course complete the work.

My attendance (tick each session you attend):

Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7	Session 8
☐	☐	☐	☐	☐	☐	☐	☐

After the course, take your proposal to your research office or ethics committee, keep improving your manuscript, and — if you want to run analyses yourself — follow the free *Clinical Data Analysis in R* course and the optional advanced modules.

SESSION 1

ASK — from clinical problem to research question

When	Weekend 1 · Saturday · Session 1 of 8
Length	3 hours, with a 15-minute break (pre-course quiz at the start)
Your group builds	Proposal Part 1 (first half) — topic, PICO, research question, FINER check, aim, objectives and hypothesis

Learning objectives

By the end of this session you will be able to:

- explain why personal experience is not enough to know whether a treatment works, and read the evidence pyramid;
- tell research, clinical audit and quality improvement (QI) apart, and say which approval each needs;
- turn a clinical problem into a focused question with **PICO(T)** (or PECO for exposures);
- tell a descriptive question from an analytical one, and a vague question from an answerable one;
- check a question with **FINER** and rewrite it when one letter fails;
- write one aim, 2–3 measurable objectives and a null (H0) and alternative (H1) hypothesis.

1.1 Before the session — prepare

- ☐ Read "Getting started" at the front of this workbook (10 minutes).
- ☐ Think of **one problem from your own unit** that puzzles you or your colleagues (a complication, a delay, a practice that varies between staff). Write it here in one sentence.

- ☐ Ask yourself: what would you need to **count or measure** to know whether the problem is

getting better or worse?

- ☐ Bring a pen and this workbook. A phone or laptop is useful for the online quiz.

1.2 Session plan

Time	Min	Segment
0:00	20	Welcome, introductions, course map, groups and roles, ground rules
0:20	15	Pre-course quiz
0:35	35	1.1 Why research? Anecdote vs evidence, evidence pyramid, research cycle, types of research, research vs audit vs QI, the journey from idea to publication
1:10	15	BREAK
1:25	40	1.2 From problem to question: where questions come from, PICO(T) with several worked examples, descriptive vs analytical questions, vague vs answerable, FINER (pair task)
2:05	20	1.3 Aims, objectives, hypotheses (null vs alternative) + quick exercise
2:25	30	Proposal builder 1: choose topic, PICO, research question, FINER check, aim & objectives
2:55	5	Wrap-up, homework

1.3 Key ideas in pictures

Experience asks the question; fair comparisons answer it

Every clinician remembers the patient who did well after a treatment. But a few memorable patients, with no comparison group, cannot tell us what would have happened anyway. Research compares **enough** patients **fairly** — the same outcome, counted the same way, in groups that differ only in the treatment.

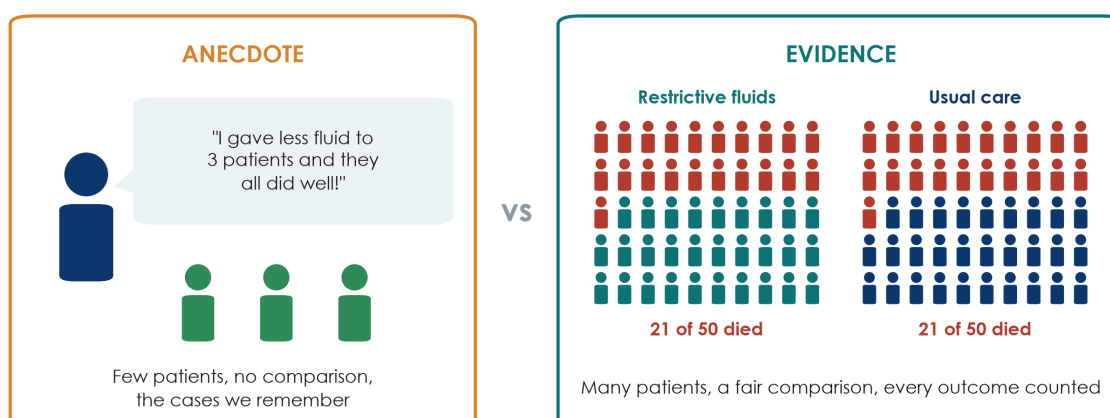


Figure 1.1: Anecdote vs evidence: three remembered patients versus a fair comparison of 50 patients per group

Take-away: Experience suggests questions. Fair comparisons in many patients answer them.

The large CLASSIC trial (1,554 adults with septic shock) found 90-day mortality of 42.3% with

restrictive fluids and 42.1% with standard care: the patients we remember "doing well on less fluid" would probably have done well either way.

The evidence pyramid ranks designs

Higher in the pyramid means less risk of bias **for questions about treatment effects**. It ranks *designs*, not individual papers: a careful cohort can be more trustworthy than a small, badly run trial, and a descriptive question ("how many?") does not need a trial at all.

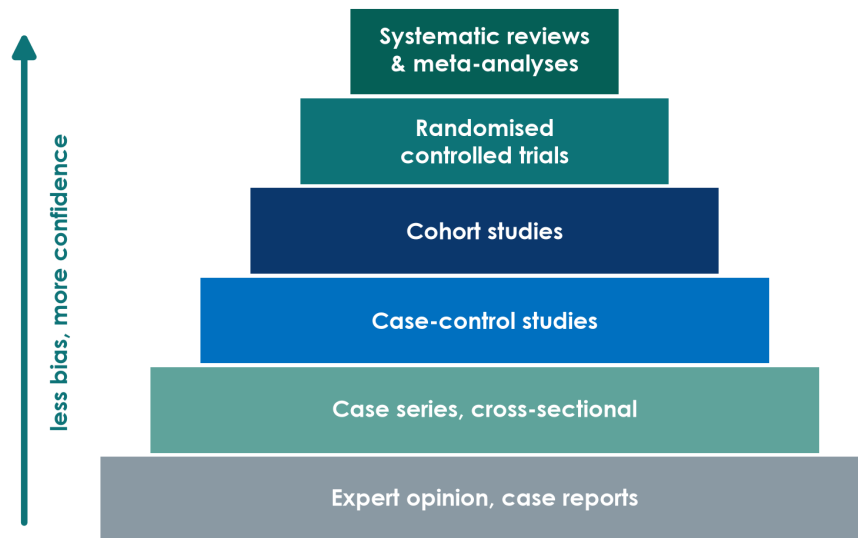


Figure 1.2: The evidence pyramid

Take-away: The pyramid ranks designs, not individual papers: quality still has to be checked.

Research, audit or quality improvement?

Three different jobs. **Research** asks *what is best?* and creates new knowledge; it needs an ethics committee. **Audit** asks *do we do what the standard says?* **Quality improvement** asks *how can we do it better here?*, often in small Plan–Do–Study–Act cycles. Audit and QI usually need local approval; when in doubt, ask the ethics committee **before** you start.

	RESEARCH	CLINICAL AUDIT	QUALITY IMPROVEMENT
Asks	What is the best practice?	Are we meeting the standard?	How do we get better, step by step?
Fluid example	Restrictive vs usual: which is better?	Is fluid balance recorded daily?	A daily checklist, tested in PDSA cycles
Results apply to	patients elsewhere (generalisable)	our unit	our unit
Ethics	Ethics committee approval: ALWAYS	Local approval; usually no committee	Local approval; ethics review if new risk

Not sure which it is? Ask the ethics committee **BEFORE** you start - journals ask for approval or a waiver.

Figure 1.3: Research, audit or quality improvement

Take-away: Research asks what is best; audit asks if we do it; QI makes it better.

PICO(T): a question you can answer

A good question names five things: **P**opulation (who?), **I**ntervention or exposure (what is given, or what are they exposed to?), **C**omparison (compared with what?), **O**utcome (what is counted?) and **T**ime (when?). For an observational question, I becomes **E** (exposure): PECO. A descriptive question ("how many?") may only need P and O.

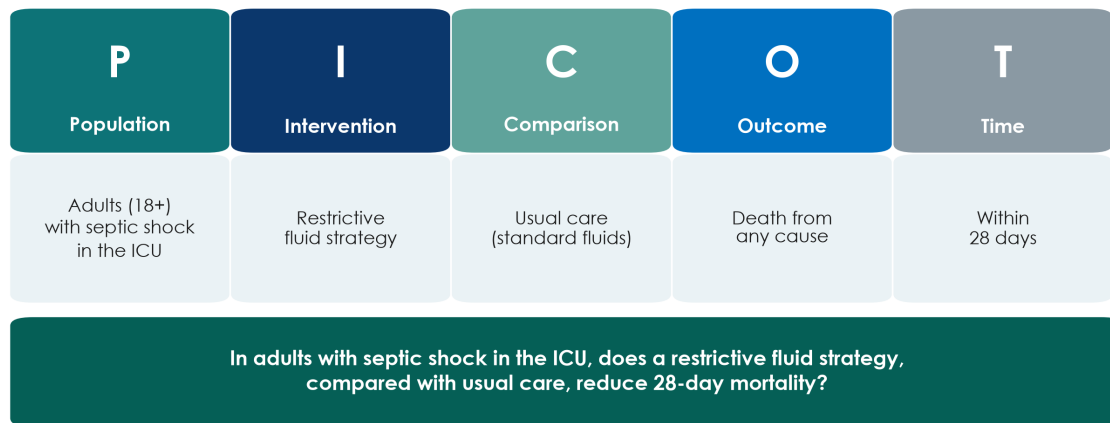


Figure 1.4: PICO(T) of our running example

Take-away: Every session we come back to this one question.

FINER: is the question worth asking — here, now, by us?

Check each letter: **F**easible, **I**nteresting, **N**ovel, **E**thical, **R**elevant. Our running example is important, but a mortality difference needs well over a thousand patients: it fails **F** for one unit. The honest move is to rewrite it — for example a pilot with fluid balance as the outcome, as Paper 1 did.

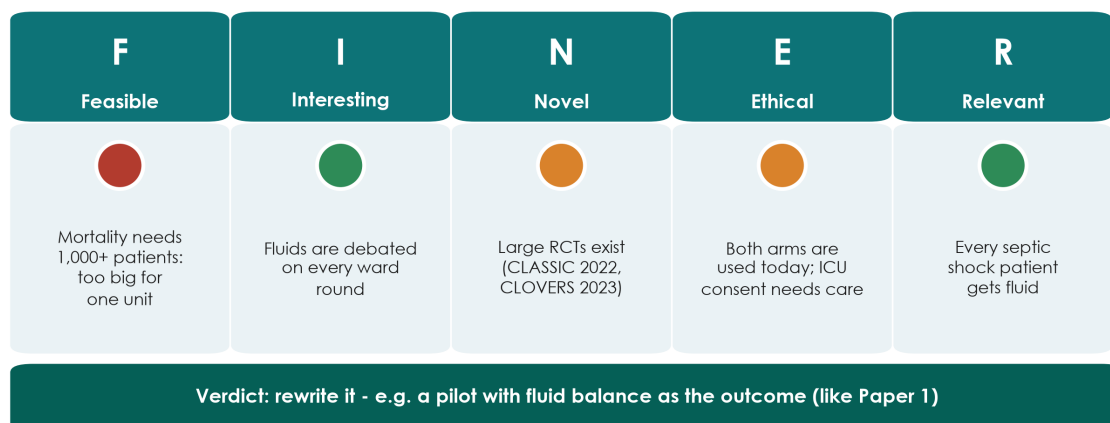


Figure 1.5: FINER traffic light for the running example

Take-away: If the question fails one letter, rewrite it before going further.

Aim, objectives, hypothesis

The **aim** is the broad *why* in one sentence. **Objectives** are the specific, measurable *what* — they start with a measurable verb (*compare, estimate, describe, determine*), never *understand* or *study*. The **null hypothesis (H0)** says there is no difference; the data can reject it or fail to reject it, but never prove it.

1.4 Key words

Key word	In one line
Evidence pyramid	Ranks study designs by how well they protect against bias
Research / audit / QI	What is best? / Do we do it? / How do we improve it?
PICO(T) / PECO	Population, Intervention (or Exposure), Comparison, Outcome, Time
Descriptive vs analytical question	Describes (how many?) vs compares or tests a cause
FINER	Feasible, Interesting, Novel, Ethical, Relevant
Aim / objective	The broad why / the specific, measurable what
H0 / H1	No difference / a difference — we test H0
Primary outcome	The one outcome the study is built to answer, with a time point

1.5 Exercises

How the exercises work today. You do not need a computer today. Write on the sheet.

- Exercise 1 (warm-up) — spot the PICO, on your own (3 minutes, segment 1.2).
- Exercises 1–2 — **pairs**, segment 1.2 (11 minutes).
- Exercise 3 — aim, objectives and hypothesis, on your own (segment 1.3, 7 minutes).
- Exercise 4 — your **group's Proposal builder 1** (28 minutes).
- Exercise 5 — homework for tomorrow (synonym list).

The answers are discussed in the room — there is no mark.

The running example (for reference)

P — Population	Adults (18+) with septic shock in the ICU
I — Intervention	Restrictive fluid strategy
C — Comparison	Usual care (standard fluids)
O — Outcome	Death from any cause
T — Time	Within 28 days

Question: *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?*

Exercise 1 — From clinical scenario to PICO(T)

warm-up on your own 3 minutes, then pairs 8 minutes

Warm-up — spot the PICO (on your own, 3 minutes)

"In adults aged 65+ after hip-fracture surgery, does a daily non-drug bundle, compared with usual care, reduce delirium in the first 7 days?"

P	I	C	O	T
---	---	---	---	---

Pair task. Pick **ONE** scenario. Fill the PICO(T) boxes, then write the question as **one sentence**.

Tips

- For an observational question, **I** becomes **E** (exposure) — "PECO".
- A descriptive question (how many? how often?) may only need **P** and **O**.
- Choose **one** primary outcome and say **when** it is measured.

Scenario A — Ventilator-associated pneumonia. Your ICU raises the head of the bed to 30–45° for ventilated patients, but compliance is patchy at night. A colleague wonders whether a twice-daily nurse checklist would reduce ventilator-associated pneumonia.

Scenario B — Pressure injuries. Several patients in your medical ward developed pressure injuries last quarter. Some staff turn patients every 2 hours, others every 4 hours. Nobody knows how often pressure injuries occur in your unit.

Scenario C — Delirium. Older patients after hip-fracture surgery often become confused. The geriatric team suggests a daily non-drug bundle (glasses and hearing aids on, orientation, sleep protection, early mobilisation).

Scenario D — Stroke door-to-needle time. Your emergency department wants to know whether a pre-hospital alert from the ambulance shortens the time from arrival to thrombolysis.

Scenario E — Medication errors. The pharmacy notices many errors when patients move from ICU to the ward. They propose a pharmacist-led medication reconciliation at transfer.

My scenario: _____**P** — who?**I / E** — what is given or what are they exposed to?**C** — compared with what?**O** — which outcome?**T** — measured when?

My question in one sentence:

Type of question (tick one): ☐ descriptive ☐ analytical — observational ☐ analytical — experimental

Exercise 2 — FINER traffic light

pairs, part of the same 8 minutes; Part B is extra practice

Part A. Rate *your* question from Exercise 1. Colour or write **G** (green — fine), **A** (amber — needs thought) or **R** (red — must change).

Letter	Test	G / A / R	Why?
F — Feasible	Enough patients, time, money, skills, data?		
I — Interesting	To you, your team and readers?		
N — Novel	Confirms, refutes or extends what is known?		
E — Ethical	Could an ethics committee approve it?		
R — Relevant	Could the answer change practice, policy or research?		

If any letter is **red**, rewrite the question here:

Part B (if time). Which FINER letter is the main problem in each question below?

1. "In all hospitalised adults in our country, does vitamin D supplementation reduce mortality over 10 years?" — Main problem: _____
2. "In ICU patients with septic shock, does withholding antibiotics for 24 hours reduce resistance?" — Main problem: _____
3. "Do nurses in our ICU wash their hands?" (already measured monthly by the infection-control audit for 5 years) — Main problem: _____
4. "In adults with community-acquired pneumonia, does the colour of the drug chart affect 28-day mortality?" — Main problem: _____

Exercise 3 — Aim, objectives and hypothesis

individual, 7 minutes, segment 1.3

Use the question from your pair task (or the delirium warm-up question).

Aim (one sentence, starting "To compare...", "To describe..." or "To estimate..."):

Objectives (2, each with a measurable verb — compare, estimate, describe, determine, measure; tick the primary one):

1. ☐ _____

2. ☐ _____

Vague-verb check: rewrite "To understand pressure injuries on our ward" as a measurable objective:

Hypothesis. Write **H0** (null: no difference) and **H1** (alternative: a difference) for your primary objective. If your question is **descriptive**, write "No hypothesis — descriptive" and say what you will *estimate* instead.

H0: _____

H1: _____

Practice — write H0 and H1 for:

a) "In ventilated ICU adults, does early mobilisation, compared with usual care, shorten ICU length of stay?"

H0: _____

H1: _____

b) "What proportion of our septic shock patients have a positive cumulative fluid balance on day 3?"

H0 / H1 needed? ☐ yes ☐ no — because: _____

Exercise 5 — Homework: synonyms for tomorrow's search

individual, about 15 minutes

PubMed searches in **English**. For each PICO word in your group's question, write 2–3 English words or phrases that authors might use.

PICO element	Our word	English synonyms (2–3)
P		
I / E		
O		

Example (pressure injury): pressure ulcer · pressure injury · bedsore · decubitus ulcer.

One local number that shows the problem (an audit result, a count, an incident):

Exit question

on a sticky note, no names

1. Which FINER letter did our running example fail, and why?

2. Rewrite as an objective: "To understand pressure injuries on our ward".

1.6 Demo companion

There is **no software demo** in Session 1. Instead, watch how the trainer turns a **corridor question** into a PICO question on the slides ("Vague vs answerable"), and note the steps:

Step	What the trainer does	My note
1	Starts from a vague question ("Are fluids bad for ICU patients?")	
2	Names the population precisely	
3	Defines the intervention and the comparison	
4	Chooses ONE outcome with a time point	
5	Writes the one-sentence question	

1.7 Proposal builder 1

Your group's first piece of the research proposal. Work on your **own** topic — not the running example.

Exercise 4 — Proposal builder 1

group, 28 minutes

Write your answers into the **Research Proposal template** (Templates/Research_Proposal_Template.docx), Part 1. This sheet is your checklist. The search, references, problem statement and working title are **tomorrow** (Proposal builder 2).

Suggested roles: chair · scribe (types into the template) · timekeeper · "devil's advocate" (checks FINER).

Time	Task	Done
0–7 min	1. Choose ONE problem from your unit: list 3 ideas, vote, pick one	☐
7–14 min	2. PICO(T) table and the research question in one sentence	☐
14–19 min	3. FINER check — traffic light for every letter; rewrite the question if any letter is red	☐
	4. Research, audit or QI? Write the approval route (ethics committee or local approval)	☐
19–25 min	5. Aim (one sentence), 2–3 objectives (measurable verbs; mark the primary), H0 / H1 (or "descriptive — no hypothesis")	☐
25–28 min	One group reads its question and aim aloud; the room suggests one improvement	☐

Our question: _____

FINER: F ☐ I ☐ N ☐ E ☐ R ☐ (G / A / R) **Type:** ☐ research ☐ audit ☐ QI

Quality checks before you stop

- ☐ Every PICO box is filled, and there is only **one** primary outcome with a time point.
- ☐ The question, aim, primary objective and hypothesis all talk about the **same** outcome.
- ☐ No FINER letter is red.
- ☐ You know whether it is research, audit or QI, and who must approve it.

1.8 After the session — homework

- **Group:** finish the PICO, question, aim and objectives if not done.
 - **Everyone:** Exercise 5 — synonym list; one local number that shows the problem.
 - **Optional:** install Zotero (free, zotero.org) on your laptop.
 - **Bring:** your group's PICO question, your synonym list, a laptop or phone per group.
- ☐ Read your group's question once more tonight: is there **one** primary outcome with a time point?
 - ☐ Note one local number that shows the size of your problem (an audit result, a count from the ward register) — it opens your problem statement tomorrow.

1.9 Extra practice

Extra practice 1.1 — Research, audit or QI?

For each project, tick one and write who must approve it.

Project	Research	Audit	QI	Approval route
a) Counting how many ventilated patients had the head of the bed at 30–45° last month, against the unit standard	☐	☐	☐	
b) Testing a new night-time checklist in three rapid PDSA cycles on one ward	☐	☐	☐	
c) Randomising patients to two dressing types to compare wound infection at 30 days	☐	☐	☐	
d) Interviewing families about their experience of ICU visiting rules	☐	☐	☐	

Extra practice 1.2 — Find the PICO

"In adults admitted to our medical ward with community-acquired pneumonia, is a switch from intravenous to oral antibiotics within 48 hours, compared with a later switch, associated with a shorter hospital stay?"

P	I / E	C	O	T
---	-------	---	---	---

Is this question descriptive, analytical-observational or analytical-experimental? _____

Extra practice 1.3 — Make it answerable

Rewrite each vague question as a one-sentence PICO(T) question.

1. "Is our new sepsis bundle any good?"

2. "Do older patients fall more at night?"

Extra practice 1.4 — Objective or not?

Tick the statements that are good, measurable objectives. Rewrite one of the others.

- ☐ To understand pain management after surgery.
- ☐ To compare the proportion of patients with moderate or severe pain at 24 hours between the two analgesia protocols.
- ☐ To estimate the incidence of catheter-associated urinary infection on our ward over 6 months.
- ☐ To look at hand hygiene.

My rewrite: _____

My notes — Session 1

SESSION 2

FIND & READ — finding, managing and reading the evidence

When	Weekend 1 · Sunday · Session 2 of 8
Length	3 hours, with a 15-minute break
Your group builds	Proposal Part 1 (complete) — search string, 5 key references in the literature matrix, problem statement, working title and keywords

Learning objectives

By the end of this session you will be able to:

- turn a PICO question into **search blocks** with synonyms, MeSH terms and Boolean operators (OR inside, AND between);
- run, document and repeat a PubMed search, and know when to use Google Scholar, trial registries and grey literature;
- save references in a reference manager and cite them in a consistent style;
- paraphrase safely and recognise patchwriting and plagiarism;
- read a paper in 15 minutes in the right order, check what an abstract must contain and spot **spin**;
- fill a literature matrix and write a three-paragraph problem statement and a working title.

2.1 Before the session — prepare

- ☐ Finish Exercise 5 of Session 1: **2–3 English synonyms** for each PICO word of your group's question.
- ☐ Bring **one local number** that shows the size of your problem (an audit result, a count from the ward register).
- ☐ Optional: install **Zotero** (free, zotero.org) on your laptop and create an empty library.
- ☐ Bring a laptop or phone per group for PubMed.
- ☐ Have Paper 1 (Boulet 2024, *Crit Care*) ready — you read its **abstract** in class.

2.2 Session plan

Time	Min	Segment
0:00	10	Recap quiz
0:10	40	2.1 Finding the evidence: search funnel, PICO → search blocks, Boolean, MeSH, search tools, PubMed vs Google Scholar, live PubMed demo
0:50	20	2.2 Managing and citing: reference managers (Zotero demo), citation styles, paraphrasing safely, grey literature & trial registries
1:10	15	BREAK
1:25	40	2.3 Reading a paper in 15 minutes: IMRaD, reading order, what an abstract must contain, spin; individual task on Paper 1's abstract
2:05	15	2.4 From literature to introduction: literature matrix, problem-statement funnel, working title & keywords
2:20	35	Proposal builder 2: search string, 5 references in the matrix, problem statement, working title
2:55	5	Wrap-up

2.3 Key ideas in pictures

The search funnel: start wide, narrow step by step, write it down

A search starts wide and narrows: from one keyword (tens of thousands of hits) to the PICO blocks combined, then to sensible filters, then to papers you screen by reading. Record the database, the date, the exact string, the filters and the number of hits, so that anyone can repeat the search.

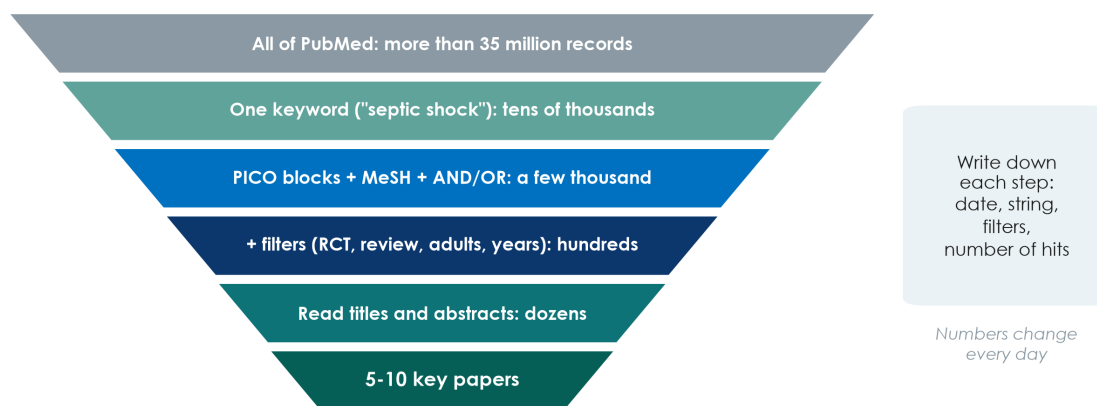


Figure 2.1: The search funnel

Take-away: A good search starts wide, narrows step by step, and is written down.

From PICO to search blocks

One block per PICO concept (the comparison is usually not searched). Inside a block, synonyms are joined with **OR**; the blocks are joined with **AND**. Each block mixes the official **MeSH** heading and free-text words in titles and abstracts, because the newest papers are not indexed with MeSH yet.

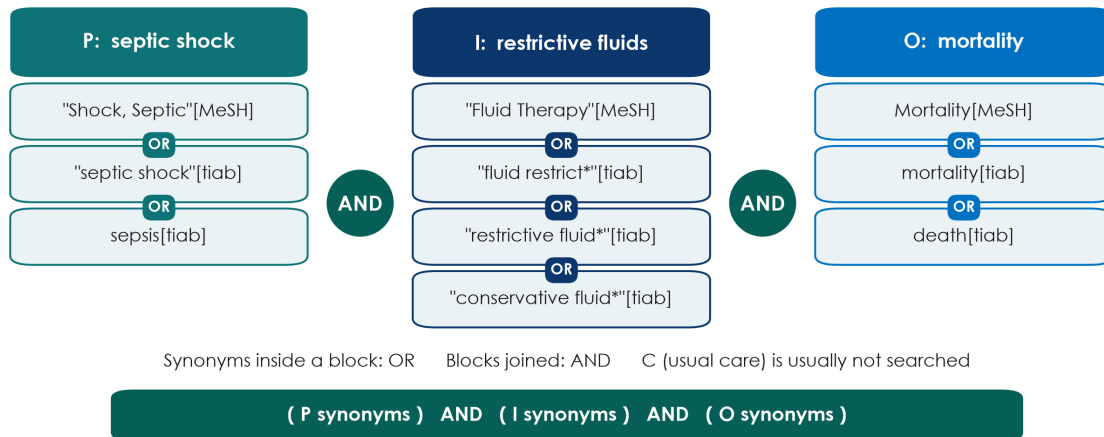


Figure 2.2: From PICO to search blocks

Take-away: One block per PICO concept: OR inside a block, AND between blocks.

Use other people's work honestly

A reference manager stores each paper once and formats every citation in any style. When you use an idea, change the **structure**, not only the words, and always cite. Copying a sentence and swapping a few words (patchwriting) is plagiarism **even with a reference**.

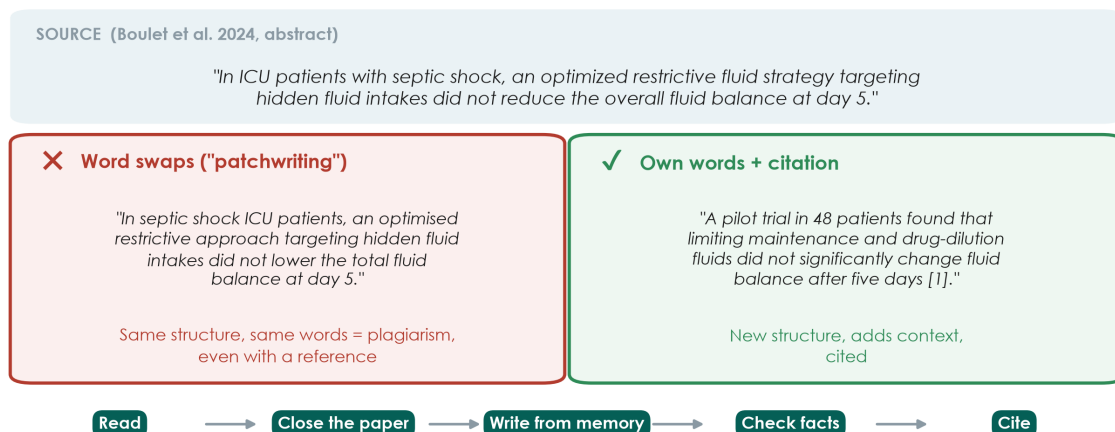
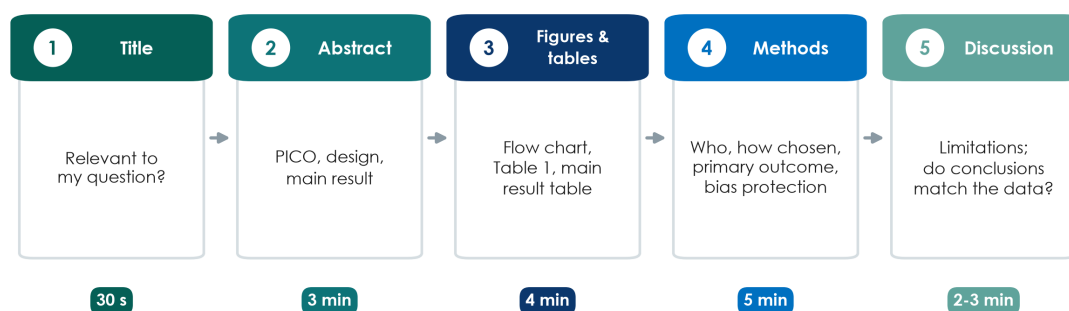


Figure 2.3: Citing vs copying: paraphrase safely

Take-away: Change the structure, not just the words — and always cite the source.

Reading a paper in 15 minutes

Every original paper has the IMRaD shape. You do not read it from the first word: title (30 s) → abstract (3 min) → figures and tables (4 min) → methods (5 min) → discussion and limitations (2–3 min). Stop early if it does not answer your question.



Stop after step 1 or 2 if it does not answer your question. Total: about 15 minutes.

Figure 2.4: The 15-minute reading order

Take-away: Title, abstract, figures and tables, methods — stop early if it is not relevant.

Spin: when the words go beyond the numbers

Read the numbers first, then check whether the words match them. "A non-significant trend" for $p = 0.42$, a headline built on a secondary outcome, or "impact" for an observational association are all spin.

	What the data say		What the words say (spin)
p-value	$p = 0.42$	→	"a non-significant trend" Paper 1, Discussion
Outcomes	primary outcome: no difference	→	headline = a positive secondary outcome
Design	observational association	→	"Impact of...", "reduces" Paper 2, title
Comparison	no between-group test	→	"patients improved" (before vs after only)

Figure 2.5: Spin: when words go beyond the data

Take-away: Read the numbers first, then check whether the words match them.

From literature to introduction

The papers you read become the first page of your proposal. A **literature matrix** gives one row per paper with the same columns; the **problem statement** follows a funnel in three short paragraphs: what is known, what is not known (the gap), and why it matters here — and your question follows naturally.



Figure 2.6: From literature to your Introduction: the problem-statement funnel

Take-away: Known, not known, why here — and the question follows naturally.

2.4 Key words

Key word	In one line
MeSH	The shared label PubMed gives to many ways of writing one idea
AND / OR / NOT	Narrows / widens / removes — OR inside a block, AND between blocks
[tiab], *	Search title and abstract; truncation (any ending)
Grey literature, trial registry	Work outside journals; public records of planned and ongoing trials
Reference manager	Stores each paper once and formats citations in any style
Patchwriting	Copying with a few words swapped — plagiarism even if cited
IMRaD	Introduction, Methods, Results and Discussion
Spin	Words that claim more than the numbers show
Literature matrix	One row per paper, the same columns for every paper

2.5 Exercises

How the exercises work today.

- Recap quiz on Session 1 — on your own (8 minutes, start of the session).
- Exercise 1 (Boolean search builder) — **on your own**, segment 2.1 (5 minutes).
- Exercise 2 (paraphrasing) — at home.
- Exercise 3 (reading Paper 1's abstract) — **on your own**, segment 2.3 (12 minutes).
- Exercise 4 — your **group's Proposal builder 2** (33 minutes).

The answers are discussed in the room — there is no mark.

Recap quiz — Session 1

individual, 8 minutes

1. Checking whether fluid balance is charted daily as the protocol says: research, audit or QI?

2. "In ICU staff, does hand-hygiene feedback vs none increase compliance at 3 months?" — P
____ I ____ C ____ O ____ T ____
3. Which FINER letter did our running example fail? _____
4. Which is an objective: "to understand delirium" or "to estimate the proportion with delirium"?

5. Write H0 for: "Does early mobilisation shorten ICU stay?" _____

Exercise 1 — Boolean search builder

individual, 5 minutes, segment 2.1

Rules: synonyms inside a block are joined with **OR**; blocks are joined with **AND**; put each block in **brackets**. Write operators in CAPITALS. Use English search words.

Warm-up — which gives MORE hits? (circle)

- septic shock AND fluids OR septic shock OR fluids
- sepsis OR sepsis NOT neonatal

Build your search from your group's PICO question (Session 1) and your synonym list. The comparison (C) is usually *not* searched.

Block	Concept	Synonyms (2–4). Mark possible MeSH terms with (M)
P		
I / E		
O		

My search string:

(____ OR ____ OR ____) AND (____ OR ____ OR ____) AND (____ OR ____)

One filter I would add, and why: _____

Swap with your neighbour. One synonym they suggest adding: _____

The search string from the live demo (copy it here to run at home):

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
AND ("Fluid Therapy"[Mesh] OR "fluid restrict*" [tiab] OR "restrictive fluid*" [tiab]
OR "conservative fluid*" [tiab])
AND ("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Field tag / symbol	Meaning
[Mesh] or [mh]	MeSH term (includes narrower terms automatically)
[tiab]	word in the title or abstract
[pt]	publication type, e.g. randomized controlled trial[pt]
"..."	exact phrase
*	any ending (truncation): restrict* = restrict, restricted, restriction, restrictive

Exercise 2 — Citing vs copying: paraphrase safely

individual, at home, 10 minutes

Method: read → close the paper → write from memory → check the facts → cite. Patchwriting (copying the sentence and swapping a few words) is plagiarism **even with a reference**.

Part A — Spot the problem. The source is the Results section of Paper 2's abstract (Hyun 2023):

"The positive fluid groups in both the D2 (HR, 2.13; 95% CI 1.48–3.06; $P < 0.001$) and D3 (HR, 1.56; 95% CI 1.10–2.22; $P = 0.012$) cohorts had significantly higher mortality rates than the negative fluid groups."

For each version, tick **OK** or **Problem**, and say why.

Version	OK	Problem	Why?
a) "The positive fluid groups in the D2 and D3 cohorts had significantly higher death rates than the negative fluid groups [2]."	☐	☐	
b) "In a matched registry cohort, a positive cumulative fluid balance on ICU days 2 and 3 was associated with higher 28-day mortality (hazard ratios 2.13 and 1.56) [2]."	☐	☐	
c) "Giving more fluid on days 2 and 3 doubles the risk of death [2]."	☐	☐	
d) "In a matched registry cohort, a positive cumulative fluid balance on ICU days 2 and 3 was associated with higher 28-day mortality." (no citation)	☐	☐	

Part B — Your turn. Paraphrase this sentence from Paper 1's abstract in your own words, with a citation:

"After 28 days, mortality and the numbers of days alive without life support were similar between groups."

My paraphrase: _____

Check: different structure? ☐ facts still correct? ☐ no stronger claim than the source? ☐ citation? ☐

Exercise 3 — Read Paper 1's abstract

individual, 12 minutes, segment 2.3

Use the **abstract only** of Paper 1: Boulet N, et al. *Crit Care* 2024;28:429 (in Papers/). 8 minutes reading, 4 minutes comparing with your neighbour.

The 15-minute reading order (for later): title (30 s) → abstract (3 min) → figures & tables (4 min) → methods (5 min) → discussion & limitations (2–3 min). Stop early if the paper does not answer your question.

Item	What the abstract says
Design (and blinding)	
P — who was included?	
I vs C — what was compared?	
Primary outcome (and when)	
Numbers — screened / randomised / analysed	
Main result — estimate with 95% CI and p	
Conclusion — one line	
Does the conclusion match the main result?	☐ yes ☐ no — why?
Any spin (words stronger than the numbers)?	☐ yes ☐ no — where?
Registration number	
Does this paper answer our running-example question (28-day mortality)?	☐ yes ☐ no — why?

Fast finishers: read the **title and abstract** of Paper 2 (Hyun 2023). Which word in the title is stronger than the design allows? Which word does the abstract conclusion use instead?

Spin checklist — words to look for

Data say	Spin says
p = 0.42 for the primary outcome	"a non-significant trend"
primary outcome: no difference	headline = a positive secondary outcome
observational association	"impact", "effect", "reduces"
change within one group only	"patients improved"

Exit question

on a sticky note, no names

1. Which gives MORE hits: septic shock AND fluids or septic shock OR fluids?

2. What was the primary outcome of Paper 1?

2.6 Demo companion

Two live demos today: **PubMed** (segment 2.1, about 10 minutes) and **Zotero** (segment 2.2, about 5 minutes). You watch; you do not search during the demo. Write **today's** numbers and the date — hit counts change every day as new papers are added, and that is part of the lesson.

Part A — PubMed. The trainer builds the running-example search step by step.

Step	What you will see	Write down
1. One keyword: <i>septic shock</i>	Tens of thousands of hits; the sort menu and the filters column	Hits: _____ Date: _____
2. The three PICO blocks joined with AND	Each block alone has many hits; combined, far fewer (a few thousand)	Hits: _____
3. <i>Search details</i> (Advanced → History)	How PubMed translated the search: MeSH with narrower terms, truncation, field tags	One thing PubMed added: _____
4. Filters: article type (RCT, systematic review, meta-analysis), 10 years, adults 19+	The count falls to the hundreds or fewer	Hits: _____ Why these filters? _____
5. One record (e.g. the CLASSIC trial)	Abstract, MeSH terms list, <i>Similar articles</i> , <i>Cited by</i> , full-text links	Two MeSH terms we could reuse: _____

Part B — Zotero. PubMed *Send to* → *Citation manager* creates a .nbib file; Zotero imports it (*File* → *Import*) with authors, title, journal, PMID and DOI — nothing typed by hand. In Word, *Add/Edit Citation* inserts the citation and *Add/Edit Bibliography* builds the list; switching from a numbered style to APA reformats everything in one click.

What to note: which step reduced the hits most? (Usually combining the blocks with AND, and the article-type filter.)

Teaching points to keep. Write down database, date, string, filters and hits. OR inside, AND between; NOT only with a reason. MeSH + free text in every block. Every filter needs a reason — never filter for *free full text*. The computer narrows; people choose by reading. Export references, never type them.

2.7 Proposal builder 2

Your group completes **Part 1 of the proposal**: the search, five key papers, the problem statement and a working title.

Exercise 4 — Proposal builder 2

group, 33 minutes

Complete Part 1 of the **Research Proposal template** (Templates/Research_Proposal_Template.docx) and the **literature review worksheet**, building on yesterday's question, aim and objectives. This sheet is your checklist.

Suggested roles: chair · scribe (types into the template) · searcher (phone or laptop — runs your PubMed string) · timekeeper.

Time	Task	Done
0–8 min	1. Run your search string (Exercise 1) in PubMed; write the date, the filters and the number of hits	☐
8–16 min	2. Screen titles and abstracts; pick 5 key papers ; export them (<i>Send to</i> → <i>Citation manager</i>) into Zotero	☐
16–22 min	3. Literature matrix — one row per paper (split the papers between members)	☐
22–28 min	4. Problem statement — three short paragraphs (about 300 words): (a) what is known, (b) what is not known — the gap, (c) why it matters <i>here</i> (your local number)	☐
28–31 min	5. Working title and keywords (see below)	☐
31–33 min	One group reads its title and its gap sentence aloud; the room suggests one improvement	☐

Working title and keywords

Working title (10–16 words; population, comparison or exposure, and design visible; no abbreviations; no "impact" or "effect" for observational designs)

Number of words

Keywords (3–6; MeSH terms **not** already in the title) 1. _____ 2. _____ 3. _____ 4. _____

Example (feasible rewrite of the running example): "Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial" (13 words). Keywords: Fluid Therapy; Water-Electrolyte Balance; Critical Care; Vasoconstrictor Agents.

Literature review worksheet — one row per paper

#	Study (first author, year, journal)	Design, size	Compared	Primary outcome: result	Key limitation	Why it matters for our question
1						
2						
3						
4						
5						

Search used for the references (paste the string, the database and the date):

Quality checks before you stop

- ☐ The problem statement ends with your question; the title names the same population and outcome as the question.
- ☐ At least one reference is a randomised trial or a systematic review, if one exists; you checked a trial registry for ongoing trials.
- ☐ References are saved in a reference manager or exported from PubMed (*Send to → Citation manager*). No sentence is copied from a source.

2.8 After the session — homework

- **Group:** finish the five key references and the problem statement — Proposal Part 1 complete, with working title and keywords.
 - **Each member:** read one paper with the 15-minute routine and fill its row in the matrix.
 - **Everyone:** read the **abstract** of Paper 2 (Hyun 2023, *Ann Intensive Care*) with the Exercise 3 grid.
 - Exercise 2 (paraphrasing) if not done in class.
 - **Optional:** look up Paper 1's registration (NCT04947904) on ClinicalTrials.gov.
- ☐ Check that every one of your five references is saved in the reference manager (or exported from PubMed) — no reference typed by hand.

2.9 Extra practice

Extra practice 2.1 — Build the blocks

Question: *"In adults after cardiac surgery, does early mobilisation, compared with usual care, reduce postoperative delirium?"* Write two search blocks (2–3 synonyms each) and join them.

Block	Synonyms joined with OR
P (cardiac surgery)	
I (early mobilisation)	
O (delirium)	

Full string: _____

Extra practice 2.2 — Which search gives more hits?

Circle the search with MORE hits in each pair, and say why in a few words.

1. delirium AND (mobilisation OR mobilization) OR delirium AND mobilisation — why? _____
2. "pressure ulcer"[tiab] OR "Pressure Ulcer"[Mesh] OR "pressure ulcer"[tiab] OR bed sore[tiab] — why? _____
3. sepsis AND fluids OR sepsis AND fluids AND randomized controlled trial[pt] — why? _____

Extra practice 2.3 — Spot the spin

For each abstract sentence, write what the numbers allow you to say.

Abstract sentence	What the data allow
"Mortality showed a favourable trend in the intervention group (24% vs 27%, $p = 0.41$)."	
"Patients receiving the bundle improved markedly over 7 days." (no control group)	
"Higher fluid intake leads to kidney injury" (registry cohort)	

Number these sentences 1–3 in the order of a problem-statement funnel (known → not known → why here).

- ☐ "In our ICU, 6 of 10 patients with septic shock had a positive fluid balance on day 3 in last year's audit."
- ☐ "Fluid overload in septic shock is associated with organ dysfunction and death in several large cohorts."
- ☐ "Whether a restrictive strategy can be delivered safely in units like ours is not known."

My notes — Session 2

SESSION 3

DESIGN — choosing the right study

When	Weekend 2 · Saturday · Session 3 of 8
Length	3 hours, with a 15-minute break
Your group builds	Proposal Part 2 (first half) — study design and justification, setting, population (inclusion and exclusion criteria)

Learning objectives

By the end of this session you will be able to:

- name the exposure and the outcome of a study, and say who decided the exposure;
- recognise the main designs — case report/series, cross-sectional, case-control, cohort, RCT, systematic review and meta-analysis — with the **three-question decision tree**;
- give one strength and one weakness of each design, and say what it may claim ("caused" or "associated with");
- recognise trial variants (cluster, crossover, factorial, non-inferiority), pilot studies, before–after and interrupted time series, and qualitative designs;
- choose a feasible design for a question, and write inclusion and exclusion criteria for your own study.

3.1 Before the session — prepare

- ☐ Re-read your group's Proposal Part 1 (question, aim, objectives, problem statement).
- ☐ Read the **abstract of Paper 2** (Hyun 2023) if you did not do it last weekend: which group was compared with which?
- ☐ For your own question, write down: what is the **exposure** (or intervention), and what is the **outcome**?

Exposure / intervention: _____ Outcome: _____

- ☐ Think: could your unit **randomise** patients for your question? Why or why not?

3.2 Session plan

Time	Min	Segment
0:00	10	Recap quiz
0:10	45	3.1 Study designs in pictures: descriptive vs analytical, case report/series, cross-sectional, case-control, cohort, RCT, systematic review/meta-analysis, evidence pyramid, decision tree, strengths & weaknesses, what each design can claim
0:55	15	Exercise: Name that design
1:10	15	BREAK
1:25	35	3.2 More designs: trial variants (parallel, crossover, cluster, factorial; superiority vs non-inferiority), pilot/feasibility, before–after & interrupted time series (QI), qualitative designs; matching design to question
2:00	20	Exercise: choose the design for 5 questions (incl. your own)
2:20	35	Proposal builder 3: design + justification, setting, population (inclusion/exclusion)
2:55	5	Wrap-up

3.3 Key ideas in pictures

One question, two designs

Papers 1 and 2 ask about the same clinical topic. In the **trial** (Paper 1), the investigators assigned the fluid strategy by chance. In the **cohort** (Paper 2), they only observed what happened to patients who happened to accumulate more or less fluid. The design decides what each paper may claim.

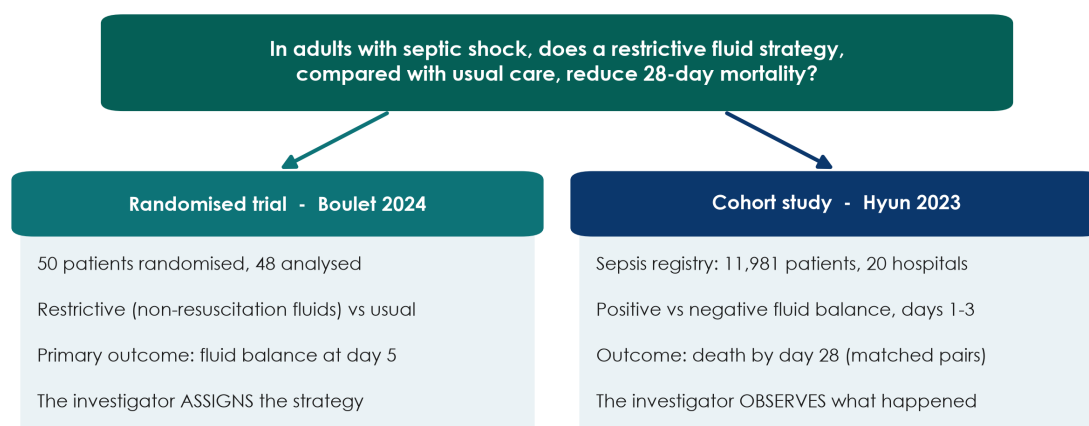


Figure 3.1: One question, two designs

Take-away: Same question, two designs: the design decides what you may claim.

The family tree of designs

Descriptive designs describe (case report, case series, cross-sectional survey); **analytical** designs compare groups. Analytical designs are **observational** (the investigator watches: case-control, cohort) or **experimental** (the investigator assigns: trials). Studies of studies — systematic reviews and meta-analyses — sit on top.

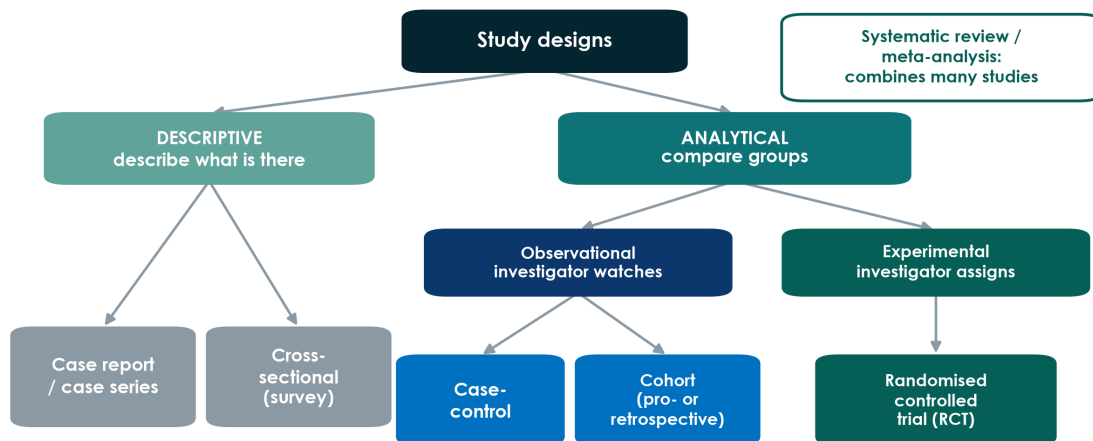


Figure 3.2: The family tree of study designs

Take-away: Descriptive designs describe; analytical designs compare groups.

Which way does time run?

A **cohort** starts from the exposure and follows people forward to the outcome. A **case-control** study starts from the outcome (cases and controls) and looks back for past exposures. A **cross-sectional** study measures both at the same moment — a snapshot that cannot show which came first.

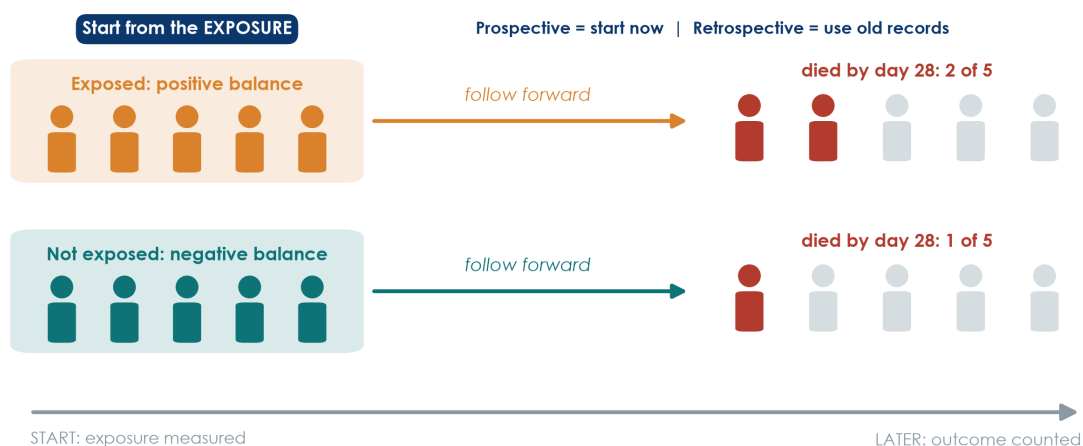


Figure 3.3: Cohort: start from the exposure, follow forward

Take-away: Cohort: exposure first, outcome later — time runs the right way.

Name the design in three questions

1. Did the investigator **assign** the exposure? If yes — was it **at random**? (RCT or non-randomised trial.)
2. If not assigned: is there a **comparison group**? (If not: case report, series or descriptive survey.)
3. If there is: in which **direction does time run**? (Cohort, case-control or cross-sectional.)

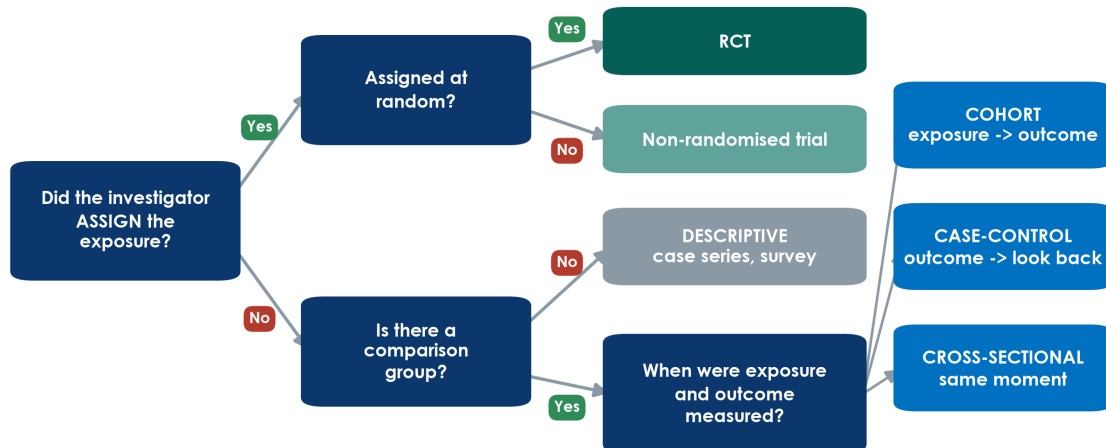


Figure 3.4: Name the design in three questions

Take-away: Assigned? Random? Which way does time run? Three questions name the design.

What may each design claim?

In an RCT, chance balances the groups for factors we know **and** factors we do not know, so a difference can be blamed on the treatment: the RCT may say "caused" (or "reduced"). A cohort can only balance what was measured: it should say "associated with".

RCT - Boulet 2024 (n = 48)	Cohort - Hyun 2023 (thousands)
CAN claim ✓ Groups alike at the start (chance) ✓ A difference can be blamed on the strategy	CAN claim ✓ Large, real patients, 28-day mortality ✓ Can say ASSOCIATED (HR 2.13 on day 2)
CANNOT claim ✗ Pilot: too small for mortality (6 vs 6 deaths) ✗ Clinicians knew the group (single-blind)	CANNOT claim ✗ Cannot prove fluid CAUSES death ✗ Sicker patients got more fluid (confounding)

Figure 3.5: What can each design claim?

Take-away: RCTs can say "caused"; cohorts should say "associated".

Match the design to the question

Start from the question, not from the design you like: *how many?* → cross-sectional; *what happens to people exposed?* → cohort; *what caused this rare outcome?* → case-control; *does the treatment work?* → RCT (or a pilot first); *did our bundle change the trend?* → interrupted time series; *why do people act this way?* → qualitative.

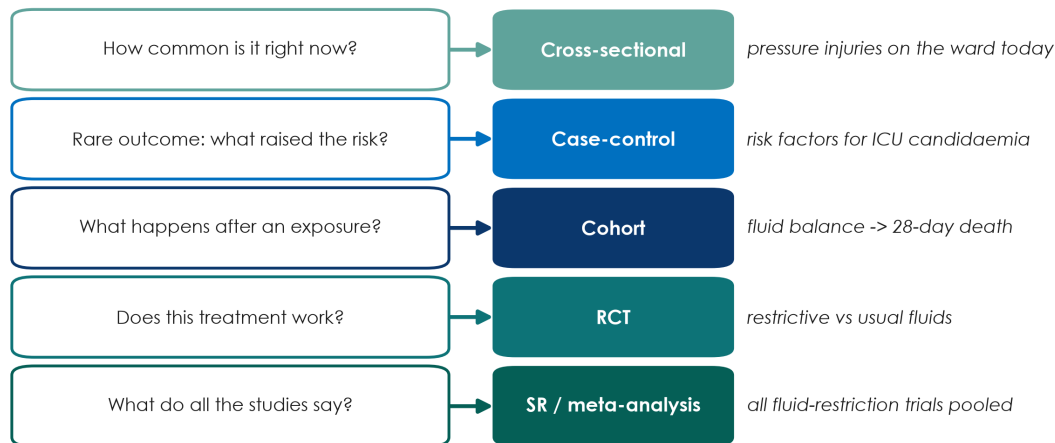


Figure 3.6: Match the design to the question

Take-away: Start from the question — the design follows from it.

Who is in your study?

Inclusion criteria describe your P and can be checked on day 1. **Exclusion criteria** remove eligible people for a stated reason (safety, the outcome cannot be measured) — they are not just the inclusions written in the negative.

3.4 Key words

Key word	In one line
Exposure / outcome	What comes first / what we count later
Observational / experimental	We watch / we assign
Randomised	Chance decides who gets what
Cross-sectional	A snapshot: exposure and outcome measured at the same moment
Case-control	Start from the outcome, look back for exposures
Cohort (prospective / retrospective)	Start from the exposure, follow forward (now, or in old records)
Cluster / crossover / factorial trial	Randomise groups / each patient gets both / test two treatments at once
Non-inferiority	"Not worse than" by more than a margin fixed in advance
Pilot study	Asks "can we do it?", not "does it work?"
Interrupted time series	Compares the trend after a change with the trend expected from before
Qualitative design	Explains why and how, with words (interviews, focus groups)

3.5 Exercises

This handout has three parts. No computer or calculator is needed.

Part	Activity	When	How
A	Name that design (8 abstracts + 1 optional)	End of section 3.1 (15 min)	Pairs
B	Choose the design for 5 questions (4 given + your own; 2 optional bonus)	After section 3.2 (20 min)	Your group
C	Proposal builder 3 — design, setting and population	End of the session (35 min)	Your group

Part A — Name that design

Instructions

1. Work in pairs. Read each short abstract.
2. Ask the three questions from the decision tree:
 - **Did the investigator ASSIGN the exposure or treatment?** If yes: **was it assigned at random?**
 - If not assigned: **is there a comparison group?**
 - If there is: **in which direction does time run?** Exposure first, then outcome (cohort); outcome first, then look back (case-control); both at the same moment (cross-sectional).
3. Write the design and **one word or phrase from the abstract that proves it.**

4. Put a star next to the abstract you found hardest.

Design choices: case report / case series · cross-sectional · case-control · cohort (prospective or retrospective) · non-randomised intervention study · randomised controlled trial (RCT) · systematic review / meta-analysis

Abstract 1. Two hundred and forty adults on mechanical ventilation were randomly allocated by a central computer to oral care with chlorhexidine twice daily or to standard oral care. Ventilator-associated pneumonia was recorded over 28 days by an assessor who did not know the group.

Design: _____ Evidence: _____

Abstract 2. On one day in March, two trained nurses examined all 412 inpatients of a hospital for pressure injuries and recorded the type of mattress each patient was lying on. Pressure injuries were present in 9% of patients on standard mattresses and 4% on pressure-relieving mattresses.

Design: _____ Evidence: _____

Abstract 3. Forty ICU patients who developed candidaemia were compared with 80 ICU patients from the same units who did not, matched on age and month of admission. The investigators then searched the records of both groups for parenteral nutrition in the 14 days before the culture date.

Design: _____ Evidence: _____

Abstract 4. All 600 adults admitted with acute ischaemic stroke who received thrombolysis were classified on arrival by door-to-needle time (under 60 minutes or 60 minutes and over). Each patient was then followed up to record functional independence at 90 days.

Design: _____ Evidence: _____

Abstract 5. Using electronic records from 2019 to 2022, the investigators identified 1,150 ICU patients and divided them into those who did and did not receive a continuous benzodiazepine infusion for 3 days or more. They then compared how many patients in each group had delirium recorded later in the ICU stay.

Design: _____ Evidence: _____

Abstract 6. We describe five patients who developed severe hypernatraemia within a week of a new fluid protocol being introduced on our unit, with their laboratory results and clinical course.

Design: _____ Evidence: _____

Abstract 7. Central-line infection rates were compared for the six months before and the six months after a unit-wide insertion-bundle education programme, which the research team introduced on all beds on the same day.

Design: _____ Evidence: _____

Abstract 8. We searched three databases for randomised trials comparing restrictive with liberal fluid strategies in adults with septic shock, assessed each trial's risk of bias, and pooled 28-day mortality across nine trials.

Design: _____ Evidence: _____

Abstract 9 (optional). Twelve hospital wards were randomly allocated, ward by ward, to a new hand-hygiene feedback programme or to usual practice. Hospital-acquired infections were counted

in all patients on those wards for six months.

Design: _____ Evidence: _____

Extra (if you finish early): for each design, write one strength and one weakness.

Part B — Choose the design for 5 questions

Instructions. For each question, name the best **feasible** design (use the decision tree and the "Match the design to the question" slide), give **one reason**, and name the design's **main weakness**. Question 5 is your group's own research question from Part 1.

#	Question	Design	Reason	Main weakness
1	How many inpatients have a pressure injury today, and is it more common on standard mattresses?			
2	Does daily chlorhexidine bathing reduce bloodstream infections? It will be introduced ward by ward, and nurses on one ward cannot bathe only half their patients.			
3	A sepsis bundle was introduced in our hospital in 2023. We have monthly sepsis mortality for 3 years before and 2 years after. Did mortality fall?			
4	Why do nurses not report near-miss medication errors?			
5	Our own question: _____			

Bonus (optional, if you finish early)

#	Question	Design	Reason
6	What are the risk factors for a rare event: Stevens–Johnson syndrome in hospital patients?		
7	A new 5-day antibiotic course is cheaper and easier than the 10-day course. We only need to show it is not worse for cure.		

For your own question (5), also answer: **at the end, will you be allowed to say "caused" or "associated with"?** _____

3.6 Demo companion

There is **no software demo** in Session 3. The slides walk both course papers through the decision tree ("The decision tree in action: Boulet and Hyun"). Note the answers as the trainer goes:

Question of the decision tree	Paper 1 (Boulet 2024)	Paper 2 (Hyun 2023)
Did the investigator assign the exposure?		
At random?		
Comparison group?		
Direction of time		
Name of the design		
May it say "caused"?		

3.7 Proposal builder 3

Part C — Proposal builder 3: design, setting and population

Instructions. As a group, complete the first three rows of Part 2 of the proposal, then copy them into the template (Templates/Research_Proposal_Template.docx). The rest of Part 2 (exposure, comparator, outcomes, biases) is built in Session 4.

Our research question (from Part 1): _____

#	Section	What to write	Our answer
1	Study design and justification	Name the design. In 3–5 sentences explain why it fits YOUR question, and why the "ideal" design is or is not feasible.	
2	Setting	Where (type of unit, how many sites), when (study period), source of patients or records. Do not name a specific hospital in the course version.	
3	Population	Inclusion criteria (3–5): who is eligible. Exclusion criteria (3–5): who is removed, and why. Exclusions should not just repeat the inclusions in the negative.	

Draw your study on a timeline (exposure → outcome; mark when each is measured):



Group self-check before you finish

- ☐ The design matches the type of question.
- ☐ It is feasible in our setting (patients, time, data).
- ☐ Inclusion criteria describe the P of our PICO and can be checked from records on day 1.
- ☐ Every exclusion criterion has a reason (safety, outcome cannot be measured ...) and is not just the opposite of an inclusion.
- ☐ We know what we will be allowed to claim: "caused" or "associated with".

3.8 After the session — homework

- ☐ Finish the first three rows of Proposal Part 2 (design and justification, setting, population) and copy them into the template.
- ☐ Draw your study on a timeline (exposure → outcome) and show it to one colleague who is not in your group: can they say which way time runs?
- ☐ Read the **Methods** of Paper 1 or Paper 2 (whichever is closer to your design) with the 15-minute routine.
- ☐ Tomorrow is about bias: write down the one thing that could make your study's answer **wrong**.

3.9 Extra practice

Extra practice 3.1 — Name the design

#	Study	Design	The words that prove it
a	All 120 nurses of a hospital answer a questionnaire on night-shift sleep and needle-stick injuries in the past month.		
b	50 patients with ventilator-associated pneumonia and 100 ventilated patients without it are compared for earlier use of stomach-acid drugs.		
c	300 adults starting dialysis are classified by vascular access type and followed for 1 year for bloodstream infection.		
d	16 ICUs are randomly allocated, unit by unit, to a daily sedation-holiday protocol or usual care.		

Extra practice 3.2 — Caused or associated?

For each finding, circle the verb the design allows.

1. RCT of 600 patients: a new dressing **reduced** / **was associated with fewer** wound infections (RR 0.62, 95% CI 0.45 to 0.85).
2. Registry cohort of 8,000 patients: early antibiotics **reduced** / **were associated with lower** mortality.
3. Case-control study: proton-pump inhibitors **cause** / **were associated with** *C. difficile* infection.

Extra practice 3.3 — Fix the exclusion criteria

A cohort study includes "adults aged 18 years or more admitted to the ICU with sepsis". Its exclusion criteria are: (a) "age under 18"; (b) "patients without sepsis"; (c) "patients expected to die within 24 hours". Which are real exclusion criteria, and which just repeat the inclusions? Write one better exclusion criterion with its reason.

SESSION 4

BIAS & EFFECT — can we trust it, how big is it?

When	Weekend 2 · Sunday · Session 4 of 8
Length	3 hours, with a 15-minute break
Your group builds	Proposal Part 2 (complete) — exposure and comparator, outcomes, main biases and how they are reduced

Learning objectives

By the end of this session you will be able to:

- tell random error from bias, and recognise selection bias, information bias and **confounding (by indication)**;
- tell confounding (a bias to remove) from effect modification (a finding to report), and spot reverse causation;
- explain how randomisation, concealment, blinding, matching and adjustment fight bias;
- calculate risk, **RR**, **RD**, **NNT** and **OR** from a 2×2 table and say each one in plain words;
- explain sensitivity, specificity and why the **PPV depends on prevalence**;
- read a forest plot, recognise heterogeneity and choose the right risk-of-bias tool for a design.

4.1 Before the session — prepare

- ☐ Re-read Proposal Part 2, rows 1–3 (design, setting, population).
- ☐ Remind yourself of fractions and percentages: 20 out of 100 = 0.20 = 20%. A calculator (or phone) helps for the 2×2 exercise.
- ☐ Think about your own question: **why** do some patients receive the exposure and others not? That reason may be a confounder.

Why some patients get the exposure: _____

4.2 Session plan

Time	Min	Segment
0:00	10	Recap quiz
0:10	40	4.1 Bias, confounding and causation: random error vs bias, selection, information, confounding (by indication), interaction, validity, Bradford Hill, reverse causation, how designs fight bias; spot-the-bias quiz
0:50	30	4.2 Measuring disease and association: prevalence vs incidence, risk, 2x2, RR / OR / RD / NNT, OR vs RR; live 2x2 calculator demo
1:20	15	BREAK
1:35	20	Exercise: two 2x2 calculations
1:55	30	4.3 Beyond single studies: diagnostic accuracy (sens/spec, PPV & prevalence, ROC), systematic reviews & forest plots, heterogeneity, risk-of-bias tools, "are the conclusions supported?" (pair task)
2:25	30	Proposal builder 4: exposure & comparator, outcomes, main biases & mitigation
2:55	5	Wrap-up

4.3 Key ideas in pictures

Bias is systematic error — size does not fix it

Random error scatters results around the truth; a bigger sample shrinks it. **Bias** pushes results away from the truth in one direction; a bigger sample only makes a wrong answer look more precise.

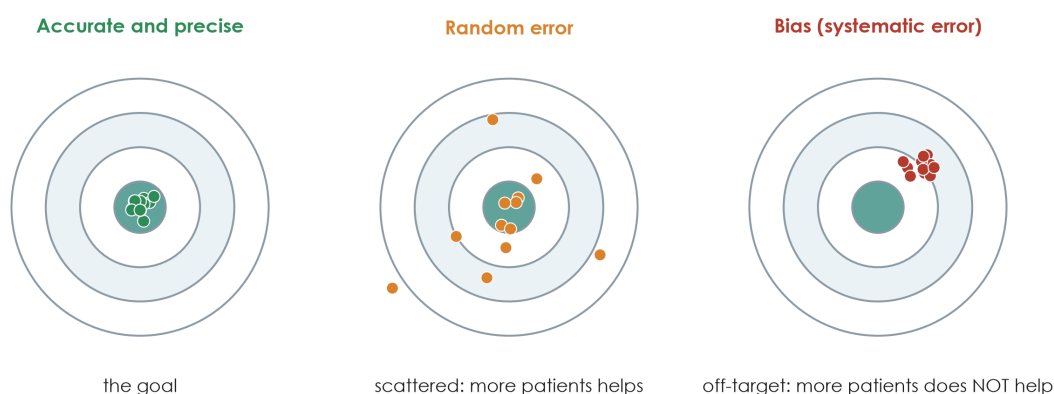


Figure 4.1: Random error versus bias

Take-away: A bigger sample shrinks random error but leaves bias untouched.

Confounding by indication: the ICU fluid example

Sicker patients (shock, high lactate, high SOFA) **receive more fluid** and **die more often**. Severity is linked to the exposure and causes the outcome: it is a **confounder**. If we compare patients with more and less fluid without taking severity into account, fluid looks more harmful than it is.

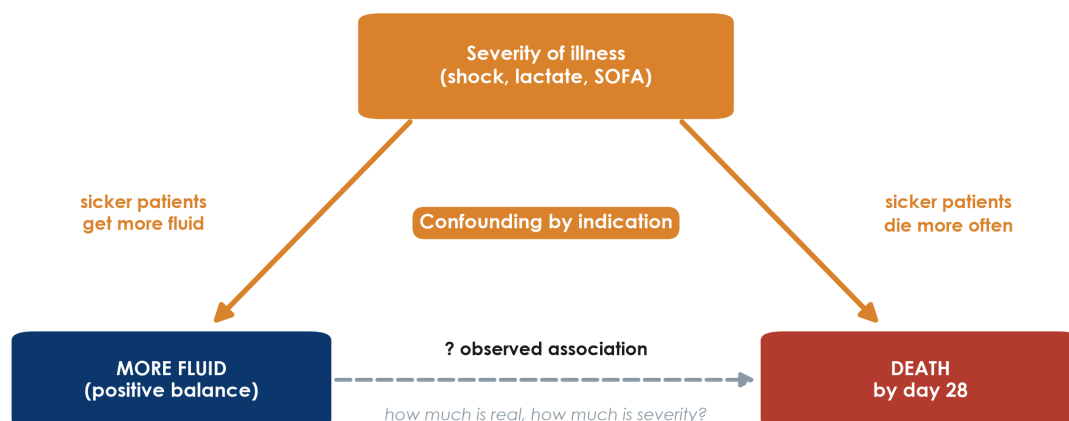


Figure 4.2: Confounding by indication

Take-away: A confounder is linked to the exposure AND causes the outcome.

Confounding or effect modification?

Split the result by a third factor. If the strata **agree with each other** but not with the crude result → confounding: report the adjusted result. If the strata **differ from each other** → effect modification: report each group separately.

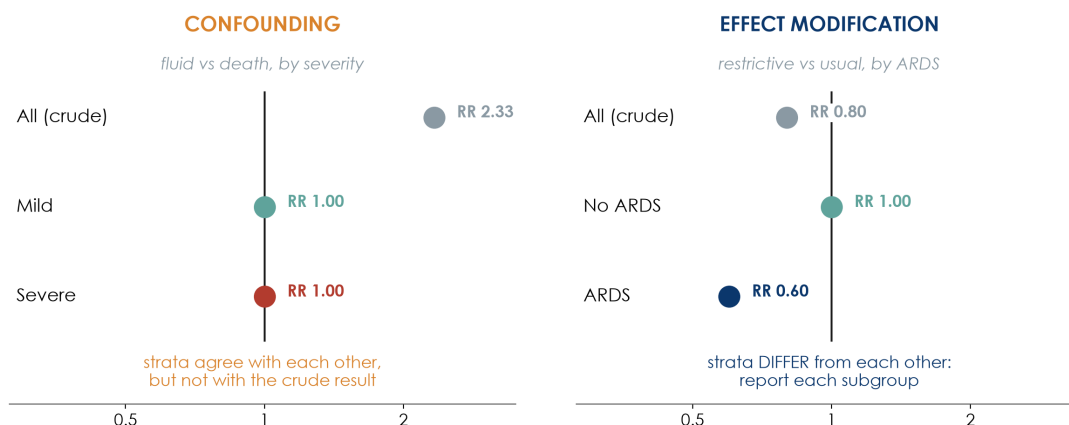


Figure 4.3: Confounding or effect modification?

Take-away: Confounding is a bias to remove; effect modification is a finding to report.

How designs fight bias

Randomisation balances what we know **and** what we do not know. Allocation **concealment** protects who gets into which group; **blinding** protects how outcomes are measured. Matching and adjustment can only balance what was measured.

The 2×2 table: one tool for every measure

Exposure (or treatment) in **rows**, exposed on top; outcome in **columns**, "outcome yes" on the left: cells a, b, c, d. Risk = $a/(a+b)$ and $c/(c+d)$. **RR** = ratio of the risks (how strong); **RD** = difference of the risks (how many patients); **NNT** = $1/|RD|$; **OR** = $(a \times d)/(b \times c)$.

The template		Worked example (hypothetical trial)	
	Died	Survived	Total
Exposed / treated	a	b	a + b
Not exposed / control	c	d	c + d
Risk in exposed = $a / (a + b)$ Risk in control = $c / (c + d)$		Risk restrictive = $20 / 100 = 20\%$ Risk usual care = $30 / 100 = 30\%$	

Figure 4.4: The 2×2 table

Take-away: Exposure in rows, outcome in columns: a, b, c, d.

The OR is close to the RR only when the outcome is **rare** (under about 10%); for common outcomes it lies further from 1 and exaggerates the effect. In a **case-control** study only the OR may be calculated.

Same test, different setting: PPV depends on prevalence

Sensitivity and specificity belong to the **test**. The positive predictive value depends on how **common** the disease is: with sensitivity 90% and specificity 80%, a positive result is right about half the time in the ICU (prevalence 20%) but fewer than 1 in 10 times in screening (prevalence 2%).

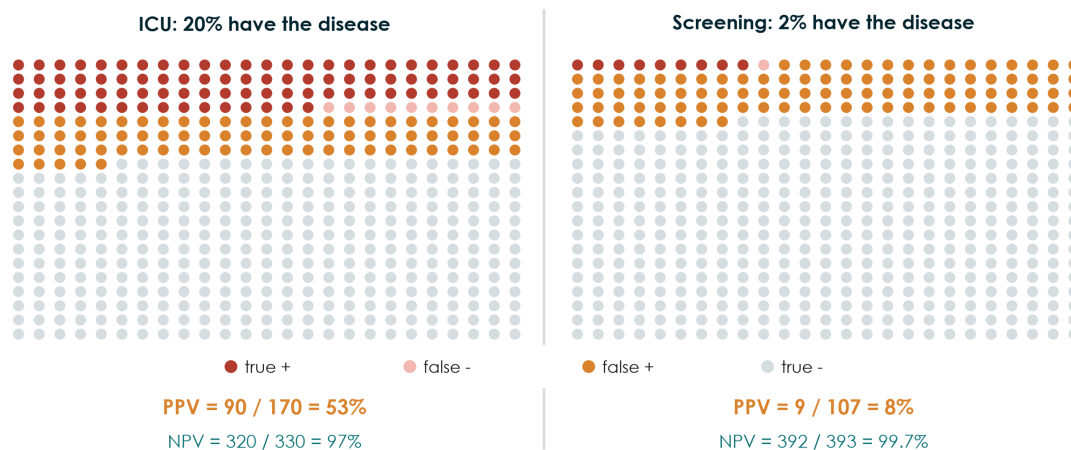


Figure 4.5: PPV depends on prevalence

Take-away: A positive result means much less when the disease is rare.

Reading a forest plot

Squares are the studies (bigger square = more weight), lines are their 95% CIs, the diamond is the pooled result, and the vertical line is "no effect" (RR = 1). Then ask whether the studies agree (heterogeneity, I^2) and how much each can be trusted (risk-of-bias tool).

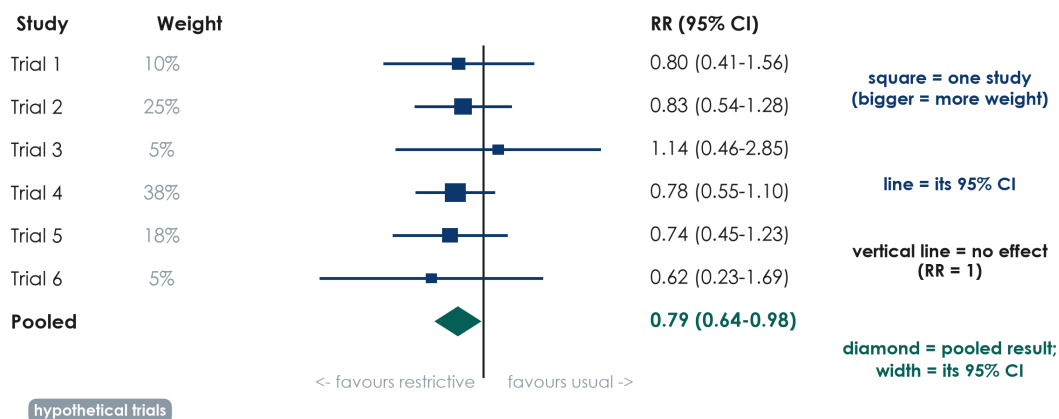


Figure 4.6: Reading a forest plot in four steps

Take-away: Squares = studies, lines = CIs, diamond = pooled result, vertical line = no effect.

4.4 Key words

Key word	In one line
Bias	Systematic error: pushes the answer one way; size does not fix it
Selection / information bias	Who got in / how things were measured
Confounding (by indication)	A third factor linked to the exposure that also causes the outcome (e.g. severity)
Effect modification	The effect differs between groups of patients — report each group
Reverse causation	The outcome (or its early stage) caused the "exposure"
Risk, RR, RD, NNT, OR	Proportion with the outcome; ratio; difference; 1/RD; ratio of odds
Sensitivity / specificity	Catches the sick / clears the healthy
PPV / NPV	Chance that a positive / negative result is right — depends on prevalence
Forest plot, I^2	Studies and their pooled result; how much the studies disagree
Risk-of-bias tool	RoB 2, ROBINS-I, Newcastle–Ottawa, QUADAS-2, AMSTAR 2 ... by design

4.5 Exercises

This handout has six parts. You will not need a computer. For the calculations, simple division is enough; the instructor checks every answer with the 2×2 calculator.

Part	Activity	When	How
A	Spot the bias (4 scenarios)	End of section 4.1 (4 min)	Whole room / alone
B	Confounding or effect modification? (3 scenarios)	Homework, or if 4.1 runs early	Alone or pairs
C	Two 2×2 calculations	After the break (20 min)	Alone, then check with a neighbour
D	PPV in two settings	Homework, or if 4.3 runs early	Alone
E	Read a forest plot and choose the risk-of-bias tool	Pair task in 4.3 (6 min)	Pairs
F	Proposal builder 4 — exposure, outcomes and biases	End of the session (30 min)	Your group

Part A — Spot the bias

Instructions. For each scenario, choose the **main** problem: **selection bias**, **recall bias**, **observer bias** or **confounding**. Then write **one practical way** to reduce it.

#	Scenario	Main bias	One way to reduce it
1	In a case-control study, mothers of babies born with a birth defect remember the medicines they took during pregnancy in more detail than mothers of healthy babies.		
2	In an unblinded trial of an early-mobilisation programme, the trial nurse who knows each patient's group decides whether the patient is "ready for discharge".		
3	A study of quality of life after ICU recruits only the survivors who come back to the follow-up clinic.		
4	In a registry, patients with higher lactate received more fluid and also died more often. The authors conclude that fluid increases mortality.		

Part B — Confounding or effect modification?

Instructions. Each study splits its result by a third factor. Compare the strata **with each other** and **with the overall (crude) result**. Decide: **confounding**, **effect modification (interaction)** or **neither**. Then say what the authors should report.

Rule of thumb: strata similar to each other but different from the crude result → confounding.
 Strata different from each other → effect modification.

#	Study (hypothetical)	Over-all RR	Stratum 1	Stratum 2	Your answer	What to report
1	Vasopressor X vs Y and death by day 28. Clinicians chose X mainly for patients with high lactate.	1.80	lactate < 4: RR 1.00	lactate ≥ 4: RR 1.00		
2	Early mobilisation and ICU delirium	0.70	age < 65: RR 0.50	age ≥ 65: RR 0.95		
3	Hand-hygiene programme and hospital infection	0.80	day shift: RR 0.80	night shift: RR 0.79		

Part C — Two 2×2 calculations

Instructions

- 1. Draw the 2×2 table first.** Exposure (or treatment) in **rows**, with the exposed/treated group on top. Outcome in **columns**, with "outcome yes" on the left.
- 2.** Calculate only what is asked. Round to two decimals.
- 3.** Write **one sentence** a colleague without research training would understand.

Reminder

Measure	Formula (from the table a, b, c, d)
Risk in exposed / treated	$a / (a + b)$
Risk in unexposed / control	$c / (c + d)$
Risk ratio (RR)	risk exposed ÷ risk control
Risk difference (RD)	risk exposed – risk control
Number needed to treat (NNT)	$1 \div RD $ (RD written as a proportion, e.g. 0.05)
Odds ratio (OR)	$(a \times d) \div (b \times c)$

Calculation 1 — a randomised trial

A trial randomised 400 adults on a medical ward to a new pressure-relieving mattress (200 patients) or a standard mattress (200 patients). A new pressure injury developed in **10** patients on the new mattress and in **20** patients on the standard mattress.

	Pressure injury: YES	Pressure injury: NO	Total
New mattress	a =	b =	
Standard mattress	c =	d =	

1. Risk with the new mattress = ____ Risk with the standard mattress = ____
2. RR = ____ 3. RD = ____ 4. NNT = ____ 5. OR = ____
3. Is the OR close to the RR here? Why? _____
4. One sentence for a colleague: _____

Calculation 2 — a case-control study

Investigators selected **40** ICU patients with candidaemia (cases) and **80** ICU patients without candidaemia (controls). They looked back for parenteral nutrition. Among the cases, **24** had received parenteral nutrition and 16 had not. Among the controls, **16** had received it and 64 had not.

	Cases (candidaemia)	Controls (no candidaemia)
Parenteral nutrition: YES	a =	b =
Parenteral nutrition: NO	c =	d =

1. OR = ____

2. Why can you **not** calculate the risk of candidaemia, the RR or the NNT from this study?

3. One sentence for a colleague: _____

Note: in a case-control study the columns are the case and control groups. The OR formula $(a \times d) \div (b \times c)$ stays the same.

Part D — PPV in two settings

Instructions. The same test has **sensitivity 90%** and **specificity 80%**. Complete each table for 500 people, then calculate PPV and NPV.

Disease present = prevalence × 500. True positives = sensitivity × disease present. True negatives = specificity × disease absent.

Setting 1 — ICU, prevalence 20%

	Disease present	Disease absent	Total
Test positive	TP =	FP =	
Test negative	FN =	TN =	
Total			500

PPV = $TP / (TP + FP)$ = ____ NPV = $TN / (TN + FN)$ = ____

Setting 2 — screening, prevalence 2%

	Disease present	Disease absent	Total
Test positive	TP =	FP =	
Test negative	FN =	TN =	
Total			500

PPV = ____ NPV = ____

1. Why is the PPV so different when the test is the same? _____
2. Calculate $LR+ = \text{sensitivity} / (1 - \text{specificity})$ = _____. Does it change between the settings?

3. Optional: repeat for a prevalence of 50%.

Part E — Read a forest plot and choose the risk-of-bias tool (pair task)

The forest plot below is the one shown in section 4.3 (six hypothetical trials of restrictive vs usual fluids; outcome death; fixed-effect pooling).

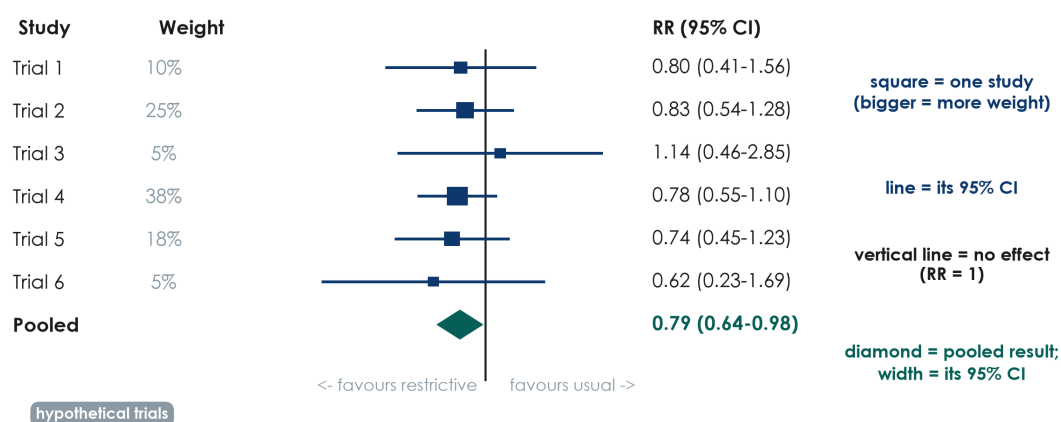


Figure 4.7: Forest plot of six hypothetical trials

Trial	Restrictive: died / total	Usual care: died / total
Trial 1	12 / 60	15 / 60
Trial 2	30 / 150	36 / 150
Trial 3	8 / 40	7 / 40
Trial 4	45 / 250	58 / 250
Trial 5	20 / 100	27 / 100
Trial 6	5 / 30	8 / 30

Reading the plot

- Which trial carries the most weight? Why? _____
- Does any single trial show a clear effect (a 95% CI that does not cross 1)? _____
- Does the pooled diamond cross RR = 1? Write the pooled result in one plain sentence.

- Trial 3 points the other way (RR 1.14). Should it worry you? Why or why not?

Choosing the tool. For each study, choose the risk-of-bias tool: **RoB 2**, **ROBINS-I**, **Newcastle–Ottawa** (or **ROBINS-E**), **QUADAS-2** or **AMSTAR 2**.

#	Study	Tool
a	Boulet 2024 — randomised trial of restrictive vs usual fluids	
b	Hyun 2023 — registry cohort, positive vs negative fluid balance	
c	Before–after study of a central-line bundle on one unit	
d	Point-of-care lactate compared with the laboratory reference test for detecting hyperlactataemia	
e	Systematic review and meta-analysis of fluid-restriction trials	
f	Cluster-randomised trial of a hand-hygiene programme (12 wards)	

4.6 Demo companion

Live demo: the 2×2 calculator (section 4.2, about 8 minutes, [Resources/tools/2x2_calculator.xlsx](#)). You watch; no laptop needed.

Step	What you will see	Write down
1. The layout	Restrictive 20 died / 80 survived; usual care 30 / 70	Treated group on top? Outcome "yes" on the left?
2. The results in words	Risks 20% vs 30%; RR 0.67; RD –10%; NNT 10; OR 0.58	RR in words: _____ NNT in words: _____
3. Make the outcome rare (2/98 vs 3/97)	RR still 0.67; RD –1%; NNT 100; OR 0.66	What changed, what did not? _____
4. Case-control warning	The sheet still shows "risks" — do not read them	Which measure is valid? _____
Add-on (4.3): sheet <i>Diagnostic</i>	Sensitivity 90%, specificity 80%; PPV 52.9% at prevalence 20%; 8.4% at 2%	Why so different? _____

After the break, the trainer types **your** two calculations (Part C) into the calculator: check your answers against the screen.

Teaching points to keep. One layout, every measure — most errors are layout errors, not arithmetic. Relative vs absolute: the same RR can mean an NNT of 10 or of 100. OR $\approx$ RR only when the outcome is rare. Case-control → OR only. Sensitivity and specificity belong to the test; PPV and NPV depend on prevalence.

4.7 Proposal builder 4

Part F — Proposal builder 4: exposure, outcomes and biases

Instructions. Complete the rest of Part 2 of your proposal (rows 1–3 were written in Session 3), then copy it into the template. For every outcome write **what**, **how** and **when**.

Our research question: _____

#	Section	What to write	Our answer
4	Exposure / intervention and comparator	What exactly is the exposure or intervention, and how is it defined or measured? Who is the comparison group?	
5	Outcomes	ONE primary outcome: what it is, how and when it is measured. 2–3 secondary outcomes.	
6	Main biases and how we reduce them	Name at least two threats (selection, information, confounding) and the step in your design that reduces each.	

Our claim at the end will be: ☐ "caused" (randomised design) ☐ "associated with" (observational design), because _____

Group self-check before you finish

- ☐ There is a comparison group, or we explain why our study is descriptive.
- ☐ There is only ONE primary outcome, with a method and a time point.
- ☐ If our design is observational, we have named the main confounders (think: confounding by indication).
- ☐ Each bias has a concrete mitigation step ("we will be careful" is not a mitigation).
- ☐ Part 2 is complete: design, setting, population (Session 3) + exposure, comparator, outcomes, biases (today).

4.8 After the session — homework

- ☐ Finish **Proposal Part 2** (design, setting, population, exposure and comparator, outcomes, biases) before Weekend 3.
- ☐ Do Parts B and D of the exercises if they were not done in class (confounding or effect modification; PPV in two settings).
- ☐ Find the 2x2 table hidden in Paper 1: deaths by day 28 were **6/24** in each group. Calculate the risk in each group and the RR.

Risk restrictive: ____ Risk usual care: ____ RR: ____

4.9 Extra practice

Extra practice 4.1 — Another 2×2

In a trial, 300 patients receive a new catheter care bundle and 300 receive usual care. Bloodstream infection occurs in **9** patients with the bundle and **18** with usual care.

	Infection: YES	Infection: NO	Total
Bundle	a =	b =	
Usual care	c =	d =	

Risk with the bundle = ____ Risk with usual care = ____ RR = ____ RD = ____ NNT = ____

One sentence for a colleague: _____

Extra practice 4.2 — Which bias?

#	Scenario	Main problem
a	Only patients who answered a satisfaction survey are analysed; 40% did not answer.	
b	Nurses who know which patients received the new mattress grade the pressure injuries.	
c	Patients prescribed sedatives were sicker, and sicker patients develop more delirium.	
d	People with early, undiagnosed cancer lose weight; a study finds weight loss "causes" cancer.	

Extra practice 4.3 — PPV at 50%

Use the test from Part D (sensitivity 90%, specificity 80%) in 500 people with a prevalence of **50%**.

	Disease present	Disease absent	Total
Test positive	TP =	FP =	
Test negative	FN =	TN =	
Total			500

PPV = ____ NPV = ____

SESSION 5

DATA — collect it well, describe it honestly

When	Weekend 3 · Saturday · Session 5 of 8 (Level 2 begins)
Length	3 hours, with a 15-minute break
Your group builds	Proposal Part 3 (first half) — variables table, mini-CRF, data dictionary, data-management plan · Manuscript — your FLUID-ICU question and Table 1 understood

Learning objectives

By the end of this session you will be able to:

- classify variables (continuous, count, binary, nominal, ordinal, time-to-event) and say why the type matters;
- write CRF items that can only be filled in one way, and a data dictionary row for each variable;
- describe good data management: validation at entry, double entry or electronic capture, audit trail, cleaning log, backups;
- explain what was wrong in the raw FLUID-ICU data and why it is cleaned with a script, never by hand;
- choose mean (SD) or median (IQR) and n (%), and the right graph for each type of data;
- read a **Table 1** and say whether the groups were alike at the start.

5.1 Before the session — prepare

- ☐ Finish Proposal Part 2 (design, setting, population, exposure, outcomes, biases).
- ☐ Read **Chapter 10** of this workbook (the course dataset FLUID-ICU, about 10 minutes): what is in the data, and what are the five group questions?
- ☐ Look at one **paper form or electronic chart** from your unit: find one item that two nurses could fill in differently.

The item: _____ Why ambiguous: _____

5.2 Session plan

Time	Min	Segment
0:00	10	Recap quiz
0:10	35	5.1 Good data by design: variable types, CRF principles, data dictionary, validation, data management (EDC, double entry, audit trail, backups), preventing missing data
0:45	25	5.2 (course dataset) Meet the course dataset: the FLUID-ICU study, its data dictionary, raw vs clean data, the data-quality problems found, live R demo of cleaning and the cleaning log; group questions assigned
1:10	15	BREAK
1:25	35	5.3 Describing and showing data: mean/median, SD/IQR, shapes, which graph for which data, good vs bad figures, publication-quality tables, Table 1
2:00	25	5.4 (course dataset) Describe the course dataset: live demo of Table 1 and figures; each group reads its results pack's Table 1 and figures (work-sheet)
2:25	30	Proposal builder 5: variables table, mini-CRF, data dictionary rows, data-management plan
2:55	5	Wrap-up

5.3 Key ideas in pictures

The type of variable decides everything that follows

The type decides the CRF box, the summary in Table 1, the graph and the statistical test. Collect numbers as exact numbers when you can (age in years, not age groups): you can always group later, never un-group.

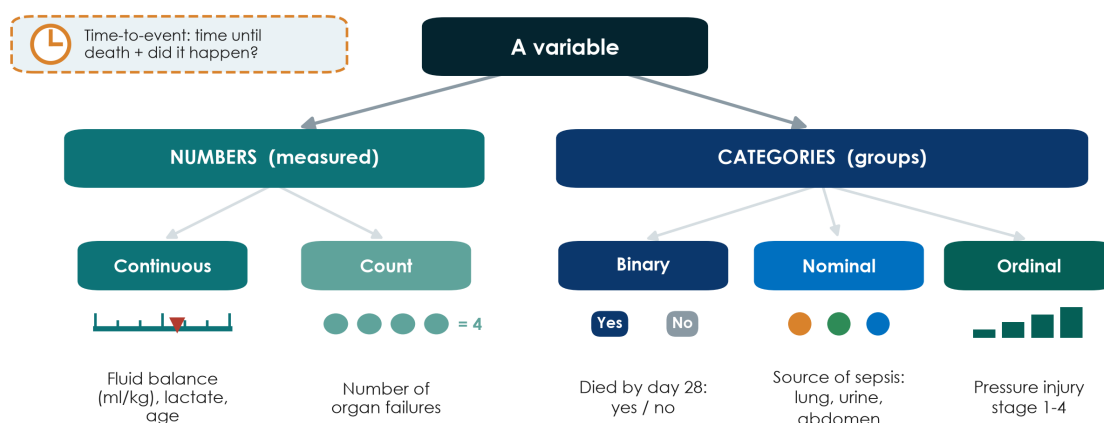


Figure 5.1: What type of variable is it?

Take-away: The variable type decides the CRF box, the summary and the test.

Good data by design: CRF, dictionary, checks

A good CRF item has one possible reading: the unit, the time point, one answer only, and an “unknown / not done” option. The **data dictionary** is the instruction manual of the dataset: one row per variable with its name, meaning, type, unit or codes and allowed values. **Validation rules** catch errors while the patient is still in the unit.

Good CRF items

Cumulative fluid balance, ICU admission to day 5 (ml/kg): ± □ □ □ □ ✓

Date of birth: DD / MM / YYYY

Main source of infection (tick ONE):

☐ Lung
☐ Urine
☐ Abdomen

☐ Other
☐ Unknown

Lactate at ICU admission (mmol/L): □ □ . □

Figure 5.2: Good vs bad CRF items

Take-away: If two colleagues can fill an item differently, rewrite it.

Meet the course dataset: from raw to clean

FLUID-ICU (simulated) arrived from four ICUs as **623 rows** with the problems real data have: duplicate records, the same thing spelled many ways, three date formats, missing-value codes such as 999 and -99, lactate in the wrong unit at one ICU, impossible values (age 210, weight 7 kg). A script fixed each problem and wrote it in the **cleaning log**; the result is **620 patients**.

Problem	Raw file (as sent)	Clean file (01_clean.R)
Sex	8 spellings: M, m, male, F, f ...	Male / Female
Yes / No	Yes, Y, yes, 1 No, N, no, 0	Yes / No
Admission date	2023-03-14 14/03/2023 14-Mar-2023	2023-03-14
Lactate, ICU C	23.4 (mg/dL)	2.6 (mmol/L)
Missing codes	999 -99	missing (NA)
Impossible	age 210 weight 7 kg output -2990 ml	missing + query to site
Duplicates	623 rows (3 IDs twice)	620 patients

Figure 5.3: Raw vs clean: what the four ICUs actually sent

Take-away: Real data arrive messy: clean them with a script, never by hand.

Describe before you compare: look at the shape first

Symmetric data → **mean (SD)**; skewed data → **median (IQR)**; categories → **n (%)**. Never mix the pairs. ICU length of stay is the classic skewed variable: a few very long stays pull the mean up (FLUID-ICU survivors: mean 7.5 vs median 6.2 days).

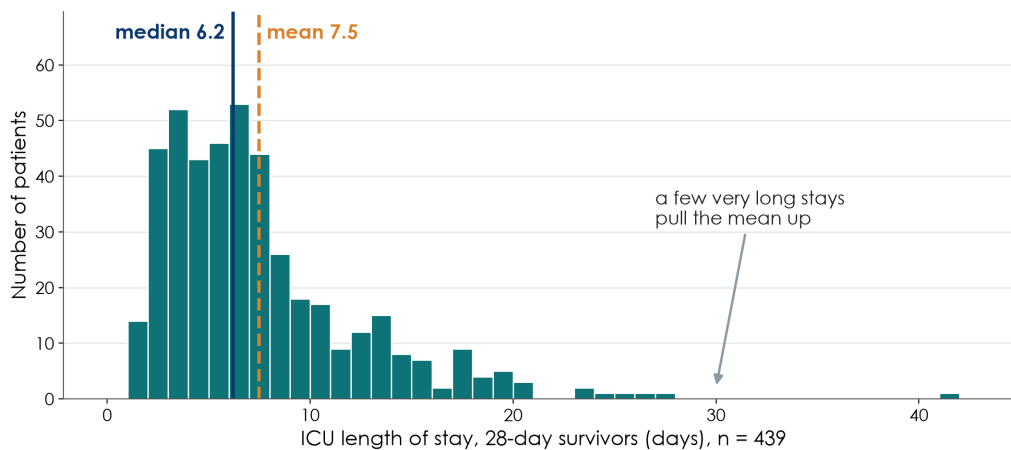


Figure 5.4: The skewed outcome: ICU stay in survivors

Take-away: For skewed data like length of stay, the median describes the typical patient.

Show the data, not just a bar

A bar with an error bar ("dynamite plot") hides how many patients there were, how they are spread and any outlier. Dots plus a box show every patient. Choose the graph from the variable type, not from the software menu.

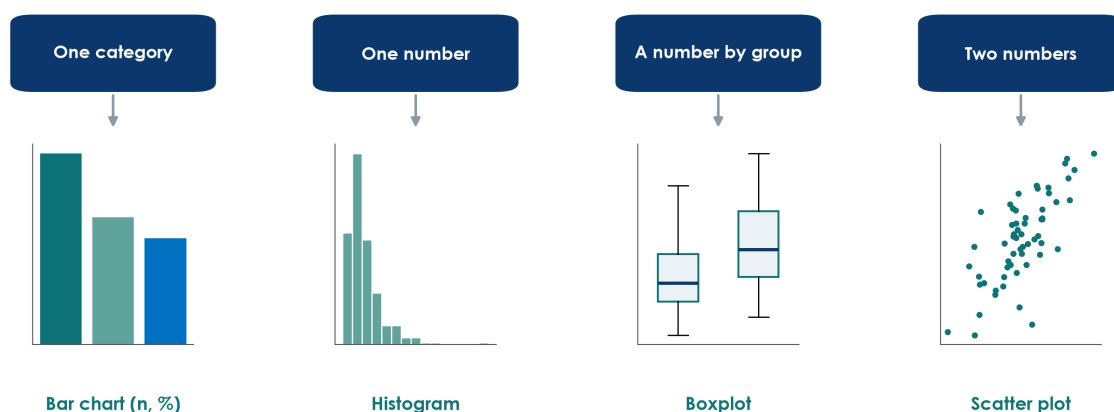


Figure 5.5: Which graph for which data?

Take-away: Choose the graph from the variable type, not from the software menu.

Table 1: were the groups alike at the start?

Table 1 describes the patients by group: who they were, and whether the groups were alike. In FLUID-ICU they were **not**: the positive-balance group was sicker (median SOFA 8 vs 6; vasopressor 55.0% vs 24.9%; mechanical ventilation 57.9% vs 39.6%). That is a warning for Session 6: confounding by indication. Table 1 has **no p-values** — it describes, it does not test.

Characteristic	Overall n = 618	Negative balance n = 245	Positive balance n = 373
Age, years, mean (SD)	62.6 (15.2)	64.6 (15.0)	61.4 (15.2)
Male sex, n (%)	375 (61%)	143 (58%)	232 (62%)
Heart failure, n (%)	97 (16%)	54 (22%)	43 (12%)
SOFA score, median (IQR)	7 (5, 9)	6 (4, 8)	8 (6, 10)
Lactate, mmol/L, median (IQR)	2.6 (1.7, 3.7)	2.2 (1.4, 3.0)	2.9 (1.9, 4.0)
Vasopressor, n (%)	266 (43%)	61 (25%)	205 (55%)
Mechanical ventilation, n (%)	313 (51%)	97 (40%)	216 (58%)

Figure 5.6: FLUID-ICU Table 1 (simulated data)

Take-away: The positive-balance group was sicker at baseline: a warning for Session 6.

5.4 Key words

Key word	In one line
Variable type	Continuous, count, binary, nominal, ordinal, time-to-event
CRF	Case report form: the questions exactly as the data collector reads them
Data dictionary	One row per variable: name, meaning, type, unit / codes, allowed values
Validation check	A rule that catches impossible or inconsistent values at entry
Audit trail / cleaning log	The record of every change: what, how many, why, who, when
Raw vs clean data	As sent (never edited by hand) vs analysis-ready (made by a script)
Mean (SD) / median (IQR)	Summaries for symmetric / skewed numbers
Skewed	A long tail on one side; mean pulled away from the median
Table 1	Baseline characteristics by group; no p-values
Dots + box	A graph that shows every patient

5.5 Exercises

Work with your neighbour or group as indicated. No calculator is needed: every exercise is about *choosing* and *interpreting*. Exercises marked **[L2]** use the course dataset FLUID-ICU (simulated) and your group's results pack in Course/Data/results_packs/. Exercise 8 becomes part of your proposal (Part 3).

#	Exercise	When	Time
1	Classify the variables	5.1 pair task	3 min
2	Fix the bad CRF items	5.1 pair task	3 min
3	Choose the right summary	5.3 check / homework	3 min
4	Judge a bad figure	homework	5 min
5	[L2] Spot the problems in the raw FLUID-ICU data	homework (after the 5.2 demo)	5 min
6	[L2] Read your group's Table 1	5.4 group worksheet	10 min
7	[L2] Your group's question	5.2 group task	4 min
8	Proposal builder 5 worksheet	proposal builder	30 min

Exercise 1 — Classify the variables

For each variable, tick **one** type.

#	Variable (ICU study of septic shock)	Continuous	Count	Binary	Nominal	Ordinal	Time-to-event
a	Age at ICU admission (years)	☐	☐	☐	☐	☐	☐
b	Sex (male / female)	☐	☐	☐	☐	☐	☐
c	Glasgow Coma Scale on admission (3–15)	☐	☐	☐	☐	☐	☐
d	Main source of sepsis (lung / urine / abdomen / skin / other)	☐	☐	☐	☐	☐	☐
e	Number of vasopressors running at 24 h (0, 1, 2, 3)	☐	☐	☐	☐	☐	☐
f	Cumulative fluid balance at day 5 (ml/kg)	☐	☐	☐	☐	☐	☐
g	Alive at day 28 (yes / no)	☐	☐	☐	☐	☐	☐
h	Time from ICU admission to death (patients alive at day 28 are censored)	☐	☐	☐	☐	☐	☐
i	Pain: none / mild / moderate / severe	☐	☐	☐	☐	☐	☐
j	Blood group (A, B, AB, O)	☐	☐	☐	☐	☐	☐

Bonus: which ONE of these would you most like to collect as an exact number rather than in groups, and why?

Exercise 2 — Fix the bad CRF items

Each item below can be filled in two different ways by two different nurses. Rewrite it so that it has **only one possible reading**. Check yourself: unit? time point? one answer only? an "unknown / not done" option?

#	Bad item	What is wrong?	Your rewritten item
a	Fluid given: _____		
b	Weight: _____		
c	Comorbidities: _____		
d	Lactate high? Y / N		

Exercise 3 — Choose the right summary

For each variable, choose the summary you would report in Table 1 and the best graph.

#	Variable (200 ICU patients)	Shape / type	Summary	Graph
a	Age (years), bell-shaped			
b	ICU length of stay (days): most 1–5 days, a few > 30 days			
c	Died in hospital (yes / no)			
d	C-reactive protein (mg/L): long right tail			
e	Source of sepsis (5 categories)			
f	Systolic blood pressure, compared between two treatment groups			

Exercise 4 — Judge a bad figure

A poster in the hospital corridor shows this figure (fluid balance at day 5, 24 patients per group):

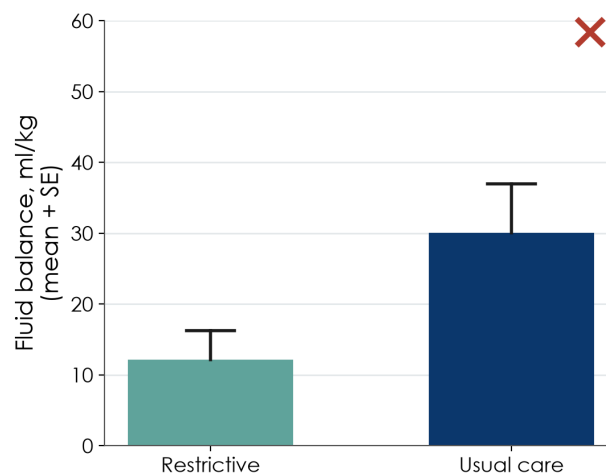


Figure 5.7: Dynamite plot

1. What does each bar show, and what does the error bar show? What is missing?

2. Name **two things the reader cannot see** in this figure.

3. Redraw it (sketch) the way you would show it in a paper. What do you gain?

A second poster shows ward mortality as bars of 31% (Ward A) and 29% (Ward B), with the y-axis running from 28% to 32%.

4. What impression does it give, and how would you fix it?

5. A third poster puts hand-hygiene compliance (%) and infection rate on one graph with two different y-axes and writes "hygiene works!". Give two reasons to be careful.

Exercise 5 — [L2] Spot the problems in the raw FLUID-ICU data

Below are six cells copied from the raw file `Course/Data/fluid_icu_raw.csv` (as the four ICUs sent it). For each, say what is wrong and what the cleaning script should do.

#	Column	Value in the raw file	What is wrong?	What should the script do?
a	sex	m, male, Female		
b	admission date	14/03/2023 in one row, 14-Mar-2023 in another		
c	lactate (ICU C)	23.4		
d	lactate	999		
e	age	210		
f	patient ID	FLU-139 appears in two identical rows		

Bonus: why must the raw file never be edited by hand? Name two things the cleaning log records for every change.

Exercise 6 — [L2] Read your group's Table 1 (results pack)

Open your group's results pack (`Course/Data/results_packs/Results_Pack_Qn.docx`, Table 1: baseline characteristics by fluid-balance group).

- How many patients are in each column? Why is the overall number 618 and not 620?

- For **three** variables, write the summary used (mean (SD), median (IQR) or n (%)) and why.

Variable	Summary used	Why this summary?
----------	--------------	-------------------

- Were the two groups alike at the start? Name the **biggest** difference between the negative- and positive-balance groups.
- Which variable has the most missing values, and how many?

5. Write **two sentences** describing the patients, as they would appear at the start of your Results section (numbers in brackets, no p-values).

6. Looking at this table, would you expect the crude association between positive fluid balance and death to be larger or smaller than the adjusted one? Why?

Exercise 7 — [L2] Your group's question (5.2)

Your answer

Group question (Q1–Q5), copied from page 1 of your results pack

P — population

E — exposure (variable name in the data dictionary) and its type

C — comparator

O — outcome (variable name) and its type

One-sentence research question

Which test will compare the groups? Which model will adjust? (guess now, check in Session 6)

5.6 Demo companion

Two short live demos in R today. You **watch and predict**: before each block the trainer asks "what do you expect?". FLUID-ICU is **simulated** — no real patient, hospital or country.

Part A — from raw to clean (segment 5.2, 8 minutes, script `Data/R/01_clean.R`).

Block	What you will see	Write down
Read the raw file	623 rows, 29 columns, everything read as text	Why 623 rows when 620 were enrolled? _____
Duplicates	3 study IDs appear twice (identical rows) → 620	
Sites, sex, yes/no	icu a, ICU-A ...; M, m, male, "Female"; Y, 1, yes → one spelling each	
Dates	2023-03-14, 14/03/2023, 14-Mar-2023 → one format	
Missing codes	maxima of 999 and minima of -99 → set to missing	
Units	lactate at ICU C about 9 times higher (mg/dL) → divided by 9.01	How was it found? _____
Impossible values	age 210, weight 7 kg, a negative fluid output → missing + query; patient kept	
Derived variables	cumulative fluid balance (ml/kg), positive balance, any AKI — calculated, never typed	
What is left	lactate missing in 93 patients	
Reproducibility	"identical to the reference clean file: TRUE"	

Part B — Table 1 and figures (segment 5.4, 6 minutes, script `Data/R/02_describe.R`).

Block	What you will see (simulated data)	Write down
Groups	Positive balance 373/618 (60.4%); 2 patients without a balance	Why 618? _____
Shape first	Age mean 62.6 (SD 15.2); lactate skewed	
Table 1	Negative n = 245, positive n = 373; SOFA 6 vs 8; vasopressor 24.9% vs 55.0%	Biggest difference: _____
Histogram	Cumulative balance at day 3: median 8.8 ml/kg (IQR -10.3 to 26.2)	
Box plot with dots	SOFA by group, medians 6 vs 8	
Outcomes, described	28-day mortality 179/620 (28.9%); ICU stay mean 7.3 vs median 6.0 days	

Why not fix the errors in Excel? Hand edits leave no trace and cannot be repeated. A script plus a cleaning log can be checked by a reviewer and re-run by anyone. **Why set age 210 to missing instead of 21?** It could be 21, 71 or 10: guessing invents data. Query the site and correct only from the source document.

5.7 Proposal builder 5

Your **own** question again — not the course dataset.

Exercise 8 — Proposal builder 5 worksheet

group, 30 minutes

Write your answers here first, then copy them into Part 3 of the group proposal template. (Your **own** question, not the course dataset.)

8.1 Variables table (8 min)

Role	Variable	How measured (unit, time point)	Type
Primary outcome			
Secondary outcome (optional)			
Exposure / intervention			
Confounder 1			
Confounder 2			
Confounder 3			
Descriptive (e.g. age, sex)			

Tip: if your design is an RCT, confounders are less of a problem (randomisation), but list the key baseline variables you will show in Table 1.

8.2 Mini-CRF: 6–10 items (10 min)

#	Question exactly as the data collector will read it	Answer format (boxes / ticks / unit)	"Unknown / not done" option?
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐

8.3 Data dictionary: 3–5 rows + one data-quality rule (6 min)

Variable name	Label (meaning)	Type	Unit / codes	Allowed values

Data-quality plan (one or two sentences): who checks the forms, how often, and what range/consistency check you will run.

8.4 Data-management plan (6 min)

Question	Our answer
Paper CRF with double data entry, or electronic data capture (e.g. REDCap / KoBoToolbox)?	
Coded study IDs? Where is the linking log kept, and who can open it?	
Validation rules at entry (two examples from your data dictionary)	
Audit trail / cleaning log: who records changes?	
Backups (3-2-1) and data-sharing statement in the consent form	

5.8 After the session — homework

- ☐ Read your group's **results pack**: Table 1 and the figures (Data/results_packs/Results_Pack_Qn.docx). Mark anything you do not understand.
- ☐ Finish Exercises 4 and 5 (bad figure; raw data problems) if not done in class.
- ☐ Finish the variables table, mini-CRF and data dictionary of Proposal Part 3.
- ☐ Read the **SAP** of FLUID-ICU (Data/SAP_FLUID-ICU.md), sections 1–5: which confounders were chosen before the analysis?

5.9 Extra practice

Extra practice 5.1 — Summary or graph?

Variable (FLUID-ICU, simulated)	Type	Summary for Table 1	Best graph
Age (years, roughly symmetric)			
SOFA score (0–24, skewed)			
Infection source (5 categories)			
Died by day 28 (yes / no)			
AKI stage (0, 1, 2, 3)			

Write the data-dictionary row for "lactate on ICU admission".

Variable name	Label (meaning)	Type	Unit / codes	Allowed values
---------------	-----------------	------	--------------	----------------

Extra practice 5.3 — Read a sentence of Table 1

Using the FLUID-ICU Table 1 figure above, complete the Results sentence (numbers in brackets, no p-values):

"Patients with a positive fluid balance were more severely ill at baseline than those with a negative balance (median SOFA _____ vs _____; vasopressor on admission _____% vs _____%)."

Extra practice 5.4 — Mean or median?

Seven patients stayed in the ICU 2, 3, 3, 4, 5, 6 and 26 days. Calculate the mean and the median. Which describes the typical patient, and why?

Mean = _____ Median = _____ Typical patient: _____

My notes — Session 5

SESSION 6

ANALYSE — from sample to answer

When	Weekend 3 · Sunday · Session 6 of 8
Length	3 hours, with a 15-minute break
Your group builds	Proposal Part 3 (complete) — sample size, analysis plan, missing-data plan · Manuscript — your group's results interpreted

Learning objectives

By the end of this session you will be able to:

- read a **95% confidence interval** and a p-value correctly, and avoid the five common p-value mistakes;
- explain type I and II errors, power, and the four ingredients of a sample-size calculation;
- choose a test from the outcome type, its shape and whether the data are paired;
- read a regression table and a forest plot: comparison, effect, 95% CI, and whether the CI crosses the "no effect" value;
- explain what "adjusted" means and why the FLUID-ICU odds ratio shrank after adjustment;
- read a Kaplan–Meier curve and a hazard ratio; tell MCAR, MAR and MNAR apart and compare complete-case analysis with multiple imputation.

6.1 Before the session — prepare

- ☐ Read your group's **results pack**: Table 1, the comparison table and the figures. Write your main result in one sentence (you may correct it later today).

- ☐ Find one sentence in Paper 1 or Paper 2 that gives an effect with a 95% CI. Copy it here:

- ☐ For your own proposal: what do you expect in the comparison group (a proportion or a mean), and where does that number come from?

6.2 Session plan

Time	Min	Segment
0:00	10	Recap quiz
0:10	40	6.1 From sample to population: CI, p-values & misinterpretations, type I/II errors, power, sample size (live demo), choosing a test, paired vs unpaired, multiple comparisons
0:50	20	6.2 (course dataset) Compare groups in the course dataset: live demo (t-test / Mann–Whitney / chi-square); groups interpret their outputs
1:10	15	BREAK
1:25	35	6.3 Relationships and models: correlation, linear and logistic regression, adjustment, interaction, overfitting, Kaplan–Meier & Cox, prediction/ML; reading regression tables & forest plots
2:00	25	6.4 (course dataset) Model the course dataset: crude vs adjusted OR (confounding by indication made visible), Kaplan–Meier + Cox HR, complete-case vs multiple imputation; the SAP that was written before the analysis
2:25	30	Proposal builder 6: sample size, analysis plan, missing-data plan
2:55	5	Wrap-up

6.3 Key ideas in pictures

The confidence interval: the range of compatible values

If we repeated the same study many times, about 95 of 100 such intervals would capture the true value — we never know which ones. Read the **estimate and its CI** first: a wide CI that crosses “no effect” means “we do not know”, not “no effect”.

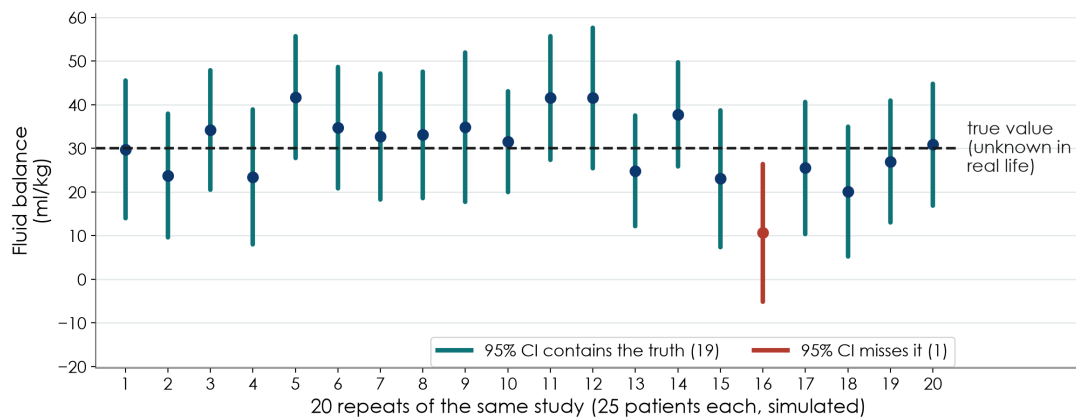


Figure 6.1: The dance of the confidence intervals

Take-away: About 95 of 100 such intervals capture the truth; we never know which ones.

The p-value in one picture

The p-value says how surprising our result would be **if there were truly no difference**. It is not the probability that the null hypothesis is true, and it says nothing about how large or important the effect is.

Errors, power and sample size

Alpha is the risk of a false alarm (type I error, usually 5%); power is the chance of detecting a real effect (usually 80–90%). Sample size depends on the effect you hope to detect, the noise, alpha and power — the software does the arithmetic, the ingredients are your clinical job.

Choosing a test

Outcome type + shape + paired or not = the test: t-test or Mann–Whitney for numbers, chi-square (or Fisher) for proportions, paired versions when the same patients are measured twice. Testing 20 things at once gives about one "significant" result by chance alone.

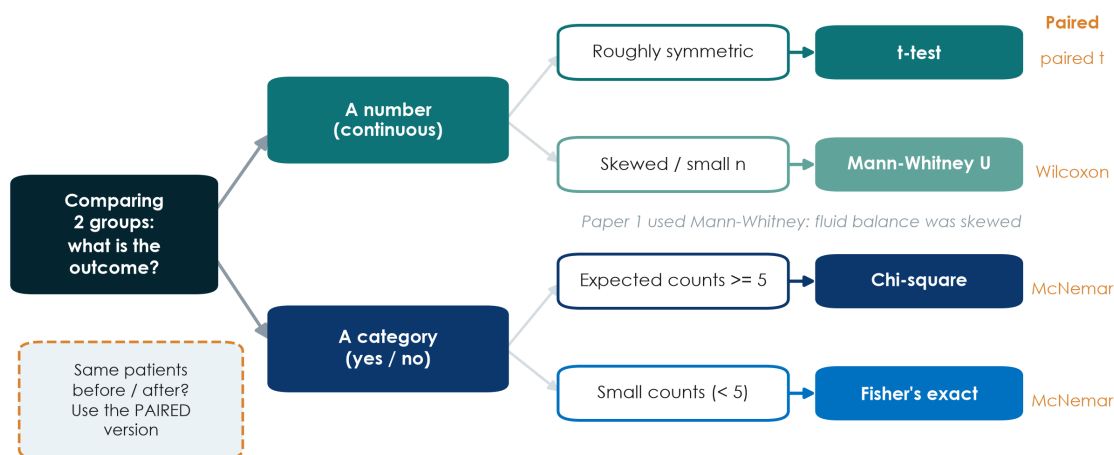


Figure 6.2: Choosing a test: one decision chart

Take-away: Outcome type + shape + paired or not = the right test.

FLUID-ICU: four outcomes, four comparisons (simulated data)

Every outcome was worse in the positive-balance group **before any adjustment**: 28-day death 134/373 (35.9%) vs 45/245 (18.4%), risk difference 17.6 percentage points (95% CI 10.7 to 24.4); AKI 61.7% vs 29.0%; ventilator-free days median 20 vs 28; ICU stay in survivors median 7.5 vs 4.8 days. But the positive-balance group was sicker.

Outcome (type)	Negative (n = 245)	Positive (n = 373)	Test	Result
28-day death (yes/no)	45 (18.4%)	134 (35.9%)	Chi-square	RD 17.6 points (10.7 to 24.4), $p < 0.001$
AKI (yes/no)	71 (29.0%)	230 (61.7%)	Chi-square	$p < 0.001$
Ventilator-free days (0-28, skewed)	median 28 (22 to 28)	median 20 (0 to 28)	Mann-Whitney	$p < 0.001$
ICU stay, survivors (days, skewed)	4.8 (3.3 to 7.1) n = 200	7.5 (5.2 to 11.2) n = 239	Mann-Whitney	difference 2.5 days (1.8 to 3.2), $p < 0.001$

n (%) or median (IQR); unadjusted comparisons. Source: 03_compare.R, key_results.md (simulated data)

Figure 6.3: FLUID-ICU: four outcomes, four comparisons

Take-away: Every outcome is worse in the positive-balance group, before any adjustment.

Which model? It depends on the outcome

A number → linear regression (a difference); yes/no → logistic regression (an odds ratio); time to an event → Cox regression (a hazard ratio). "Adjusted" means comparing patients who are alike on the confounders. Read each row: comparison, effect, 95% CI, does the CI include the null value (0 for differences, 1 for ratios)?

Confounding by indication, made visible

In FLUID-ICU the crude odds ratio for death with a positive balance was **2.49 (95% CI 1.70 to 3.70)**. After adjusting for the confounders chosen in the SAP (age, SOFA, lactate, vasopressor, ventilation, chronic kidney disease) it shrank to **1.62 (1.02 to 2.59)**: much of the crude difference came from severity. A modest association remained.

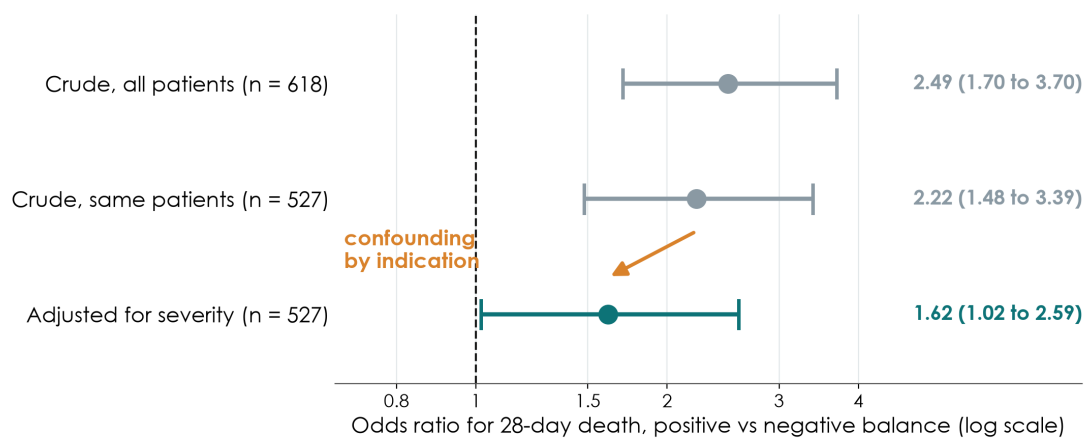


Figure 6.4: Crude vs adjusted odds ratio (simulated data)

Take-away: Crude OR 2.49 shrinks to 1.62 after adjusting for severity: still above 1.

Survival: Kaplan–Meier and Cox

Each step down is a death; each tick is a censored patient (still alive when follow-up ended); always read the numbers at risk. In FLUID-ICU, day-28 survival was 81.6% (negative balance) vs 64.1% (positive); log-rank $p < 0.001$ (unadjusted). The adjusted Cox hazard ratio was 1.46 (1.00 to 2.13), $p = 0.053$ — the same story as the logistic model ($p = 0.044$): read the estimates and CIs, not the side of 0.05.

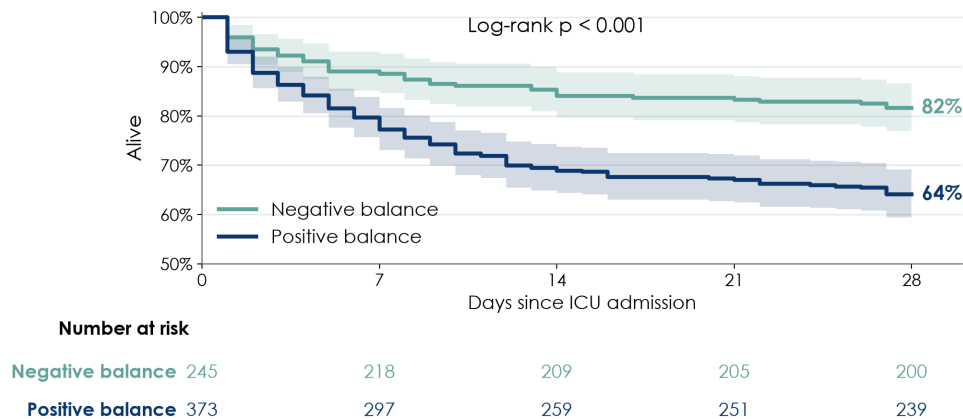


Figure 6.5: Kaplan–Meier survival to day 28 by fluid balance (simulated data)

Take-away: Read the steps, the ticks, the gap, and always the numbers at risk.

Missing data: why it is missing matters

MCAR: missing by pure chance. **MAR:** missing for a reason we recorded (e.g. severity). **MNAR:** missing because of the value itself. In FLUID-ICU lactate was missing in 93/620 (15.0%), mostly in less sick patients — not MCAR, MAR plausible. Multiple imputation kept all 620 patients and changed the answer only modestly.

MCAR: Missing Completely At Random			MAR: Missing At Random			MNAR: Missing Not At Random		
Night adm.	Lactate	Died	Night adm.	Lactate	Died	Night adm.	Lactate	Died
no	2.1	no	no	2.1	no	no	2.1	no
yes	?	no	yes	?	no	yes	3.4	no
no	6.8	yes	no	6.8	yes	no	?	(6.8) yes
no	?	no	no	1.8	no	no	1.8	no
yes	4.2	no	yes	?	no	yes	4.2	no
no	9.1	yes	no	9.1	yes	no	?	(9.1) yes
yes	2.6	no	yes	?	no	yes	2.6	no
no	7.5	yes	no	7.5	yes	no	?	(7.5) yes
Tube dropped, analyser down: unrelated to anything			Not measured at night: depends on something we RECORDED			Highest values missing (sickest patients, off-scale results)		
Complete-case analysis is unbiased (just less precise)			Fixable: multiple imputation using the recorded variable			No guaranteed fix: sensitivity analysis		

Figure 6.6: Why is it missing? MCAR, MAR and MNAR

Take-away: The reason data are missing decides how dangerous the gaps are.

6.4 Key words

Key word	In one line
95% CI	The range of values compatible with the data
p-value	How surprising the data would be if there were no difference
Type I / type II error, power	False alarm / missed effect / chance of detecting a real effect
t-test / Mann–Whitney / chi-square	Compare means / skewed numbers / proportions
Paired	The same patients measured twice
Adjusted OR	Odds ratio comparing patients alike on the confounders
Hazard ratio, censoring	Ratio of death rates over time; followed until a time, outcome unknown after
SAP	Statistical analysis plan, written before the analysis
MCAR / MAR / MNAR	Missing by chance / for a recorded reason / because of the value itself
Multiple imputation	Fills each gap many times with plausible values; keeps every patient

6.5 Exercises

No calculator is needed. Exercises marked **[L2]** use the course dataset FLUID-ICU (simulated) and your group's results pack; your answers to Exercises 4 and 8 become the core of your manuscript's Results (Session 7). Exercise 9 completes Part 3 of your proposal.

#	Exercise	When	Time
1	True or false: CIs, p-values and multiple tests	6.1 / homework	5 min
2	Choose the test	6.1 check / homework	5 min
3	Match r to the scatter plots	homework	3 min
4	[L2] Interpret your group's comparison	6.2 group worksheet	7 min
5	Read a regression table	6.3 check (pairs)	5 min
6	Read a Kaplan–Meier curve	homework	5 min
7	Why is it missing? MCAR, MAR or MNAR	homework	5 min
8	[L2] Interpret your group's model	6.4 group worksheet	7 min
9	Proposal builder 6 worksheet	proposal builder	30 min

Figures for Exercises 3 and 6 are in [Resources/figures/en/sat3/](#) (Vietnamese versions in [.../vi/sat3/](#)).

Exercise 1 — True or false: confidence intervals, p-values and multiple tests

Circle **T** (true) or **F** (false). Be ready to explain each "false" in one sentence.

#	Statement	
1	Paper 2 reports HR 2.13 (95% CI 1.48 to 3.06). The data are compatible with a true hazard ratio anywhere from about 1.5 to about 3.1.	T / F
2	Paper 1 found $p = 0.42$ for fluid balance, so the restrictive strategy has no effect on fluid balance.	T / F
3	$p = 0.03$ means there is a 3% probability that the null hypothesis (no difference) is true.	T / F
4	If the 95% CI for a difference between two means includes 0, the p-value is greater than 0.05.	T / F
5	Other things being equal, a larger study gives a narrower confidence interval.	T / F
6	$p < 0.001$ proves that the effect is large and clinically important.	T / F
7	If we repeated the same study many times, about 95% of the 95% CIs would contain the true value.	T / F
8	A "negative" result ($p > 0.05$) in a small pilot trial could be a type II error.	T / F
9	Alpha (usually 5%) is the risk we accept of concluding there is an effect when there really is none.	T / F
10	$p = 0.06$ shows "a trend towards significance".	T / F
11	If you test 20 different outcomes and none has a real effect, you will usually still find at least one with $p < 0.05$.	T / F
12	Lactate measured in the same 40 patients at admission and 6 hours later should be compared with an unpaired t-test.	T / F

Exercise 2 — Choose the test

Use the decision chart from the slides (outcome type → shape or counts → paired or not). Write the test and one reason.

#	Scenario	Test	Why?
a	Mean systolic blood pressure (roughly bell-shaped) in 60 patients on drug A vs 60 on drug B.		
b	ICU length of stay (skewed) in the restrictive-fluid group vs the usual-care group, 24 patients each.		
c	28-day mortality (yes/no) in 200 patients with a positive fluid balance vs 200 with a negative balance.		
d	Acute kidney injury needing dialysis: 2 of 24 patients in one group vs 5 of 24 in the other.		
e	Serum lactate at ICU admission and 6 hours later in the same 40 patients (the changes are roughly symmetric).		
f	Pressure injury (yes/no) in the same 80 patients assessed before and after a new turning protocol.		

Exercise 3 — Match r to the scatter plots

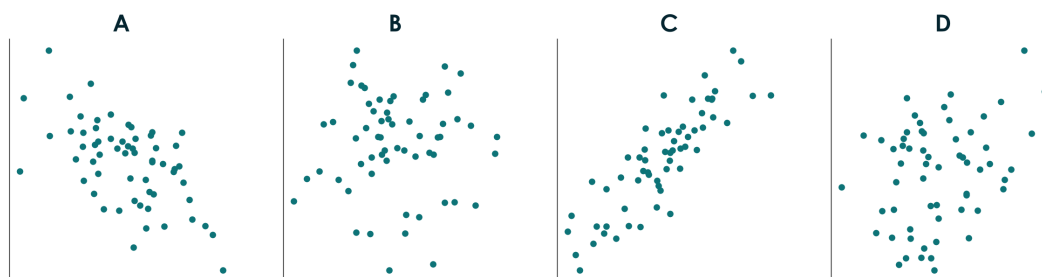


Figure 6.7: Four scatter plots A to D

Match each plot (A–D) to one correlation coefficient: $r = 0.86$, $r = 0.30$, $r = 0.01$, $r = -0.58$.

Plot	r
A	
B	
C	
D	

Bonus: a bedside glucose meter and the laboratory have $r = 0.97$ in 50 patients. Can the meter replace the laboratory? What would you look at instead?

Exercise 4 — [L2] Interpret your group's comparison (6.2)

Use the comparison table in your results pack.

Your answer
Outcome and its type
Test used, and why this test
Result in the negative-balance group (n (%)) or median (IQR))
Result in the positive-balance group
Effect with 95% CI (difference, risk difference, ratio or Hodges–Lehmann difference)
p-value
One reason this unadjusted comparison could be biased

Your Results sentence (no "significant" on its own, no causal words):

Exercise 5 — Read a regression table

A (mock) cohort study of 600 adults admitted to a medical ICU asked which factors are associated with **hospital-acquired delirium** (yes/no) during the ICU stay.

Table 3. Multivariable logistic regression: factors associated with delirium

Variable	Adjusted OR	95% CI	p-value
Benzodiazepine infusion (yes vs no)	2.10	1.40 – 3.20	< 0.001
Age (per 10 years)	1.30	1.12 – 1.51	< 0.001
Mechanical ventilation (yes vs no)	1.60	0.95 – 2.70	0.08
Female sex (vs male)	0.85	0.60 – 1.20	0.36
Pre-existing dementia (yes vs no)	3.40	1.90 – 6.10	< 0.001
Sleep protocol: eye masks + earplugs (yes vs no)	0.62	0.41 – 0.94	0.02

Adjusted for all variables in the table and for APACHE II score.

Authors' conclusion: *"Benzodiazepines cause delirium in ICU patients. Mechanical ventilation showed a trend towards more delirium ($p = 0.08$). Our sleep protocol reduces delirium by 38%."*

Questions:

1. What type of outcome is this, and why was logistic regression used?

2. What is the "no effect" value for the numbers in this table?

3. For each row, does the 95% CI include the no-effect value? Mark ✓ (excludes) or × (includes).

4. Write **one plain-language sentence** for the benzodiazepine row.

5. Interpret the age row: what does "1.30 per 10 years" mean for a patient 20 years older?

6. Which rows are compatible with no association?

7. Find **three** problems in the authors' conclusion.

Exercise 6 — Read a Kaplan–Meier curve

A (simulated) trial randomised 200 adults with septic shock to a restrictive fluid strategy or usual care and followed them for 28 days.

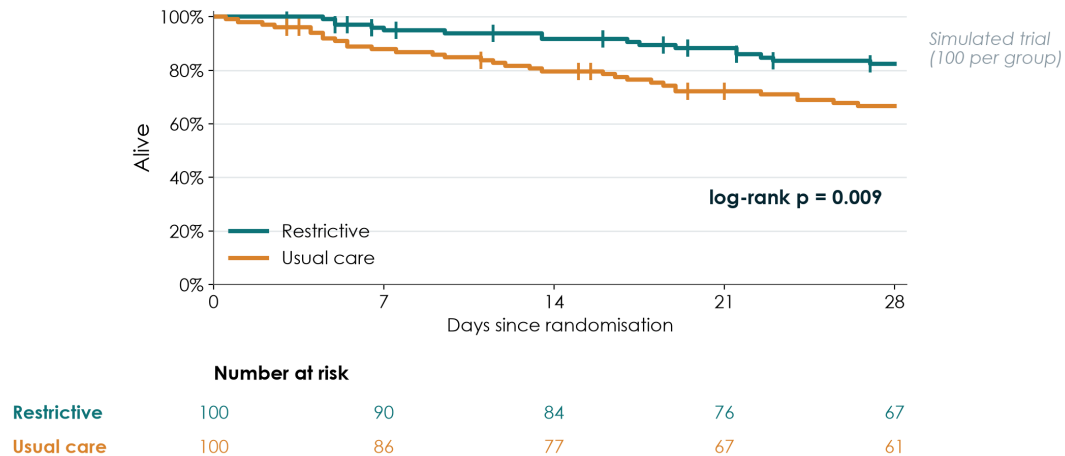


Figure 6.8: Kaplan-Meier curve

1. Approximately what proportion of patients in each group were alive at day 28?

2. What do the small vertical ticks on the curves mean?

3. How many patients in the usual-care group were still being followed (at risk) at day 21? Why does the number fall even in a group where few patients die?

4. What does "log-rank $p = 0.009$ " tell you? What does it **not** tell you?

5. Which model would give a hazard ratio with a 95% CI, and what assumption would you check by looking at the curves?

6. Would you be happy if the authors had shown this figure **without** the numbers at risk? Why?

Exercise 7 — Why is it missing? MCAR, MAR or MNAR

For each scenario, choose the most likely mechanism (**MCAR**, **MAR** or **MNAR**) and say what it means for a complete-case analysis.

#	Scenario	Mechanism	Consequence for complete-case analysis
a	12 blood samples for lactate were lost when a courier box was dropped.		
b	Weight is missing more often in patients admitted at night; admission time is recorded for everyone.		
c	Pain scores are missing for the patients in most pain, who could not complete the questionnaire.		
d	Day-28 status is missing for patients transferred to another hospital, often because they were deteriorating.		
e	A form version used in the first month did not include the "smoking" question.		
f	Older patients are less likely to return a quality-of-life questionnaire at 3 months; age is recorded for all.		

Bonus: for scenario (d), describe one **sensitivity analysis** you could report.

Exercise 8 — [L2] Interpret your group's model: crude vs adjusted, complete case vs imputation (6.4)

Use the regression table and the missing-data table in your results pack.

Part A — crude vs adjusted

Estimate	95% CI	n
Crude		
Adjusted		

1. Which confounders were in the adjusted model? Were they chosen before the analysis (look at the SAP)?

2. How much did the estimate move after adjustment, and in which direction? What does that tell you about confounding by indication?

3. Does the adjusted 95% CI include the "no effect" value (1 for a ratio, 0 for a difference)?

Part B — complete case vs multiple imputation

4. How many patients were in the complete-case model, and how many in the multiple-imputation model? Who was left out of the complete-case analysis?

5. Did multiple imputation change the answer? Is the result robust?

Part C — (Q1 only) Kaplan–Meier and Cox

6. Survival at day 28 in each group; the log-rank p ; the adjusted Cox HR with its 95% CI. Why do the logistic ($p = 0.044$) and Cox ($p = 0.053$) results tell the same story?

Your two Results sentences (crude, then adjusted, with CIs; "associated with", not "caused"):

6.6 Demo companion

Three short demos. Participants **watch and predict**; nobody calculates. All FLUID-ICU numbers are from the simulated dataset.

Part A — sample size (6.1, 8 minutes; online calculator and `Resources/tools/sample_size_calculator.xlsx`).

Step	Inputs	Result to note
Paper 1 (two means)	70 vs 40 ml/kg, SD 35, alpha 5%, power 80%	22 per group; +10% for deaths → 50 in total (as in Paper 1)
Our question: mortality (two proportions)	40% vs 30%, alpha 5%, power 80%	356 per group (376 with continuity correction); 792 in total after 10% loss
Change one input	power 90% → 477; 40% vs 35% → 1,471; 40% vs 25% → 152; alpha 1% → 530 (per group)	Which change costs most patients? _____

Part B — comparing two groups (6.2, 6 minutes; `Data/R/03_compare.R`). Before each block, name the test yourself.

Outcome (type)	My test	Trainer's test	Result (simulated)
Age (symmetric)		Welch t-test	mean 64.6 vs 61.4 years
SOFA, lactate (skewed / score)		Mann–Whitney	SOFA 6 vs 8; lactate 2.2 vs 2.9 mmol/L
28-day death (binary)		chi-square	45/245 (18.4%) vs 134/373 (35.9%); RD 17.6 points (10.7 to 24.4); RR 1.96
Ventilator-free days		Mann–Whitney	median 28 vs 20 days
ICU stay in survivors (skewed)		Mann–Whitney (not t-test)	median 4.8 vs 7.5; Hodges–Lehmann 2.5 days (1.8 to 3.2)

Part C — models and missing data (6.4, with the slides; `Data/R/04_models.R`, `05_missing.R`). Note on your group's worksheet (Exercise 8): the crude and adjusted estimates, the *n* in each model, the Kaplan–Meier and Cox results (Q1), and complete case vs multiple imputation (Q4).

Questions people ask. *Why did the OR go down after adjustment?* Sicker patients received more fluid and died more often; adjustment compares patients of similar severity. *The Cox *p* is 0.053 — so no effect?* No: HR 1.46 and OR 1.62 have overlapping CIs; the evidence is the same. *Isn't imputation making up data?* It fills each gap many times with plausible values and widens the CI for the uncertainty; valid under MAR, always compared with complete case. *Why not adjust for AKI?* It happens after the fluid and may be on the causal pathway.

6.7 Proposal builder 6

Exercise 9 — Proposal builder 6 worksheet

group, 30 minutes

Your **own** question again. Copy the answers into Part 3 of the proposal template.

9.1 Sample-size statement (8 min)

Input	Your value	Source (paper, audit, pilot, clinical judgement)
Expected value in the comparison group (proportion or mean)		
Smallest difference worth detecting		
SD (only for a numerical outcome)		
Alpha (two-sided)	5%	convention
Power	80% / 90%	
Expected drop-out		

Sentence (use the spreadsheet [Resources/tools/sample_size_calculator.xlsx](#) or ask the facilitator):

Assuming ____ in the control group and ____ in the intervention group, ____ patients per group give ____% power at a two-sided alpha of 5%; allowing ____% loss, we will recruit ____ patients in total.

9.2 Plain-language analysis plan (10 min)

Step	What we will do
Describe the patients (Table 1)	Numerical symmetric variables as ____; skewed as ____; categories as ____
Compare the primary outcome	Test: ____ because the outcome is ____
Report the effect	Effect measure (difference / RR / OR / HR) with 95% CI: ____
Deal with confounding (observational designs only)	We will adjust for: ____ using ____ (e.g. logistic regression)
Missing data	We will report how many values are missing for each variable and prevent it by: ____

Confounder	Why it is a confounder (linked to exposure AND outcome, not caused by the exposure)
------------	---

Question	Our answer
Which variables are most likely to be missing, and why? (MCAR / MAR / MNAR thinking)	
How will we prevent it? (form design, training, weekly checks, chasing outcome status)	
Main analysis: complete case or multiple imputation? Why?	
One sensitivity analysis	
How will we report missingness? (n missing per variable, flow diagram)	

6.8 After the session — homework

- ☐ Finish **Proposal Part 3** (variables, CRF, data management, sample size, analysis plan, missing-data plan).
- ☐ Read **Papers 1 and 2** in full with the 15-minute routine (needed for Session 8, "Be the reviewer").
- ☐ Start your group's **6-slide presentation** (Chapter 11): agree who presents what.
- ☐ Write your group's **key numbers list** for the manuscript: the main effect with 95% CI (crude and adjusted) and 3–5 supporting results, copied from your results pack.

6.9 Extra practice

Extra practice 6.1 — Does the CI include the null?

Result	Null value	CI includes the null?	One plain sentence
Adjusted OR 1.62 (95% CI 1.02 to 2.59)			
Difference 13.2 ml/kg (95% CI –15.2 to 41.6)			
HR 1.17 (95% CI 0.85 to 1.60)			
Mean difference –0.61 days (95% CI –1.00 to –0.21)			

Extra practice 6.2 — Choose the test

Scenario	Test
Proportion with pressure injury on two different wards (60 and 70 patients)	
Pain score (0–10, skewed) before and after a new protocol in the same 30 patients	
Time to death within 28 days in two groups, with censoring	

In a cohort, the crude OR for death with early intubation is 2.8 (95% CI 1.9 to 4.1); adjusted for severity it is 1.1 (95% CI 0.7 to 1.7). Explain in two sentences what happened and what the authors may conclude.

SESSION 7

PROTECT & WRITE — ethics, integrity, Methods and Results

When	Weekend 4 · Saturday · Session 7 of 8
Length	3 hours, with a 15-minute break
Your group builds	Proposal Part 4 — ethics and consent, registration, confidentiality, feasibility, publication plan · Manuscript — Methods + Results (about 600 words) with Table 1 and one figure or table

Learning objectives

By the end of this session you will be able to:

- describe what a protocol contains (SPIRIT) and the principles of the Declaration of Helsinki and Good Clinical Practice;
- follow the ethics-committee journey and choose a consent pathway, including consent in the ICU (legal representative, deferred consent, waiver);
- say when a study must be registered, and how to de-identify data;
- recognise research misconduct (fabrication, falsification, plagiarism) and questionable practices (duplicate and salami publication, gift authorship), and apply the ICMJE authorship criteria and CRediT roles;
- use AI tools in research honestly;
- write Methods and Results in the right order and style, with tables that stand alone, using STROBE.

7.1 Before the session — prepare

- ☐ Finish Proposal Part 3 and the first slides of your presentation.
- ☐ Re-read your group's **results pack** and your key-numbers list from Session 6.
- ☐ Read the **abstracts of Papers 1 and 2** again (for tomorrow's "Be the reviewer").
- ☐ Find out: which **ethics committee** would review a study in your unit, and how long does it usually take?

Committee: _____ Usual time: _____

7.2 Session plan

Time	Min	Segment
0:00	10	Recap quiz
0:10	30	7.1 Protocol, ethics and consent: protocol (SPIRIT), Helsinki & GCP, ethics committee journey, consent incl. ICU consent, registration, confidentiality, feasibility
0:40	20	7.2 Research integrity: FFP, duplicate/salami, authorship & CRediT, conflicts of interest, data sharing, AI use
1:00	15	Proposal builder 7: ethics, registration, feasibility, publication plan
1:15	15	BREAK
1:30	35	7.3 Writing the paper (I): writing order, Methods (incl. statistical paragraph), Results (participants first, numbers with CIs, no interpretation), stand-alone tables & figures, STROBE for the course-dataset paper
2:05	50	(course dataset) Manuscript workshop 1: each group drafts Methods + Results (~600 words) with Table 1 and one figure/table from its results pack
2:55	5	Wrap-up

7.3 Key ideas in pictures

The protocol and the ethics journey

The protocol is the building plan of the study — your four proposal parts are its skeleton. The ethics committee reviews it **before** any data are collected: no approval letter, no data collection, not even “just a few patients”. Trials must be **registered before the first patient**.

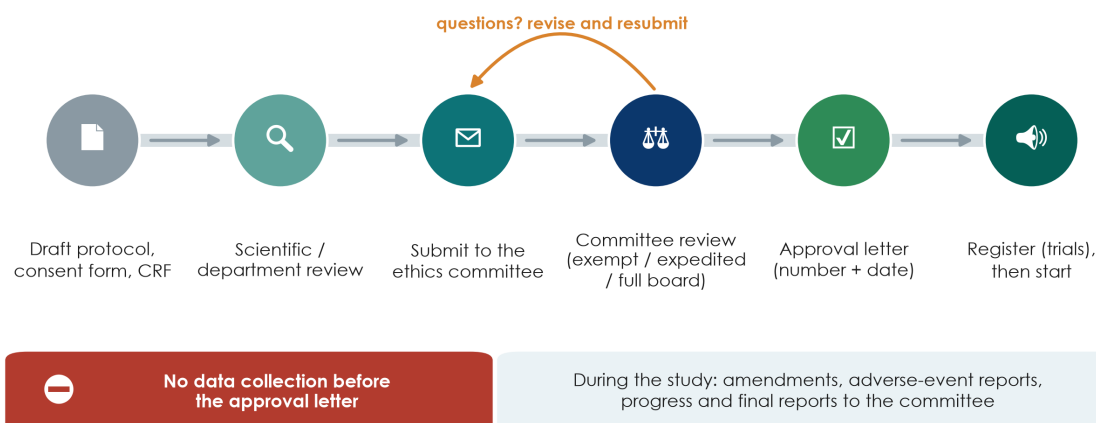


Figure 7.1: The ethics-committee journey

Take-away: No approval letter, no data collection — not even “just a few patients”.

Consent in the ICU

Informed consent needs information, understanding, voluntariness and capacity. Many ICU patients cannot decide: the order is the patient (if able), then the legal representative, then — only if the committee approved it — **deferred consent** (enrol in the emergency, ask as soon as possible). Routine-data cohorts may be granted a **waiver**.

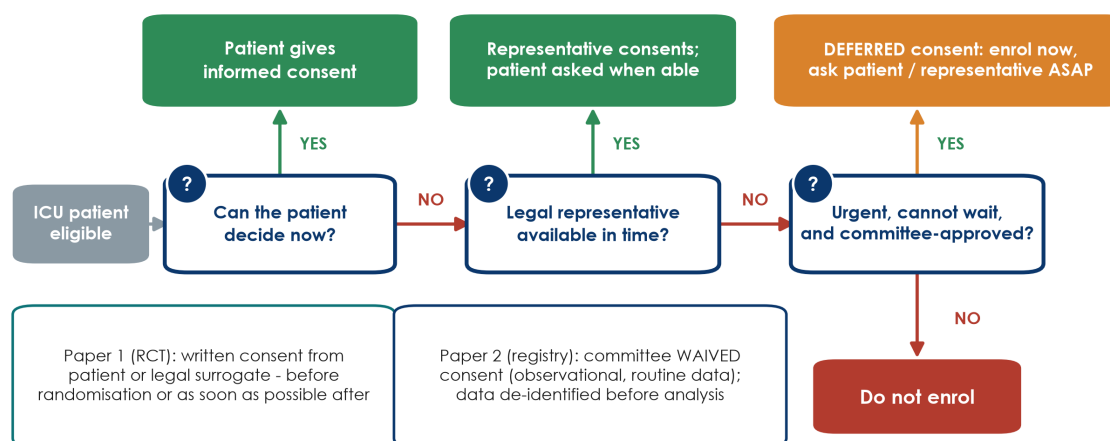


Figure 7.2: Consent in the ICU: when the patient cannot decide

Take-away: Patient, then legal representative, then deferred consent — and only if approved.

Confidentiality: de-identify before you analyse

Names and hospital numbers stay on the ward. Each patient gets a study code; the key file (code ↔ identity) is locked and kept separately; only coded data travel.

Research integrity

Fabrication (making up data), **falsification** (changing or leaving out data to fit the story) and **plagiarism** are research misconduct. Missing data are reported as missing — never invented, never quietly removed. An honest error is not misconduct, but it must be corrected openly.

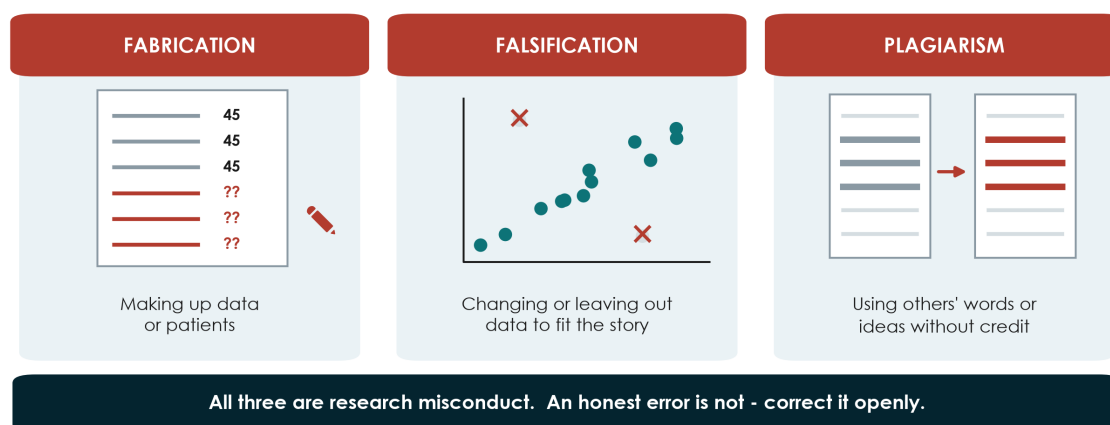


Figure 7.3: Fabrication, falsification, plagiarism

Take-away: Missing data are reported as missing — never invented, never quietly removed.

Who is an author? Who did what?

Every author meets **all four** ICMJE criteria: substantial contribution; drafting or critically revising; approving the final version; being accountable for all of it. Others are acknowledged. **CRedit** roles say who did what. AI tools cannot be authors: use them like a junior assistant, check everything, disclose it, and keep patient data out.

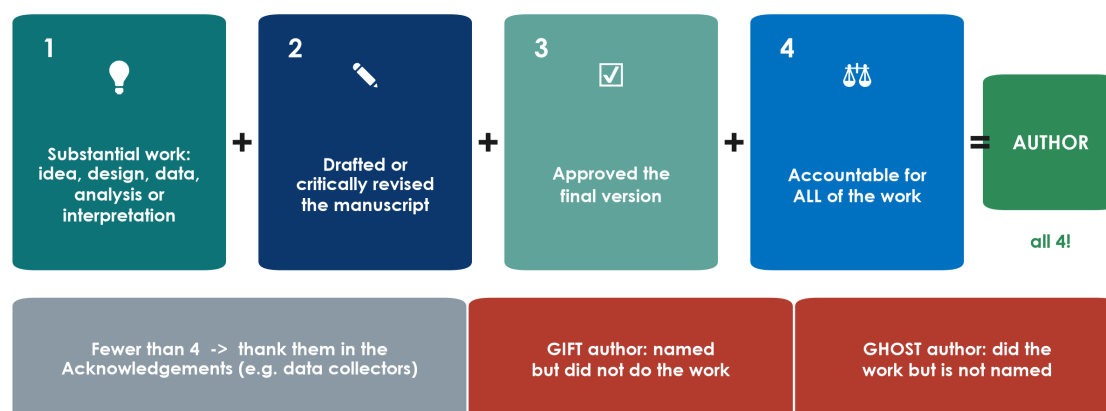


Figure 7.4: Who is an author? The four ICMJE criteria

Take-away: All four criteria, for every author — decide the list at the start.

Writing the paper (I): Methods and Results

Write in a different order from the one readers read: tables first, then Methods and Results, then Discussion and Introduction, the abstract and the title last. **Methods** give enough detail to repeat the study, with a statistical paragraph that names a method for every comparison. **Results** report, they do not interpret: participants first, then each outcome as an effect with its 95% CI.

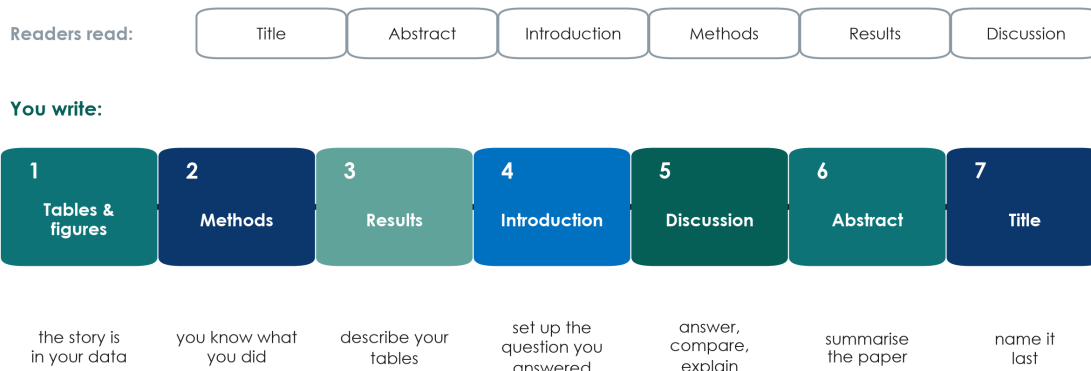


Figure 7.5: Write in a different order from the one readers read

Take-away: Start with your tables; write the title last.

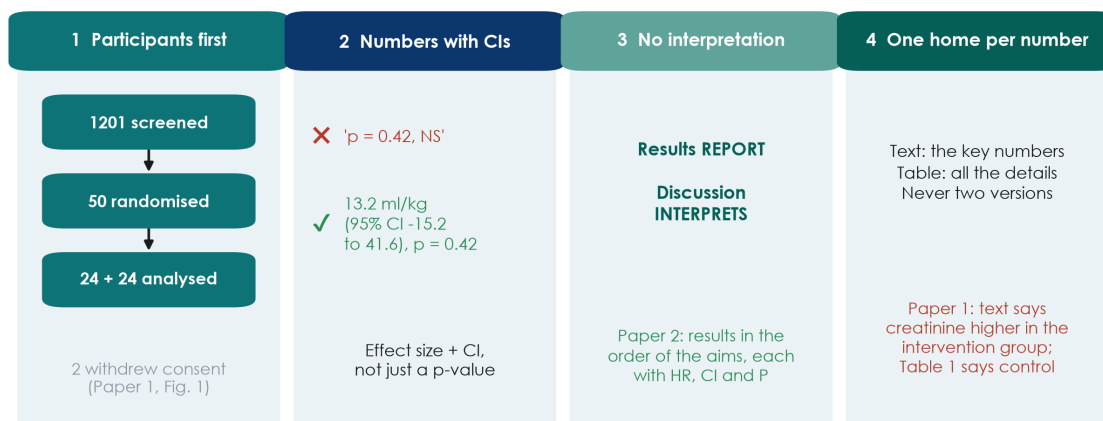


Figure 7.6: Results: report, do not interpret

Take-away: Participants first, then each outcome as an effect size with its 95% CI.

7.4 Key words

Key word	In one line
Protocol (SPIRIT)	The study's building plan, written and approved before the first patient
Declaration of Helsinki / GCP	Why we protect participants / how to run a study well
Deferred consent	Enrol first in an emergency, ask consent as soon as possible (committee-approved)
Registration	Public record of a study before patient 1 (mandatory for trials)
De-identification	Replacing names and numbers by study codes; key file kept apart
FFP	Fabrication, falsification, plagiarism — research misconduct
Salami / duplicate publication	One study cut into many papers / the same results published twice
ICMJE criteria / CRediT	Who may be an author / who did what
Conflict of interest	Anything that could be seen to influence your judgement — declare it
STROBE	Reporting checklist for observational studies: Methods items 4–12, Results 13–17

7.5 Exercises

Part	Exercise	When	Time
A	Integrity scenarios: green, amber or red?	0:55	5 min (3 in class, rest at home)
B	Proposal builder 7 — ethics, registration, feasibility, publication plan	1:00	15 min, finish at home
C	[L2] Manuscript workshop 1 — Methods + Results of your FLUID-ICU paper	2:05	50 min

Part A — Integrity scenarios: green, amber or red?

5 minutes, 0:55; three in class, the rest at home

For each scenario, decide as a group and hold up a colour:

- **GREEN** — acceptable as described.
- **AMBER** — acceptable **only if** something is fixed or disclosed (say what).
- **RED** — research misconduct or a serious breach; do not do it.

Write one sentence explaining your colour.

#	Scenario	Colour	Why?
1	Three patients' day-3 fluid charts are lost. A resident "remembers roughly" what their balances were and enters values so the dataset is complete.		
2	After seeing the results, the team removes two patients with "strange" fluid values because they weaken the association. The paper does not mention it.		
3	The head of department did not take part in the study but asks to be last author "because it is my department".		
4	(a) A nurse uses an AI chatbot to improve the English of the Discussion she wrote, checks every sentence, and states this in the Acknowledgements. (b) A colleague uploads the study spreadsheet (names and hospital numbers included) to a free public chatbot and asks it to "run the statistics".	(a)	
		(b)	
5	A cohort study of ventilated patients is published as three papers — mortality, length of stay and ventilator days — in three journals, none mentioning the others.		
6	Two paragraphs of a published review are copied word for word into your Introduction. The review is cited at the end of the paragraph, but there are no quotation marks.		

7.6 Demo companion

There is **no software demo** in Session 7. The slides show two worked examples for FLUID-ICU question 1 (simulated data) — a model **Methods** section and a model **Results** section. As you watch, tick which STROBE item each sentence answers:

Sentence in the model	STROBE item(s)	✓
Design, setting (4 ICUs) and dates	4, 5	☐
Who was included and excluded	6	☐
Exposure, outcome and confounders defined	7	☐
Data sources (ICU charts, CRF)	8	☐
How confounding by indication was addressed	9	☐
Study size ("all eligible patients in the cohort")	10	☐
Statistical methods, missing data, software	12	☐
Participant numbers at each step	13	☐
Crude AND adjusted estimates with 95% CI	16	☐

7.7 Proposal builder 7 and manuscript workshop 1

Two products today: before the break you finish **Part 4 of the proposal**; after the break your group writes the **Methods and Results** of its FLUID-ICU paper.

Part B — Proposal builder 7: ethics, registration, feasibility and publication plan

15 minutes, 1:00

Complete as a group in 15 minutes (B1–B3 about 8 minutes, B4 about 7 minutes); finish at home tonight. Slides 5 (ethics & feasibility) and 6 (publication plan) of tomorrow's presentation come from this sheet.

B1. Ethics and consent

Question	Your answer
Which ethics committee will review the study? What type of review do you expect (exempt / expedited / full)? The committee decides.	
Who are the participants, and can they decide for themselves when they become eligible?	
Who gives consent? (patient / legal representative / deferred consent / waiver requested) — use the ICU consent flowchart	
Who takes consent, when and where?	
What are the risks and burdens for participants, and how are they minimised?	
Does the study need registration (trials: yes, before the first patient - ClinicalTrials.gov or another WHO registry; systematic reviews: PROSPERO; observational: recommended)?	

B2. Confidentiality

Question	Your answer
Which direct identifiers will you collect (if any), and why?	
How will each patient get a study code?	
Where is the key file (code ↔ identity) kept, and who can open it?	
Where will the de-identified data be stored (institution-approved, password-protected)?	

B3. Feasibility check (tick and comment)

✓	Question	Comment
☐	Time: realistic timeline (ethics 1–3 months, recruitment, analysis, writing)?	
☐	Money: budget headings named (staff time, materials/tests, publication fee)?	
☐	People: who collects data (nights, weekends), who analyses, protected time?	
☐	Patients: how many eligible patients did your unit see last year? How many can you realistically enrol?	
☐	Data: are the key variables recorded routinely? How much is likely missing?	

Biggest feasibility risk: _____ One fix: _____

B4. Publication plan (show it on slide 6 of your presentation)

Item	Your answer
Working title (from your PICO: design, I vs C, P, O; 10–16 words)	
Target journal (and a back-up) — checked in DOAJ / NLM Catalog, fees and licence known	
Reporting guideline (CONSORT / STROBE / STARD / PRISMA / SPIRIT / CARE)	
Registration needed? Where? (from B1)	
Authors (all four ICMJE criteria) and their CRediT roles	
Conflicts of interest and funding to declare	
Data-sharing statement (what, where, under which conditions)	
Conference where you would present an abstract first	

Structured-abstract skeleton — one line each; Results are the results you *expect to report* (format, not numbers).

Section	One line
Background	
Methods	
Results (expected)	e.g. "28-day mortality: n/N per group, RR (95% CI)"
Conclusion	Write the sentence you would use if the result is positive — and the one if it is not.

Part C — [L2] Manuscript workshop 1: draft your Methods and Results

50 minutes

What you are doing

Each group writes the **Methods** and **Results** of a short STROBE-guided original article answering its **FLUID-ICU question**, using **only** its results pack. Nobody calculates anything: every number comes from the pack.

Group	Question (results pack)
1	Q1 — Positive fluid balance by day 3 and 28-day mortality (Course/Data/results_packs/Q1/)
2	Q2 — Cumulative fluid balance and acute kidney injury (Q2/)
3	Q3 — Cumulative fluid balance and ventilator-free days (Q3/)
4	Q4 — Initial lactate and 28-day mortality; complete-case vs multiple imputation (Q4/)
5	Q5 — Fluid balance and ICU length of stay among survivors (Q5/)

FLUID-ICU is a **simulated teaching dataset**. In your ethics sentence write: *"This analysis used a simulated teaching dataset; no ethics approval or consent was required."*

Roles (agree in the first minute)

Role	Who	Job
Methods writers	2	Design & setting, participants, variables, statistical analysis
Results writers	2	Participants, Table 1 sentence, main result (crude + adjusted), one secondary/sensitivity result
Tables & figures	1	Table 1 + one figure or table from the pack: title (what, in whom, when), n per column, units, footnotes
Number checker	1	Compares every number in the text with the pack; checks STROBE items 4–17

Timing

Minute	Task
0–5	Read your results pack together. Write the main result in ONE sentence (effect size + 95% CI).
5–25	Methods (~300 words) — use the headings below.
25–40	Results (~300 words) — participants first; numbers with CIs; no interpretation.
40–47	Table 1 + one figure/table with title, legend and footnotes.
47–50	Tick STROBE items; check every number; save and share the file.

Methods — headings and checklist (STROBE items in brackets)

- **Study design and setting** (4, 5): prospective multicentre cohort; 4 ICUs; dates as given in the pack.
- **Participants** (6): who was included and excluded; how many.
- **Variables** (7): exposure (how fluid balance was calculated and over which days), outcome (definition and time point), confounders adjusted for.
- **Data sources** (8): routine ICU charts / study CRF.
- **Bias** (9): how confounding by indication was addressed (adjustment for severity).
- **Study size** (10): "all eligible patients in the cohort were included" (no new sample size for a fixed dataset).
- **Statistical analysis** (12): summary statistics; the test for each comparison; the regression model(s) and which variables; how missing data were handled; software (R, version from the pack); significance level / reporting of 95% CIs.

Results — checklist

- ☐ Participants first: number analysed (and excluded, with reasons) (13)
- ☐ Table 1 summarised in one or two sentences — how the groups differed at baseline (14)
- ☐ Outcome numbers per group (15)
- ☐ Main result: **crude AND adjusted** estimate with 95% CI (16)
- ☐ One further analysis (secondary outcome, sensitivity, imputation) (17)
- ☐ No interpretation ("because", "suggests", "important") — move it to the Discussion
- ☐ No number without its unit and, for effects, its 95% CI

Deliverable

Methods + Results (~600 words) + Table 1 + one figure or table, saved in the shared folder as FLUID-ICU_Q[n]_draft1.

7.8 After the session — homework

- ☐ Finish **Proposal Part 4** and your **6 slides** tonight; send the file to the course lead before Session 8 (GroupN_slides.pptx or PDF).
- ☐ Re-read your group's Methods + Results (FLUID-ICU_Q[n]_draft1): is every number identical to the results pack?
- ☐ Read Papers 1 and 2 again, at least the abstract, Table 1 and the Discussion: tomorrow you appraise one of them.
- ☐ Rehearse the presentation once **with a timer** (6 minutes).

7.9 Extra practice

Extra practice 7.1 — Methods, Results or Discussion?

Where does each sentence belong? Write M, R or D.

Sentence	M / R / D
"Of 620 patients enrolled, 618 had a day-3 fluid balance and were analysed."	
"This may be because sicker patients received more fluid."	
"Odds ratios were estimated with logistic regression adjusted for age, SOFA score, lactate, vasopressor use, mechanical ventilation and chronic kidney disease."	
"Our findings agree with a large registry cohort (Paper 2)."	
"Death by day 28 occurred in 134/373 (35.9%) patients with a positive balance and 45/245 (18.4%) with a negative balance."	

Extra practice 7.2 — Author or acknowledgement?

Person	Author?	Why?
The nurse who collected the data on nights but did not see the manuscript		
The resident who designed the CRF, analysed the data, wrote the first draft and approved the final version		
The head of department who provided the ward but did not take part		
A statistician who planned the analysis, revised the Methods and approved the final paper		

Write the ethics sentence for your FLUID-ICU manuscript (simulated data), and then one for a real observational study in your unit that used routine records.

SESSION 8

PUBLISH — finish the paper, get it out, present

When	Weekend 4 · Sunday · Session 8 of 8
Length	3 hours, with a 15-minute break (presentations and post-course quiz at the end)
Your group builds	Manuscript — Introduction, Discussion, Abstract and Title (complete draft) · Proposal — presented (6 minutes + 2 minutes of feedback)

Learning objectives

By the end of this session you will be able to:

- write an Introduction as a three-paragraph funnel and a Discussion in six moves;
- choose verbs that match the design (**calibrated language**) and write a structured abstract and an informative title;
- choose the right reporting guideline (via EQUATOR) and a trustworthy journal, and recognise a predatory journal;
- describe the submission package, the peer-review journey and how to respond to reviewers point by point;
- appraise a published paper with the **10-question checklist** and say what each design may claim;
- present your group's proposal in 6 minutes and pitch your manuscript in 1 minute.

8.1 Before the session — prepare

- ☐ Your **6 slides** are finished and sent to the course lead; bring a copy on a USB stick.
- ☐ Rehearse the presentation with a timer: between 5 and 6 minutes.
- ☐ Re-read your group's Methods + Results draft; bring your references from Sessions 1–2 (for the Introduction).
- ☐ Read the **abstract, Table 1 and Discussion** of your appraisal paper: Groups 1–3 read Paper 1 (RCT), Groups 4–5 read Paper 2 (cohort).
- ☐ Prepare your 1-minute **manuscript pitch**: question → main result with 95% CI → one-sentence

conclusion.

8.2 Session plan

Time	Min	Segment
0:00	5	Recap
0:05	25	8.1 Writing the paper (II): Introduction funnel, Discussion six moves, calibrated language, conclusion, abstract, title & keywords
0:30	30	(course dataset) Manuscript workshop 2: Introduction, Discussion, Abstract and Title → complete draft
1:00	15	BREAK
1:15	25	8.2 Getting published: reporting guidelines (EQUATOR demo), choosing a journal, open access & licences, predatory journals, submission package, peer review, responding to reviewers, rejection & resubmission, proofs, conference abstracts
1:40	20	8.3 Be the reviewer: simplified appraisal of Papers 1–2 (jigsaw, 10-question checklist) — training for the manuscript peer review after the course
2:00	45	Group presentations: own-topic research proposal + publication plan (5 groups × 6 min + 2 min feedback) + 5-min summary of the manuscript work
2:45	15	Post-course quiz, evaluation, next steps (peer-review swap of manuscripts as homework)

8.3 Key ideas in pictures

Introduction: a funnel in three paragraphs

What is known (with references) → what is not known (the gap) → the aim, which matches the primary outcome and uses "associated" for an observational design. About 350 words.

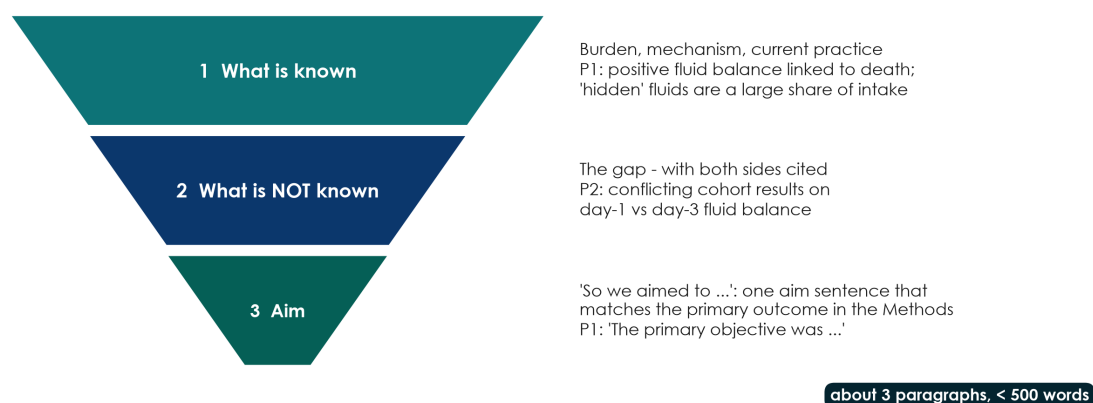


Figure 8.1: Introduction: a funnel in three paragraphs

Take-away: Known, unknown, aim — and the aim matches the primary outcome.

Discussion: six moves

1 **Answer** the question in the first sentence, with the main number; 2 **compare** with other studies and say why results may differ; 3 **explain** (mechanism, plausibility — briefly); 4 **limitations**, each with its likely direction; 5 **implications** for practice (cautious) and research; 6 **conclusion** — the same message and verb as the abstract and the title.

1 Answer First sentence answers the primary question <i>P1 p.8: the protocol 'did not lead to a relative 40% reduction'</i>	2 Compare With other studies - and WHY they differ <i>P1: vs CLASSIC and POINCARÉ-2, which fluids each targeted</i>	3 Explain Mechanism and plausibility, briefly <i>P2: four-phase fluid model in one paragraph</i>
4 Limitations Specific: mechanism, direction and a fix <i>P1: control-group contamination; stepped-wedge design as the fix</i>	5 Implications For practice and research - labelled, not speculative <i>P1: 'would probably be advantageous' = speculation without data</i>	6 Conclusion Same message as the abstract and the title <i>P1: conclusion matches the abstract almost word for word</i>

Figure 8.2: Discussion: six moves

Take-away: Answer first; limitations with their direction; conclusion = abstract = title.

Why the odds ratio shrank — say it in the Discussion

Report both the crude and the adjusted estimate and explain the gap: in FLUID-ICU (simulated) the crude OR of 2.49 fell to 1.62 after adjustment, because sicker patients received more fluid. Multiple imputation gave 1.70. Keep the verb at "associated".

Calibrated language: climb the verb ladder only as far as the design allows

was observed in (case series, survey) → *is associated with* (cohort, case-control) → *may contribute to* (a cautious explanation, in the Discussion) → *reduced / did not reduce* (a randomised trial) → *reduces / causes* (consistent large trials and a meta-analysis). No "trend" for $p = 0.42$; no "impact" or "effect" for an association.

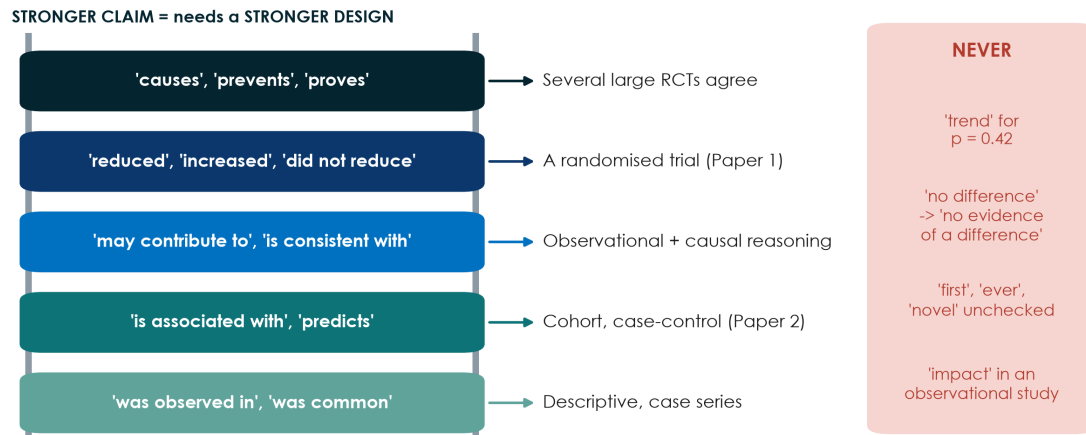


Figure 8.3: Calibrated language: match the verb to the design

Take-away: Climb the ladder only as far as your design allows.

The structured abstract and the title

Background, Methods, Results, Conclusions in ≤ 250 words; every result with a number and a 95% CI, copied from the Results — the abstract must contain no new numbers. The title states the question and the design, without sales words ("novel", "first", "impact").

Getting published

Choose the **reporting guideline** by design (CONSORT for trials, STROBE for observational studies, STARD, PRISMA, SPIRIT, CARE — all at equator-network.org). Choose a journal your readers read and that is honestly run: check DOAJ, the NLM Catalog and Think.Check.Submit. A journal that finds you by flattering e-mail deserves a second check.

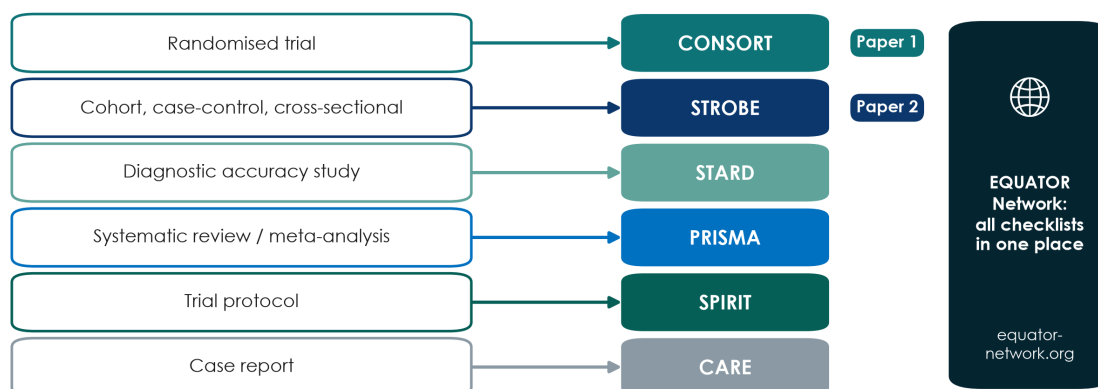


Figure 8.4: Reporting guidelines: choose by design

Take-away: Name your guideline in Part 4 — and use it as a writing checklist.

✉ From: editor.office.ijams@freemail.com

Dear Esteemed Doctor,

We read your wonderful work. Submit to the International Journal of Advanced Medical and All Sciences!

Peer review in 72 hours, publication in 1 week.

Impact Factor 7.8 (Global Index Factor).

Fee details after acceptance.

- 1 Flattering mass e-mail from a free address
- 2 Vague, huge scope; name mimics a real journal
- 3 Promises impossibly fast peer review
- 4 Fake or unknown 'impact' metrics
- 5 Fees hidden until after acceptance

✓ Check before you submit:
 Think.Check.Submit, DOAJ, MEDLINE (NLM Catalog) - or ask a librarian

Figure 8.5: Spot the predatory journal

Take-away: If a journal finds you by flattering e-mail, check it twice.

Peer review and responding to reviewers

Most papers are revised before acceptance. Thank the reviewers, answer **every** point, show exactly what changed (page and line), and disagree politely, with evidence. Never answer just "Done".

Be the reviewer: what may each design claim?

In the jigsaw, Groups 1–3 appraise Paper 1 (randomised pilot trial) and Groups 4–5 Paper 2 (propensity-matched cohort) with the same 10 questions (printed in full in Chapter 9). Randomisation earns the word "caused"; observation earns "associated".

Paper 1 - randomised trial (n = 48)	Paper 2 - matched cohort (n = 2516)
Chance decided the groups: known AND unknown factors balanced Day-5 balance: difference 13.2 ml/kg (95% CI -15.2 to 41.6), p = 0.42	Doctors decided the fluid: only MEASURED factors balanced Day-2 positive balance: 28-day death HR 2.13 (95% CI 1.48 to 3.06)
MAY say: 'did not reduce' / 'caused'	MAY say: 'is associated with'
But: too small to be sure - no 'trend', no 'proof of no effect'	NOT: 'impact', 'did not impact', 'show' - sicker patients get more fluid

Figure 8.6: What may each design claim?

Take-away: Randomisation earns the word "caused"; observation earns "associated".

8.4 Key words

Key word	In one line
Introduction funnel	Known → not known → aim, in three paragraphs
Six moves	Answer, compare, explain, limitations, implications, conclusion
Calibrated language	Verbs as strong as the design allows ("associated", not "caused")
Structured abstract	Background, Methods, Results, Conclusions; ≤ 250 words; numbers with CIs
Reporting guideline	CONSORT, STROBE, STARD, PRISMA, SPIRIT, CARE (EQUATOR Network)
Predatory journal	Takes fees without real peer review or indexing
Open access, licence	Free to read; what others may re-use (e.g. CC BY)
Preprint	Public manuscript posted before peer review
Point-by-point response	A table answering every reviewer comment with the change made
10-question checklist	The course's simplified appraisal tool ("Be the reviewer")

8.5 Exercises

Part	Exercise	When	Time
A	Writing exercises: fix the spin, verb ladder, results sentence, title from PICO	warm-up / homework	15 min
B	[L2] Manuscript workshop 2 — Introduction, Discussion, Abstract, Title	0:30	30 min
C	Getting published: spot the predatory journal; responding to reviewers	at home	25 min
D	Student appraisal pack — "Be the reviewer"	1:40	20 min

Part A — Writing exercises (section 8.1; warm-up or homework)

Use these to practise before or after Manuscript workshop 2: A2 and A3 can be done in class if a group finishes early; the rest are homework.

A1. Fix the spin

Rewrite each sentence so it says only what the data and the design allow.

#	Original sentence (from the papers)	Your rewrite
a	Paper 1 (p.10), primary outcome $p = 0.42$, difference 13.2 ml/kg (95% CI -15.2 to 41.6): the authors base the case for a larger trial on "the non-significant trend in fluid balance reduction".	
b	Paper 2 (p.4): "a positive fluid balance on the first ICU day did not impact the 28-day mortality".	
c	Paper 2 (title): "Impact of a cumulative positive fluid balance during the first three ICU days in patients with sepsis ..."	
d	Paper 1 (p.10): the protocol obtained "the lowest fluid balance on Day 5 ever published".	

A2. Climb the verb ladder

Choose the strongest verb each design allows: **was observed in** · **is associated with** · **may contribute to** · **reduced / did not reduce** · **causes**.

#	Study	Strongest honest verb
a	A case series of 12 ICU patients who developed pressure injuries under a new mattress	
b	A cohort of 3,000 septic patients: positive fluid balance and mortality, adjusted for severity	
c	A single randomised trial of 48 patients: restrictive protocol and day-5 fluid balance	
d	Five large randomised trials and a meta-analysis, all consistent	
e	A cross-sectional survey of hand-hygiene compliance on two wards	

A3. Write the results sentence (number + CI, no adjectives)

Write one sentence for each, in the style "In [whom], [outcome] was [effect] ([measure] [value], 95% CI [lower] to [upper])".

1. Paper 2, Day-3 cohort: 28-day mortality, positive vs negative fluid balance, HR 1.56, 95% CI 1.10 to 2.22, P = 0.012.
2. Paper 1: death by day 28, 6/24 (restrictive) vs 6/24 (control), relative risk 1.0, 95% CI 0.38 to 2.66.
3. Paper 2, Day-1 cohort: HR 1.17, 95% CI 0.85 to 1.60, P = 0.354.

A4. A title from your PICO

Formula: **[Design] of [Intervention / exposure] vs [Comparator] in [Population]: effect on / association with [Outcome]**. 10–16 words, design stated, no "impact", "novel", "first".

Our PICO: P _____ I/E _____ C _____ O _____

Working title: _____

Word count: ____ Design named? ☐ Calibrated verb (association for observational designs)? ☐

Part C — Getting published (section 8.2, at home)

C1. Spot the predatory journal

1. Circle the red flags in this (invented) invitation. How many can you find?

From: editorial.board.gjcmr@freemail.com **Subject:** Invitation to publish in our esteemed journal!!

Dear Respected Professor/Doctor,

Greetings! We have gone through your excellent article and are delighted to invite you to

submit to the *Global Journal of Clinical Medicine, Nursing, Engineering and Management Research*. We publish all areas of science.

Our journal has an Impact Factor of 9.2 (International Universal Index). We guarantee peer review within 48 hours and publication in 5 days. Processing charges will be communicated after acceptance. Special discount this week only!

Kindly submit your manuscript by replying to this e-mail.

2. Which journal would you trust? For each (invented) profile, write *Trust* / *Check further* / *Avoid* and one reason.

Journal profile	Your verdict	Reason
J1: Indexed in MEDLINE (confirmed in the NLM Catalog), editor names verifiable at universities, fee (APC) shown on the website, median 9 weeks to first decision, uses CONSORT/STROBE checklists.		
J2: Says "indexed in Google Scholar and Index Copernicus", promises acceptance in 72 hours, no fee information, editorial board photos without affiliations.		
J3: Open access, listed in DOAJ, small society journal, no impact factor yet, clear peer-review and retraction policy.		
J4: Its name differs by one word from a well-known journal; the website address is different from the real journal's; it e-mailed you first.		

3. Three checks you can always do: Think.Check.Submit checklist · DOAJ (open-access journals) · NLM Catalog ("currently indexed in MEDLINE") — and ask your librarian.

C2. Responding to reviewers

Imagine your group's study has been completed and submitted. Write a response row for each comment (thank, answer, say exactly what changed). One of them deserves a polite rebuttal.

#	Reviewer comment	Our response	Change made (page, line)
R1.1	"The title says your intervention <i>improves</i> outcomes, but this is an observational study."		
R1.2	"Please report the main result with a confidence interval, not only the p-value."		
R1.3	"Text and Table 2 give different numbers of patients analysed (212 vs 208)."		
R2.1	"Please add 1-year mortality." (Your protocol followed patients to day 28 only.)		

Tone checklist: ☐ thanked the reviewer ☐ answered every point ☐ said where the manuscript changed ☐ disagreed only with evidence, politely

Part D — Student appraisal pack: "Be the reviewer"

20 minutes, 1:40

What you are doing

Two published papers answer our running question — **fluids in septic shock** — with different designs:

	Paper	Design	Who appraises it
Paper 1	Boulet N, et al. <i>Crit Care</i> 2024;28:429 — restrictive strategy targeting non-resuscitative fluids (OPTIFLUID)	Randomised controlled pilot trial, 48 patients	Groups 1, 2, 3
Paper 2	Hyun D-G, et al. <i>Ann Intensive Care</i> 2023;13:105 — cumulative positive fluid balance in the first three ICU days	Propensity-score-matched cohort from a sepsis registry, 2516 patients	Groups 4, 5

You do **not** need to understand every statistic. Answer each question **Yes / No / Can't tell** and write **the page** where you found the answer. "Can't tell" is a good answer when the paper does not say.

Page numbers are the journal's own page numbers printed at the top of each page ("Page X of 13" for Paper 1, "Page X of 9" for Paper 2). Use the original (unannotated) PDF or printout.

Step by step (10 minutes of group work)

Minute	What to do
0–1	Choose a timekeeper and a speaker . Split into three pairs: pair A = questions 1–3, pair B = 4–6, pair C = 7–10.
1–7	Each pair reads only the pages listed for its questions (table below) and fills in its rows of the grid.
7–9	The group goes through all 10 questions together and agrees the answers.
9–10	Write your one-sentence verdict ("This paper shows / suggests ... but ...").

Then the plenary (8 minutes): each group reports for 1 minute on the questions the instructor assigns, and we compare what each design may claim.

Tip: read the papers' abstracts at home the evening before (Session 7 homework) — the 10 minutes then go much further. **After the course** you use the **same 10 questions** to review another group's FLUID-ICU manuscript (peer-review swap, 2 weeks).

Where to look

Question	Paper 1 (RCT) — pages	Paper 2 (cohort) — pages
1 Question	Abstract (p.1); last paragraph of Background (p.3)	Abstract (p.1); end of Background (p.2)
2 Design	Title (p.1); "Trial design and oversight" (p.3)	Title (p.1); Methods (pp.2–3)
3 Participants	"Patients" (p.3); flow chart Fig. 1 (p.5); Discussion limitations (p.10)	"Data source and patient selection" (p.2); Results first paragraph (p.3)
4 Comparison groups	"Randomization and blinding" (p.3); Table 1 (p.6)	"Propensity score matching" (pp.2–3); Table 1 (pp.5–6); text p.4
5 Bias	"Randomization and blinding" (p.3); Discussion (p.8)	Methods (pp.2–3); Discussion limitations (pp.7–8)
6 Outcome	"Outcomes" (p.4)	"Propensity score matching and outcomes" (p.3)
7 Numbers	"Primary outcome" (p.6); Table 3 (p.10)	Abstract (p.1); "Outcomes" (p.4)
8 Words vs design	Abstract conclusion (p.1); Discussion (pp.8, 10); Conclusions (p.11)	Title (p.1); Abstract conclusion (p.1); Discussion (pp.4, 7); Conclusions (p.8)
9 Limitations	Discussion (pp.8–10)	Discussion (pp.7–8)
10 Use	Your judgement — whole paper	Your judgement — whole paper

Answer grid

Group number: ____ Paper: 1 / 2 Speaker: _____

#	Question (simplified appraisal checklist)	Yes	No	Can't tell	Page	Notes (one line)
1	Question — Can you write the study's PICO in one line?	☐	☐	☐		P: ... I/E: ... C: ... O: ...
2	Design — What is the design, and is it the right one for the question?	☐	☐	☐		
3	Participants — Who was included, and who was left out? Would your patients have been included?	☐	☐	☐		
4	Comparison groups — Were the groups similar at the start (Table 1)? How was this achieved (randomisation / matching / adjustment)?	☐	☐	☐		
5	Bias — Who knew which group patients were in (blinding)? Could that change the outcome?	☐	☐	☐		
6	Outcome — Was there ONE pre-specified primary outcome, measured the same way in both groups?	☐	☐	☐		
7	Numbers — What is the main result as an effect size with 95% CI? Is the CI narrow or wide?	☐	☐	☐		Effect: 95% CI:
8	Words vs design — Do the conclusions match the design ("associated" vs "caused"; no "trend" for $p = 0.42$)?	☐	☐	☐		Words found:
9	Limitations — Are the main limitations stated honestly? Name one the authors missed.	☐	☐	☐		Missed:
10	Use — Would this change practice in your unit? Why / why not?	☐	☐	☐		

Our verdict in one sentence: _____

One thing we would copy into our own proposal: _____

Hints if you are stuck

- **Q1:** the PICO is usually in the last paragraph of the Introduction ("we aimed to ..."). Is the paper's outcome the same as in our running question (28-day mortality)?
- **Q4:** a randomised trial makes groups similar by chance; a cohort can only match or adjust for what was measured. Find the sentence that says which.
- **Q5:** who decided how much fluid each patient received — and did they know the group?
- **Q7:** look for "difference" or "HR" followed by brackets. Does the interval include 0 (for a

difference) or 1 (for a ratio)?

- **Q8:** search for the words *impact, show, trend, ever, regardless*.

8.6 Demo companion

Live demo: find a reporting checklist and check a journal (section 8.2, 3 minutes). You watch; write down the three web addresses.

Time	What you will see	What to note
0:00	EQUATOR Network home page: a search box and the main guidelines	Address: _____
0:30	CONSORT 2025: the checklist and the participant flow diagram (Paper 1 has one on page 5: 1,201 screened, 50 randomised)	
1:00	STROBE: checklists for cohort, case-control and cross-sectional studies — your FLUID-ICU paper follows the cohort version	
1:20	DOAJ: a journal record with licence, fees and type of peer review	Address: _____
1:50	NLM Catalog: "currently indexed for MEDLINE" — or not	
2:30	Think.Check.Submit: the checklist to use before you submit anywhere	Address: _____

Questions people ask. *Is Google Scholar an index?* No — it lists almost anything, including predatory journals. *Is a journal without an impact factor bad?* Not necessarily; fake "impact factors" from unknown agencies are the red flag. *Do I have to fill in the checklist?* Many journals require it at submission, with page numbers — use it while writing.

8.7 Manuscript workshop 2 and presentations

Part B — [L2] Manuscript workshop 2: complete the draft

30 minutes

Start from yesterday's file (FLUID-ICU_Q[n]_draft1). Today you add the **Introduction, Discussion, Abstract and Title** — and the draft is complete.

Roles and timing

Minute	Who	Task
0–8	2 people	Introduction — three paragraphs (known → not known → aim), ~350 words, references from Sessions 1–2 + Papers 1–2
0–15	3 people	Discussion — six moves (answer, compare, explain, limitations with direction, implications, conclusion), ~600 words
0–15	1 person	Number and verb checker: reads Methods + Results again; prepares the abstract numbers
15–23	all	Abstract — Background / Methods / Results / Conclusions, ≤ 250 words, numbers copied from the Results
23–27	all	Title (10–16 words, design stated, calibrated verb) + 3–6 key-words (MeSH, not already in the title)
27–30	checker	Verb check: every "cause", "impact", "effect of", "show" → "associated with"; every number identical in abstract and Results

Discussion — six moves checklist

- ☐ **1 Answer:** first sentence answers the question with the main number (adjusted estimate + 95% CI)
- ☐ **2 Compare:** with Paper 2 and at least one other study — and why results may differ
- ☐ **3 Explain:** one short paragraph on mechanism / plausibility
- ☐ **4 Limitations:** observational design (residual confounding — especially severity), simulated/teaching data, missing data, single time point... each with its likely *direction*
- ☐ **5 Implications:** for practice (cautious) and research (e.g. a randomised trial such as Paper 1)
- ☐ **6 Conclusion:** one sentence, same verb and message as the abstract and title

Group-specific reminders

- **Q1:** crude and adjusted ORs differ — explain confounding by indication in move 3/4.
- **Q2:** AKI may also *cause* fluid accumulation (reverse causation) — say so.
- **Q3:** ventilator-free days are skewed and include death as zero — explain what the measure means.
- **Q4:** compare complete-case and multiple-imputation estimates; say what the missing-data assumption (MAR) means for your conclusion.
- **Q5:** survivors only — those who died are excluded, so say who the result applies to.

Deliverable

A complete draft (~2,000 words, Template: Templates/Manuscript_Template_IMRaD.docx) saved as FLUID-ICU_Q[n]_draft2. After the course: peer-review swap (2 weeks) using the 10-question checklist (Part D) and a response letter (Templates/Response_to_Reviewers_Template.docx).

Your presentation: timing card

Slide	Content	Time	Who presents
1	Title & team	0:20	
2	Problem & PICO	1:00	
3	Design & participants	1:15	
4	Data & analysis plan	1:15	
5	Ethics & feasibility	1:00	
6	Publication plan & impact	1:10	

Green card at 4:00 (2 minutes left), yellow at 5:00 (1 minute left), red at 6:00 (stop). Full guide in Chapter 11.

Our 1-minute manuscript pitch:

"In 600 simulated ICU patients with sepsis, we asked whether _____ was associated with _____. The main result was _____ (95% CI _____ to _____). _____"

8.8 After the session — homework

- ☐ **By 23:59 today:** the lead author submits the draft manuscript with the STROBE checklist (GroupN_Manuscript_QN_draft.docx) — required.
- ☐ **Within 1 week:** review another group's draft with the 10-question checklist and the writing checks (peer-review swap: Group 1 → 2 → 3 → 4 → 5 → 1) and send the report.
- ☐ **Within 2 weeks:** send the revised manuscript (tracked changes) and the response letter (Templates/Response_to_Reviewers_Template.docx), and the revised written proposal.
- ☐ Do Part C (predatory journal, responding to reviewers) of the exercises if not done in class.
- ☐ Take your proposal to your research office or ethics committee.

8.9 Extra practice

Extra practice 8.1 — Calibrate the verb

Rewrite each sentence so that the verb matches the design.

1. Cohort: "A positive fluid balance increases mortality in sepsis."

2. Pilot RCT, $p = 0.42$ for the primary outcome: "The restrictive protocol showed a trend towards a lower fluid balance."

Extra practice 8.2 — Order the Discussion

Number the six moves (1–6) for these sentences from a FLUID-ICU (simulated) Discussion.

Sentence	Move
"Residual confounding by severity may explain part of the remaining association, which would bias it away from no effect."	
"A positive fluid balance by day 3 was associated with higher 28-day mortality (adjusted OR 1.62, 95% CI 1.02 to 2.59)."	
"This agrees with a large registry cohort, although our adjusted estimate is smaller."	
"A randomised trial is needed before practice changes."	
"Fluid accumulation may cause tissue oedema and organ dysfunction."	
"In this simulated cohort, a positive fluid balance was associated with higher 28-day mortality after adjustment for severity."	

Reviewer: "The authors should state that fluid restriction will save lives." Write a polite response (your study is an observational cohort).

Name three red flags of a predatory journal, and two places where you can check a journal.

Templates and tools

The course comes with Word templates and practical tools for every step from question to publication. They are in the course folder (Templates/ and Resources/), each .docx with a matching .pdf for printing. This chapter tells you, for each one, **what it is for, when you use it, where to find it** and **the headings you will fill**, so that your group can prepare before the session. The most-used checklists are printed in full.

Tips for the Word templates. Replace every [grey placeholder] with your own details (Ctrl + H → find [steps through them). To delete all the grey guidance at once: Home → Styles → right-click **Guidance** → Select All → Delete. For a consent form, cover letter or manuscript, remove the Neudata header and use your institution's letterhead or the journal's format.

9.1 Your toolkit at a glance

Resource	File
Research proposal template	Templates/Research_Proposal_Template.docx
Study protocol template	Templates/Study_Protocol_Template.docx
Informed consent template	Templates/Informed_Consent_Template.docx
CRF + data dictionary template	Templates/CRF_Template.docx
Literature review worksheet	Templates/Literature_Review_Worksheet.docx
Statistical analysis plan template (+ worked example Data/SAP_FLUID-ICU.md)	Templates/Statistical_Analysis_Plan_Template.docx
Manuscript (IMRaD) template	Templates/Manuscript_Template_IMRaD.docx
Cover letter and response-to-reviewers templates	Templates/Cover_Letter_Template.docx, Templates/Response_to_Reviewers_Template.docx
Conference abstract & poster guide	Templates/Conference_Abstract_Template.docx
Critical appraisal checklist (10 questions)	Resources/Critical_Appraisal_Checklist.pdf
Risk-of-bias tool chooser	Resources/Risk_of_Bias_Tool_Chooser.pdf
Writing & submission checklist	Resources/Writing_and_Submission_Checklist.pdf
Sample ethics application, sample grant concept note (fictional)	Resources/
2x2 and sample-size calculators	Resources/tools/
Course dataset, data dictionary, re-sults packs	Data/

Resource	File
English–Vietnamese glossary, further reading	References/

9.2 Planning the study

9.2.1 Research proposal template

File	Templates/Research_Proposal_Template.docx
For	Your group's proposal and publication plan (Group product 1), about 6–8 pages when filled
When	Part 1 in Sessions 1–2 · Part 2 in Sessions 3–4 · Part 3 in Sessions 5–6 · Part 4 in Session 7 · presented in Session 8

Headings to fill. Every section has guidance, a running-example illustration and a placeholder.

Part	Sections
1 — Ask (S1–S2)	1.1 Working title · 1.2 Problem statement and rationale · 1.3 PICO · 1.4 Research question · 1.5 FINER check · 1.6 Aim, objectives and hypothesis · 1.7 Five key references
2 — Design (S3–S4)	2.1 Study design and justification · 2.2 Setting · 2.3 Study population · 2.4 Intervention or exposure, and comparator · 2.5 Outcomes · 2.6 Main threats of bias and how they are reduced
3 — Measure (S5–S6)	3.1 Variables table · 3.2 Mini-CRF (6–10 items) · 3.3 Data dictionary (key rows) · 3.4 Data management and quality plan · 3.5 Sample size statement · 3.6 Analysis plan in plain language · 3.7 Missing-data plan
4 — Protect and publish (S7)	4.1 Ethics and consent · 4.2 Registration · 4.3 Confidentiality · 4.4 Feasibility (Gantt chart, budget headings, resources and recruitment) · 4.5 Publication plan (target journal and reporting guideline, authorship and CRediT roles, working title, structured-abstract skeleton)

9.2.2 Literature review worksheet

File	Templates/Literature_Review_Worksheet.docx
For	Turning the PICO into search terms and keeping track of what you found
When	Session 2 (2.1 finding the evidence; 2.4 literature matrix; Proposal Part 1)

Steps to fill: Step 1 — from PICO to search terms (synonyms, MeSH, Boolean; worked example included) · Step 2 — search log (database, date, string, filters, hits) · Step 3 — evidence matrix (two worked rows from the course papers) · Step 4 — what is known, and the gap · Step 5 — five key references (Vancouver style).

9.2.3 Case report form (CRF) and data dictionary

File	Templates/CRF_Template.docx
For	A data-collection form built on good CRF design, with example items for the running example; a data dictionary follows on a landscape page
When	Session 5 (5.1 Good data by design; Proposal Part 3)

Forms in the template: Form 1 — eligibility and enrolment · Form 2 — demographics and baseline (ICU admission) · Form 3 — severity in the first 24 hours (SOFA and lactate) · Form 4 — daily fluid input and output (days 1–5) · Form 5 — outcome at day 28 · sign-off · data dictionary.

9.2.4 Statistical analysis plan (SAP)

File	Templates/Statistical_Analysis_Plan_Template.docx; worked example for the course dataset: Data/SAP_FLUID-ICU.md
For	The analysis decided before data lock
When	Session 6 (6.4; Proposal Part 3)

14 sections: 1 study summary and objectives · 2 outcomes and their definitions · 3 analysis populations · 4 descriptive analysis (Table 1) · 5 primary analysis · 6 secondary analyses · 7 confounders and adjustment · 8 subgroups and interaction (pre-specified) · 9 multiple comparisons · 10 missing data · 11 sample size · 12 software · 13 table and figure shells · 14 version history and sign-off before data lock.

9.2.5 Study protocol

File	Templates/Study_Protocol_Template.docx
For	A SPIRIT-style protocol for an observational study or a simple trial — the next step after your proposal
When	Session 7 (7.1 Protocol, ethics and consent); take-home

Sections: administrative information (version history, signature, abbreviations) · 1 synopsis · 2 background and rationale · 3 objectives and hypotheses · 4 study design · 5 participants (setting, eligibility, screening and recruitment) · 6 interventions or exposures · 7 outcomes · 8 participant timeline (schedule of assessments) · 9 sample size · 10 allocation and blinding (trials) · 11 data collection and management · 12 statistical methods · 13 monitoring and safety · 14 ethics and consent · 15 confidentiality and data protection · 16 dissemination, authorship and data sharing · 17 funding and conflicts of interest · 18 references · 19 appendices.

9.2.6 Informed consent

File	Templates/Informed_Consent_Template.docx
For	A plain-language participant information sheet and consent form
When	Session 7 (7.1); take-home

Part A — information sheet (13 questions): invitation · why is this study being done? · why have I been asked? · do I have to take part? · what will happen if I take part? · possible risks or discomforts · possible benefits · what if something goes wrong or new information becomes available? · will my information be kept confidential? · what will happen to the results? · will I be paid, or will it cost me anything? · who has reviewed this study? · who can I contact? — plus guidance on research in emergencies and intensive care. **Part B — consent form:** participant · person taking consent · impartial witness · legally authorised representative · continuing (deferred) consent.

9.3 Writing and publishing

9.3.1 Manuscript template (IMRaD)

File	Templates/Manuscript_Template_IMRaD.docx
For	Your group's FLUID-ICU manuscript (Group product 2), about 2,000 words
When	Session 7 (Methods + Results) and Session 8 (the rest of the draft)

Headings: title page · abstract (structured) and keywords · Introduction (three-paragraph funnel) · Methods (study design and setting; participants; intervention/exposure and comparator; outcomes; data collection; sample size; statistical analysis; ethics) · Results (participants; primary outcome; secondary outcomes; harms) · Discussion (six moves) · Conclusion · Declarations (ethics, consent, registration, funding, competing interests, data availability, AI use, CRediT grid) · References · Tables and figures (Table 1 shell, Table 2 shell, flow-diagram shell) · reporting-guideline reminder.

9.3.2 Cover letter, response to reviewers, conference abstract

File	For	Key parts	When
Templates/Cover_Letter_Template.docx	A one-page letter to the editor	The question and why it matters; what you did and the main finding (calibrated); why this journal; standard statements (originality, ethics, registration, reporting checklist, competing interests); optional suggested reviewers	Session 8 (8.2)
Templates/Response_to_Reviewers_Template.docx	Replying to peer review	Letter to the editor; worked fictional rows (two agreements, one polite disagreement); point-by-point tables (comment / response / change made with page and line) for the editor and two reviewers	Session 8; peer-review swap after the course
Templates/Conference_Abstract_Template.docx	A 250–300-word conference abstract and a poster	Word budget: Background 40–50, Methods 70–80, Results 90–110, Conclusions 30–40 words; poster layout guide (three columns, one key message, title ≥ 85 pt, body ≥ 24 pt)	Session 8 (8.2)

9.4 Appraisal and reporting tools

9.4.1 Critical appraisal checklist — “Be the reviewer” (10 questions)

Resources/Critical_Appraisal_Checklist.pdf · introduced in Session 4 (4.3), used on Papers 1–2 in Session 8 (8.3) and for the manuscript peer-review swap.

How to use it. Read the abstract, then Table 1, the primary outcome result, and the first and last paragraphs of the Discussion — you do not need to read every word. For each question tick **Yes**, **No** or **Can’t tell**, and write **where** in the paper you found the answer. “Can’t tell” is a valid answer: it often means poor reporting. There is no total score; questions 2, 4, 5 and 7 matter most for deciding whether to trust the result. Finish with a one-sentence verdict.

Paper: _____ Reviewer / group: _____

Design (from Q2): ☐ RCT ☐ Cohort ☐ Case-control ☐ Cross-sectional ☐ Other

#	Question	Yes	No	Can’t tell	Where in the paper
1	Question — Can you write the study’s PICO in one line?	☐	☐	☐	
2	Design — What is the design, and is it the right one for the question?	☐	☐	☐	
3	Participants — Who was included, and who was left out? Would your patients have been included?	☐	☐	☐	
4	Comparison groups — Were the groups similar at the start (Table 1)? How was this achieved (randomisation / matching / adjustment)?	☐	☐	☐	
5	Bias — Who knew which group patients were in (blinding)? Could that change the outcome?	☐	☐	☐	
6	Outcome — Was there ONE pre-specified primary outcome, measured the same way in both groups?	☐	☐	☐	
7	Numbers — What is the main result as an effect size with 95% CI? Is the CI narrow or wide?	☐	☐	☐	
8	Words vs design — Do the conclusions match the design (“associated” vs “caused”; no “trend” for $p = 0.42$)?	☐	☐	☐	
9	Limitations — Are the main limitations stated honestly? Name one the authors missed.	☐	☐	☐	
10	Use — Would this change practice in your unit? Why / why not?	☐	☐	☐	

Main strength	Main weakness	One-sentence verdict

9.4.2 Risk-of-bias tool chooser

Resources/Risk_of_Bias_Tool_Chooser.pdf · Session 4 (4.3) and Session 8 (8.3).

How to choose. Step 1: identify the design (the decision tree from Session 3). Step 2: pick the tool in the table. Appraise each outcome you care about, not the whole paper at once. Step 3: judge each domain (low / some concerns / high risk), then ask whether the conclusions are supported by the data. In the course we use the simplified 10-question checklist; use these full tools when you appraise evidence formally (e.g. for a systematic review).

Study design	Tool	What it checks	Free source
Randomised controlled trial	RoB 2	Bias from randomisation, deviations from intended interventions, missing outcome data, outcome measurement and selection of the reported result	Cochrane — riskofbias.info
Non-randomised study of an intervention	ROBINS-I	Confounding, selection of participants, classification of interventions, deviations, missing data, measurement and reporting — judged against an ideal "target trial"	Cochrane — riskofbias.info
Cohort and case-control (exposures)	Newcastle–Ottawa Scale or ROBINS-E	NOS: stars for selection of groups, comparability (confounding) and outcome/exposure ascertainment. ROBINS-E: the ROBINS-I domains, for exposures	ohri.ca ; riskofbias.info
Diagnostic accuracy study	QUADAS-2	Patient selection, index test, reference standard, flow and timing — risk of bias AND applicability	bristol.ac.uk (QUADAS)
Systematic review / meta-analysis	AMSTAR 2	16 items: protocol registered in advance, comprehensive search, risk-of-bias assessment of included studies, appropriate meta-analysis, publication bias	amstar.ca
Qualitative study	CASP Qualitative Checklist	10 questions: aims, method and design, recruitment, data collection, researcher relationship, ethics, rigour of analysis, findings, value	casp-uk.net

Other designs: prediction models → PROBAST; cross-sectional studies and case series → JBI critical appraisal checklists (jbi.global).

Reporting guideline or risk-of-bias tool?

Reporting guidelines (CONSORT, STROBE ...) tell authors what to **report**. Risk-of-bias tools judge how **trustworthy** the result is. They are not the same thing.

9.4.3 Writing and submission checklist

Resources/Writing_and_Submission_Checklist.pdf · Sessions 7–8; take-home. Tick each item before you press "Submit". Ask a co-author who did not write the section to check it against the tables.

Title and keywords

- ☐ Title states population, intervention/exposure, outcome and design; no abbreviations; within the limit
- ☐ 3–6 keywords (MeSH where possible) that do not just repeat the title

Abstract

- ☐ Structured as the journal requires; within the word limit
- ☐ Primary outcome reported first, with effect size and 95% CI; numbers identical to the Results
- ☐ Conclusion matches the design (no spin, no causal words for observational data); registration number given

Introduction and Methods

- ☐ Introduction is a funnel: known → unknown → aim; ends with the aim or hypothesis
- ☐ Methods allow replication: design, setting, dates, eligibility, intervention/exposure, outcomes
- ☐ Sample size calculation, statistical methods, handling of missing data and software are described
- ☐ Ethics approval (committee, number) and consent procedure stated

Results and numbers

- ☐ Participant flow first (flow diagram with numbers and reasons at each step)
- ☐ Every percentage has its numerator and denominator; denominators add up
- ☐ Effect sizes with 95% CIs; exact p-values ($p = 0.042$, not $p < 0.05$; $p < 0.001$ allowed)
- ☐ Decimal places consistent; units given; each number in one place (text OR table)
- ☐ No interpretation in the Results; results of all pre-specified outcomes reported

Tables and figures

- ☐ Each stands alone: informative title, units, n per column, footnotes for abbreviations
- ☐ Numbers in tables match the text and the abstract
- ☐ Figures show the data (no 3-D, no truncated axes); resolution and format as required

Discussion and conclusion

- ☐ Six moves: answer, compare, explain, limitations, implications, conclusion
- ☐ Limitations are specific, with their likely direction of bias
- ☐ Language calibrated: "associated with" for observational studies; no "trend" for $p = 0.42$

Reporting guideline, declarations and references

- ☐ Correct guideline chosen (CONSORT, STROBE, STARD, PRISMA, other via EQUATOR); checklist completed with page numbers; flow diagram included
- ☐ Ethics, consent, registration, funding, competing interests (ICMJE forms) completed
- ☐ Data availability statement; use of AI disclosed; CRediT author contributions
- ☐ All authors meet the ICMJE criteria and approved the final version; ORCID iDs added
- ☐ Journal style used consistently; every citation in the list and vice versa; each reference checked against the original; no retracted papers; DOIs added

Journal fit, cover letter and submission package

- ☐ Scope, readership and article type fit; word, table and figure limits respected
- ☐ Journal is indexed (e.g. PubMed/MEDLINE, DOAJ) and not predatory; fees (APC) and licence known
- ☐ One-page cover letter to the named editor: question, main finding, why this journal
- ☐ Statements of originality, no simultaneous submission, preprint disclosed
- ☐ Title page, blinded manuscript (if required), figures, supplementary files, checklist, suggested reviewers

After acceptance: proofs

- ☐ Proofs checked within the deadline: author names, affiliations, numbers, tables, figures
- ☐ Only corrections, no new content; licence and open-access choices confirmed
- ☐ Paper shared: institution, repository (as allowed), colleagues and patients; registry updated with results

9.5 Calculators and worked examples

Tool	File	What it does	Session
2x2 calculator	Resources/tools/2x2_calculator.xlsx	Sheet <i>Association</i> : risks, RR, RD, NNT/NNH, odds and OR from a, b, c, d, each with a plain-words line (case-control warning: read only the OR). Sheet <i>Diagnostic</i> : sensitivity, specificity, PPV, NPV, and the PPV in a setting with another prevalence. Only the yellow cells can be edited.	S4
Sample-size calculator	Resources/tools/sample_size_calculator.xlsx	Sample size for two proportions (sheet 2_Proportions) and two means (sheet 3_Means), with drop-out and a ready-made sentence for your proposal	S6
Sample ethics application (fictional)	Resources/Sample_Ethics_Application.pdf	A completed application for a prospective cohort (FLUID-BAL: fluid balance and 28-day mortality): sections A–L (identification, lay summary, background, objectives, design and methods, risks and benefits, consent, data protection, conflicts and funding, budget, dissemination, declaration), with a waiver/deferred-consent justification and "why this is a good answer" notes	S7
Sample grant concept note (fictional)	Resources/Sample_Grant_Concept_Note.pdf	A completed 2-page concept note for the same study, with funder tips in the margin	S7
The two course papers	Papers/	Paper 1 (RCT) and Paper 2 (cohort), original PDFs	S2–S8

9.5.1 The two course papers

	Paper 1	Paper 2
Citation	Boulet N, et al. <i>Crit Care</i> 2024;28:429	Hyun D-G, et al. <i>Ann Intensive Care</i> 2023;13:105
Design	Single-blind, multicentre, parallel-group pilot RCT ; 3 ICUs; 48 patients analysed (24 per arm); registered NCT04947904	Propensity-score-matched cohort from a prospective multicentre sepsis registry (20 hospitals); 483, 373 and 392 matched pairs for the day-1, 2 and 3 cohorts
Primary outcome	Cumulative fluid balance (ml/kg) at day 5; 28-day mortality secondary	28-day mortality
Main result	Median 6.9 vs 35.0 ml/kg (difference 13.2 ml/kg, 95% CI –15.2 to 41.6; p = 0.42); mortality similar	Positive balance associated with higher 28-day mortality at day 2 (HR 2.13, 1.48–3.06) and day 3 (HR 1.56, 1.10–2.22), not at day 1 (HR 1.17, 0.85–1.60)
Reporting guideline	CONSORT (not named in the paper)	STROBE (not named in the paper)
Risk-of-bias tool	RoB 2	Newcastle–Ottawa (or ROBINS-E)

CHAPTER 10

The course dataset — FLUID-ICU

Simulated data — not real patients

FLUID-ICU is an **invented** study. Every value was produced by a computer program with a fixed random seed. It does not describe any real patient, hospital, ICU or country. The data were built so that the classic lessons show up — confounding by indication, missing data, a skewed outcome and an interaction worth discussing. Any manuscript written from it must say that the data are simulated, and must never be submitted as real research.

10.1 What FLUID-ICU is

FLUID-ICU is a simulated **multicentre prospective cohort** of **620 adults with sepsis** admitted to four ICUs ("ICU A" to "ICU D") from January 2023 to December 2024 (ICU A n = 191; ICU B n = 164; ICU C n = 147; ICU D n = 118). It carries the course's running example into Weekends 3–4, in its observational form:

In adults with sepsis in the ICU, is a positive cumulative fluid balance by day 3 associated with higher 28-day mortality?

The data contain demographics; comorbidities (diabetes, chronic kidney disease, heart failure, hypertension); infection source; severity (SOFA, APACHE II, lactate, mean arterial pressure, vasopressor, mechanical ventilation); daily fluid in and out on days 1–3 and weight; the derived cumulative fluid balance at day 3 (ml/kg) and positive balance (yes/no); acute kidney injury (KDIGO); renal replacement therapy; ventilator-free days; ICU length of stay; and 28-day mortality with the time to death or censoring. Clinical ranges follow the two course papers.

You never need to run any software. The trainers run the analyses live in R in Sessions 5–6, and each group receives a **results pack** with every number, table and figure it needs.

File (in)	What it is	Used in
fluid_icu_raw.csv / .xlsx	The data as the four ICUs "sent" it: 623 rows with realistic errors	S5 cleaning demo
fluid_icu_clean.csv / .rds	Analysis-ready data: 620 patients, 32 variables	All analyses
data_dictionary.md	Every variable: label, type, units or codes, allowed range, missing code, notes	S5
cleaning_log.md	Every problem found in the raw file and what was done	S5
SAP_FLUID-ICU.md	The statistical analysis plan, written "before the analysis"	S6, S7
results_packs/Q1 ... Q5	One pack per group question (also as Results_Pack_QN.docx)	S5–S8
R/	The analysis scripts (optional, for those who know R)	Demos

10.2 Who is in the data: from 623 rows to 527 complete cases

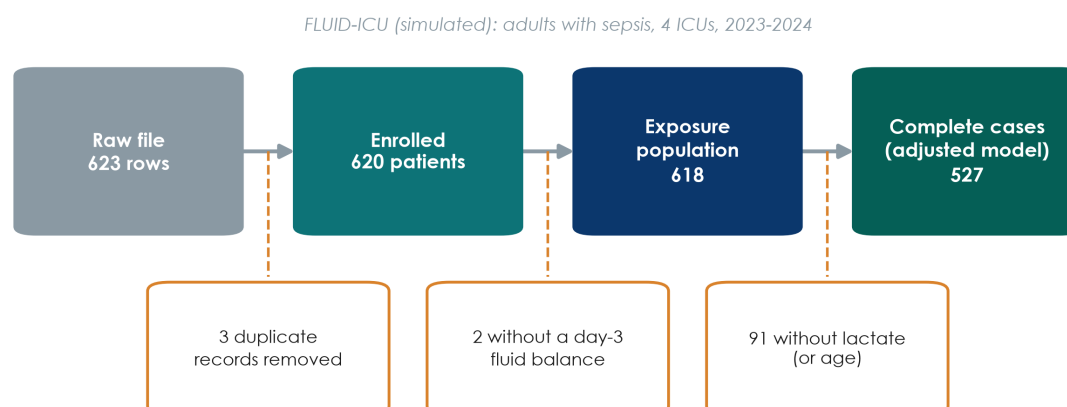


Figure 10.1: Participant flow in FLUID-ICU (simulated)

Take-away: 623 raw rows → 620 patients (3 duplicates removed) → 618 with a day-3 fluid balance → 527 complete cases for the adjusted models (91 without lactate or age).

Step	Patients	Why the number changes
Raw file	623 rows	3 patients were entered twice (identical rows)
Enrolled	620	All analyses of description; Q4
Exposure analysis	618	2 patients without a day-3 balance (weight or output missing after cleaning)
Complete-case adjusted models	527	91 without lactate (or age)
Survivors at day 28 (Q5)	439	Patients who died are excluded from the length-of-stay question

10.3 The data dictionary

Pre-specified confounders (SAP): **age, SOFA, lactate, vasopressor, mechanical ventilation, chronic kidney disease**. Pre-specified effect modifier: **heart failure**. Main exposure: `cum_fluid_balance_d3_ml_kg` and `positive_balance_d3`. Primary outcome: `died_28d` (with `time_to_event_28d`). DERIVED variables are calculated by the cleaning script and do not exist in the raw file.

Variable	Label	Type	Units / codes	Allowed range
<code>study_id</code>	Study identifier	Character (ID)	FLU-001 ... FLU-620	Unique
<code>site</code>	Recruiting ICU	Categorical (4 levels)	ICU A / ICU B / ICU C / ICU D	4 levels
<code>admission_date</code>	Date of ICU admission (enrolment)	Date	YYYY-MM-DD	2023-01-01 to 2024-12-31
<code>age</code>	Age	Numeric (integer)	Years	18 to 110

Variable	Label	Type	Units / codes	Allowed range
sex	Sex	Categorical (2 levels)	Female / Male	2 levels
weight_kg	Body weight on admission	Numeric	kg (1 decimal)	30 to 250
diabetes	Diabetes mellitus	Binary	No / Yes	No, Yes
ckd	Chronic kidney disease	Binary	No / Yes	No, Yes
heart_failure	Chronic heart failure	Binary	No / Yes	No, Yes
hypertension	Hypertension	Binary	No / Yes	No, Yes
infection_source	Source of infection	Categorical (5 levels)	Respiratory / Urinary / Abdominal / Skin/soft tissue / Other	5 levels
sofa	SOFA score on ICU admission	Numeric (integer score)	Points	0 to 24
apache_ii	APACHE II score (first 24 h)	Numeric (integer score)	Points	0 to 71
lactate_mmol_l	Blood lactate on admission	Numeric	mmol/L (1 decimal)	0.1 to 30
map_mmhg	Mean arterial pressure on admission	Numeric (integer)	mmHg	20 to 200
vasopressor	Vasopressor on admission	Binary	No / Yes	No, Yes
mech_vent	Invasive mechanical ventilation on admission	Binary	No / Yes	No, Yes
fluid_in_d1_ml	Total fluid input, ICU day 1	Numeric (integer)	ml	0 to 20000
fluid_out_d1_ml	Total fluid output, ICU day 1	Numeric (integer)	ml	0 to 20000
fluid_in_d2_ml	Total fluid input, ICU day 2	Numeric (integer)	ml	0 to 20000
fluid_out_d2_ml	Total fluid output, ICU day 2	Numeric (integer)	ml	0 to 20000
fluid_in_d3_ml	Total fluid input, ICU day 3	Numeric (integer)	ml	0 to 20000
fluid_out_d3_ml	Total fluid output, ICU day 3	Numeric (integer)	ml	0 to 20000
cum_fluid_balance_d3_ml_kg	Cumulative fluid balance, ICU days 1-3	Numeric (DERIVED)	ml/kg (1 decimal)	-150 to 250
positive_balance_d3	Positive cumulative fluid balance at day 3	Binary (DERIVED)	No / Yes	No, Yes
aki_stage	Acute kidney injury, maximum KDIGO stage, ICU days 1-7	Ordinal (integer)	0 = none, 1, 2, 3	0 to 3
aki_any	Any acute kidney injury (KDIGO stage ≥ 1)	Binary (DERIVED)	No / Yes	No, Yes
rrt	Renal replacement therapy by day 28	Binary	No / Yes	No, Yes
vent_free_days_28	Ventilator-free days to day 28	Numeric (integer)	Days	0 to 28
icu_los_days	ICU length of stay	Numeric	Days (1 decimal)	> 0
died_28d	Death from any cause by day 28	Binary	No / Yes	No, Yes
time_to_event_28d	Time from ICU admission to death or censoring	Numeric (integer)	Days	1 to 28

10.4 How the raw data were cleaned

The raw file is never edited. Every change is made by the script `R/01_clean.R` and written in the cleaning log; the script ends by checking that its output is identical to the reference clean file.

Step	Problem found	How many	What we did	Why
1	Column names typed like a spreadsheet: spaces, brackets, capitals (Weight (kg), MAP (mmHg), VFD 28)	29 columns	Converted to short lower-case names with <code>clean_names()</code> ; renamed 4 (e.g. <code>map_mm_hg</code> to <code>map_mmhg</code> , <code>vfd_28</code> to <code>vent_free_days_28</code>)	Names without spaces are easy to type and the same in every script
2	Duplicate records: the same patient entered twice	3 IDs (FLU-139, FLU-274, FLU-574): 623 rows for 620 patients	Checked that the two copies were identical in every column, then kept one copy	Each patient must be counted once
3	ICU name typed in different ways (<code>icu a</code> , <code>ICU-A</code> , <code>ICUA</code> , trailing space)	14 rows	Standardised to <code>ICU A</code> , <code>ICU B</code> , <code>ICU C</code> , <code>ICU D</code>	Otherwise R sees 11 "different" ICUs
4	Admission date in three formats: <code>2023-03-14</code> , <code>14/03/2023</code> , <code>14-Mar-2023</code>	365 / 164 / 94 rows	Read each format explicitly and converted all to one date (YYYY-MM-DD); 0 dates failed	Mixed formats are read wrongly or not at all; 03/04 could be March or April
5	Sex coded 9 ways (Male, M, m, male, Female, F, f, female, Female)	200 rows not Male/Female	Recoded to Female / Male	One variable, one coding
6	Yes/No variables mixed (Yes, No, Y, N, yes, no, 1, 0 (ICU B sends 1/0))	8 variables; about 1,900 cells not Yes/No	Recoded to No / Yes; checked that no value was lost	1 and Yes must mean the same thing
7	All columns read as text	all	Converted numbers to numbers (whole numbers as integers)	Needed for calculations
8	Missing-value codes used as if they were numbers: lactate 999 and -99, APACHE II -99, MAP 999	Lactate 21 (12 x 999, 9 x -99), APACHE II 6, MAP 4	Set to missing (NA)	A lactate of 999 would wreck the mean; missing is missing
9	Units: ICU C reported lactate in mg/dL , the other ICUs in mmol/L (ICU C median 23.8 vs about 2.5 elsewhere)	126 values (all ICU C lactates)	Divided ICU C values by 9.01 (1 mmol/L = 9.01 mg/dL), 1 decimal; ICU C median became 2.65	One unit for one variable; found by comparing medians by ICU
10	Impossible values (outside the allowed range in the data dictionary)	3: FLU-529 age 210; FLU-277 weight 7 kg; FLU-282 day-2 output -2990 ml	Set to missing and raised a query with the site. The patient stays in the study	We cannot guess the true value (21? 71?); deleting the patient would bias the sample
10b	Range checks on all other variables (SOFA, APACHE II, lactate, MAP, AKI stage, VFD, ICU stay, follow-up time; survivors followed to day 28)	0 problems	None	Documents that the checks were done

Step	Problem found	How many	What we did	Why
11	Infection source as free text	5 categories	Fixed category order (Respiratory, Urinary, Abdominal, Skin/soft tissue, Other)	Same order in every table
12	Derived variables needed (not in the raw file)	3 new variables	Computed <code>cum_fluid_balance_d3_ml_kg = (fluid in d1-3 - fluid out d1-3) / weight</code> ; <code>positive_balance_d3 (> 0 ml/kg)</code> ; <code>aki_any (KDIGO stage >= 1)</code>	Derived values are calculated by the script, never typed by hand

10.5 What the data show (simulated): the numbers you will meet

All numbers below come from `Data/key_results.md`. Use only the numbers in your own results pack in your manuscript.

	Value
Age, mean (SD)	62.6 (15.2) years
Male sex	376/620 (60.6%)
SOFA score, median (IQR)	7 (5 to 9)
Positive fluid balance by day 3	373/618 (60.4%)
Cumulative balance at day 3, median (IQR)	8.8 (−10.3 to 26.2) ml/kg
Lactate missing	93/620 (15.0%)
28-day mortality	179/620 (28.9%)
AKI (KDIGO ≥ 1); renal replacement therapy	302/620 (48.7%); 79/620 (12.7%)

Table 1 (negative vs positive balance). The positive-balance group was sicker: SOFA median 6 (4 to 8) vs 8 (6 to 10); vasopressor 61/245 (24.9%) vs 205/373 (55.0%); mechanical ventilation 97/245 (39.6%) vs 216/373 (57.9%); lactate median 2.2 (1.4 to 3.0) vs 2.9 (1.9 to 4.0) mmol/L. Heart failure was more common in the negative-balance group, 54/245 (22.0%) vs 43/373 (11.5%) — clinicians give less fluid in heart failure.

Question	Crude result	Adjusted result
Q1 Positive balance and 28-day death	134/373 (35.9%) vs 45/245 (18.4%); OR 2.49 (95% CI 1.70 to 3.70)	OR 1.62 (1.02 to 2.59), n = 527; Cox HR 1.46 (1.00 to 2.13)
Q2 Balance and AKI (per 10 ml/kg)	OR 1.37 (1.27 to 1.47)	OR 1.27 (1.16 to 1.40)
Q3 Balance and ventilator-free days (per 10 ml/kg)	−1.48 days (−1.84 to −1.13)	−0.61 days (−1.00 to −0.21)
Q4 Lactate and 28-day death (per 1 mmol/L)	Complete case 1.38 (1.23 to 1.55)	Complete case 1.39 (1.24 to 1.57), n = 527 vs multiple imputation 1.45 (1.29 to 1.64), n = 620
Q5 Positive balance and ICU stay in survivors	Median 7.5 vs 4.8 days; ratio of geometric means 1.50 (1.34 to 1.68)	Ratio 1.24 (1.10 to 1.39), n = 366

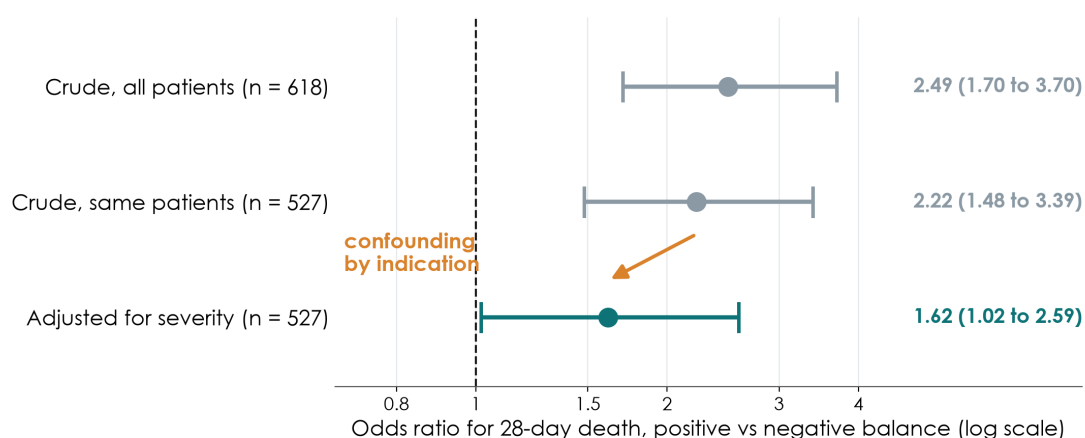


Figure 10.2: Crude vs adjusted: confounding by indication made visible (simulated data)

Take-away: Crude OR 2.49 shrinks to 1.62 after adjusting for severity: much of the crude difference was confounding by indication.

10.6 The five group questions

Questions are assigned in Session 5. By default Group *N* takes Question *N*.

Q	Research question	Exposure	Outcome	Main analysis in the results pack
1	Is a positive fluid balance by day 3 associated with higher 28-day mortality? (the primary question)	Positive cumulative fluid balance by day 3 (yes/no)	28-day mortality	Crude and adjusted logistic regression (OR); Kaplan–Meier curves and Cox model (HR)
2	Is cumulative fluid balance associated with acute kidney injury?	Cumulative fluid balance at day 3 (ml/kg)	AKI (KDIGO)	Crude and adjusted logistic regression
3	Is cumulative fluid balance associated with fewer ventilator-free days?	Cumulative fluid balance at day 3 (ml/kg)	Ventilator-free days to day 28	Linear regression and a non-parametric comparison
4	Is initial lactate associated with 28-day mortality, and does the answer change with how missing lactate is handled?	Initial lactate (mmol/l)	28-day mortality	Logistic regression; complete-case analysis vs multiple imputation
5	Among survivors, is fluid balance associated with a longer ICU stay?	Cumulative fluid balance at day 3 (ml/kg)	ICU length of stay in survivors	Median (IQR); linear regression on the log scale or non-parametric tests

10.7 How to read your results pack

Your pack is `Data/results_packs/QN/results_pack_QN.md` (or `Results_Pack_QN.docx` for printing). Every pack has the same sections — read them in this order:

Section	What it gives you	What you do with it
1. The question	Your research question in one sentence	Copy it into the aim of your Introduction
2. Design and data	Design, setting, exposure, outcome, analysis sample, confounders, software, participant flow	The skeleton of your Methods (STROBE items 4–12)
3. Table 1	Baseline characteristics by group, with missing values	Your Table 1 and the first Results sentences; read it first: were the groups alike?
4. Main results	The comparison table, crude and adjusted models, figures (e.g. Kaplan–Meier, forest plot)	Your key numbers: the main effect with 95% CI, crude AND adjusted
5. Other analyses	Interaction, sensitivity analyses, missing-data comparison (depending on the question)	One further result for the Results (STROBE item 17)
6. What this means	Plain-language interpretation	The first move of your Discussion
7. What this does NOT mean	The traps: causal words, “proof”, over-reading p-values	Your verb check
8. Limitations to discuss	Residual confounding, reverse causation, missing data and more	The limitations paragraph of your Discussion
STROBE items	Which items each section helps you write	Your STROBE checklist

Reading a results-pack row. “Positive fluid balance (vs negative) — Crude OR 2.49 (1.70 to 3.70), $p < 0.001$ — Adjusted OR 1.62 (1.02 to 2.59), $p 0.044$ ” means: *before* adjustment the odds of death were about 2.5 times higher with a positive balance; *after* comparing patients of similar severity they were about 1.6 times higher, and the data are compatible with anything from almost no increase (1.02) to about 2.6 times. Write: “was associated with higher 28-day mortality (adjusted OR 1.62, 95% CI 1.02 to 2.59)”.

Step	When	What your group does	Output
------	------	----------------------	--------

10.8 The manuscript workflow

Step	When	What your group does	Output
1. Understand the data	Session 5 (5.2, 5.4)	Receive your question; read the data dictionary, the cleaning log and your Table 1 and figures (work-sheet)	Notes: who is in the cohort, how variables are defined, what was cleaned
2. Interpret the results	Session 6 (6.2, 6.4)	Read your group comparisons and regression or Kaplan–Meier/Cox outputs, the crude vs adjusted estimates and the missing-data sensitivity analysis	A “key numbers” list: the main effect with 95% CI, plus 3–5 supporting results
3. Methods + Results	Session 7 (manuscript workshop 1)	Draft the Methods (design, setting, participants, variables, data sources, bias, statistical methods incl. missing data) and the Results (participants first, then descriptive, main and other analyses)	About 600 words + Table 1 + one further table or figure
4. Introduction, Discussion, Abstract, Title	Session 8 (manuscript workshop 2)	Write the Introduction (funnel), the Discussion (six moves, calibrated language), the structured abstract and the title	Complete draft
5. Final tidy and submission	By 23:59 on the day of Session 8	Fill the STROBE checklist with page numbers, check the numbers, add the AI-use statement, submit	Draft manuscript (required)
6. Peer-review swap	Within 1 week after Session 8	Review another group’s draft (peer_review_instructions.md)	Review report
7. Response and revision	Within 2 weeks after Session 8	Respond point by point and revise your draft	Response letter + revised manuscript

Rules for every group. Use only the numbers in your results pack. Write “associated with”, never “caused” or “led to”, and no “trend towards significance”. Say how much was missing and how it was handled. In the Methods write: *“This analysis used a simulated teaching dataset; no ethics approval or consent was required.”* Add an AI-use statement at the end.

Our group question: Q ____ Our main result (effect, 95% CI): _____

CHAPTER 11

The assignments

Your group produces two things during the course. The **research proposal** is about **your own question** — planning a study that has not been done. The **manuscript** is about **someone else's data** (FLUID-ICU) — reporting a study that has been done. This chapter summarises the briefs, the presentation, the peer review and the marking rubrics. The full documents are in Assignment/.

What is expected

Attend the sessions (at least 6 of the 8), **present your group's proposal** in Session 8 and **submit your group's manuscript draft** (by 23:59 on the day of Session 8). All marks are **formative**: they are there to help you improve.

11.1 Group product 1: the research proposal

Output: one written proposal with a publication plan (about 6–8 pages, `Templates/Research_Proposal_Template.docx`) and one 6-minute presentation (6 slides) in Session 8.

A good topic is a problem you see often, that matters to patients, and that your unit could realistically study in 12–24 months with its own patients and records. Please **do not reuse the running example** (fluids in septic shock).

11.1.1 What to write, part by part

Part 1 — Sessions 1–2 (Ask; Find and read)

Section	What we expect
Working title	Built from your PICO: names the population, intervention/exposure, outcome and (ideally) the design, e.g. "Restrictive fluids and 28-day mortality in septic shock: a randomised trial". You will refine it in Part 4.
Problem statement & rationale	What is the problem, how big is it, why does it matter here, what is not yet known (½ page).
PICO	Population, Intervention/exposure, Comparison, Outcome (and Time) — one line each.
Research question	One sentence built from the PICO.
FINER check	One line each: Feasible, Interesting, Novel, Ethical, Relevant — and what you changed because of it.
Aim, objectives, hypothesis	One broad aim; 2–3 specific, measurable objectives; one testable hypothesis (if the study compares groups).
5 key references	From your PubMed search in Session 2 (literature review worksheet / literature matrix), stored in Zotero or another reference manager; Vancouver style; at least 3 from the last 10 years.

Part 2 — Sessions 3–4 (Design; Bias and effect)

Section	What we expect
Study design & justification	Name the design and say why it fits your question (and why a stronger design is or is not possible).
Setting	Type of unit/hospital, period of the study (no real hospital name needed).
Population	Inclusion and exclusion criteria, written so two people would select the same patients.
Exposure/intervention & comparator	What exactly is given or measured, and what it is compared with.
Outcomes	One primary outcome (with how and when it is measured) + 1–3 secondary outcomes.
Main biases & how they are reduced	At least 2 specific threats (selection, information, confounding) and the step that reduces each (randomisation, blinding, standard definitions, matching, adjustment...).

Part 3 — Sessions 5–6 (Data; Analyse)

Section	What we expect
Variables table	Outcome, exposure and 3–6 confounders/other variables, each with its type (binary, categorical, ordinal, continuous) and unit.
Mini-CRF & data dictionary	A data collection form of 6–10 items, and matching data-dictionary rows (name, definition, type, unit/codes, allowed range).
Data management & quality plan	How data will be captured (paper CRF or electronic data capture, e.g. REDCap or KoBoToolbox), coded, checked (range and consistency checks, double entry or spot checks), stored securely with backups, and who can change them (audit trail).
Sample size statement	One or two sentences: expected outcome rates (from the literature), the difference you want to detect, power, significance level and the resulting number — or, for a descriptive study, the precision you want. Show the tool you used.
Analysis plan in plain language	How each variable will be summarised (mean/SD, median/IQR, n%), which test or model for the primary comparison, which confounders you will adjust for, and how you will avoid multiple-comparison problems (one pre-specified primary outcome).
Missing-data plan	How you will prevent missing data, how you will report it (how much, for which variables), and what you will do about it (e.g. complete-case analysis, plus multiple imputation as a sensitivity analysis if a statistician is available).

Part 4 — Session 7 (Protect and write)

Section	What we expect
Ethics & consent	Is it research, audit or QI? Ethics committee approval route, who consents and how (including patients who lack capacity, if relevant), risks and benefits.
Registration	Where and when the study will be registered (e.g. ClinicalTrials.gov or a WHO ICTRP primary registry for trials, before the first participant; PROSPERO for systematic reviews), or why registration is not required.
Confidentiality	De-identification (study codes), who holds the key, where data are stored, who can access them, how long they are kept.
Feasibility	Timeline (simple Gantt chart), budget headings (no figures needed), people and resources, expected recruitment per month and how you know it is achievable.

Section	What we expect
Publication plan — target journal & reporting guideline	One target journal (checked with Think. Check. Submit.; scope, open-access fee) and a back-up journal; the reporting guideline for your design (CONSORT, STROBE, STARD, PRISMA, SQUIRE...), found via EQUATOR.
Publication plan — authorship & CRediT roles	Who will be an author (ICMJE criteria), in what order, and each person's CRediT role (e.g. conceptualisation, methodology, data curation, formal analysis, writing – original draft); who will be acknowledged instead.
Publication plan — working title	The final working title (from Part 1, refined).
Publication plan — structured-abstract skeleton	About 200 words: Background (1–2 sentences), Methods (design, setting, participants, primary outcome, analysis), <i>expected</i> Results (what you will report, with placeholders, e.g. "28-day mortality was X% vs Y% (RR ..., 95% CI ...)") — no invented numbers), Conclusion (what each possible result would mean).
Dissemination	How results will be fed back to the unit (meeting, poster, conference abstract) and to patients if relevant.

11.1.2 Deadlines

When	What is due
End of Session 1 (weekend 1, Saturday)	Topic agreed; PICO, research question, FINER check, aim and objectives drafted.
End of Session 2 (weekend 1, Sunday)	Part 1 complete: problem statement, working title and 5 key references.
End of Session 4 (weekend 2)	Part 2 complete. Finish any gaps before weekend 3.
End of Session 6 (weekend 3)	Part 3 complete. Homework before weekend 4: start the 6 slides (presentation_guide.md) and think about the target journal and authors.
End of Session 7 (weekend 4, Saturday)	Part 4 complete (ethics, registration, feasibility, publication plan). Homework that evening: finish the 6 slides.
Before Session 8 starts	Slides sent to the course lead (one file per group, GroupN_slides.pptx or PDF) so all presentations run from one laptop.
Session 8 (weekend 4, Sunday)	Presentation of the proposal and publication plan (6 min + 2 min feedback).
2 weeks after Session 8	Final written proposal (Parts 1–4, revised after the feedback) submitted for written feedback — the same deadline as the manuscript response letter.

Submission. One Word file per group, named GroupN_Proposal.docx (PDF also accepted), by e-mail to the course lead or in the shared course folder. Cover page: group number, working title, names and professions of all members, and each member's group role. Add a short **AI-use**

statement if you used AI tools; never include real patient data, names or record numbers; write in your own words and cite every source.

11.2 The presentation in Session 8

6 minutes, 6 slides on your proposal and publication plan, then **2 minutes** of structured feedback; at the end of the block, a **1-minute manuscript pitch** per group (not scored). The whole block takes 45 minutes: 5 groups × (6 + 2) minutes = 40 minutes, plus 5 × 1 minute of manuscript pitches.

11.2.1 The six slides and their timing

Slide	Title	What to show	Time
1	Title & team	Working title; group number; names, professions and roles; one photo or icon of the clinical problem (optional).	0:20
2	Problem & PICO	The problem in one sentence with one number (how common or how serious); the PICO as a 4-row table; the research question; the hypothesis.	1:00
3	Design & participants	Design and why; setting; key inclusion/exclusion criteria; a simple flow picture (eligible → enrolled → groups → outcome); the main bias and how you reduce it.	1:15
4	Data & analysis plan	Primary outcome (how and when measured); 4–6 key variables with types; how data are captured and checked; sample size statement; which summary and which test or model; how you will handle missing data.	1:15
5	Ethics & feasibility	Consent approach; registration; confidentiality; a simple timeline (Gantt bar); expected recruitment per month; the biggest practical risk and your plan for it.	1:00
6	Publication plan & impact	Target journal and the reporting guideline you will follow; working title; authors with their CRediT roles; one line of the abstract skeleton (e.g. the Conclusion you would write for each possible result); what will change for patients or the unit.	1:10
Total			6:00

11.2.2 Timekeeping cards

Card	Shown at	Meaning
Green	4:00	2 minutes left — you should be on slide 5.
Yellow	5:00	1 minute left — move to your final slide.
Red	6:00	Time is up — finish your sentence and stop.

At **6:30** the chair stops the presentation politely, even mid-slide.

11.2.3 The two-minute feedback

Time	Who	What
0:00–0:45	One peer group (rotating: Group 2 comments on Group 1, Group 3 on Group 2 ... Group 1 on Group 5)	Two stars and a wish: two specific things that were strong, one specific suggestion.
0:45–1:45	Instructor	One strength, the most important improvement for the written proposal or publication plan (usually design, outcome definition, feasibility or choice of journal/guideline), and one question if time allows.
1:45–2:00	Presenting group	One-sentence reply or "thank you, we will take that on board". No defending.

Tips. One message per slide (the title can be the message); at most ~6 short lines per slide, 24-point font or larger; pictures before words (a PICO table, a flow diagram, a timeline bar); numbers with units and sources; be honest about weaknesses; calibrated language on slide 6. Rehearse at least twice **with a timer** — most groups overrun on slide 2. Prepare for three questions: *Why this design? Can you really recruit that many patients? Why this journal?*

The 1-minute manuscript pitch. Say three things: (1) the question — "In 600 simulated ICU patients with sepsis, we asked whether ... was associated with ..."; (2) the main result with its 95% CI, crude and adjusted if relevant; (3) a one-sentence conclusion in calibrated language, and the next step (peer-review swap).

11.3 Group product 2: the manuscript

Output: a STROBE-guided original article of about **2,000 words** (Templates/Manuscript_Template_IMRaD.docx), a review of another group's draft, and a response-to-reviewers letter. The question, the data and the results pack are described in Chapter 10.

11.3.1 Length and format

Part	Target	Notes
Title page	—	Title, group authors with CRediT roles, word count, the simulated-data statement
Structured abstract	≤ 250 words	Background, Methods, Results (main effect with 95% CI), Conclusions
Introduction	about 350 words	Three paragraphs: what is known → what is not known → the aim
Methods	about 550 words	STROBE items 4–12; include the statistical-analysis paragraph and how missing data were handled
Results	about 450 words	Participants first; numbers with denominators and 95% CIs; no interpretation
Discussion	about 650 words	Main finding → comparison with literature (incl. Paper 2) → possible explanations → strengths and limitations → implications → conclusion
Main text total	about 2,000 words	Introduction to Discussion ($\pm 10\%$); abstract, tables and references not counted
Tables and figures	Table 1 + up to 2 more	Each must stand alone (title, units, n, abbreviations explained)
References	10–20	Vancouver style, managed in Zotero or another reference manager
STROBE checklist	Completed	With the page number for each item, or "not applicable"

11.3.2 Roles

Role	Responsibilities
Lead author (coordinator)	Keeps the single master file, sets internal deadlines, writes the title page, submits the draft, coordinates the response to reviewers.
Methods writer	Drafts the Methods from the data dictionary, cleaning log and SAP; makes sure the statistical paragraph matches what the results pack actually did.
Results and tables writer	Drafts the Results; builds Table 1 and the other table or figure from the results pack; checks every number against the pack.
Introduction and references lead	Drafts the Introduction; manages the references in a reference manager; finds 2–3 papers for the Discussion comparison.
Discussion writer	Drafts the Discussion and Conclusion in calibrated language; leads the limitations paragraph.
STROBE and consistency checker	Completes the STROBE checklist; checks that numbers agree between abstract, text and tables; checks word counts, the AI-use statement and the simulated-data statement.

Authorship: all group members who meet the ICMJE criteria are authors, with a CRediT statement;

the instructors who produced the analysis are acknowledged, not listed as authors.

11.3.3 Files and deadlines

File	Due
GroupN_Manuscript_QN_draft.docx (draft + completed STROBE checklist as an appendix)	23:59 on the day of Session 8
GroupN_Review_of_GroupM.docx (review report)	1 week after Session 8
GroupN_Manuscript_QN_revised.docx (with tracked changes) and GroupN_Response_to_Reviewers.docx (using Course/Templates/Response_to_Reviewers_Template.docx)	2 weeks after Session 8

Review pairs rotate: Group 1 reviews Group 2, Group 2 reviews Group 3, ... Group 5 reviews Group 1.

Simulated data and AI use

Say on the title page and in the Methods that the data are simulated ("This manuscript uses simulated data created for training; it reports no real patients"). End the manuscript with an **AI-use statement** ("AI use: none", or which tool was used for what, and that the authors checked every sentence and reference). AI tools must not produce results, invent references or write the interpretation; they cannot be authors. **Never paste another group's draft into an AI tool** — it is confidential.

11.4 The peer-review swap (after the course)

Plan about **2–3 hours**: about 45 minutes reading alone, a 60-minute group meeting (online is fine), and 30–45 minutes to write the report.

When	Who	What
Day of Session 8, 23:59	Each group	Submits its draft manuscript (GroupN_Manuscript_QN_draft.docx)
Within 2 days	Course lead	Sends each group the draft it will review (without changes)
Within 1 week after Session 8	Reviewing group	Sends its review report (GroupN_Review_of_GroupM.docx) to the course lead, who forwards it to the authors
Within 2 weeks after Session 8	Authors	Send the revised manuscript (tracked changes) and the response letter to the course lead

How to review. (1) Read alone first and write in one sentence what the paper found. (2) One member checks the main numbers against the authors' results pack. (3) Apply the 10-question appraisal checklist (Chapter 9) as a group. (4) Apply the writing checks below. (5) Sort your comments into *major* (would change the conclusions or the reader's trust; about 3–6) and *minor* (clarity, wording, formatting; about 5–10). (6) Write the report with the template below and agree it as a group.

11.4.1 Writing checks

Check	Yes / No	Page / line
Title names population, exposure, outcome and design		
Structured abstract $\leq$ 250 words, with the main effect and 95% CI		
Introduction ends with a clear aim		
Methods: variables defined, statistical methods (including missing data) described, simulated-data statement present		
Results: participants first; numbers with denominators and 95% CIs; no interpretation		
Numbers agree between abstract, text, tables and the results pack		
Tables and figures stand alone (title, units, n, abbreviations)		
Discussion: main finding, comparison, explanations, limitations, implications, conclusion		
STROBE checklist completed with page numbers		
References consistent (Vancouver); about 2,000 words; CRediT and AI-use statements present		

11.4.2 Tone rules for reviewers

Comment on the paper, not the people · be specific (page and line, the problem, why it matters) · end every major comment with a suggestion · start with what works · do not rewrite the paper · stay within the data · keep the draft confidential (no AI tools) · be kind and honest.

11.4.3 The review report

Manuscript reviewed: Group ____, Question ____, working title: _____

Reviewing group: Group ____ **Date:** _____

1. Summary (3–4 sentences): what the paper asks, what it did and what it found, in your own words.

2. Strengths (2–3 points)

3. Major comments (numbered; each with page/line, the problem, why it matters and a suggestion)

4. Minor comments (numbered; page/line and suggestion)

5. Recommendation: ☐ Accept as it is ☐ Minor revision ☐ Major revision ☐ Reject (explain why)

11.4.4 How authors respond

Read the review twice, then wait a day. Use `Templates/Response_to_Reviewers_Template.docx`; thank the reviewers in one sentence; answer **point by point** — the comment (copied), your reply, what you changed with page and line; make the changes with tracked changes on; disagree politely, with reasons; never answer just "Done".

Reviewer comment	Response	Change in manuscript
M1. The Results give only the adjusted OR; readers cannot see how much adjustment changed the estimate.	Thank you. We now report the crude OR alongside the adjusted OR and comment on the difference in the Discussion.	Results, p. 6, lines 4–7; Discussion, p. 8, lines 15–18; Table 2 updated.

11.5 The marking rubrics (summary)

Levels: **Excellent** — complete, correct and specific; **Good** — mostly complete and correct, minor gaps; **Needs work** — present but vague, incomplete or partly incorrect; **Missing** — absent or wholly incorrect. A simple, feasible study planned well scores higher than an ambitious study planned vaguely.

11.5.1 Proposal rubric (100 points: written 70 + presentation 30)

#	Criterion	Points	Excellent looks like
1	Problem statement, rationale & working title	6	Problem sized with a number and a source; gap and local importance; title built from the PICO
2	PICO, research question & FINER	6	Specific PICO; question follows from it; honest FINER check that changed something
3	Aim, objectives, hypothesis & references	6	One aim, 2–3 measurable objectives, testable hypothesis; 5 relevant, correctly cited references
4	Design & justification	6	Design correctly named and matched; why a stronger design is or is not possible
5	Setting, population, exposure & outcomes	6	Reproducible criteria; one primary outcome with how and when
6	Bias & how it is reduced	6	Two specific threats for <i>this</i> study with a concrete step for each
7	Variables, mini-CRF, dictionary & data management	6	Correct types and units; clear CRF; checks, storage and audit trail
8	Sample size, analysis & missing-data plan	6	Assumptions with sources; test matches the outcome; named confounders; missing-data plan
9	Ethics, registration, confidentiality & feasibility	6	Consent route (incl. lack of capacity); registration; de-identification; realistic Gantt and recruitment

#	Criterion	Points	Excellent looks like
10	Publication plan	6	Checked journal + back-up; right guideline; ICMJE/CRedit; abstract skeleton without invented results
11	Coherence & clarity (whole document)	10	Parts agree (same question, outcome, title); plain language; template used
12	Presentation: structure & content	6	All 6 slides; key decisions visible
13	Clarity & plain language	6	Understandable to every profession; one message per slide
14	Visual design	6	Readable from the back; pictures and tables instead of text blocks
15	Timekeeping	6	Finished between 5:00 and 6:00
16	Questions & teamwork	6	Direct, honest answers; several members contribute

Points per level: criteria 1–10 and 12–16: 6 / 4 / 2 / 0; criterion 11: 10 / 7 / 3 / 0. **Bands:** 80–100 strong (ready for a full protocol); 60–79 good (revise flagged sections); 40–59 promising (rethink question or design); below 40 major revision.

Common problems: outcome without a time point · an RCT where randomisation is unethical (or a cohort where an RCT is feasible) · "we will adjust for confounders" without naming them · a sample size with no assumptions · recruitment the unit cannot deliver · "consent will be obtained" for unconscious patients without saying from whom · CONSORT for an observational study, or STROBE for a trial · a journal chosen from an e-mail invitation · no registration for a trial · missing data ignored · invented numbers or causal claims in the abstract skeleton · gift authorship.

11.5.2 Manuscript rubric (100 points: manuscript 90 + peer review and response 10)

#	Criterion	Points	Excellent looks like
1	Title and abstract	10	Population, exposure, outcome and design in the title; structured abstract $\leq$ 250 words with the main effect and 95% CI
2	Introduction	10	Three-paragraph funnel; aim matches the group's question; about 350 words
3	Methods	15	STROBE items: design, setting, participants, variables, data sources and cleaning, bias, study size, statistics (crude and adjusted, missing data, software), simulated-data statement
4	Results	15	Participants first; crude AND adjusted effects with 95% CI; denominators; no interpretation; numbers match the pack
5	Tables and figures	10	Table 1 + 1–2 more that stand alone; the right figure type
6	Discussion	15	Six moves, including Paper 2 and honest limitations
7	Calibrated language	5	"Associated with" throughout; no "trend"; conclusion no stronger than the design
8	STROBE adherence	5	Checklist completed with page numbers and true to the text
9	References, length and presentation	5	10–20 references in Vancouver style; about 2,000 words; CRediT and AI-use statements
10	Review given and response to reviewers	10	Specific, constructive review; point-by-point response; changes really made

Points per level: 10-point criteria 10 / 7 / 3 / 0; 15-point criteria 15 / 10 / 5 / 0; 5-point criteria 5 / 3 / 1 / 0. **Mark the writing, not the analysis** — the instructors ran the analyses.

Common problems: only the adjusted OR (or only the crude) · "positive fluid balance increased mortality" (causal language from a cohort) · p-values without effect sizes, "p = 0.000", percentages without denominators · Methods and Results that do not match each other or the pack · missing lactate not mentioned · mean $\pm$ SD for skewed outcomes (Q3, Q5) · hazard ratio read as a risk ratio; Kaplan–Meier ticks read as deaths (Q1) · simulated-data or AI-use statement missing · a Discussion that repeats the Results or ignores Paper 2.

APPENDIX A

Answers to the exercises and extra practice

This appendix gives model answers to every exercise of the eight sessions, followed by the answers to the extra practice. **Try each exercise yourself first.** These are *model* answers: many other answers are acceptable when they are specific, consistent and could be acted on. Numbers about FLUID-ICU come from the course's key results and are **simulated**.

A.1 Session 1 · ASK

These are **model** answers. Many other answers are acceptable: what matters is that your answer is specific, consistent and could be acted on.

Quick quiz (segment 1.1) — Rank the evidence

Answer (weakest → strongest): A (expert opinion) → B (registry cohort) → C (randomised trial, CLASSIC) → D (systematic review / meta-analysis of trials).

Source	Design	Real example
A	Expert opinion	—
B	Observational cohort	Similar to Paper 2: Hyun 2023, <i>Ann Intensive Care</i> — positive cumulative fluid balance on ICU days 2–3 associated with higher 28-day mortality (day 2 cohort HR 2.13, 95% CI 1.48–3.06)
C	Randomised controlled trial	CLASSIC: Meyhoff et al., <i>N Engl J Med</i> 2022 — 1,554 ICU adults with septic shock; 90-day mortality 42.3% restrictive vs 42.1% standard
D	Systematic review / meta-analysis	Pooled analyses of fluid-restriction trials

Discussion points

- Why might B and C disagree? In a cohort, sicker patients receive more fluid, so fluid can look harmful even if it is not — **confounding by indication** (taught in Session 4). Randomisation removes this.
- What could make C weaker than B? Very small size, many losses to follow-up, poor allocation concealment, an outcome that does not matter to patients, or a population unlike ours.
- The pyramid ranks **designs**, not individual papers. Quality must still be checked (Session 8).

Check your understanding (segment 1.1) — types of research

Item	Answer
A Counting ICU pressure injuries this year	Quantitative, primary
B Interviewing families about visiting rules	Qualitative, primary
C Re-analysing 3 years of discharge records	Quantitative, secondary (data already exist; still needs ethics review)
D Hand-hygiene survey + nurse focus groups	Mixed methods, primary

Check your understanding (segment 1.1) — research, audit or QI?

Project	Answer	Ethics route
Checking whether fluid balance is charted daily, against the unit protocol	Audit	Local/clinical-governance approval; usually no ethics committee
Introducing a daily fluid-balance checklist and refining it in PDSA cycles	Quality improvement	Local approval; ethics review if it adds risk or goes beyond routine care
Randomising patients to restrictive or usual fluids	Research	Ethics committee approval + trial registration before the first patient
Interviewing nurses about why they hesitate to stop fluids	Research (qualitative)	Ethics committee approval (consent, confidentiality)
Reviewing 3 years of ICU records to see if fluid balance is linked to mortality	Research (secondary data, observational)	Ethics committee approval or a formal waiver of consent

Key point: rules differ between institutions. When in doubt, ask the ethics committee **before** starting; journals ask for approval or a waiver at submission.

Check your understanding (segment 1.2) — spot the PICO (Exercise 1 warm-up)

P	I	C	O	T
Adults 65+ after hip-fracture surgery	Daily non-drug delirium bundle	Usual care	Delirium (screened daily, e.g. 4AT or CAM)	First 7 days after surgery

Common error: labelling "hip fracture" as the outcome — it has already happened and defines who is eligible (P).

Exercise 1 — Model PICO(T) answers

Scenario A — Ventilator-associated pneumonia (experimental)

P Adults on invasive mechanical ventilation for ≥ 48 h in the ICU	
I	Twice-daily nurse checklist for head-of-bed elevation 30–45°
C	Usual care (no checklist)
O	Ventilator-associated pneumonia (using a stated definition)
T	During ventilation, up to ICU day 28

In adults ventilated for at least 48 hours in the ICU, does a twice-daily nurse checklist for head-of-bed elevation, compared with usual care, reduce the incidence of ventilator-associated pneumonia by ICU day 28? Type: analytical — experimental (could be a before–after study or a cluster trial; designs are Session 3). FINER watch-point: **F** — VAP definitions vary and VAP is not very common; the group must state the definition and estimate how many cases occur per year.

Scenario B — Pressure injuries (descriptive first)

The honest first question is descriptive, because nobody knows the size of the problem: *What is the incidence of hospital-acquired pressure injuries (stage 2 or worse) among adults admitted to our medical ward over 6 months?*

P Adults admitted to the medical ward	
O	New pressure injury, stage 2 or worse
T	During the admission, over a 6-month period

Type: descriptive. No hypothesis needed — we **estimate** incidence (with a 95% CI). A later analytical question: *does 2-hourly vs 4-hourly repositioning reduce new pressure injuries in at-risk adults?*

Scenario C — Delirium (experimental)

P Adults aged 65+ after hip-fracture surgery	
I	Daily multicomponent non-drug delirium-prevention bundle
C	Usual care
O	Incident delirium (screened daily with a validated tool, e.g. 4AT or CAM)
T	During the first 7 postoperative days

In adults aged 65 and over after hip-fracture surgery, does a daily non-drug prevention bundle, compared with usual care, reduce incident delirium in the first 7 postoperative days? FINER watch-point: **E** — consent from patients who may become confused; **F** — staff time to deliver the bundle every day.

Scenario D — Stroke door-to-needle time (observational or before–after)

P	Adults with acute ischaemic stroke receiving intravenous thrombolysis in our emergency department
E	Pre-hospital ambulance alert
C	No pre-alert
O	Door-to-needle time (minutes)
T	From hospital arrival to start of thrombolysis

In adults with acute ischaemic stroke who receive thrombolysis in our emergency department, is a pre-hospital ambulance alert associated with a shorter door-to-needle time than no alert? Type: analytical — observational (PECO). Note: the outcome is continuous (minutes), so the hypothesis is about a difference in median or mean time.

Scenario E — Medication errors (experimental or before–after)

P	Adults transferred from ICU to a general ward
I	Pharmacist-led medication reconciliation at transfer
C	Usual transfer process
O	Patients with at least one medication discrepancy
T	Within 48 hours of transfer

In adults transferred from the ICU to a ward, does pharmacist-led medication reconciliation at transfer, compared with the usual process, reduce the proportion of patients with at least one medication discrepancy within 48 hours? Watch-point: "medication error" must be defined (discrepancy vs harmful error); one primary outcome only.

What good looks like (Exercise 1)

- All boxes filled; the comparison is named, not "nothing".
- One primary outcome, **measurable**, with a **time point**.
- The sentence reads naturally and contains every PICO element.
- Common faults to correct: two outcomes in one question ("mortality and length of stay"); outcome without time; population too broad ("all patients"); intervention not defined ("better care").

Exercise 2 — FINER

Part A: answers vary. Push pairs to justify each colour. A red **F** is the commonest and the most useful to discuss.

Part B — model answers

#	Main problem	Why
1	Feasible	Whole country, 10 years, mortality — far beyond any department's resources. (Large vitamin D trials already exist too — Novel.)
2	Ethical	Withholding antibiotics in septic shock causes harm; there is no equipoise. An ethics committee would refuse it.
3	Novel	The answer is already measured monthly. This is an audit question, not a research question — use the existing data.
4	Relevant (and Interesting)	No plausible mechanism and no impact on practice.

Running example (slide): F red (a mortality difference needs well over a thousand patients — CLASSIC 1,554, CLOVERS 1,563); N amber (large trials exist, no mortality difference found); E amber (equipoise exists, but consent in

septic shock is difficult); I and R green. **Feasible rewrites:** a pilot with cumulative fluid balance at day 5 as the primary outcome (as Paper 1 did); a descriptive study of fluid balance in our unit; an observational study of fluid balance and organ support.

Exercise 3 — Aim, objectives and hypothesis

Model answer — delirium example

- **Aim:** To compare a daily non-drug delirium-prevention bundle with usual care in adults aged 65 and over after hip-fracture surgery.
- **Objectives:** 1 (primary) To compare the incidence of delirium in the first 7 postoperative days between groups. 2 To compare length of hospital stay between groups.
- **Hypothesis:** H0 — the incidence of delirium in the first 7 days is the same in both groups. H1 — it differs.
- **Descriptive version (no hypothesis):** To estimate the proportion of patients aged 65+ who develop delirium in the first 7 days after hip-fracture surgery.

Vague-verb check: "To understand pressure injuries on our ward" → "To estimate the incidence of new stage 2 or worse pressure injuries among adults admitted to our ward over 6 months."

a) Early mobilisation

- H0: ICU length of stay is the **same** with early mobilisation and with usual care.
- H1: ICU length of stay **differs** between early mobilisation and usual care (two-sided).

b) Proportion with positive fluid balance — No hypothesis needed. This is descriptive: we *estimate* a proportion with its 95% confidence interval.

What good looks like

- H0 always states "no difference / no association".
- H1 is usually two-sided ("differs"), even when we expect benefit, because the intervention could also harm.
- The hypothesis uses the **same** outcome and time point as the primary objective.
- Correct the misconception: data can **reject** H0 or **fail to reject** it; we never "prove H0". "No significant difference" is not "no difference" (Paper 1, n = 48, could not detect a mortality difference — it was not powered for it).

Exercise 4 — Proposal builder 1: what good looks like

A worked model using the **feasible rewrite** of the running example (a pilot). Groups choose their own topics; use this as the quality benchmark, not as the answer. The references, problem statement and title are added tomorrow (Session 2, Exercise 4).

Type of project: research (randomised; changes care) → ethics committee approval and prospective trial registration.

PICO(T): P adults with septic shock within 24 h of starting vasopressors · I restrictive fluid protocol · C usual care · O cumulative fluid balance (ml/kg) · T at day 5.

Question: *In adults with septic shock in our ICU, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5?*

FINER: F green/amber (pilot of about 50 patients over 12–18 months) · I green · N amber → green (local setting, different protocol) · E amber (deferred or surrogate consent — Session 7) · R green.

Aim: To compare a restrictive fluid protocol with usual care in adults with septic shock in our ICU. **Objectives:**

1. (Primary) To compare cumulative fluid balance (ml/kg) at day 5 between groups.
2. To estimate recruitment rate and protocol adherence (feasibility).
3. To describe 28-day mortality and kidney injury needing dialysis in each group (exploratory).

Hypothesis: H0 — cumulative fluid balance at day 5 is the same in both groups. H1 — it differs.

What good looks like (self-check)

Element	Good	Needs work
Topic	One focused problem from the unit	Three problems merged; "improve care in the ICU"
Question	All PICO(T) elements; one primary outcome with time point	Two outcomes; no comparison; vague population
FINER	Each letter justified; red letters fixed	Ticked without thought
Research / audit / QI	Correctly classified; approval route noted	"It's only an audit" for a randomised project
Aim/objectives	One aim; 2–3 measurable objectives; primary marked	"To understand...", "to study..."; more than 4 objectives
Hypothesis	Matches primary objective; H0 and H1; or correctly "descriptive"	Hypothesis about a different outcome

Exercise 5 — Synonym list (homework)

Model for the running example: **P** septic shock · sepsis · vasopressor-dependent sepsis; **I** fluid restriction · restrictive fluid · conservative fluid · de-resuscitation; **O** mortality · death · survival. For the pressure-injury example: pressure ulcer · pressure injury · bedsore · decubitus ulcer; repositioning · turning · position change.

Exit question — answers

- 1. Feasible** — a mortality outcome needs well over a thousand patients (Novel is also amber because large trials already exist).
- 2.** e.g. *"To estimate the incidence of new stage 2 or worse pressure injuries among adults on our ward over 6 months."*

Extra practice: answers

Extra practice 1.1 — Research, audit or QI?

Project	Answer	Approval route
a) Head-of-bed position against the unit standard	Audit	Local / clinical-governance approval; usually no ethics committee
b) Night-time checklist in PDSA cycles	QI	Local approval; ethics review if it adds risk or goes beyond routine care
c) Randomising patients to two dressings	Research (experimental)	Ethics committee approval + trial registration before the first patient
d) Interviewing families about visiting rules	Research (qualitative)	Ethics committee approval (consent, confidentiality)

Extra practice 1.2 — Find the PICO

P: adults admitted to the medical ward with community-acquired pneumonia · E: switch from intravenous to oral antibiotics within 48 hours · C: a later switch · O: length of hospital stay (days) · T: during the admission. Type: **analytical — observational** (the question says "associated with"; nobody assigns the switch time).

Extra practice 1.3 — Make it answerable (examples)

1. *"In adults presenting with sepsis to our emergency department, does the sepsis bundle, compared with care before the bundle, shorten the time from arrival to the first antibiotic dose?"*
2. *"In inpatients aged 65 and over on our medical wards, is the rate of falls per 1,000 bed-days higher at night (22:00–06:00) than during the day, over 6 months?"* (descriptive comparison: no intervention).

Extra practice 1.4 — Objective or not?

Good objectives: the **second** (compare the proportion with moderate or severe pain at 24 hours) and the **third** (estimate the incidence over 6 months). "To understand..." and "To look at..." have no measurable verb. Rewrite, for example: *"To estimate the proportion of patients with moderate or severe pain (score ≥ 4 out of 10) 24 hours after surgery on our ward."*

A.2 Session 2 · FIND & READ

These are **model** answers. Many other answers are acceptable: what matters is that your answer is specific, consistent and could be acted on.

Recap quiz — Session 1: answers

1. **Audit** — checking practice against an agreed standard.
2. **P** ICU staff · **I** hand-hygiene feedback · **C** no feedback · **O** hand-hygiene compliance · **T** 3 months.
3. **Feasible** — a mortality outcome needs more than 1,000 patients.
4. **"To estimate the proportion with delirium"** — a measurable verb; "understand" is not measurable.
5. **H0**: ICU length of stay is the same with early mobilisation and with usual care.

If many miss 4 or 5, spend one extra minute: objectives use measurable verbs; H0 always states "no difference".

Check your understanding (segment 2.1) — AND, OR or NOT?

1. **AND**. 2. **OR**. 3. **Fewer** (AND narrows). 4. It removes every paper that mentions children anywhere — including good adult studies that mention children once.

Check your understanding (segment 2.2) — OK or not OK?

1. **Not OK** — patchwriting is plagiarism even with a citation. 2. **OK** (but rare in medical writing). 3. **Not OK** — someone else's finding needs a citation. 4. **Not OK** — you may repeat an error; read and cite the original.

Check your understanding (segment 2.3) — spot the spin

1. Causal word for an observational design → *"was associated with lower pneumonia"*.
2. $p = 0.31$ is no evidence of a difference → *"no significant difference; the confidence interval was wide"*.
3. Headline is a secondary outcome → lead with the primary result.
4. Before–after change without a comparison group → *"mortality was lower after implementation; without a control group we cannot attribute this to the protocol"*.

Exercise 1 — Boolean search builder

Warm-up: septic shock OR fluids gives more hits (either term). sepsis gives more hits than sepsis NOT neonatal (NOT removes papers).

Model string for the running example (as in the demo):

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
AND ("Fluid Therapy"[Mesh] OR "fluid restrict*" [tiab] OR "restrictive fluid*" [tiab]
OR "conservative fluid*" [tiab])
AND ("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Sensible extra synonyms for the I block: de-resuscitation[tiab], deresuscitation[tiab], "fluid balance"[tiab], "Water-Electrolyte Balance"[Mesh]. Sensible filters (each with a reason): article type *Randomized Controlled Trial* + *Systematic Review* (treatment question); *Adult: 19+ years* (our population); publication date *10 years* (current practice). Avoid *Free full text* (loses key papers).

Model string for Scenario C (delirium)

```
(Delirium[Mesh] OR delirium[tiab] OR "acute confusion"[tiab])
AND ("Hip Fractures"[Mesh] OR "hip fracture*" [tiab])
AND (prevent*[tiab] OR "multicomponent" [tiab] OR "non-pharmacological" [tiab] OR
bundle[tiab])
```

Common faults

- Missing brackets: A OR B AND C is read differently from (A OR B) AND C.
- AND inside a block (septic shock AND sepsis) — synonyms need OR.
- A block for the comparison ("usual care") — adds nothing and loses papers.
- NOT used to remove a topic the paper only mentions in passing.
- Search words in the local language: PubMed indexes English titles and abstracts.

Exercise 2 — Paraphrasing: model answers

Part A

Version	Verdict	Why
a)	Problem — patchwriting	Same structure and almost the same words as the source; one word swapped ("mortality" → "death"). Plagiarism even though it is cited.
b)	OK	New structure, adds context (matched registry cohort, 28-day mortality), keeps the calibrated word "associated", gives numbers and the citation.
c)	Problem — distorts the source	"Giving more fluid ... doubles the risk" turns an observational <i>association</i> into causation, and "doubles" overstates (HR 2.13 only for day 2; 1.56 for day 3). Citation does not fix misrepresentation.
d)	Problem — no citation	Good wording, but presenting someone else's finding without a citation is plagiarism.

Part B — model paraphrase "Mortality and days alive without organ support at day 28 did not differ between the two groups, although the pilot trial was too small to detect such differences [1]." What good looks like: new structure; facts unchanged (28 days, mortality, days alive without life support, similar); no stronger claim (do **not** write "restriction is safe"); citation. Adding "the trial was too small" is accurate context from the paper (48 patients; not powered for mortality).

Exercise 3 — Paper 1's abstract: answer key

Item	Answer (from the abstract)
Design (and blinding)	Multicentre, randomised, controlled pilot trial; single-blind — patients and statisticians blinded, clinicians not
P	Adults 18–85 years with septic shock, in the first 24 h of vasopressor infusion, in three ICUs; excluded: AKI needing renal replacement, end-stage CKD, severe malnutrition
I vs C	Optimised fluid restriction (reducing maintenance fluids and fluids for drug dilution during the first 7 days) vs standard fluid strategy
Primary outcome	Cumulative fluid balance (ml/kg) in the first 5 days — not mortality
Numbers	1,201 screened; 50 randomised; 2 withdrew (control); 48 analysed (24 per group)
Main result	Median cumulative balance at day 5: 6.9 vs 35.0 ml/kg; absolute difference 13.2 ml/kg (95% CI –15.2 to 41.6), $p = 0.42$
Conclusion	"...did not reduce the overall fluid balance at day 5"
Matches the result?	Yes — the CI includes zero; the conclusion says "did not reduce"
Spin in the abstract?	No in the abstract conclusion. (Spin appears in the <i>Discussion</i> , which calls $p = 0.42$ a "non-significant trend".) Watch-point: the fluid <i>volumes</i> (89.7 vs 114.3 ml/kg, $p = 0.0506$) are a secondary result, and the abstract reports four decimals for a borderline p — mild emphasis, not a claim.
Registration	ClinicalTrials.gov NCT04947904 (registered 1 July 2021, before recruitment began in September 2021)
Answers our 28-day mortality question?	No . It is a small pilot powered for fluid balance; 28-day mortality was a secondary outcome and was similar between groups — "no evidence of a difference", not "no difference"

Fast finishers (Paper 2): the title says "**Impact**" (causal); the abstract conclusion correctly says "**associated**".

Common errors to correct in the debrief

- Writing "mortality" as the primary outcome.
- Copying only the p -value and not the estimate with its 95% CI.
- Calling the trial "double-blind" — clinicians, who prescribe the fluids, were not blinded.
- Reading the medians (6.9 vs 35.0) as "five times less fluid balance, so it works" — the CI crosses zero.

Exercise 4 — Proposal builder 2: what good looks like

Continuing the worked model from Session 1 (the feasible rewrite of the running example).

Search (documented): PubMed, date, the string from Exercise 1, filters (article type RCT + systematic review; adults; 10 years), number of hits; plus a check of ClinicalTrials.gov for ongoing trials.

Working title: *Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial* (13 words). **Keywords:** Fluid Therapy; Water-Electrolyte Balance; Critical Care; Vasoconstrictor Agents (MeSH terms not already in the title).

Problem statement ($\approx$ 300 words, three paragraphs)

1. *Known*: septic shock is a common cause of ICU death; early fluid resuscitation restores perfusion, but continued fluid often produces a positive fluid balance, which observational studies associate with higher mortality (e.g. Hyun 2023).
2. *Not known*: large trials of restrictive fluids (CLASSIC 2022; CLOVERS 2023) found no mortality difference, and a pilot targeting hidden "non-resuscitative" fluids (Boulet 2024) did not significantly reduce fluid balance at day 5. Whether a restrictive protocol can reduce fluid balance in *our* setting is unknown.
3. *Why here*: a 3-month audit in our ICU found that most septic shock patients were fluid-positive by day 3 (state the number from your own audit), and practice varies between consultants.

Five key references (model)

#	Reference	Why included
1	Boulet N, et al. <i>Crit Care</i> 2024;28:429 — pilot RCT, n = 48	Closest design and outcome; primary outcome not significantly different
2	Hyun DG, et al. <i>Ann Intensive Care</i> 2023;13:105 — matched cohort	Positive balance on days 2–3 associated with higher 28-day mortality
3	Meyhoff TS, et al. (CLASSIC) <i>N Engl J Med</i> 2022;386:2459–70	Large RCT, septic shock, no difference in 90-day mortality
4	Shapiro NI, et al. (CLOVERS) <i>N Engl J Med</i> 2023;388:499–510	Large RCT, sepsis-induced hypotension, restrictive vs liberal, no difference in 90-day death before discharge
5	Evans L, et al. Surviving Sepsis Campaign guidelines 2021. <i>Crit Care Med</i> 2021;49:e1063–e1143	Current guideline; shows the remaining uncertainty

What good looks like (self-check)

Element	Good	Needs work
Search	String, database, date, filters and hits recorded; brackets correct	"We googled it"; no date; AND inside a block
References	5, relevant, include the best available design, exported to a reference manager	Websites, textbooks only, typed by hand, unread
Literature matrix	Same columns for every paper; limitation column filled	Copied abstract sentences
Problem statement	Known → gap → local reason, each fact cited, ends with the question	A literature dump, or no local reason
Title & key-words	10–16 words; P, I/E, C and design visible; calibrated words; keywords add MeSH not in the title	"Fluids in the ICU"; abbreviations; "impact of" for an observational design
Integrity	Own words, every claim cited	Sentences pasted from abstracts

Exit question — answers

1. **OR** gives more hits.
2. Paper 1's primary outcome was **cumulative fluid balance at day 5 (ml/kg)** — not mortality.

Extra practice: answers

Extra practice 2.1 — Build the blocks (one good answer)

- P: ("Cardiac Surgical Procedures"[Mesh] OR "cardiac surgery"[tiab] OR "heart surgery"[tiab])
- I: ("Early Ambulation"[Mesh] OR "early mobilis*"[tiab] OR ambulation[tiab])
- O: ("Delirium"[Mesh] OR delirium[tiab] OR "acute confusion"[tiab])

Full string: the three blocks joined with **AND**. The comparison (usual care) is not searched.

Extra practice 2.2 — Which search gives more hits?

1. The **first**: OR adds the American spelling, so the block is wider.
2. The **second**: the MeSH term and a synonym are added with OR.
3. The **first**: adding a publication-type filter with AND narrows the search.

Extra practice 2.3 — Spot the spin

Sentence	What the data allow
"Favourable trend" (24% vs 27%, p = 0.41)	"Mortality was 24% vs 27%; the difference was compatible with chance" — report the difference with its 95% CI; no "trend"
"Patients improved markedly" (no control)	"Scores changed over 7 days; without a control group the change cannot be attributed to the bundle"
"Leads to kidney injury" (cohort)	"Higher fluid intake was associated with kidney injury" — and kidney injury may itself cause fluid accumulation (reverse causation)

Extra practice 2.4 — Order the problem statement

1 = "Fluid overload ... is associated with organ dysfunction and death..." (known) · 2 = "Whether a restrictive strategy can be delivered safely in units like ours is not known" (the gap) · 3 = the local audit sentence (why it matters here).

A.3 Session 3 · DESIGN

#	Design	Evidence in the abstract	Why (the three questions)
---	--------	--------------------------	---------------------------

Part A — Name that design

#	Design	Evidence in the abstract	Why (the three questions)
1	Randomised controlled trial (RCT)	"randomly allocated by a central computer"	Assigned: yes. Random: yes. The blinded assessor and central allocation are also good practice (blinding, allocation concealment).
2	Cross-sectional (analytical)	"On one day", "examined all 412 inpatients"	Not assigned; comparison of mattress types; exposure and outcome measured at the same moment. We cannot tell whether the mattress came before the injury, or whether patients with injuries were moved onto special mattresses (reverse causation), which would hide part of the benefit.
3	Case-control (matched)	"who developed candidaemia were compared with ... who did not", "searched the records ... before"	Not assigned; people selected by OUTCOME, then looked back for exposure. Only an OR can be calculated.
4	Prospective cohort	"classified on arrival", "followed up ... at 90 days"	Not assigned (door-to-needle time is what happened, not what the researchers chose); grouped by exposure, then followed forward.
5	Retrospective cohort	"divided them into those who did and did not receive" ... "then compared how many ... had delirium recorded later"	Uses old records (so many people say "case-control"), but patients were selected and grouped by EXPOSURE, and the outcome came later. Selection by exposure = cohort.
6	Case series	"We describe five patients"	No comparison group. Raises a hypothesis (protocol → hypernatraemia) but cannot test it.

#	Design	Evidence in the abstract	Why (the three questions)
7	Non-randomised intervention study (before–after / quasi-experimental)	"before and after a ... programme, which the research team introduced"	The investigators assigned the intervention (to everyone, at one time), but not at random. Rates may change for other reasons over time (seasonality, new staff, other initiatives) — a weak design for cause. With many monthly measurements before and after, the same project could be analysed as an interrupted time series, which is much stronger.
8	Systematic review with meta-analysis	"searched three databases", "assessed each trial's risk of bias", "pooled"	A study of studies: systematic search + appraisal (review) + pooling of numbers (meta-analysis).
9 (optional)	Cluster randomised trial	"wards were randomly allocated, ward by ward"	Assigned: yes; random: yes; but the unit of randomisation is the WARD, not the patient. Needs more patients than an individual trial, and the analysis must account for clustering.

- Abstracts **5**, **7** and **9** cause most discussion. For 5, ask: "How were people chosen — by what happened to them later, or by what they received?" For 7, ask: "Who decided the intervention? Was chance involved?"
- Accept "cross-sectional" for abstract 2 without "analytical"; the point is the single time point.
- If a pair writes "retrospective study" alone, push for the full name: retrospective *what*?

Part B — Choose the design for 5 questions

#	Best feasible design	Reason	Main weakness
1	Cross-sectional (point-prevalence survey)	"How many today" = prevalence; one visit per patient, cheap and fast	Exposure and outcome measured together — cannot tell whether the mattress came first (patients with injuries may be moved onto special mattresses)
2	Cluster randomised trial (randomise wards)	The intervention is delivered at ward level; individual randomisation would contaminate	Needs more patients and wards; the analysis must account for clustering; few clusters give chance imbalance
3	Interrupted time series (quasi-experimental)	The change already happened to everyone; many monthly points before and after allow comparison with the pre-existing trend	Other changes at the same time (co-interventions); no randomisation — strengthen with a control series

#	Best feasible design	Reason	Main weakness
4	Qualitative — in-depth interviews or focus groups	A "why" question about experience and behaviour; numbers cannot answer it	Findings are not generalisable as numbers; depends on interviewer skill; small purposive sample
5	Depends on the group	Check it against the question type and FINER feasibility	Must be named honestly; observational designs → "associated with"
6 (bonus)	Case-control	Rare outcome; start from the few cases and compare their past drug exposures with controls	Recall/record bias; choosing controls; only an OR
7 (bonus)	Non-inferiority RCT	We only need to show "not unacceptably worse"; margin fixed in advance	Needs a justified margin; poor adherence can make arms look falsely similar

- The most common error is choosing an RCT for everything. For question 2, accept "RCT" only if the group explains that wards (not patients) are randomised.
- For question 3, a group that answers "before–after study" is half right — ask how to make it stronger (more time points, compare with the trend, control series).
- For the group's own question, push for **feasibility**: one unit, no funding, 3–6 months often means a retrospective cohort, cross-sectional study or ITS rather than a trial.

Part C — Proposal builder 3: what a good answer looks like

Use this model to coach groups; do not hand it out before they have tried.

Section	Model answer (observational version of the running question)
Design + justification	Retrospective cohort. Our question is about the effect of a fluid strategy on death; an RCT is the ideal design but is not feasible for one ICU without funding. Existing charts record daily fluid input/output, severity scores and outcomes, so a cohort can estimate the association between positive fluid balance and death, and generate the effect size for a future trial. We will say "associated with", not "caused".
Setting	Adult mixed ICU of one teaching hospital; patients admitted over 3 calendar years; data from the electronic chart and ICU register.
Population	Inclusion: age ≥ 18 ; septic shock (Sepsis-3) on ICU admission; vasopressor started within 24 h of admission; ICU stay ≥ 72 h. Exclusion: chronic dialysis; transfer from another ICU after > 24 h; treatment-limitation decision within 24 h; missing fluid charts on day 1–3.

Common problems to avoid

1. RCT chosen for a question that cannot be randomised (harm, prognosis, risk factors).
2. Exclusion criteria that repeat inclusion criteria in the negative ("age < 18 ").
3. Overambitious setting (five hospitals, 2,000 patients) for a group with no funding — point them back to FINER.
4. No data source named for a retrospective design (which register, which charts?).

Extra practice: answers

Extra practice 3.1 — Name the design

#	Design	Proof
a	Cross-sectional survey	Exposure (night shifts, sleep) and outcome (injuries in the past month) measured at the same time
b	Case-control	Chosen by outcome (with / without VAP), then looking back for earlier exposure
c	Prospective cohort	Classified by exposure (access type), then followed forward for 1 year
d	Cluster randomised trial	Whole ICUs randomly allocated, unit by unit

Extra practice 3.2 — Caused or associated?

1. **reduced** (a randomised trial; the CI excludes 1).
2. **were associated with lower** (cohort).
3. **were associated with** (case-control).

Extra practice 3.3 — Fix the exclusion criteria

(a) and (b) simply repeat the inclusion criteria in the negative: they are not exclusion criteria. (c) is a real exclusion with a reason (the outcome is almost certain and fluid management is not comparable), but "expected to die" should be defined, e.g. *"comfort care only at ICU admission"*. Another good criterion: *"chronic dialysis before admission — fluid balance and kidney outcomes cannot be interpreted"*.

A.4 Session 4 · BIAS & EFFECT

Part A — Spot the bias

#	Main bias	Why	One way to reduce it (any of these)
1	Recall bias (a type of information bias)	Cases search their memory harder than controls, so exposures look more common in cases.	Use exposure data recorded before the outcome was known (prescription or pharmacy records); ask cases and controls the same structured questions; blind interviewers to case/control status.
2	Observer bias (information bias)	A soft, judgement-based outcome assessed by someone who knows the group.	Blinded outcome assessor; objective, pre-defined discharge criteria; choose a harder outcome (e.g. actual days to discharge).
3	Selection bias (including loss to follow-up)	Survivors who return to clinic are probably healthier and more motivated than those who do not; the sample no longer represents all ICU survivors.	Contact every survivor (phone/post), record and report non-responders and compare them with responders, plan follow-up from the start.
4	Confounding (by indication)	Lactate (severity) leads to more fluid AND independently predicts death. The crude association overstates (or creates) the effect of fluid.	Randomise; or measure severity at baseline (lactate, SOFA, vasopressor dose) and restrict, match or adjust; state "associated with", not "increases".

Part B — Confounding or effect modification?

#	Answer	Why	What to report
1	Confounding (by indication: lactate)	Both strata give RR 1.00 — they agree with each other but not with the crude RR 1.80. Lactate drove the choice of drug and predicts death.	The lactate-adjusted (or stratified) estimate, RR about 1.0. The crude 1.80 is misleading.
2	Effect modification (by age)	The strata differ from each other: RR 0.50 under 65, 0.95 at 65 and over.	Each age group separately (ideally a pre-specified subgroup with a test for interaction). Do not report only the average 0.70.
3	Neither	Strata agree with each other (0.80, 0.79) and with the crude result (0.80).	The overall RR 0.80; shift neither confounds nor modifies the effect.

Part C — Two 2×2 calculations

Calculation 1 — pressure-relieving mattress trial

	Pressure injury: YES	NO	Total
New mattress	a = 10	b = 190	200
Standard mattress	c = 20	d = 180	200

Measure	Working	Answer
Risk, new mattress	10 / 200	0.05 (5%)
Risk, standard mattress	20 / 200	0.10 (10%)
RR	0.05 / 0.10	0.50
RD	0.05 – 0.10	–0.05 (5 percentage points fewer)
NNT	1 / 0.05	20
OR	$(10 \times 180) / (190 \times 20) = 1800 / 3800$	0.47

- **Is OR close to RR?** Yes, fairly (0.47 vs 0.50), because the outcome is uncommon (5–10%). With rarer outcomes they converge further.
- **Model sentence:** "With the new mattress, pressure injuries were halved (5% vs 10%): about 1 injury prevented for every 20 patients who use it."

Common errors: dividing by the wrong total (using 30 or 400 as denominators); writing RD as 5 instead of 0.05 and then NNT = 0.2; table drawn with outcome in rows.

Calculation 2 — candidaemia case-control study

	Cases	Controls
Parenteral nutrition: YES	a = 24	b = 16
Parenteral nutrition: NO	c = 16	d = 64

- **OR** = $(24 \times 64) / (16 \times 16) = 1536 / 256 = \mathbf{6.0}$
- **Why no risk, RR or NNT?** The investigators chose how many people with and without candidaemia to include (40 and 80, a 1:2 ratio). The proportion with candidaemia in the study is fixed by design, not by nature, so "risk" calculated from the rows is meaningless. The OR is valid because it does not depend on the sampling ratio.
- **Model sentence:** "ICU patients with candidaemia had about six times the odds of having received parenteral nutrition, compared with similar ICU patients without candidaemia. Because candidaemia is rare, this OR approximates the relative risk — but an observational study cannot show that the nutrition caused the infection (sicker patients receive parenteral nutrition)."

Part D — PPV in two settings

Setting 1 — ICU, prevalence 20% (500 people)

	Disease present	Disease absent	Total
Test positive	TP = 90	FP = 80	170
Test negative	FN = 10	TN = 320	330
Total	100	400	500

$$\text{PPV} = 90 / 170 = \mathbf{52.9\%} \quad \text{NPV} = 320 / 330 = \mathbf{97.0\%}$$

Setting 2 — screening, prevalence 2% (500 people)

	Disease present	Disease absent	Total
Test positive	TP = 9	FP = 98	107
Test negative	FN = 1	TN = 392	393
Total	10	490	500

PPV = $9 / 107 = 8.4\%$ · NPV = $392 / 393 = 99.7\%$

- Why different?** With a rare disease, even a small false-positive rate (20%) applied to the many healthy people produces far more false positives (98) than true positives (9). Sensitivity and specificity did not change; the population did.
- LR+ = 0.90 / 0.20 = 4.5** in both settings. Likelihood ratios do not depend on prevalence.
- Prevalence 50%:** TP 225, FN 25, FP 50, TN 200 → PPV $225 / 275 = 81.8\%$, NPV $200 / 225 = 88.9\%$.

These match the slide "Same test, different setting" (4.3) and the prevalence table on the Diagnostic sheet of the 2x2 calculator.

Part E — Read a forest plot and choose the risk-of-bias tool

Numbers are computed by inverse-variance (fixed-effect) pooling of the log risk ratios in `figs_sat2.py`.

Trial	RR (95% CI)	Weight
Trial 1	0.80 (0.41–1.56)	10%
Trial 2	0.83 (0.54–1.28)	25%
Trial 3	1.14 (0.46–2.85)	5%
Trial 4	0.78 (0.55–1.10)	38%
Trial 5	0.74 (0.45–1.23)	18%
Trial 6	0.62 (0.23–1.69)	5%
Pooled	0.79 (0.64–0.98)	$I^2 = 0\%$

- Most weight:** Trial 4 (38%) — the largest trial with the most deaths (103), so the narrowest CI.
- Any single trial clear?** No. Every individual CI crosses 1.
- Pooled:** the diamond does not cross 1. *"Across these six trials, a restrictive strategy was associated with about 21% lower relative risk of death (RR 0.79, 95% CI 0.64 to 0.98); the upper limit is close to no effect."* (For randomised trials "reduced" is acceptable; the point is to give the CI.)
- Trial 3:** No great concern. It is small (5% weight, 15 deaths) with a very wide CI that overlaps all the others; $I^2 = 0\%$ means the spread is compatible with chance.

Choosing the tool

#	Study	Tool
a	Boulet 2024 (RCT)	RoB 2
b	Hyun 2023 (cohort, exposure)	Newcastle–Ottawa (or ROBINS-E)
c	Before–after bundle study	ROBINS-I (non-randomised study of an intervention)
d	Point-of-care vs laboratory lactate	QUADAS-2 (diagnostic accuracy)
e	Systematic review / meta-analysis	AMSTAR 2
f	Cluster-randomised hand-hygiene trial	RoB 2 (its cluster-randomised version)

Common error: choosing ROBINS-I for Hyun. Accept it with discussion — fluid balance is partly a treatment decision — but the usual choice for an exposure cohort is Newcastle–Ottawa or ROBINS-E.

Part F — Proposal builder 4: what a good answer looks like

Use this model to coach groups; do not hand it out before they have tried. (Rows 1–3 — design, setting, population — are in `s3_solutions.md`.)

Section	Model answer (observational version of the running question)
Exposure + comparator	Exposure: positive cumulative fluid balance (ml/kg) at 72 h after ICU admission. Comparator: zero or negative cumulative balance at 72 h.
Outcomes	Primary: all-cause death by day 28 after ICU admission (from the hospital record). Secondary: days alive and free of mechanical ventilation by day 28; new renal-replacement therapy; ICU length of stay.
Biases + mitigation	Confounding by indication (sicker patients receive more fluid): record SOFA, lactate, vasopressor dose, age and comorbidity at admission and adjust or match. Selection bias (requiring 72 h of ICU stay excludes early deaths): state it as a limitation and report how many were excluded and why. Information bias (fluid charts incomplete): pre-specified definitions, two data collectors with a double-entry check on a 10% sample.

Claim: "associated with" (observational design).

Common problems to avoid

1. No comparator ("we will study patients who receive X").
2. Three "primary" outcomes, or a primary outcome without a time point.
3. Observational design with no mention of confounding.
4. A bias named with no concrete mitigation ("we will be careful").

Extra practice: answers

Extra practice 4.1 — Another 2×2

$a = 9$, $b = 291$, $c = 18$, $d = 282$. Risk with the bundle $9/300 = 0.03$; with usual care $18/300 = 0.06$; **RR 0.50**; **RD -0.03** (3 percentage points fewer); **NNT = $1/0.03 \approx 33$** (round up: treat about 34 patients). *"With the bundle, bloodstream infections were halved (3% vs 6%): about one infection is prevented for every 33–34 patients who receive it."*

Extra practice 4.2 — Which bias?

a) **Selection bias** (non-response; responders differ from non-responders). b) **Observer (information) bias** — blind the assessor or use photographs graded blind. c) **Confounding by indication** (severity). d) **Reverse causation** (the early cancer caused the weight loss).

Extra practice 4.3 — PPV at 50%

Disease present 250, absent 250. $TP = 0.90 \times 250 = 225$; $FN = 25$; $TN = 0.80 \times 250 = 200$; $FP = 50$. $PPV = 225 / 275 = 0.82$ (**81.8%**); $NPV = 200 / 225 = 0.89$ (**88.9%**). The same test: the more common the disease, the higher the PPV and the lower the NPV.

A.5 Session 5 · DATA

Exercise numbers follow `Practicals/s5_exercises.md`. All FLUID-ICU numbers come from `Course/Data/key_results.md` (simulated data).

Exercise 1 — Classify the variables

#	Variable	Answer	Explanation
a	Age (years)	Continuous	Measured on a scale; keep the exact number on the CRF (or collect date of birth).
b	Sex	Binary	Two categories. Coding 1/2 does not make it numerical.
c	Glasgow Coma Scale	Ordinal	Ordered scores (3–15) with unequal steps. Often <i>analysed</i> like a number, which is acceptable if stated, but it is ordinal by nature.
d	Source of sepsis	Nominal	Several categories, no natural order.
e	Number of vasopressors	Count	Whole numbers 0–3. Accept "ordinal" if the pair argues it well; not continuous.
f	Cumulative fluid balance (ml/kg)	Continuous	Can be negative or positive; any value on the scale. The Paper 1 primary outcome.
g	Alive at day 28	Binary	Yes/no.
h	Time to death, censored at day 28	Time-to-event	Two pieces of information: how long, and whether the event happened. Analysed with Kaplan–Meier / Cox.
i	Pain none–severe	Ordinal	Ordered categories.
j	Blood group	Nominal	No order.

Bonus: any continuous variable (age, fluid balance, lactate). You can always group later; you cannot recover the exact value from a group.

Common errors: GCS called continuous; time-to-death called continuous (ignores censoring); number of vasopressors called continuous.

Exercise 2 — Fix the bad CRF items (model answers)

#	Bad item	Problem	Model rewrite
a	Fluid given: ____	No unit, no time window, no definition (crystalloid? blood? drug diluents?)	"Total IV fluid received from ICU admission to 24 h (ml), including crystalloids, colloids and drug diluents: [] [] [] [] ml ☐ not recorded"
b	Weight: ____	No unit; measured or estimated? when?	"Body weight at ICU admission (kg): [] [] [] [] ☐ measured ☐ estimated ☐ not available"
c	Comorbidities: ____	Free text: impossible to analyse, spelling varies	Tick list: "Comorbidities present at admission (tick ALL that apply): ☐ diabetes ☐ chronic kidney disease ☐ heart failure ☐ COPD ☐ cirrhosis ☐ cancer ☐ none ☐ unknown"
d	Lactate high? Y / N	Opinion ("high" by whose cut-off?), loses the value, no time point	"Serum lactate at ICU admission (mmol/L): [] [] [] [] ☐ not done" (define "high" later, in the analysis)

Accept any version that passes the four checks (unit, time point, one reading, unknown option). Praise pairs who notice that collecting the *number* is better than collecting a yes/no.

Exercise 3 — Choose the right summary

#	Variable	Shape / type	Summary	Graph
a	Age, bell-shaped	Symmetric continuous	Mean (SD)	Histogram
b	ICU length of stay	Right-skewed	Median (IQR)	Histogram or boxplot
c	Died in hospital	Binary	n (%)	Bar chart
d	CRP	Right-skewed	Median (IQR)	Histogram / boxplot
e	Source of sepsis	Nominal	n (%) per category	Bar chart (not 3-D pie)
f	SBP by treatment group	Symmetric, two groups	Mean (SD) per group	Side-by-side boxplots

Exercise 4 — Judge a bad figure

- Each bar is the **mean** fluid balance per group; the error bar is the **standard error (SE)**, shown in one direction only. Missing: the individual patients, the spread (SD or IQR), the shape (skew), the n per group on the figure, and what the error bar represents.
- Any two of: the data are **skewed**; **one extreme patient** (160 ml/kg) pulls the usual-care mean up; the groups **overlap** a lot; the median; the number of patients; negative values.
- Dots (one per patient) plus a boxplot per group (see the "dots + box" slide), y-axis labelled with units, n under each group. You gain: shape, spread, outliers and overlap become visible, and the reader can judge the comparison.
- The truncated axis makes a 2-percentage-point difference (31% vs 29%) look like a three-fold difference. Fix: start bars at 0 and label the values (or show the difference with its 95% CI).
- (i) The author chose the two scales, so the lines can be made to cross or move together wherever wanted; (ii) two trends over time moving together is **correlation, not causation** (other changes may have happened at the same time). Better: two panels sharing the time axis, and an appropriate design (e.g. interrupted time series, Session 3).

Exercise 5 — [L2] Spot the problems in the raw FLUID-ICU data

(See Course/Data/cleaning_log.md for the full log.)

#	What is wrong	What the script does
a	One variable coded many ways (and a leading space)	Recode to Female / Male
b	Dates in different formats (03/04: March or April?)	Read each format explicitly and convert to one date format (YYYY-MM-DD)
c	ICU C reported lactate in mg/dL, not mmol/L (about 9 times larger; found by comparing medians by ICU)	Divide ICU C values by 9.01 (1 mmol/L = 9.01 mg/dL); confirm with the site laboratory
d	999 is a missing-value code, not a lactate	Set to missing (NA)
e	Impossible age (outside the allowed range)	Set to missing, send a query to the site; keep the patient
f	Duplicate record	Check the two rows are identical, keep one copy (623 rows → 620 patients)

Bonus: editing by hand is invisible and not reproducible; a script plus a log lets anyone check and repeat every change. The log records the problem, how many records, what was done, and why (plus queries sent).

Exercise 6 — [L2] Read your group's Table 1

All numbers from Course/Data/key_results.md (simulated data).

1. Negative balance $n = 245$, positive $n = 373$, overall 618. Two of the 620 enrolled patients have no computable day-3 fluid balance (an impossible weight or fluid output was set to missing during cleaning), so they cannot be placed in a group.
2. Expected answers: age — mean (SD), roughly symmetric (64.6 vs 61.4 years); SOFA — median (IQR), a skewed score (6 (4 to 8) vs 8 (6 to 10)); lactate — median (IQR), right-skewed (2.2 (1.4 to 3.0) vs 2.9 (1.9 to 4.0) mmol/L; vasopressor, ventilation, heart failure, sex — n (%).
3. Severity: SOFA 8 vs 6; vasopressor 205/373 (55.0%) vs 61/245 (24.9%); mechanical ventilation 216/373 (57.9%) vs 97/245 (39.6%). (Heart failure goes the other way: 43/373 (11.5%) vs 54/245 (22.0%) — clinicians give less fluid in heart failure.)
4. Lactate: 91 missing among the 618 (93 of all 620 enrolled, 15.0%).
5. Model sentences: "Of 620 enrolled patients, 618 had a computable day-3 fluid balance, of whom 373 (60.4%) had a positive balance. Patients with a positive balance were more severely ill at admission (median SOFA 8 vs 6; vasopressor 55.0% vs 24.9%; mechanical ventilation 57.9% vs 39.6%)."
6. Larger: the positive-balance group was sicker, and sicker patients die more, so the crude association mixes fluid with severity (confounding by indication). Confirmed in Session 6: crude OR 2.49 vs adjusted 1.62.

Exercise 7 — [L2] Your group's question

Q	Outcome (type)	Exposure (type)	Planned comparison / model
Q1	Death by day 28 (binary; also time-to-event)	Positive balance by day 3 (binary)	Chi-square; logistic regression; Kaplan–Meier + Cox
Q2	AKI, KDIGO ≥ 1 (binary)	Cumulative balance, ml/kg (continuous, per 10 ml/kg)	Chi-square by group; logistic regression
Q3	Ventilator-free days to day 28 (count 0–28, skewed, piled at 0 and 28)	Cumulative balance (continuous)	Mann–Whitney; linear regression
Q4	Death by day 28 (binary)	Lactate at admission, mmol/L (continuous, 15% missing)	Logistic regression; complete case vs multiple imputation
Q5	ICU length of stay in survivors (continuous, right-skewed)	Positive balance (binary)	Mann–Whitney; linear regression on log scale

Example (Q1): "In adults with sepsis admitted to the ICU, is a positive cumulative fluid balance by day 3, compared with a negative balance, associated with higher 28-day mortality?"

Exercise 8 — Proposal builder 5 (what a good answer looks like)

Variables table

Role	Variable	How measured	Type
Primary outcome	Death by day 28	Hospital records / phone call at day 28	Binary
Secondary outcome	Cumulative fluid balance at day 5	Fluid charts, ml/kg of admission weight	Continuous
Intervention	Restrictive vs usual fluid strategy	Randomised allocation	Binary
Baseline / confounders	Age, SOFA score, lactate at admission, source of sepsis	CRF at ICU admission	Continuous, ordinal-like score, continuous, nominal

Sample size: 40% vs 30% mortality, alpha 5% two-sided, power 80% → 356 per group (spreadsheet sheet 2); $(1 - 0.10) \rightarrow 396$ per group, 792 in total.

Analysis plan: describe with mean (SD) / median (IQR) / n (%); compare 28-day mortality with a chi-square test; report the risk ratio and risk difference with 95% CIs; fluid balance compared with Mann–Whitney and reported as median (IQR) per group. (Observational designs: add "adjusted for age, SOFA and lactate using logistic regression".)

Quick self-check:

Element	Good enough	Needs work
Variables table	Outcome, exposure and ≥ 3 confounders, all with a type	Confounders that are actually outcomes; no types
Mini-CRF	6–10 items, units, time points, unknown option	> 15 items; free text; no units
Data dictionary	3–5 rows with codes and allowed ranges	Names with spaces, no codes
Sample size	4 inputs with a stated source	"All patients in one year"
Analysis plan	Summary + test + effect measure with 95% CI	"Data will be analysed with SPSS"

Data management (good enough): EDC with range and logic checks at entry (or paper with double entry of key variables); coded study IDs with the linking log kept separately and locked; audit trail/cleaning log; raw export read-only, dated versions, encrypted 3-2-1 backups; data-sharing statement in the consent form.

Extra practice: answers

Extra practice 5.1 — Summary or graph?

Variable	Type	Summary	Graph
Age	Continuous, symmetric	Mean (SD)	Histogram, or dots + box by group
SOFA score	Ordinal score, skewed	Median (IQR)	Dots + box
Infection source	Nominal (5 categories)	n (%)	Sorted bar chart (no 3-D pie)
Died by day 28	Binary	n (%)	Usually in a table; bar chart if shown
AKI stage	Ordinal (0–3)	n (%) per stage	Stacked or ordered bars

Extra practice 5.2 — The dictionary row

lactate_mmol_l | Blood lactate on ICU admission | Numeric (continuous) | mmol/L, 1 decimal | 0.1 to 30; empty if not measured (never 999 or –99).

Extra practice 5.3 — Read Table 1

"...(median SOFA **8** vs **6**; vasopressor on admission **55.0%** vs **24.9%**)" — positive vs negative balance (simulated data).

Extra practice 5.4 — Mean or median?

Sum = 49 days, mean = **7.0 days**; median = **4 days** (the 4th of 7 ordered values). The **median** describes the typical patient: one 26-day stay pulls the mean up.

A.6 Session 6 · ANALYSE

Exercise numbers follow Practicals/s6_exercises.md. All FLUID-ICU numbers come from Course/Data/key_results.md (simulated data).

Exercise 1 — True or false

#	Answer	One-line explanation
1	T	That is the practical reading of a 95% CI: the range of values compatible with the data.
2	F	Absence of evidence is not evidence of absence. The CI (–15.2 to 41.6 ml/kg) is wide: the pilot cannot tell.
3	F	p is calculated <i>assuming</i> the null is true; it is the probability of data at least this extreme under the null, not the probability of the null.
4	T	A 95% CI that includes the null value corresponds to $p > 0.05$ (two-sided), and vice versa.
5	T	More patients → more precision → narrower CI.
6	F	A tiny p can accompany a trivial effect in a large study. Judge size from the estimate and CI.
7	T	This is the strict definition: the 95% refers to the method over repeated studies (the "dance of the CIs").
8	T	A small study has low power; a real effect can be missed (missed fire).
9	T	Alpha = type I error rate = risk of a false alarm.
10	F	p-values do not "trend". Report the estimate and CI and call it what it is: not statistically significant.
11	T	With 20 independent tests and no real effect, the chance of at least one $p < 0.05$ is about 64% (the jelly-bean problem).
12	F	Same patients measured twice: use a paired test (paired t-test, or Wilcoxon signed-rank if skewed).

Exercise 2 — Choose the test

#	Test	Why
a	Independent (unpaired) t-test	Numerical, roughly symmetric outcome, two independent groups.
b	Mann–Whitney U	Numerical but skewed, small groups. (Same choice as Paper 1 for fluid balance.)
c	Chi-square test	Binary outcome, two independent groups, large expected counts.
d	Fisher's exact test	Binary outcome, very small counts (expected counts < 5).
e	Paired t-test	Same patients measured twice; differences roughly symmetric. (Wilcoxon signed-rank if skewed.)
f	McNemar test	Binary outcome measured twice in the same patients (paired categorical).

Exercise 3 — Match r to the scatter plots

Plot	r	Why
A	−0.58	Points fall from left to right, moderate scatter
B	0.01	Round cloud, no straight-line trend
C	0.86	Tight rising band
D	0.30	Weak rising trend, lots of scatter

(The plotted data were simulated with exactly these correlations.)

Bonus: **no**, not on correlation alone. High r means the two methods rise and fall together, not that they give the same value. Look at **agreement**: a Bland–Altman plot (difference against average), the mean difference (bias) and the 95% limits of agreement, and decide whether those differences are clinically acceptable.

Exercise 4 — [L2] Interpret your group's comparison

Numbers from Course/Data/key_results.md (all unadjusted; simulated data).

Q	Test	Result	Model sentence
Q1	Chi-square (binary outcome, two large groups)	134/373 (35.9%) vs 45/245 (18.4%); RD 17.6 points (95% CI 10.7 to 24.4); RR 1.96; $p < 0.001$	"Twenty-eight-day mortality was 35.9% with a positive and 18.4% with a negative balance (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4)."
Q2	Chi-square	AKI 230/373 (61.7%) vs 71/245 (29.0%); $p < 0.001$; gradient 29%, 54%, 63%, 75% across balance groups	"AKI occurred in 61.7% of patients with a positive and 29.0% with a negative balance."
Q3	Mann–Whitney (count, piled at 0 and 28)	Median 20 (0 to 28) vs 28 (22 to 28) days; $p < 0.001$	"Patients with a positive balance had fewer ventilator-free days (median 20 vs 28)."
Q4	Mann–Whitney (lactate skewed)	Median lactate 3.1 (2.2 to 4.5) in non-survivors vs 2.3 (1.6 to 3.4) mmol/L in survivors; mortality < 2 : 19%, 2–4: 31%, > 4 : 50%	"Non-survivors had higher admission lactate (median 3.1 vs 2.3 mmol/L)."
Q5	Mann–Whitney (skewed, survivors)	Median 7.5 (5.2 to 11.2) vs 4.8 (3.3 to 7.1) days; Hodges–Lehmann difference 2.5 days (95% CI 1.8 to 3.2); $p < 0.001$	"Among survivors, ICU stay was longer with a positive balance (median 7.5 vs 4.8 days; difference 2.5 days, 95% CI 1.8 to 3.2)." Note: the t-test difference in means (2.7 days) is the wrong summary for this skewed outcome.

Reason for bias (all groups): confounding by indication — the positive-balance group was sicker at baseline (Table 1).

Exercise 5 — Read a regression table

The table is mock data; p-values are consistent with the CIs (computed from the log-scale standard error).

1. **Binary outcome** (delirium yes/no) → logistic regression, which reports odds ratios.
2. 1 (a ratio). An OR of 1 means equal odds.
3. CI excludes 1 (✓) or includes 1 (×):

Row	OR (95% CI)	Excludes 1?
Benzodiazepine	2.10 (1.40–3.20)	✓
Age per 10 y	1.30 (1.12–1.51)	✓
Mechanical ventilation	1.60 (0.95–2.70)	×
Female sex	0.85 (0.60–1.20)	×
Dementia	3.40 (1.90–6.10)	✓
Sleep protocol	0.62 (0.41–0.94)	✓

4. Model sentence: "After adjusting for age, sex, ventilation, dementia, sleep protocol and illness severity, patients who received a benzodiazepine infusion had about twice the odds of developing delirium (OR 2.10, 95% CI 1.40 to 3.20)."
5. The OR multiplies for each 10 years: 20 years older → $1.30 \times 1.30 \approx 1.69$ times the odds (all else equal).
6. **Mechanical ventilation** and **female sex** (CIs include 1).
7. Problems in the conclusion:
 - **Causal language** ("cause", "reduces") from an observational cohort: should be "associated with". Confounding by indication is likely (sicker or more agitated patients receive benzodiazepines).
 - **"Trend"** for $p = 0.08$: the CI (0.95–2.70) includes 1; it is not statistically significant. Report the estimate and CI.
 - **"Reduces delirium by 38%"** confuses odds with risk (OR 0.62 is a 38% lower odds, not risk; delirium is common, so the risk reduction is smaller) and again implies causation. The sleep protocol was not randomised.

Exercise 6 — Read a Kaplan–Meier curve (simulated trial; values computed from the plotted data)

1. About **82%** alive in the restrictive group and about **67%** in the usual-care group at day 28 (read where each curve ends). Accept $\pm 3\%$.
2. Each tick is a **censored** patient: follow-up stopped (lost to follow-up, transferred, withdrew) without death being observed. They count as alive up to that day.
3. **67** usual-care patients were at risk at day 21. The number falls because of **deaths and censoring**; in the restrictive group 16 died but 33 left the risk set ($100 \rightarrow 67$ at day 28) because 17 were censored before day 28.
4. The log-rank test gives evidence that the two survival curves differ ($p = 0.009$: results this different would be unusual if the strategies had the same survival). It does **not** give the size of the effect, its 95% CI, or any adjustment for confounders (and in a trial, it is not "proof" of causation without a good design).
5. **Cox regression** gives the hazard ratio with a 95% CI. Check the **proportional-hazards** assumption: curves that cross or converge strongly suggest the HR is not constant over time.
6. No: without numbers at risk the reader cannot see how many patients support each part of the curve; the right-hand end may rest on very few patients. Reporting guidelines and reviewers expect them.

Exercise 7 — Why is it missing? MCAR, MAR or MNAR

#	Mechanism	Consequence for complete-case analysis
a	MCAR — an accident unrelated to the patients	Unbiased, just less precise (fewer patients)
b	MAR — depends on admission time, which is recorded	Possibly biased if night admissions differ; multiple imputation using admission time can correct it
c	MNAR — depends on the missing value itself (the worst pain)	Biased (pain looks better than it is); no guaranteed fix: sensitivity analysis
d	MNAR (deteriorating patients are transferred); at best MAR if the reason for transfer is recorded	Biased (mortality underestimated); chase day-28 status by phone/records; sensitivity analysis
e	MCAR (missing by design for one month, unrelated to patients) — unless patients in the first month differed	Usually unbiased; check whether early patients differed
f	MAR — depends on age, recorded for all	Biased towards younger patients' quality of life; multiple imputation including age

Bonus (d): report the main result plus a **best-case** (all transferred patients alive) and **worst-case** (all died) analysis, or vary the assumed mortality among transferred patients; if the conclusion holds under all scenarios, it is robust.

Exercise 8 — [L2] Interpret your group's model

Numbers from Course/Data/key_results.md (simulated data).

Q	Crude	Adjusted (complete case, n = 527 unless stated)	Missing-data sensitivity
Q1	OR 2.49 (1.70 to 3.70), n = 618; same 527 patients 2.22 (1.48 to 3.39)	OR 1.62 (1.02 to 2.59), p = 0.044; Cox HR crude 2.17 (1.55 to 3.05) → adjusted 1.46 (1.00 to 2.13), p = 0.053; PH test p = 0.10	MI (m = 20, n = 620): OR 1.70 (1.09 to 2.64), p = 0.020
Q2	OR per 10 ml/kg 1.37 (1.27 to 1.47)	1.27 (1.16 to 1.40), p < 0.001	—
Q3	−1.48 (−1.84 to −1.13) days per 10 ml/kg	−0.61 (−1.00 to −0.21) days, p = 0.003	—
Q4	OR per 1 mmol/L: CC 1.38 (1.23 to 1.55) vs MI 1.44 (1.28 to 1.61)	Adjusted (age, sex, diabetes, CKD, HF): CC 1.39 (1.24 to 1.57), n = 527 vs MI 1.45 (1.29 to 1.64), n = 620; + SOFA: CC 1.21 (1.05 to 1.39) vs MI 1.23 (1.07 to 1.42)	Lactate missing 93/620 (15.0%), mostly less sick patients (SOFA 5 vs 8; mortality 19.4% vs 30.6%): not MCAR, MAR plausible
Q5	Ratio of geometric means 1.50 (1.34 to 1.68)	1.24 (1.10 to 1.39), p < 0.001 (n = 366 survivors with complete data)	—

1. Q1 confounders (pre-specified in the SAP): age, SOFA, lactate, vasopressor, mechanical ventilation, CKD. AKI and dialysis were deliberately NOT adjusted for (they occur after day 3 — outcomes or mediators).
2. In every question the adjusted estimate is closer to "no effect" than the crude one: part of the crude association was severity (confounding by indication).

3. All adjusted CIs above exclude the null, but Q1's lower limit is close to 1 (1.02; Cox 1.00): a modest, imprecise association.
4. Complete case 527 of 620; the left-out patients were mostly those without lactate, who were less sick.
5. MI changed the estimates only slightly (Q1 1.62 → 1.70; Q4 1.39 → 1.45): the conclusions are robust to how missing lactate was handled.
6. (Q1) Day-28 survival 81.6% (negative) vs 64.1% (positive); log-rank $p < 0.001$ (unadjusted). Adjusted OR 1.62 ($p = 0.044$) and HR 1.46 ($p = 0.053$) are in the same direction and of similar size with overlapping CIs; whether p falls just below or above 0.05 does not change the interpretation.

Model sentences (Q1): "Positive fluid balance was associated with higher 28-day mortality (crude OR 2.49, 95% CI 1.70 to 3.70). After adjustment for age, SOFA, lactate, vasopressor use, mechanical ventilation and chronic kidney disease, the association was smaller (adjusted OR 1.62, 95% CI 1.02 to 2.59)."

Exercise 9 — Proposal builder 6 (what a good answer looks like)

Sample size (running example): 40% vs 30% mortality, alpha 5% two-sided, power 80% → 356 per group (spreadsheet sheet 2); $/(1 - 0.10) \rightarrow 396$ per group, 792 in total.

Analysis plan: describe with mean (SD) / median (IQR) / n (%); compare 28-day mortality with a chi-square test; report RR and risk difference with 95% CIs. Observational designs: adjusted OR (logistic) or HR (Cox) with confounders chosen in advance, about 1 model term per 10 events.

Missing-data plan (good enough): names the variables likely to be missing and why; prevention; main handling (complete case if little missing and plausibly MCAR, otherwise multiple imputation); one sensitivity analysis (e.g. complete case vs MI, or best/worst case for a missing outcome); reporting (n missing per variable, flow diagram). FLUID-ICU's SAP section 10 is a model.

Element	Good enough	Needs work
Sample size	4 inputs with a stated source	"All patients in one year"
Analysis plan	Summary + test/model + effect measure with 95% CI	"Data will be analysed with SPSS"
Confounders	Named, justified, within events-per-variable	Stepwise selection; mediators adjusted
Missing data	Expected gaps + prevention + handling + sensitivity	Nothing, or "missing data will be excluded"

Extra practice: answers

Extra practice 6.1 — Does the CI include the null?

Result	Null	Includes?	Plain sentence
OR 1.62 (1.02 to 2.59)	1	No	Associated with higher odds of death; compatible with a very small to a large increase
13.2 ml/kg (−15.2 to 41.6)	0	Yes	Too imprecise to tell; no clear difference
HR 1.17 (0.85 to 1.60)	1	Yes	No clear association at day 1
−0.61 days (−1.00 to −0.21)	0	No	Each 10 ml/kg more is associated with about 0.6 fewer ventilator-free days

Extra practice 6.2 — Choose the test

Chi-square test (Fisher's exact test if expected counts are small) · Wilcoxon signed-rank test (paired, skewed) · Kaplan–Meier curves with the log-rank test; Cox regression for a hazard ratio.

Extra practice 6.3 — Crude or adjusted?

Severity confounded the crude association: sicker patients were intubated earlier **and** died more often. After adjustment the OR is about 1 and its CI includes 1, so there is **no clear association** once severity is taken into account. The authors cannot conclude that early intubation is harmful (and residual confounding is still possible).

A.7 Session 7 · PROTECT & WRITE

Part A — Integrity scenarios

#	Colour	Why	What to do instead
1	RED — fabrication	Values "remembered" are invented data.	Record the values as missing, report the number missing, and (Level 2) handle them in the analysis. Prevent it next time: a data-collection timetable and source-document checks.
2	RED — falsification	Removing data after seeing that they weaken the result, without saying so, changes the findings.	Pre-specify rules for implausible values in the data-quality plan (Saturday 3); check the values against the source; if exclusion is justified, report it and show a sensitivity analysis with and without the patients.
3	RED — gift authorship	Seniority is not a contribution. All four ICMJE criteria are needed.	Thank the head of department in the Acknowledgements — or invite a real contribution (e.g. critical revision) so that all four criteria are met. Agree the author list in writing at the protocol stage.
4a	GREEN (with disclosure)	Language editing of the author's own text, checked by the author and disclosed, is accepted by ICMJE and COPE. The author remains responsible.	Check the target journal's AI policy; disclose tool, version and purpose.
4b	RED — breach of confidentiality	Identifiable patient data must never be pasted into a public AI tool; the analysis is also not accountable or reproducible.	De-identify; use institution-approved software; the analysis is done and checked by a named person.
5	AMBER → RED — salami slicing / redundant publication	Splitting one cohort into undisclosed separate papers distorts the evidence and can double-count patients.	One main paper with all pre-specified outcomes; any genuine secondary paper must cross-reference the main one and add something new.

#	Colour	Why	What to do instead
6	RED (amber if minor and fixed before submission) — plagiarism	Copying words verbatim without quotation marks is plagiarism even with a citation.	Paraphrase in your own words and cite, or quote with quotation marks (rare in clinical writing). Similarity-check software will flag it.

Example on the slide (a colleague lent an ultrasound machine and asks for authorship): **AMBER/RED** — providing equipment is not authorship; thank them in the Acknowledgements.

Part B — Proposal builder 7: what a good Part 4 looks like (example: observational ICU study)

B1–B3. Ethics, confidentiality, feasibility

Ethics and consent. The protocol will be submitted to the institutional research ethics committee before any data collection. Data are collected from routine ICU charts with no change in care; we will request a waiver of individual consent because most patients lack capacity on admission and the study carries no more than minimal risk. If the committee requires consent, it will be sought from the patient when able or from the legally authorised representative.

Confidentiality. Each patient receives a study code (FLU-001 ...). Names, hospital numbers and dates of birth are recorded only on a key file stored on the institution's password-protected drive, accessible to the principal investigator and one data manager. The analysis file contains the study code, age, admission month and study variables only.

Feasibility. The unit admitted about 180 adults with septic shock last year; we expect to enrol 120 eligible patients in 12 months. Two nurses per shift have been trained on the CRF; data are checked weekly. Budget headings: staff time for data entry, printing, open-access fee.

Dissemination. Target journal: a critical-care nursing journal indexed in MEDLINE; back-up: a general critical-care journal. Reporting guideline: STROBE.

Common errors to correct: "consent will be obtained" with no who/when/how; assuming retrospective studies need no ethics review; recruitment numbers copied from the sample-size calculation instead of the unit's admissions; no named place for the key file.

B4. Example publication plan (same observational ICU study)

Item	Example
Working title	"Cumulative fluid balance and 28-day mortality in adults with septic shock: a retrospective cohort study"
Target journal	A critical-care or critical-care-nursing journal indexed in MEDLINE (checked in the NLM Catalog), APC and licence known; back-up: a general critical-care open-access journal listed in DOAJ
Reporting guideline	STROBE (cohort) — not CONSORT
Registration	Not mandatory for an observational study; recommended (e.g. Clinical-Trials.gov)
Authors and CRediT	PI: Conceptualization, Methodology, Writing – original draft; two nurses: Investigation, Data curation; pharmacist: Investigation, Writing – review & editing; statistician: Formal analysis; head of department: Supervision (only if all four ICMJE criteria are met — otherwise Acknowledgements)
COI / funding	None declared / internal departmental support
Data sharing	De-identified dataset and data dictionary available from the corresponding author under a data-use agreement (if the ethics committee and consent allow)
Conference	National critical-care congress abstract after the first analysis

Abstract skeleton. Background: positive fluid balance may be linked to death in septic shock; local data are lacking; we aimed to assess its association with 28-day mortality. Methods: retrospective cohort, adults with septic shock, one ICU, 12 months; exposure = cumulative fluid balance at day 3; outcome = 28-day mortality; logistic regression adjusted for severity. Results (expected): "n patients; 28-day mortality x% vs y%; adjusted OR (95% CI)". Conclusion (if positive): "A positive day-3 fluid balance was associated with higher 28-day mortality"; (if not): "We found no evidence of an association; the CI does not exclude a clinically relevant effect".

Common errors: CONSORT named for a cohort; "impact" or "effect" in the title; an author list that includes people who only collected data; expected results written as numbers ("mortality fell by 30%").

Part C — [L2] Manuscript workshop 1: model Methods and Results for Q1

All numbers are from `Course/Data/key_results.md` and `Course/Data/results_packs/Q1/results_pack_Q1.md` (**SIMULATED DATA — NOT REAL PATIENTS**). Show these to groups only AFTER the workshop, or use them to coach. Other groups (Q2–Q5) take the same structure and their own pack.

Model Methods (~300 words)

Study design and setting. We analysed FLUID-ICU, a multicentre prospective cohort of consecutive adults with sepsis admitted to four intensive care units (ICUs A–D) between January 2023 and December 2024. FLUID-ICU is a simulated teaching dataset; no ethics approval or consent was required.

Participants. All adults with sepsis enrolled in the cohort were eligible. Patients without a cumulative fluid balance on day 3 were excluded from the exposure analysis.

Variables. The exposure was a positive cumulative fluid balance by ICU day 3, defined as total fluid input minus total fluid output over ICU days 1–3 greater than 0 ml/kg, compared with a negative or zero balance. The primary outcome was death from any cause within 28 days of ICU admission; for survival analysis, time to death was censored at day 28. Confounders were pre-specified in the statistical analysis plan: age, SOFA score, lactate, vasopressor use, mechanical ventilation and chronic kidney disease. Heart failure was pre-specified as a possible effect modifier.

Data sources and bias. Data were collected from ICU charts on a standard case report form. To address confounding by indication — sicker patients receive more fluid — we adjusted for markers of illness severity and comorbidity.

Study size. All eligible patients in the cohort were included; no separate sample-size calculation was performed.

Statistical analysis. Baseline characteristics are presented as mean (SD), median (IQR) or n (%), by fluid-balance group. 28-day mortality was compared with the chi-square test, and crude and adjusted odds ratios (ORs) were estimated by logistic regression. Survival to day 28 was described with Kaplan–Meier curves (log-rank test) and analysed with Cox regression using the same covariates; the proportional-hazards assumption was checked. Effect modification by heart failure was tested with an interaction term. The main analysis used complete cases; missing lactate was handled by multiple imputation (20 data sets) in a sensitivity analysis. Estimates are reported with 95% confidence intervals (CIs). Analyses used R 4.6 (packages gtsummary, survival, mice).

Model Results (~300 words)

Participants. Of 620 patients enrolled, 618 had a day-3 fluid balance and were included in the analysis; 527 had complete data for the adjusted model (91 without lactate or age). Mean age was 62.6 years (SD 15.2), 376 of 620 (60.6%) were men, and the median SOFA score was 7 (IQR 5 to 9).

Descriptive data. A positive balance by day 3 occurred in 373 of 618 patients (60.4%). These patients were sicker at baseline (median SOFA 8 [6 to 10] vs 6 [4 to 8]; vasopressors 205/373 [55.0%] vs 61/245 [24.9%]; mechanical ventilation 216/373 [57.9%] vs 97/245 [39.6%]) and less often had heart failure (43/373 [11.5%] vs 54/245 [22.0%]) (Table 1). Lactate was missing in 93 of 620 patients (15.0%).

Outcome data. By day 28, 134 of 373 patients (35.9%) with a positive balance and 45 of 245 (18.4%) with a negative balance had died (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4). Kaplan–Meier survival at day 28 was 64.1% and 81.6%, respectively (log-rank $p < 0.001$; Figure 1).

Main results. The crude OR for death with a positive balance was 2.49 (95% CI 1.70 to 3.70); after adjustment it was 1.62 (95% CI 1.02 to 2.59; $p = 0.044$; $n = 527$) (Table 3). The crude hazard ratio (HR) was 2.17 (95% CI 1.55 to 3.05) and the adjusted HR 1.46 (95% CI 1.00 to 2.13; $p = 0.053$).

Other analyses. With multiple imputation ($n = 620$) the adjusted OR was 1.70 (95% CI 1.09 to 2.64). The association differed by heart failure (p for interaction = 0.031): adjusted OR 4.39 (95% CI 1.65 to 12.38) with heart failure ($n = 90$) and 1.29 (95% CI 0.77 to 2.19) without ($n = 437$).

Checklist (any group)

Check	Typical fix
Statistical paragraph names a method for every result	Add the test/model and the covariates
Results start with participants and flow	Add "Of 620 enrolled, 618 ... 527 ..." (Q1)
Crude AND adjusted estimates (STROBE 16)	Add the missing one, with n
No interpretation in Results	Move "because sicker patients..." to the Discussion
Every effect has a 95% CI	Copy it from the pack
Q4: complete-case AND imputed	CC OR 1.39 (1.24 to 1.57) vs MI 1.45 (1.29 to 1.64) per mmol/L (adjusted for age, sex, diabetes, CKD, heart failure)
Q5: medians/IQR, survivors only	439 survivors; median 7.5 vs 4.8 days
Simulated data stated	Add the course ethics sentence

Extra practice: answers

Extra practice 7.1 — Methods, Results or Discussion?

R · D · M · D · R.

Extra practice 7.2 — Author or acknowledgement?

Night nurse: **acknowledgement** (did not draft, revise or approve the paper — unless invited to do so and to take responsibility). Resident: **author** (all four ICMJE criteria). Head of department: **acknowledgement** — listing them would be gift authorship. Statistician: **author**.

Extra practice 7.3 — Ethics sentences

FLUID-ICU: "This analysis used a simulated teaching dataset; no ethics approval or consent was required." Real records study: "The study was approved by [ethics committee] (approval no. [...]), which waived individual consent because only routinely collected, de-identified data were used."

A.8 Session 8 · PUBLISH

Part A — Writing exercises

A1. Fix the spin (model rewrites)

#	Model rewrite	Why
a	"The point estimate favoured the restrictive strategy, but the 95% CI was wide (−15.2 to 41.6 ml/kg) and included no difference; a larger trial would be needed to detect a smaller effect."	$p = 0.42$ is not a trend; report the estimate and its uncertainty (annotation #41).
b	"A positive fluid balance on the first ICU day was not associated with 28-day mortality (HR 1.17, 95% CI 0.85 to 1.60)."	"Impact" is causal; the cohort shows association only (#27).
c	"Positive cumulative fluid balance during the first three ICU days and 28-day mortality in sepsis: a propensity score-matched cohort study"	Title must match the observational design (#1).
d	"To our knowledge, this is among the lowest day-5 fluid balances reported in septic shock trials [refs]" — or delete.	Superlatives need protection and a citation (#40).

A2. Verb ladder

a — **was observed in** (case series) · b — **is associated with** (cohort) · c — **did not reduce** (one RCT; add "in this pilot" because it is small) · d — **reduces / causes** (consistent large RCTs and meta-analysis) · e — **was observed / was ...** % (descriptive cross-sectional).

A3. Results sentences

- "In the Day-3 matched cohort, a positive cumulative fluid balance was associated with higher 28-day mortality (HR 1.56, 95% CI 1.10 to 2.22; $P = 0.012$)."
- "By day 28, 6 of 24 patients had died in each group (relative risk 1.0, 95% CI 0.38 to 2.66)." — and in the Discussion: "the trial was too small to assess mortality".
- "In the Day-1 cohort, a positive fluid balance was not associated with 28-day mortality (HR 1.17, 95% CI 0.85 to 1.60; $P = 0.354$)."

A4. Title from PICO — running example

"Restrictive vs usual fluids and 28-day mortality in septic shock: a randomised controlled trial" (14 words) — or, for an observational group proposal: "Cumulative fluid balance and 28-day mortality in adults with septic shock: a retrospective cohort study". Check: 10–16 words, design named, no "impact", observational = "and" / "association between".

Part B — [L2] Manuscript workshop 2: model Introduction, Discussion, Abstract and Title for Q1

All numbers from Course/Data/key_results.md and results pack Q1 (**SIMULATED DATA**). References are placeholders [n]; groups use their Session 1–2 references and Papers 1–2.

Model title and keywords

Positive fluid balance by day 3 and 28-day mortality in adults with sepsis: a multicentre cohort study (16 words; design named; calibrated). Keywords: Fluid Therapy; Water-Electrolyte Balance; Critical Care; Mortality; Cohort Studies.

Model Introduction (~300 words, three paragraphs)

1. *Known.* Intravenous fluids are a cornerstone of early sepsis treatment, but fluid given beyond the resuscitation phase accumulates. A positive cumulative fluid balance has been associated with higher mortality in several observational studies, including a large propensity-matched cohort in which a positive balance on ICU day 2 was associated with higher 28-day mortality (HR 2.13) [Paper 2].
2. *Not known.* Because sicker patients receive more fluid, it remains uncertain how much of this association reflects illness severity (confounding by indication) rather than fluid itself [refs]. Randomised trials of restrictive strategies, such as a recent pilot targeting non-resuscitative fluids [Paper 1], have been small and inconclusive.
3. *Aim.* We therefore examined whether a positive cumulative fluid balance by ICU day 3 was associated with 28-day mortality in adults with sepsis, before and after adjustment for severity.

Model Discussion (~500 words, six moves)

1. **Answer.** In this multicentre cohort, a positive fluid balance by ICU day 3 was associated with higher 28-day mortality after adjustment for severity (adjusted OR 1.62, 95% CI 1.02 to 2.59).
2. **Compare.** The direction agrees with Paper 2 (day-2 HR 2.13, 95% CI 1.48 to 3.06), although our adjusted estimate was smaller; differences in case mix, timing of the balance and adjustment sets may explain this.
3. **Explain.** Most of the crude association (OR 2.49) disappeared after adjustment, consistent with confounding by indication: patients with a positive balance were sicker (median SOFA 8 vs 6; vasopressors 55.0% vs 24.9%). Fluid accumulation may contribute to organ oedema, but our data cannot show this.
4. **Limitations.** Residual confounding by unmeasured severity is likely and would exaggerate the association; reverse causation (patients who deteriorate accumulate fluid) acts in the same direction. Patients who died before day 3 had less time to accumulate fluid. The adjusted Cox HR (1.46, 95% CI 1.00 to 2.13) had a lower limit at 1. Lactate was missing in 15%, but multiple imputation gave a similar estimate (OR 1.70). The association looked stronger in heart failure (adjusted OR 4.39 vs 1.29; *p* for interaction 0.031) but was based on 90 patients and should be treated as a hypothesis. The data are simulated.
5. **Implications.** These findings do not justify restricting fluid in clinical practice; a randomised trial of fluid strategies after the resuscitation phase is needed.
6. **Conclusion.** A positive cumulative fluid balance by ICU day 3 was modestly associated with higher 28-day mortality; most of the crude association reflected severity of illness.

Model abstract (~190 words)

Background: A positive fluid balance is associated with death in sepsis, but sicker patients receive more fluid. We examined the association after adjustment for severity. **Methods:** Multicentre prospective cohort (simulated teaching data) of 620 adults with sepsis in four ICUs, 2023–2024. Exposure: positive cumulative fluid balance by ICU day 3. Outcome: 28-day mortality. Logistic and Cox regression adjusted for age, SOFA, lactate, vasopressors, mechanical ventilation and chronic kidney disease; multiple imputation for missing lactate. **Results:** Of 618 patients analysed, 373

(60.4%) had a positive balance; 134/373 (35.9%) vs 45/245 (18.4%) died. The crude OR was 2.49 (95% CI 1.70 to 3.70) and the adjusted OR 1.62 (95% CI 1.02 to 2.59); with imputation 1.70 (1.09 to 2.64). The adjusted HR was 1.46 (95% CI 1.00 to 2.13). **Conclusions:** A positive fluid balance by day 3 was modestly associated with 28-day mortality; most of the crude association reflected illness severity. A randomised trial is needed before fluid strategies are changed.

Group-specific Discussion points (from the packs)

Group	Answer sentence (numbers from key_results.md)	Key limitation
Q2 AKI	Adjusted OR 1.27 per 10 ml/kg (95% CI 1.16 to 1.40)	Reverse causation: oliguric AKI causes fluid accumulation
Q3 VFD	Adjusted -0.61 days per 10 ml/kg (95% CI -1.00 to -0.21)	VFD combines death and ventilation; skewed outcome
Q4 Lactate	Adjusted OR per mmol/L 1.39 (CC, n = 527) vs 1.45 (MI, n = 620)	Missing not at random cannot be excluded; MAR plausible (missing in less sick patients)
Q5 ICU stay	Adjusted ratio of geometric means 1.24 (95% CI 1.10 to 1.39), survivors (n = 366 adjusted)	Survivors only — conditioning on survival; skewed outcome

Part C — Getting published

C1. Spot the predatory journal

1. Red flags in the invitation (at least 8):

1. Generic flattering greeting ("Respected Professor/Doctor"), unsolicited mass e-mail.
2. Free e-mail address, not a publisher domain.
3. Vague claim to have read "your excellent article" (no title).
4. Absurdly broad scope (medicine, nursing, engineering, management; "all areas of science").
5. A fake metric ("International Universal Index" impact factor — only Clarivate issues the Journal Impact Factor).
6. Guaranteed review in 48 h and publication in 5 days.
7. Fees hidden until after acceptance.
8. Pressure tactics ("this week only"), multiple exclamation marks, unusual grammar.
9. Submission by replying to an e-mail instead of an editorial system.

2. Journal profiles:

Journal	Verdict	Reason
J1	Trust	Indexing verified at source, transparent fees and timelines, verifiable editors, uses reporting checklists.
J2	Avoid	Google Scholar / Index Copernicus are not quality indexes; 72-h acceptance; hidden fees; unverifiable board.
J3	Trust / check further	DOAJ listing and clear policies are good signs; no impact factor is normal for a new or small journal. Check it publishes work like yours.
J4	Avoid	A "hijacked" or look-alike journal. Always reach the real journal through its publisher's website or the NLM Catalog, not through an e-mail link.

C2. Responding to reviewers (model rows)

#	Model response	Change
R1.1	"Thank you. We agree that our design supports association, not causation. We have changed the title to '... and ...: a cohort study' and replaced 'improves' with 'was associated with' throughout."	Title p.1; abstract; Discussion p.x
R1.2	"Thank you. The main result is now reported as RR 0.78 (95% CI 0.61 to 0.99) in the abstract, Results and Table 2."	Abstract; Results p.x
R1.3	"Thank you for spotting this. 212 patients were included and 4 were excluded from the adjusted model because of missing lactate; the flow diagram and text now show 212 → 208 with reasons."	Fig. 1; Results p.x
R2.1	(polite rebuttal) "We agree 1-year mortality is important. Our protocol, approved by the ethics committee, followed patients to day 28 only, so these data were not collected. We now mention this as a limitation and suggest it for future studies."	Limitations p.x

Marking points: every comment answered; thanks without grovelling; concrete location of every change; no new data invented; disagreement supported by a reason (protocol, ethics approval).

Extra practice: answers

Extra practice 8.1 — Calibrate the verb

1. "A positive fluid balance was associated with higher mortality in sepsis."
2. "The point estimate favoured the restrictive protocol, but the 95% CI was wide and included no difference ($p = 0.42$); a larger trial would be needed."

Extra practice 8.2 — Order the Discussion

Answer (1): "A positive fluid balance by day 3 was associated with higher 28-day mortality (adjusted OR 1.62...)". Compare (2): "This agrees with a large registry cohort...". Explain (3): "Fluid accumulation may cause tissue oedema...". Limitations (4): "Residual confounding by severity...". Implications (5): "A randomised trial is needed...". Conclusion (6): "In this simulated cohort..."

Extra practice 8.3 — Answer the reviewer

"We thank the reviewer. Because this is an observational cohort, our data show an association and cannot show that restricting fluid saves lives. We have therefore kept 'associated with' and added to the Discussion that a randomised trial is needed to test a restrictive strategy (page ..., lines ...)."

Extra practice 8.4 — Red flags

Any three: a flattering, unsolicited e-mail (often from a free-mail address); a scope covering "all areas of science"; a fake impact factor from an unknown index; guaranteed peer review in days; fees hidden until after acceptance; submission by replying to the e-mail; a name that mimics a well-known journal. Check the journal in **DOAJ**, the **NLM Catalog** (currently indexed for MEDLINE) and with **Think.Check.Submit** — and ask your librarian.

APPENDIX B

Glossary — English–Vietnamese

Key terms grouped by the session in which they are taught, with the Vietnamese term and a plain-language definition in both languages. Sessions 5–8 include the terms needed for the course dataset and the group manuscript. The definitions are written for this course; they are not formal textbook definitions. Abbreviations widely used in Vietnamese practice (PICO, RCT, CRF, OR, RR, NNT, GCP ...) are kept in English.

Các thuật ngữ chính, xếp theo buổi học mà chúng được giảng dạy, kèm thuật ngữ tiếng Việt và định nghĩa bằng ngôn ngữ đơn giản. Các chữ viết tắt thường dùng (PICO, RCT, CRF, OR, RR, NNT, GCP...) được giữ nguyên tiếng Anh.

Session 1 — ASK · Buổi 1 — ĐẶT CÂU HỎI

Term · Thuật ngữ	Definition · Định nghĩa
Research cycle Chu trình nghiên cứu	The repeating path from a clinical problem to a question, a study, results, publication and changed practice — which raises new questions. <i>Con đường lặp lại từ vấn đề lâm sàng đến câu hỏi, nghiên cứu, kết quả, công bố và thay đổi thực hành — từ đó nảy sinh câu hỏi mới.</i>
Evidence-based medicine Y học dựa trên bằng chứng	Combining the best research evidence with clinical expertise and patient values to make care decisions. <i>Kết hợp bằng chứng nghiên cứu tốt nhất với kinh nghiệm lâm sàng và giá trị, mong muốn của người bệnh để ra quyết định chăm sóc.</i>
Levels of evidence (evidence pyramid) Mức độ bằng chứng (tháp bằng chứng)	A ranking of study types by how well they protect against bias; systematic reviews of trials at the top, expert opinion at the bottom. <i>Thứ bậc các loại nghiên cứu theo khả năng hạn chế sai lệch; tổng quan hệ thống các thử nghiệm ở đỉnh, ý kiến chuyên gia ở đáy.</i>
Quantitative / qualitative / mixed-methods research Nghiên cứu định lượng / định tính / phương pháp hỗn hợp	Quantitative research measures and counts; qualitative research explores experiences and meanings through words (interviews, observation); mixed methods combines both. <i>Nghiên cứu định lượng đo lường và đếm; nghiên cứu định tính tìm hiểu trải nghiệm và ý nghĩa qua lời nói (phỏng vấn, quan sát); phương pháp hỗn hợp kết hợp cả hai.</i>
Primary vs secondary research Nghiên cứu sơ cấp và nghiên cứu thứ cấp	Primary research collects new data; secondary research analyses existing data or studies (e.g. a records review or a systematic review). <i>Nghiên cứu sơ cấp thu thập dữ liệu mới; nghiên cứu thứ cấp phân tích dữ liệu hoặc nghiên cứu có sẵn (ví dụ hồi cứu hồ sơ bệnh án, tổng quan hệ thống).</i>
Clinical audit Đánh giá thực hành lâm sàng (audit lâm sàng)	Measuring current practice against an agreed standard to see whether the standard is met. <i>Đo lường thực hành hiện tại so với một tiêu chuẩn đã thống nhất để xem tiêu chuẩn đó có được đáp ứng hay không.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Quality improvement (QI) and PDSA cycle Cải tiến chất lượng (QI) và chu trình PDSA	Making and testing local changes to improve care, often in small Plan–Do–Study–Act cycles. <i>Thực hiện và thử nghiệm các thay đổi tại đơn vị để cải thiện chăm sóc, thường theo các chu trình nhỏ Lập kế hoạch – Thực hiện – Nghiên cứu – Hành động (PDSA).</i>
Research question Câu hỏi nghiên cứu	The single, specific question a study is designed to answer. <i>Câu hỏi duy nhất, cụ thể mà nghiên cứu được thiết kế để trả lời.</i>
PICO(T) PICO(T)	A frame for a clinical question: Population, Intervention (or exposure), Comparison, Outcome, (Time). <i>Khung xây dựng câu hỏi lâm sàng: Đối tượng (P), Can thiệp hoặc phơi nhiễm (I), So sánh (C), Kết cục (O), (Thời gian – T).</i>
Population Đối tượng nghiên cứu (quần thể)	The patients the question is about, e.g. adults with septic shock in the ICU. <i>Nhóm người bệnh mà câu hỏi hướng đến, ví dụ người lớn bị sốc nhiễm khuẩn tại ICU.</i>
Intervention Can thiệp	A treatment or action deliberately given, e.g. a restrictive fluid strategy. <i>Phương pháp điều trị hoặc hành động được chủ động áp dụng, ví dụ chiến lược hạn chế dịch.</i>
Exposure Phơi nhiễm (yếu tố tiếp xúc)	A factor people have or receive that is not assigned by the researcher, e.g. a large fluid balance. <i>Yếu tố mà người bệnh có hoặc nhận được nhưng không do nhà nghiên cứu chỉ định, ví dụ cân bằng dịch dương nhiều.</i>
Comparison (comparator) Nhóm so sánh (nhóm chứng)	What the intervention is compared with, e.g. usual care or placebo. <i>Đối tượng hoặc phương pháp dùng để so sánh với can thiệp, ví dụ chăm sóc thường quy hoặc giả dược.</i>
Outcome Kết cục	The result that is measured to answer the question, e.g. death by day 28. <i>Kết quả được đo lường để trả lời câu hỏi, ví dụ tử vong trong 28 ngày.</i>
FINER criteria Tiêu chí FINER	A check that a question is Feasible, Interesting, Novel, Ethical and Relevant. <i>Bộ tiêu chí kiểm tra câu hỏi có Khả thi, Thú vị, Mới, Đạo đức và Phù hợp (thiết thực) hay không.</i>
Aim Mục tiêu tổng quát	The broad purpose of the study in one sentence. <i>Mục đích chung của nghiên cứu, viết trong một câu.</i>
Objective Mục tiêu cụ thể	A specific, measurable step towards the aim, e.g. "to compare 28-day mortality between groups". <i>Bước cụ thể, đo lường được để đạt mục tiêu tổng quát, ví dụ "so sánh tỷ lệ tử vong 28 ngày giữa hai nhóm".</i>
Hypothesis Giả thuyết nghiên cứu	A testable prediction about the result, naming the groups and the outcome. <i>Dự đoán có thể kiểm định về kết quả, nêu rõ các nhóm và kết cục.</i>
Null hypothesis Giả thuyết không (H0)	The starting assumption that there is no difference or no effect. <i>Giả định ban đầu rằng không có khác biệt hoặc không có tác động.</i>

Session 2 — FIND & READ · Buổi 2 — TÌM VÀ ĐỌC Y VĂN

Term · Thuật ngữ	Definition · Định nghĩa
Literature review Tổng quan tài liệu	A structured summary of what published studies already show about the question. <i>Bản tóm tắt có hệ thống những gì các nghiên cứu đã công bố cho thấy về câu hỏi.</i>
Search strategy Chiến lược tìm kiếm	The planned combination of terms, operators and filters used to search a database. <i>Sự kết hợp có kế hoạch giữa từ khóa, toán tử và bộ lọc để tìm kiếm trong cơ sở dữ liệu.</i>

Term · Thuật ngữ	Definition · Định nghĩa
MeSH (Medical Subject Headings) Tiêu đề chủ đề y khoa (MeSH)	PubMed's controlled vocabulary: standard terms that group articles on the same topic, whatever words the authors used. <i>Bộ từ vựng chuẩn của PubMed, gom các bài báo cùng chủ đề dù tác giả dùng từ ngữ khác nhau.</i>
Boolean operators (AND, OR, NOT) Toán tử Boolean (AND, OR, NOT)	Words that combine search terms: AND narrows, OR widens, NOT excludes. Các từ dùng để kết hợp từ khóa: AND thu hẹp, OR mở rộng, NOT loại trừ.
Grey literature Tài liệu xám	Research not published in journals: reports, theses, conference abstracts, trial registries. <i>Tài liệu nghiên cứu không đăng trên tạp chí: báo cáo, luận văn, tóm tắt hội nghị, cơ sở dữ liệu đăng ký thử nghiệm.</i>
Reference manager Phần mềm quản lý tài liệu tham khảo	Software (e.g. Zotero, Mendeley, EndNote) that stores references and inserts citations in the chosen style. <i>Phần mềm (ví dụ Zotero, Mendeley, EndNote) lưu trữ tài liệu tham khảo và chèn trích dẫn theo kiểu đã chọn.</i>
Citation style (Vancouver) Kiểu trích dẫn (Vancouver)	The format for citing and listing references; most medical journals use the numbered Vancouver style. <i>Quy cách trích dẫn và liệt kê tài liệu tham khảo; đa số tạp chí y khoa dùng kiểu Vancouver đánh số.</i>
Paraphrasing Diễn giải bằng lời của mình	Expressing someone else's idea in your own words and sentence structure, with a citation. <i>Trình bày ý tưởng của người khác bằng từ ngữ và cấu trúc câu của chính mình, kèm trích dẫn nguồn.</i>
IMRaD Cấu trúc IMRaD	The standard structure of a research paper: Introduction, Methods, Results and Discussion. <i>Cấu trúc chuẩn của bài báo khoa học: Đặt vấn đề, Phương pháp, Kết quả và Bàn luận.</i>
Structured abstract Tóm tắt có cấu trúc	A short summary of a paper under fixed headings, usually Background, Methods, Results and Conclusions. <i>Bản tóm tắt ngắn của bài báo theo các mục cố định, thường là Đặt vấn đề, Phương pháp, Kết quả và Kết luận.</i>
Spin Trình bày sai lệch kết quả ("spin")	Reporting that makes results look more favourable than they are, e.g. stressing secondary outcomes when the primary outcome was negative. <i>Cách trình bày khiến kết quả có vẻ tích cực hơn thực tế, ví dụ nhấn mạnh kết cục phụ khi kết cục chính không có khác biệt.</i>
Problem statement Đặt vấn đề	A short description of the clinical problem, how big it is, and what is not yet known. <i>Mô tả ngắn gọn vấn đề lâm sàng, mức độ quan trọng và những điều còn chưa biết.</i>
Keywords Từ khóa	3–6 terms (ideally MeSH terms) chosen so that others can find your paper. <i>3–6 thuật ngữ (nên dùng thuật ngữ MeSH) giúp người khác tìm thấy bài báo của bạn.</i>

Session 3 — DESIGN · Buổi 3 — THIẾT KẾ NGHIÊN CỨU

Term · Thuật ngữ	Definition · Định nghĩa
Descriptive study Nghiên cứu mô tả	A study that describes who has a condition, where and when, without comparing groups to test a cause. <i>Nghiên cứu mô tả ai mắc bệnh, ở đâu, khi nào, không so sánh các nhóm để kiểm định nguyên nhân.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Analytical study Nghiên cứu phân tích	A study that compares groups to test whether an exposure or treatment is linked to an outcome. <i>Nghiên cứu so sánh các nhóm để kiểm định mối liên quan giữa phơi nhiễm hoặc điều trị và kết cục.</i>
Case report / case series Báo cáo ca bệnh / loạt ca bệnh	A description of one or a few patients with an unusual condition or response. <i>Mô tả một hoặc một số người bệnh có tình trạng hoặc đáp ứng bất thường.</i>
Cross-sectional study Nghiên cứu cắt ngang	Exposure and outcome are measured at the same point in time, like a snapshot. <i>Phơi nhiễm và kết cục được đo cùng một thời điểm, giống như một "bức ảnh chụp nhanh".</i>
Case-control study Nghiên cứu bệnh–chứng	People with the outcome (cases) and without it (controls) are compared for past exposures. <i>So sánh người có kết cục (nhóm bệnh) và không có kết cục (nhóm chứng) về các phơi nhiễm trong quá khứ.</i>
Cohort study (prospective / retrospective) Nghiên cứu thuần tập (tiền cứu / hồi cứu)	People grouped by exposure are followed over time to see who develops the outcome; data may be collected forwards or from existing records. <i>Người tham gia được chia nhóm theo phơi nhiễm và theo dõi theo thời gian để xem ai xuất hiện kết cục; dữ liệu có thể thu thập về phía trước hoặc từ hồ sơ có sẵn.</i>
Randomised controlled trial (RCT) Thử nghiệm lâm sàng ngẫu nhiên có đối chứng (RCT)	Participants are allocated by chance to the intervention or the comparator, then followed for the outcome. <i>Người tham gia được phân bổ ngẫu nhiên vào nhóm can thiệp hoặc nhóm chứng, sau đó được theo dõi kết cục.</i>
Randomisation Phân bổ ngẫu nhiên	Allocating participants by chance so that groups are similar in both measured and unmeasured factors. <i>Chia nhóm người tham gia một cách ngẫu nhiên để các nhóm tương đồng về cả yếu tố đo được và không đo được.</i>
Blinding (masking) Làm mù	Keeping patients, staff or assessors unaware of which group a patient is in. <i>Giữ cho người bệnh, nhân viên hoặc người đánh giá không biết người bệnh thuộc nhóm nào.</i>
Systematic review Tổng quan hệ thống	A review that finds, appraises and summarises all studies on a question using a pre-set, reproducible method. <i>Tổng quan tìm kiếm, thẩm định và tổng hợp tất cả nghiên cứu về một câu hỏi theo phương pháp định trước, có thể lặp lại.</i>
Meta-analysis Phân tích gộp	Statistical combination of the results of several studies into one overall estimate. <i>Kỹ thuật thống kê gộp kết quả của nhiều nghiên cứu thành một ước lượng chung.</i>
Crossover trial Thử nghiệm bắt chéo	Each participant receives both treatments in turn, separated by a washout period; suits stable, chronic conditions. <i>Mỗi người tham gia lần lượt nhận cả hai phương pháp điều trị, cách nhau một giai đoạn thải trừ; phù hợp với bệnh mạn tính, ổn định.</i>
Cluster randomised trial Thử nghiệm ngẫu nhiên theo cụm	Groups such as wards or hospitals, not individual patients, are randomised. <i>Phân bổ ngẫu nhiên theo nhóm (ví dụ khoa phòng hoặc bệnh viện) thay vì theo từng người bệnh.</i>
Factorial trial Thử nghiệm giai thừa	Tests two or more interventions at once by randomising to every combination (e.g. A, B, A + B, neither). <i>Đánh giá cùng lúc hai hay nhiều can thiệp bằng cách phân bổ ngẫu nhiên vào mọi tổ hợp (ví dụ A, B, A + B, không can thiệp).</i>
Superiority vs non-inferiority trial Thử nghiệm vượt trội và thử nghiệm không kém hơn	A superiority trial asks whether a treatment is better; a non-inferiority trial asks whether it is not worse by more than a pre-set margin (useful if it is cheaper, simpler or safer). <i>Thử nghiệm vượt trội hỏi liệu điều trị có tốt hơn không; thử nghiệm không kém hơn hỏi liệu điều trị có kém hơn quá một ngưỡng định trước hay không (hữu ích khi điều trị rẻ hơn, đơn giản hơn hoặc an toàn hơn).</i>

Term · Thuật ngữ	Definition · Định nghĩa
Pilot / feasibility study Nghiên cứu thí điểm / nghiên cứu khả thi	A small study to test whether a larger study can be done (recruitment, procedures), not to test whether the treatment works. <i>Nghiên cứu nhỏ để kiểm tra liệu một nghiên cứu lớn hơn có thực hiện được không (tuyển chọn, quy trình), không nhằm đánh giá hiệu quả điều trị.</i>
Quasi-experimental (before–after) study Nghiên cứu bán thực nghiệm (trước–sau)	An intervention is introduced without randomisation and outcomes are compared before and after; common in QI. <i>Cán thiệp được áp dụng không qua phân bổ ngẫu nhiên và kết cục được so sánh trước và sau; thường gặp trong cải tiến chất lượng.</i>
Interrupted time series Chuỗi thời gian ngắt quãng	Many measurements before and after a change, to see whether the level or trend shifts beyond the existing trend. <i>Nhiều lần đo trước và sau một thay đổi, để xem mức hoặc xu hướng có thay đổi vượt ra ngoài xu hướng sẵn có hay không.</i>
Qualitative study (interviews, focus groups) Nghiên cứu định tính (phỏng vấn sâu, thảo luận nhóm tập trung)	Explores why and how — experiences, beliefs, barriers — through words rather than numbers. <i>Tìm hiểu “tại sao” và “như thế nào” — trải nghiệm, niềm tin, rào cản — qua lời nói thay vì con số.</i>

Session 4 — BIAS & EFFECT · Buổi 4 — SAI LỆCH VÀ ĐO LƯỜNG TÁC ĐỘNG

Term · Thuật ngữ	Definition · Định nghĩa
Bias Sai lệch (sai số hệ thống)	A systematic error in design or conduct that pushes results away from the truth. <i>Sai số có hệ thống trong thiết kế hoặc thực hiện, làm kết quả lệch khỏi sự thật.</i>
Selection bias Sai lệch chọn lựa	Bias from the way participants are chosen or lost, so the groups are not comparable. <i>Sai lệch do cách chọn hoặc mất người tham gia, khiến các nhóm không còn so sánh được.</i>
Information bias (recall, observer) Sai lệch thông tin (nhớ lại, người quan sát)	Bias from measuring exposure or outcome differently between groups, e.g. cases remember past exposures better. <i>Sai lệch do đo phơi nhiễm hoặc kết cục khác nhau giữa các nhóm, ví dụ nhóm bệnh nhớ phơi nhiễm trong quá khứ rõ hơn.</i>
Confounding (confounder) Nhiều (yếu tố gây nhiễu)	A mixing of effects: a third factor linked to both exposure and outcome distorts their apparent relationship. <i>Sự pha trộn tác động: một yếu tố thứ ba liên quan đến cả phơi nhiễm và kết cục làm sai lệch mối liên quan quan sát được.</i>
Confounding by indication Nhiều do chỉ định	Sicker patients receive more of a treatment, so the treatment looks harmful because of the illness, not the treatment. <i>Người bệnh nặng hơn được điều trị nhiều hơn, nên điều trị có vẻ gây hại nhưng thực ra do mức độ bệnh.</i>
Matching Ghép cặp	Choosing comparison participants with similar values of key factors (e.g. age, severity) to reduce confounding. <i>Chọn người so sánh có đặc điểm chính tương tự (ví dụ tuổi, mức độ nặng) để giảm nhiễu.</i>
Effect modification (interaction) Thay đổi tác động (tương tác)	The effect of an exposure differs between subgroups (e.g. it works in younger but not older patients); it should be reported, not “adjusted away”. <i>Tác động của một phơi nhiễm khác nhau giữa các phân nhóm (ví dụ có hiệu quả ở người trẻ nhưng không ở người cao tuổi); cần được báo cáo, không “hiệu chỉnh để loại bỏ”.</i>
Internal validity Giá trị nội tại	Whether the result is true for the people studied, i.e. free of important bias and confounding. <i>Kết quả có đúng với chính những người được nghiên cứu hay không, tức là không bị sai lệch và nhiễu đáng kể.</i>

Term · Thuật ngữ	Definition · Định nghĩa
External validity (generalisability) Giá trị ngoại suy (tính khái quát hóa)	Whether the result applies to other patients and settings, such as your own unit. <i>Kết quả có áp dụng được cho người bệnh và bối cảnh khác hay không, ví dụ đơn vị của bạn.</i>
Bradford Hill criteria Tiêu chí Bradford Hill	Points to weigh when judging whether an association is causal: strength, consistency, temporality, dose–response, plausibility and others. <i>Các điểm cần cân nhắc khi đánh giá một mối liên quan có phải quan hệ nhân quả hay không: độ mạnh, tính nhất quán, trình tự thời gian, quan hệ liều–đáp ứng, tính hợp lý và các tiêu chí khác.</i>
Reverse causation Quan hệ nhân quả ngược	The outcome (or early disease) causes the exposure, not the other way round. <i>Kết cục (hoặc bệnh ở giai đoạn sớm) gây ra phơi nhiễm, chứ không phải ngược lại.</i>
Prevalence Tỷ lệ hiện mắc	The proportion of people who have a condition at one point in time (all existing cases). <i>Tỷ lệ người đang mắc bệnh tại một thời điểm (tất cả các ca hiện có).</i>
Incidence Tỷ lệ mới mắc	The proportion of people free of a condition who develop it over a period (new cases). <i>Tỷ lệ người chưa mắc bệnh xuất hiện bệnh trong một khoảng thời gian (ca mới).</i>
Risk Nguy cơ	The number with the outcome divided by the number in the group, e.g. 30 deaths / 150 patients = 0.20. <i>Số người có kết cục chia cho tổng số người trong nhóm, ví dụ 30 tử vong / 150 người bệnh = 0,20.</i>
2×2 table Bảng 2×2	A table of exposure (yes/no) by outcome (yes/no) used to calculate risks, RR, OR and NNT. <i>Bảng phơi nhiễm (có/không) theo kết cục (có/không), dùng để tính nguy cơ, RR, OR và NNT.</i>
Relative risk (RR) Nguy cơ tương đối (RR)	Risk in one group divided by risk in the other; 1 means no difference. <i>Nguy cơ ở nhóm này chia cho nguy cơ ở nhóm kia; bằng 1 nghĩa là không khác biệt.</i>
Odds ratio (OR) Tỷ số chênh (OR)	Odds of the outcome in one group divided by odds in the other; close to the RR when the outcome is rare. <i>Số chênh của kết cục ở nhóm này chia cho số chênh ở nhóm kia; gần bằng RR khi kết cục hiếm gặp.</i>
Risk difference (RD) Hiệu số nguy cơ (RD)	Risk in one group minus risk in the other (absolute difference). <i>Nguy cơ ở nhóm này trừ nguy cơ ở nhóm kia (chênh lệch tuyệt đối).</i>
Number needed to treat (NNT) Số người cần điều trị (NNT)	How many patients must be treated to prevent one extra bad outcome; $NNT = 1 \div \text{risk difference}$. <i>Số người bệnh cần được điều trị để ngăn ngừa thêm một kết cục xấu; $NNT = 1 \div \text{hiệu số nguy cơ}$.</i>
Sensitivity Độ nhạy	The proportion of people with the disease whom a test correctly identifies as positive. <i>Tỷ lệ người có bệnh được xét nghiệm phát hiện đúng là dương tính.</i>
Specificity Độ đặc hiệu	The proportion of people without the disease whom a test correctly identifies as negative. <i>Tỷ lệ người không có bệnh được xét nghiệm xác định đúng là âm tính.</i>
Positive / negative predictive value (PPV, NPV) Giá trị tiên đoán dương / âm (PPV, NPV)	The chance that a positive (or negative) test result is correct; unlike sensitivity and specificity, they depend on prevalence. <i>Xác suất kết quả xét nghiệm dương (hoặc âm) là đúng; khác với độ nhạy và độ đặc hiệu, chúng phụ thuộc vào tỷ lệ hiện mắc.</i>
ROC curve and AUC Đường cong ROC và diện tích dưới đường cong (AUC)	A plot of sensitivity against $1 - \text{specificity}$ at every cut-off; the AUC (0.5 = chance, 1 = perfect) summarises how well a test separates sick from well. <i>Đồ thị độ nhạy theo ($1 - \text{độ đặc hiệu}$) tại mọi ngưỡng cắt; AUC (0,5 = ngẫu nhiên, 1 = hoàn hảo) tóm tắt khả năng phân biệt người có bệnh và không có bệnh.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Likelihood ratio (LR) Tỷ số khả dĩ (LR)	How much a test result changes the odds of disease; LR+ above 10 or LR– below 0.1 change decisions strongly. <i>Mức độ một kết quả xét nghiệm làm thay đổi khả năng mắc bệnh; LR+ trên 10 hoặc LR– dưới 0,1 làm thay đổi quyết định rõ rệt.</i>
PRISMA flow diagram Sơ đồ PRISMA	The diagram in a systematic review showing how many records were found, screened, excluded and included. <i>Sơ đồ trong tổng quan hệ thống cho biết số tài liệu được tìm thấy, sàng lọc, loại trừ và đưa vào.</i>
Forest plot Biểu đồ rừng (forest plot)	A chart showing effect estimates and their CIs as squares and lines, with a vertical line at no effect. <i>Biểu đồ trình bày các ước lượng tác động và khoảng tin cậy bằng ô vuông và đoạn thẳng, có một đường dọc tại giá trị không tác động.</i>
Heterogeneity (I^2) Tính không đồng nhất (I^2)	How much study results differ beyond chance; an I^2 above about 50–75% means substantial heterogeneity. <i>Mức độ kết quả các nghiên cứu khác nhau vượt quá yếu tố ngẫu nhiên; I^2 trên khoảng 50–75% là không đồng nhất đáng kể.</i>
Pooled estimate Ước lượng gộp	The single combined effect from a meta-analysis, shown as the diamond on a forest plot. <i>Ước lượng tác động chung từ phân tích gộp, thể hiện bằng hình thoi trên biểu đồ rừng.</i>
Risk-of-bias tools Công cụ đánh giá nguy cơ sai lệch	Structured checklists for judging bias, chosen by design: RoB 2 (RCTs), ROBINS-I (non-randomised studies of interventions), Newcastle–Ottawa (cohort, case-control), QUADAS-2 (diagnostic accuracy), AMSTAR 2 (systematic reviews). <i>Bảng kiểm có cấu trúc để đánh giá sai lệch, chọn theo thiết kế: RoB 2 (RCT), ROBINS-I (nghiên cứu can thiệp không ngẫu nhiên), Newcastle–Ottawa (thuần tập, bệnh–chứng), QUADAS-2 (độ chính xác chẩn đoán), AMSTAR 2 (tổng quan hệ thống).</i>

Session 5 — DATA · Buổi 5 — DỮ LIỆU

Term · Thuật ngữ	Definition · Định nghĩa
Variable Biến số	Any characteristic recorded for each patient, e.g. age, sex, fluid balance, death. <i>Bất kỳ đặc điểm nào được ghi nhận cho từng người bệnh, ví dụ tuổi, giới, cân bằng dịch, tử vong.</i>
Binary variable Biến nhị phân	A variable with two categories, e.g. died / alive. <i>Biến có hai giá trị, ví dụ tử vong / sống.</i>
Nominal variable Biến định danh	Categories with no natural order, e.g. blood group. <i>Biến phân loại không có thứ tự tự nhiên, ví dụ nhóm máu.</i>
Ordinal variable Biến thứ hạng	Categories with a natural order but unequal steps, e.g. pressure injury stage 1–4. <i>Biến phân loại có thứ tự nhưng khoảng cách giữa các bậc không bằng nhau, ví dụ loét từ độ 1–4.</i>
Continuous variable Biến liên tục	A measured number that can take any value in a range, e.g. fluid balance in ml/kg. <i>Số đo có thể nhận bất kỳ giá trị nào trong một khoảng, ví dụ cân bằng dịch tính bằng ml/kg.</i>
Case report form (CRF) Phiếu thu thập dữ liệu (CRF)	The form on which study data are recorded for each participant. <i>Biểu mẫu dùng để ghi dữ liệu nghiên cứu của từng người tham gia.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Data dictionary Từ điển dữ liệu	A table defining every variable: name, meaning, type, unit or codes, allowed values. <i>Bảng định nghĩa từng biến: tên, ý nghĩa, loại biến, đơn vị hoặc mã hóa, giá trị cho phép.</i>
Data validation (range and consistency checks) Kiểm tra tính hợp lệ của dữ liệu (kiểm tra khoảng giá trị và tính nhất quán)	Checks that catch impossible or contradictory values, e.g. age 200, or discharge before admission. <i>Các bước kiểm tra phát hiện giá trị vô lý hoặc mâu thuẫn, ví dụ tuổi 200, hoặc ngày ra viện trước ngày nhập viện.</i>
Electronic data capture (EDC) Thu thập dữ liệu điện tử (EDC)	Entering study data directly into a secure system (e.g. REDCap, KoBoToolbox) with built-in checks. <i>Nhập dữ liệu nghiên cứu trực tiếp vào hệ thống bảo mật (ví dụ REDCap, KoBoToolbox) có sẵn các bước kiểm tra.</i>
Double data entry Nhập liệu kép	Two people enter the same data independently and the differences are checked. <i>Hai người nhập cùng một bộ dữ liệu một cách độc lập và các điểm khác biệt được kiểm tra lại.</i>
Audit trail Nhật ký thay đổi dữ liệu (audit trail)	An automatic record of who changed which value, when and why. <i>Bản ghi tự động cho biết ai đã thay đổi giá trị nào, khi nào và vì sao.</i>
Missing data Dữ liệu bị thiếu	Values that should have been recorded but were not; they can bias results and reduce power. <i>Giá trị lẽ ra phải được ghi nhận nhưng không có; có thể gây sai lệch kết quả và giảm lực thống kê.</i>
Raw data vs clean data Dữ liệu thô và dữ liệu đã làm sạch	Raw data are kept exactly as collected and never edited; clean data are a separate, corrected copy produced by a documented script. <i>Dữ liệu thô được giữ đúng như khi thu thập và không bao giờ bị chỉnh sửa; dữ liệu sạch là một bản sao riêng đã được sửa lỗi bằng một quy trình (script) có ghi chép.</i>
Data cleaning Làm sạch dữ liệu	Finding and fixing errors in a dataset (duplicates, impossible values, inconsistent codes, wrong units) before analysis, in a documented and repeatable way. <i>Phát hiện và sửa lỗi trong bộ dữ liệu (bản ghi trùng, giá trị vô lý, mã hóa không thống nhất, sai đơn vị) trước khi phân tích, theo cách có ghi chép và lặp lại được.</i>
Cleaning log Nhật ký làm sạch dữ liệu	A record of every problem found and every change made, with the reason (e.g. "ID 0412: weight 7 kg → missing; implausible"). <i>Bản ghi mọi vấn đề phát hiện được và mọi thay đổi đã thực hiện, kèm lý do (ví dụ "ID 0412: cân nặng 7 kg → để trống; giá trị vô lý").</i>
Duplicate record Bản ghi trùng lặp	The same patient entered more than once; duplicates must be found and removed so that each patient is counted once. <i>Cùng một người bệnh được nhập nhiều lần; cần phát hiện và loại bỏ để mỗi người bệnh chỉ được tính một lần.</i>
Sentinel value (missing-value code) Giá trị quy ước cho dữ liệu thiếu (ví dụ 999)	A number such as 999 or –99 used to mean "missing"; it must be recoded as missing before analysis or it will distort the results. <i>Một con số như 999 hoặc –99 được dùng để biểu thị "thiếu dữ liệu"; phải mã hóa lại thành giá trị thiếu trước khi phân tích, nếu không sẽ làm sai lệch kết quả.</i>
Implausible value Giá trị vô lý (ngoài ngưỡng có thể)	A value outside the possible range (e.g. age 200, weight 7 kg in an adult); checked against source records, then corrected or set to missing. <i>Giá trị nằm ngoài ngưỡng có thể xảy ra (ví dụ tuổi 200, cân nặng 7 kg ở người lớn); cần đối chiếu hồ sơ gốc, sau đó sửa lại hoặc để trống.</i>
Derived variable Biến dẫn xuất	A variable calculated from others, e.g. cumulative fluid balance (ml/kg) from daily fluid in, fluid out and body weight. <i>Biến được tính từ các biến khác, ví dụ cân bằng dịch tích lũy (ml/kg) từ lượng dịch vào, dịch ra hằng ngày và cân nặng.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Mean Trung bình	The sum of values divided by their number; pulled by extreme values. <i>Tổng các giá trị chia cho số giá trị; bị ảnh hưởng bởi các giá trị cực đoan.</i>
Median Trung vị	The middle value when data are ordered; not affected by a few extreme values. <i>Giá trị ở giữa khi sắp xếp dữ liệu theo thứ tự; ít bị ảnh hưởng bởi vài giá trị cực đoan.</i>
Standard deviation (SD) Độ lệch chuẩn (SD)	How spread out values are around the mean; used with symmetric (normal) data. <i>Mức độ phân tán của các giá trị quanh trung bình; dùng cho dữ liệu đối xứng (phân phối chuẩn).</i>
Interquartile range (IQR) Khoảng tứ phân vị (IQR)	The range of the middle 50% of values (25th to 75th percentile); used with the median. <i>Khoảng chứa 50% giá trị ở giữa (từ bách phân vị 25 đến 75); dùng kèm trung vị.</i>
Normal distribution Phân phối chuẩn	A symmetric, bell-shaped spread of values around the mean. <i>Phân bố các giá trị đối xứng hình chuông quanh giá trị trung bình.</i>
Skewed distribution Phân phối lệch	An asymmetric spread with a long tail on one side, e.g. length of stay. <i>Phân bố không đối xứng, có một đuôi dài về một phía, ví dụ thời gian nằm viện.</i>
Table 1 Bảng 1 (đặc điểm nền)	The table that describes the baseline characteristics of the participants, usually by group. <i>Bảng mô tả các đặc điểm ban đầu của người tham gia, thường theo từng nhóm.</i>
Dynamite plot Biểu đồ cột kèm thanh sai số ("dynamite plot")	A bar showing only a mean with an error bar; it hides the data — prefer dot plots or box plots. <i>Biểu đồ cột chỉ thể hiện giá trị trung bình kèm thanh sai số; che giấu dữ liệu thực — nên dùng biểu đồ chấm hoặc biểu đồ hộp.</i>
Box plot Biểu đồ hộp	Shows the median, IQR and range of a variable; good for skewed data and for comparing groups. <i>Thể hiện trung vị, khoảng tứ phân vị và khoảng giá trị của một biến; phù hợp cho dữ liệu lệch và để so sánh các nhóm.</i>
Results pack Gói kết quả phân tích	The set of tables, figures, regression outputs and plain-language notes prepared for each group's dataset question and used to write the manuscript. <i>Bộ bảng, hình, kết quả hồi quy và ghi chú bằng ngôn ngữ đơn giản được chuẩn bị cho câu hỏi của từng nhóm, dùng để viết bản thảo.</i>

Session 6 — ANALYSE · Buổi 6 — PHÂN TÍCH

Term · Thuật ngữ	Definition · Định nghĩa
Sample Mẫu	The group of people actually studied, chosen to represent a larger population. <i>Nhóm người thực sự được nghiên cứu, được chọn để đại diện cho một quần thể lớn hơn.</i>
Confidence interval (95% CI) Khoảng tin cậy (KTC 95%)	The range of values for the true effect that is compatible with the data; narrow = precise. <i>Khoảng giá trị của tác động thực phù hợp với dữ liệu; khoảng hẹp nghĩa là ước lượng chính xác.</i>
P-value Giá trị p	If there were truly no effect, the probability of a result at least as extreme as the one observed. <i>Nếu thực sự không có tác động, đây là xác suất thu được kết quả ít nhất cực đoan như kết quả quan sát.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Statistical vs clinical significance Ý nghĩa thống kê và ý nghĩa lâm sàng	A result can be unlikely to be due to chance (statistical) yet too small to matter to patients (clinical). <i>Một kết quả có thể ít khả năng do ngẫu nhiên (ý nghĩa thống kê) nhưng quá nhỏ để có giá trị với người bệnh (ý nghĩa lâm sàng).</i>
Type I error Sai lầm loại I	Concluding there is an effect when there is none (a false alarm). <i>Kết luận có tác động trong khi thực tế không có (báo động giả).</i>
Type II error Sai lầm loại II	Missing a real effect, often because the study is too small (a missed fire). <i>Bỏ sót một tác động có thật, thường do cỡ mẫu quá nhỏ (bỏ lỡ đám cháy thật).</i>
Power Lực thống kê (power)	The chance that a study detects a real effect of a given size; usually set at 80% or 90%. <i>Khả năng nghiên cứu phát hiện được một tác động có thật với độ lớn nhất định; thường đặt ở mức 80% hoặc 90%.</i>
Sample size calculation Tính cỡ mẫu	Working out how many participants are needed to answer the question with enough power or precision. <i>Xác định số người tham gia cần thiết để trả lời câu hỏi với đủ lực thống kê hoặc độ chính xác.</i>
t-test Kiểm định t	Compares the means of a continuous, roughly normal outcome between two groups. <i>So sánh trung bình của một biến liên tục, xấp xỉ phân phối chuẩn, giữa hai nhóm.</i>
Chi-square test Kiểm định khi bình phương (χ^2)	Compares proportions of a categorical outcome between groups. <i>So sánh tỷ lệ của một biến phân loại giữa các nhóm.</i>
Non-parametric test (e.g. Mann–Whitney) Kiểm định phi tham số (ví dụ Mann–Whitney)	Compares groups using ranks; used for skewed or ordinal data. <i>So sánh các nhóm dựa trên thứ hạng; dùng cho dữ liệu lệch hoặc biến thứ hạng.</i>
Paired vs unpaired data Dữ liệu ghép cặp và không ghép cặp	Paired data come from the same patients measured twice (or from matched pairs) and need paired tests. <i>Dữ liệu ghép cặp là cùng người bệnh được đo hai lần (hoặc các cặp được ghép) và cần dùng kiểm định ghép cặp.</i>
Multiple comparisons So sánh bội (nhiều phép so sánh)	Doing many tests increases the chance of a false-positive finding; pre-specify one primary outcome or adjust (e.g. Bonferroni). <i>Thực hiện nhiều kiểm định làm tăng khả năng có kết quả dương tính giả; cần xác định trước một kết cục chính hoặc hiệu chỉnh (ví dụ Bonferroni).</i>
Correlation coefficient (r) Hệ số tương quan (r)	A number from -1 to $+1$ showing how closely two continuous variables follow a straight line; correlation is not agreement and not causation. <i>Con số từ -1 đến $+1$ cho biết hai biến liên tục bám theo một đường thẳng chặt chẽ đến đâu; tương quan không phải là sự phù hợp và không phải là quan hệ nhân quả.</i>
Regression (linear / logistic) Hồi quy (tuyến tính / logistic)	A model that relates an outcome to several factors at once; linear for continuous outcomes, logistic for binary outcomes. <i>Mô hình liên hệ một kết cục với nhiều yếu tố cùng lúc; hồi quy tuyến tính cho kết cục liên tục, hồi quy logistic cho kết cục nhị phân.</i>
R² Hệ số xác định (R ²)	The proportion of the variation in the outcome that a linear model explains. <i>Tỷ lệ biến thiên của kết cục được mô hình tuyến tính giải thích.</i>
Log transformation Biến đổi logarit	Taking the logarithm of a skewed variable (e.g. ICU length of stay) so that a linear model fits better; results are then reported as ratios. <i>Lấy logarit của một biến lệch (ví dụ thời gian nằm ICU) để mô hình tuyến tính phù hợp hơn; kết quả khi đó được báo cáo dưới dạng tỷ số.</i>
Adjusted estimate Ước lượng đã hiệu chỉnh	An effect estimate that accounts for other measured factors (confounders) in the model. <i>Ước lượng tác động đã tính đến các yếu tố khác đo được (yếu tố gây nhiễu) trong mô hình.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Crude vs adjusted estimate Ước lượng thô và ước lượng đã hiệu chỉnh	The crude estimate ignores other factors; the adjusted estimate accounts for measured confounders. A large gap between them signals confounding. <i>Ước lượng thô không tính đến các yếu tố khác; ước lượng đã hiệu chỉnh tính đến các yếu tố gây nhiễu đã đo. Chênh lệch lớn giữa hai ước lượng là dấu hiệu của nhiễu.</i>
Interaction term Số hạng tương tác	A term added to a regression model to test whether the effect of one variable differs across levels of another (e.g. fluid balance × chronic kidney disease). <i>Thành phần được thêm vào mô hình hồi quy để kiểm định xem tác động của một biến có khác nhau theo các mức của biến khác hay không (ví dụ cân bằng dịch × bệnh thận mạn).</i>
Propensity score matching Ghép cặp theo điểm xu hướng	Pairing exposed and unexposed patients who had a similar probability of exposure (based on measured factors), to make groups comparable in an observational study, as in Paper 2. <i>Ghép cặp người có và không có phơi nhiễm có xác suất phơi nhiễm tương tự (dựa trên các yếu tố đã đo), để các nhóm so sánh được trong nghiên cứu quan sát, như trong Bài báo 2.</i>
Overfitting (events per variable) Quá khớp (số biến cố trên mỗi biến)	A model with too many variables for the number of outcome events fits noise and fails on new patients; stepwise selection makes this worse. <i>Mô hình có quá nhiều biến so với số biến cố sẽ khớp cả nhiễu ngẫu nhiên và không dùng được cho người bệnh mới; chọn biến từng bước (stepwise) làm tình trạng này tệ hơn.</i>
Hazard ratio (HR) Tỷ số nguy cơ tức thời (HR)	Compares the rate at which an event (e.g. death) happens over time in two groups; 1 means no difference. <i>So sánh tốc độ xảy ra một biến cố (ví dụ tử vong) theo thời gian giữa hai nhóm; bằng 1 nghĩa là không khác biệt.</i>
Censoring Dữ liệu bị kiểm duyệt (censoring)	When a patient's follow-up ends before the event happens (study ends, transfer, loss to follow-up). <i>Khi việc theo dõi người bệnh kết thúc trước khi biến cố xảy ra (nghiên cứu kết thúc, chuyển viện, mất theo dõi).</i>
Kaplan–Meier curve Đường cong Kaplan–Meier	A step-shaped graph of the proportion of patients still free of the event over time, allowing for censoring. <i>Đồ thị dạng bậc thang thể hiện tỷ lệ người bệnh chưa xảy ra biến cố theo thời gian, có tính đến dữ liệu bị kiểm duyệt.</i>
Log-rank test Kiểm định log-rank	Compares Kaplan–Meier curves between groups. <i>So sánh các đường cong Kaplan–Meier giữa các nhóm.</i>
Cox regression (proportional hazards) Hồi quy Cox (nguy cơ tỷ lệ)	A model for time-to-event outcomes that gives adjusted hazard ratios; it assumes the HR stays constant over time. <i>Mô hình cho kết cục dạng thời gian đến biến cố, cho ra tỷ số nguy cơ tức thời đã hiệu chỉnh; giả định HR không đổi theo thời gian.</i>
Prediction model Mô hình dự báo (tiền lượng)	A model that estimates an individual patient's risk of an outcome from several predictors. <i>Mô hình ước tính nguy cơ xảy ra một kết cục cho từng người bệnh dựa trên nhiều yếu tố dự báo.</i>
Machine learning Học máy	Computer methods that learn patterns from data to make predictions; judged like any prediction model (TRIPOD+AI). <i>Các phương pháp máy tính học quy luật từ dữ liệu để đưa ra dự báo; được đánh giá như mọi mô hình dự báo (TRIPOD+AI).</i>
Calibration and discrimination Hiệu chuẩn và khả năng phân biệt	Calibration: predicted risks match observed risks. Discrimination: the model ranks patients who have the event above those who do not (AUC). <i>Hiệu chuẩn: nguy cơ dự báo khớp với nguy cơ quan sát được. Khả năng phân biệt: mô hình xếp người có biến cố cao hơn người không có biến cố (AUC).</i>
External validation Thẩm định ngoài	Testing a model on patients from a different time or place than those used to build it. <i>Kiểm tra mô hình trên người bệnh ở thời điểm hoặc địa điểm khác với dữ liệu dùng để xây dựng mô hình.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Statistical analysis plan (SAP) Kế hoạch phân tích thống kê (SAP)	A detailed, dated plan of all analyses, written before the data are examined. <i>Kế hoạch chi tiết, có ghi ngày, cho tất cả các phân tích, được viết trước khi xem dữ liệu.</i>
MCAR / MAR / MNAR Thiếu hoàn toàn ngẫu nhiên / thiếu ngẫu nhiên / thiếu không ngẫu nhiên	Why data are missing: completely at random (unrelated to anything); at random (explained by data that were recorded); not at random (related to the missing value itself). <i>Lý do dữ liệu bị thiếu: hoàn toàn ngẫu nhiên (không liên quan đến yếu tố nào); ngẫu nhiên (giải thích được bằng dữ liệu đã ghi nhận); không ngẫu nhiên (liên quan đến chính giá trị bị thiếu).</i>
Complete-case analysis Phân tích các trường hợp đầy đủ	Analysing only patients with no missing values; simple, but it can be biased and wastes data. <i>Chỉ phân tích những người bệnh không thiếu giá trị nào; đơn giản nhưng có thể sai lệch và lãng phí dữ liệu.</i>
Multiple imputation Quy gán đa bội	Filling in missing values several times from a model that uses other variables, then combining the results so the uncertainty is reflected. <i>Điền các giá trị bị thiếu nhiều lần dựa trên mô hình sử dụng các biến khác, rồi gộp kết quả để phản ánh độ không chắc chắn.</i>
Sensitivity analysis Phân tích độ nhạy	Repeating the analysis under different assumptions to see whether the conclusion changes. <i>Lặp lại phân tích với các giả định khác nhau để xem kết luận có thay đổi hay không.</i>

Session 7 — PROTECT & WRITE (I) · Buổi 7 — BẢO VỆ VÀ VIẾT BÀI (I)

Term · Thuật ngữ	Definition · Định nghĩa
Protocol Đề cương nghiên cứu	The full written plan of a study: why, who, what, how and how it will be analysed. <i>Bản kế hoạch đầy đủ của nghiên cứu: lý do, đối tượng, nội dung, phương pháp và cách phân tích.</i>
SPIRIT Hướng dẫn SPIRIT	A guideline listing what a trial protocol should contain. <i>Hướng dẫn liệt kê các nội dung cần có trong đề cương thử nghiệm lâm sàng.</i>
Declaration of Helsinki Tuyên bố Helsinki	The World Medical Association's ethical principles for medical research involving people. <i>Các nguyên tắc đạo đức của Hiệp hội Y khoa Thế giới cho nghiên cứu y học trên con người.</i>
Good Clinical Practice (GCP) Thực hành tốt thử nghiệm lâm sàng (GCP)	The international standard for designing, running and reporting clinical trials so participants are protected and data are reliable. <i>Tiêu chuẩn quốc tế về thiết kế, thực hiện và báo cáo thử nghiệm lâm sàng nhằm bảo vệ người tham gia và đảm bảo dữ liệu tin cậy.</i>
Ethics committee Hội đồng đạo đức trong nghiên cứu y sinh học	An independent group that reviews and approves research to protect participants. <i>Hội đồng độc lập xem xét và phê duyệt nghiên cứu nhằm bảo vệ người tham gia.</i>
Informed consent Sự đồng ý tham gia nghiên cứu sau khi được cung cấp thông tin	A person's voluntary agreement to take part after understanding the purpose, risks, benefits and their right to withdraw. <i>Sự đồng ý tự nguyện tham gia nghiên cứu sau khi hiểu rõ mục đích, nguy cơ, lợi ích và quyền rút khỏi nghiên cứu.</i>
Legal representative (surrogate) Người đại diện hợp pháp	A person allowed by law to give consent for a patient who cannot decide, e.g. a sedated ICU patient. <i>Người được pháp luật cho phép đồng ý thay cho người bệnh không thể tự quyết định, ví dụ người bệnh ICU đang dùng an thần.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Deferred consent Đồng ý trì hoãn	In emergencies approved by an ethics committee, consent is sought after enrolment, as soon as possible. <i>Trong tình huống cấp cứu được hội đồng đạo đức chấp thuận, sự đồng ý được xin sau khi đưa vào nghiên cứu, càng sớm càng tốt.</i>
Vulnerable participants Đối tượng dễ bị tổn thương	People less able to protect their own interests, e.g. critically ill, children, people with impaired capacity. <i>Những người ít có khả năng tự bảo vệ quyền lợi, ví dụ người bệnh nặng, trẻ em, người suy giảm năng lực hành vi.</i>
Confidentiality Bảo mật thông tin	Protecting participants' personal information from people who should not see it. <i>Bảo vệ thông tin cá nhân của người tham gia khỏi những người không có quyền tiếp cận.</i>
De-identification Loại bỏ thông tin định danh	Removing names and other identifiers and replacing them with a study code. <i>Xóa tên và các thông tin nhận dạng khác, thay bằng mã số nghiên cứu.</i>
Study registration (ClinicalTrials.gov, WHO ICTRP, PROSPERO) Đăng ký nghiên cứu (ClinicalTrials.gov, WHO ICTRP, PROSPERO)	Publicly recording a study's design and primary outcome before it starts; ICMJE journals require it for trials before the first participant is enrolled. <i>Công bố công khai thiết kế và kết cục chính của nghiên cứu trước khi bắt đầu; các tạp chí theo ICMJE yêu cầu thử nghiệm lâm sàng phải đăng ký trước khi tuyển người tham gia đầu tiên.</i>
Feasibility Tính khả thi	Whether the study can realistically be done with the available time, money, staff and patients. <i>Khả năng thực hiện được nghiên cứu với thời gian, kinh phí, nhân lực và người bệnh hiện có.</i>
Gantt chart Biểu đồ Gantt	A timeline chart showing each study task as a bar across months. <i>Biểu đồ tiến độ thể hiện mỗi công việc của nghiên cứu bằng một thanh ngang theo các tháng.</i>
Research misconduct (fabrication, falsification, plagiarism — FFP) Hành vi sai trái trong nghiên cứu (bịa đặt, làm sai lệch, đạo văn)	Inventing data, changing data or results, or using others' work or words without credit. <i>Bịa ra dữ liệu, sửa đổi dữ liệu hoặc kết quả, hoặc sử dụng công trình, câu chữ của người khác mà không ghi nguồn.</i>
Plagiarism Đạo văn	Presenting someone else's words, ideas or data as your own, without proper citation. <i>Trình bày câu chữ, ý tưởng hoặc dữ liệu của người khác như của mình mà không trích dẫn đúng.</i>
Duplicate and salami publication Xuất bản trùng lặp và xuất bản "cắt lát"	Publishing the same results twice, or slicing one study into many small papers. <i>Công bố cùng một kết quả hai lần, hoặc chia nhỏ một nghiên cứu thành nhiều bài báo nhỏ.</i>
Authorship (ICMJE criteria) Quyền tác giả (tiêu chí ICMJE)	Authors must contribute substantially, draft or revise the work, approve the final version and take responsibility for it. <i>Tác giả phải đóng góp đáng kể, viết hoặc chỉnh sửa bản thảo, phê duyệt bản cuối và chịu trách nhiệm về công trình.</i>
Gift and ghost authorship Tác giả "danh dự" và tác giả "ma"	Gift: listed without contributing. Ghost: contributed substantially but not listed. Both are unacceptable. <i>Tác giả "danh dự": có tên nhưng không đóng góp. Tác giả "ma": đóng góp đáng kể nhưng không có tên. Cả hai đều không được chấp nhận.</i>
CRedit (contributor roles) Phân loại vai trò đóng góp CRedit	A standard list of 14 contributor roles (e.g. conceptualisation, data curation, writing) used to state who did what. <i>Danh mục chuẩn gồm 14 vai trò đóng góp (ví dụ lên ý tưởng, quản lý dữ liệu, viết bản thảo) dùng để nêu rõ ai đã làm gì.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Conflict of interest Xung đột lợi ích	A financial or personal interest that could influence, or appear to influence, the research. <i>Lợi ích tài chính hoặc cá nhân có thể ảnh hưởng, hoặc bị xem là ảnh hưởng, đến nghiên cứu.</i>
Funding statement Tuyên bố nguồn tài trợ	Declaring who paid for the study and what role the funder had. <i>Nêu rõ ai tài trợ cho nghiên cứu và nhà tài trợ có vai trò gì.</i>
Data-sharing statement Tuyên bố chia sẻ dữ liệu	A statement on whether, how and with whom the study data will be shared. <i>Tuyên bố về việc dữ liệu nghiên cứu có được chia sẻ hay không, bằng cách nào và với ai.</i>
AI disclosure Công khai việc sử dụng trí tuệ nhân tạo (AI)	Stating how AI tools were used in a study or manuscript; AI cannot be an author. <i>Nêu rõ cách sử dụng công cụ AI trong nghiên cứu hoặc bản thảo; AI không thể là tác giả.</i>
STROBE checklist Bảng kiểm STROBE	The 22-item checklist for reporting observational studies; used to write and check the course-dataset manuscript. <i>Bảng kiểm 22 mục để báo cáo nghiên cứu quan sát; dùng để viết và kiểm tra bản thảo từ bộ dữ liệu của khóa học.</i>
Participant flow diagram Sơ đồ dòng người tham gia	A diagram showing how many patients were screened, excluded (with reasons) and analysed. <i>Sơ đồ cho biết số người bệnh được sàng lọc, bị loại (kèm lý do) và được đưa vào phân tích.</i>

Session 8 — PUBLISH · Buổi 8 — CÔNG BỐ

Term · Thuật ngữ	Definition · Định nghĩa
Calibrated language Ngôn ngữ chừng mực, tương xứng với bằng chứng	Words that match the strength of the evidence: "associated with" for observational studies; "caused" only where the design supports it. <i>Từ ngữ tương xứng với mức độ bằng chứng: "có liên quan với" cho nghiên cứu quan sát; chỉ dùng "gây ra" khi thiết kế nghiên cứu cho phép.</i>
Limitation Hạn chế của nghiên cứu	A weakness in design or conduct that may affect the results; good papers state them honestly. <i>Điểm yếu trong thiết kế hoặc thực hiện có thể ảnh hưởng đến kết quả; bài báo tốt nêu rõ một cách trung thực.</i>
Reporting guideline (CONSORT, STROBE, STARD, PRISMA) Hướng dẫn báo cáo (CONSORT, STROBE, STARD, PRISMA)	A checklist of what to report for each study type: trials, observational studies, diagnostic studies, systematic reviews. <i>Bảng kiểm các nội dung cần báo cáo cho từng loại nghiên cứu: thử nghiệm, nghiên cứu quan sát, nghiên cứu chẩn đoán, tổng quan hệ thống.</i>
Open access Truy cập mở	Articles free for anyone to read online, often paid for by an author fee. <i>Bài báo được mọi người đọc miễn phí trên mạng, thường do tác giả trả phí xuất bản.</i>
Article processing charge (APC) Phí xử lý bài báo (APC)	A fee that some open-access journals charge authors when a paper is accepted. <i>Khoản phí một số tạp chí truy cập mở thu của tác giả khi bài báo được chấp nhận.</i>
Licence (CC BY) Giấy phép sử dụng (CC BY)	The terms under which others may reuse a published article; CC BY allows reuse with attribution. <i>Điều khoản cho phép người khác sử dụng lại bài báo đã xuất bản; CC BY cho phép sử dụng lại với điều kiện ghi nguồn.</i>
Predatory journal Tạp chí săn mồi	A journal that charges authors but skips real peer review and editorial checks. <i>Tạp chí thu phí tác giả nhưng bỏ qua bình duyệt thực chất và kiểm tra biên tập.</i>

Term · Thuật ngữ	Definition · Định nghĩa
Cover letter Thư gửi ban biên tập (cover letter)	A one-page letter to the editor saying what the paper found, why it matters to the journal's readers, and the required declarations. <i>Thư một trang gửi biên tập viên, nêu phát hiện chính, tầm quan trọng với độc giả của tạp chí và các cam kết bắt buộc.</i>
ORCID Mã định danh ORCID	A free, permanent identifier that links a researcher to their publications. <i>Mã định danh miễn phí, cố định, liên kết nhà nghiên cứu với các công trình của họ.</i>
Preprint Bản tiền ấn phẩm (preprint)	A manuscript posted publicly (e.g. on medRxiv) before peer review. <i>Bản thảo được đăng công khai (ví dụ trên medRxiv) trước khi được bình duyệt.</i>
Peer review Bình duyệt (phản biện đồng cấp)	Evaluation of a manuscript by independent experts before a journal decides to publish it. <i>Việc các chuyên gia độc lập đánh giá bản thảo trước khi tạp chí quyết định đăng.</i>
Major / minor revision Yêu cầu sửa chữa lớn / nhỏ	Editorial decisions asking for substantial or small changes before a paper can be accepted. <i>Quyết định của tạp chí yêu cầu chỉnh sửa nhiều hoặc ít trước khi bài báo có thể được chấp nhận.</i>
Response to reviewers Thư phản hồi ý kiến phản biện	A polite, point-by-point reply to each reviewer comment, saying what was changed and where. <i>Thư trả lời lịch sự, theo từng ý, cho mỗi nhận xét của người phản biện, nêu rõ đã sửa gì và ở đâu.</i>
Point-by-point response Phản hồi theo từng ý	Answering every reviewer comment separately: the comment, your reply, and the change made (with page and line). <i>Trả lời riêng từng nhận xét của người phản biện: nhận xét, câu trả lời và thay đổi đã thực hiện (kèm số trang và dòng).</i>
Proofs Bản in thử	The typeset version of an accepted paper, which the authors check for errors before publication. <i>Bản dàn trang của bài báo đã được chấp nhận, tác giả kiểm tra lỗi trước khi xuất bản.</i>
Conference abstract and poster Tóm tắt và poster hội nghị	A short summary submitted to a scientific meeting and presented as a talk or a poster. <i>Bản tóm tắt ngắn gửi đến hội nghị khoa học, được trình bày dưới dạng báo cáo miệnng hoặc poster.</i>
Dissemination Phổ biến kết quả nghiên cứu	Sharing results with colleagues, patients and policymakers — through publication, presentations and local feedback. <i>Chia sẻ kết quả với đồng nghiệp, người bệnh và nhà hoạch định chính sách qua công bố, báo cáo và phản hồi tại đơn vị.</i>
Critical appraisal Thẩm định phê bình (đánh giá y văn)	Systematically judging whether a study's methods and results are trustworthy and useful for your patients. <i>Đánh giá có hệ thống xem phương pháp và kết quả của một nghiên cứu có đáng tin cậy và hữu ích cho người bệnh của mình hay không.</i>
Peer-review swap Trao đổi bình duyệt chéo giữa các nhóm	The after-course exercise in which each group reviews another group's draft manuscript and then responds to the review of its own draft. <i>Bài tập sau khóa học, trong đó mỗi nhóm bình duyệt bản thảo của một nhóm khác và sau đó phản hồi các nhận xét về bản thảo của nhóm mình.</i>

Further reading and free resources

Weekend 1 (Sessions 1–2) — Ask, find and read

Resource	Who publishes it	Why read it
* PubMed User Guide and PubMed online training (Help pages) — pubmed.ncbi.nlm.nih.gov	US National Library of Medicine	How to build a search with MeSH, Boolean operators and filters — repeats the live demo at your own pace.
MeSH Browser — meshb.nlm.nih.gov	US National Library of Medicine	Find the official MeSH term for your topic and see its narrower and broader terms.
Cochrane Handbook for Systematic Reviews of Interventions , Chapters 1–4 (starting a review; scope and questions; eligibility criteria; searching for studies) — training.cochrane.org/handbook	Cochrane	The gold standard on turning a question into PICO and searching systematically. Read the “Key points” box at the start of each chapter.
Oxford CEBM Levels of Evidence — cebm.ox.ac.uk	Centre for Evidence-Based Medicine, University of Oxford	A one-page table of which study type answers which kind of question best.
“How to read a paper” series, <i>BMJ</i> 1997 (first article: “Getting your bearings”)	T. Greenhalgh, <i>BMJ</i>	Classic short articles on reading research critically; the first ones help with framing questions.
<i>Designing Clinical Research</i> , chapters on the research question (<i>not free — check your library</i>)	S.B. Hulley and colleagues	The book that introduced the FINER criteria.
Zotero (free reference manager) — zotero.org	Corporation for Digital Scholarship (open source)	Store references, insert citations and switch to Vancouver style with one click. Its documentation has short start-up guides.
“Misrepresentation and distortion of research in biomedical literature”, <i>PNAS</i> 2018;115(11):2613–2619 (open access)	I. Boutron and P. Ravaud	What spin is, how common it is, and how to spot it in abstracts.
“Reporting and interpretation of randomized controlled trials with statistically nonsignificant results for primary outcomes”, <i>JAMA</i> 2010;303(20):2058–2064	I. Boutron and colleagues	The classic study of spin in trial reports — useful background for segment 1.4.

SQUIRE 2.0 guidelines (Standards for QQuality Improvement Reporting Excellence) — squire-statement.org	SQUIRE group (via EQUATOR)	How quality-improvement work is reported; helps separate QI from research and audit.
---	----------------------------	--

Weekend 2 (Sessions 3–4) — Design, bias and measures of effect

Resource	Who publishes it	Why read it
* <i>Epidemiology for the Uninitiated</i> (online book)	D. Coggon, G. Rose, D.J.P. Barker, <i>BMJ</i>	Short, plain-language chapters on measuring disease, comparing rates, case-control and cohort studies.
Catalogue of Bias — catalogof-bias.org	Centre for Evidence-Based Medicine, University of Oxford	One page per type of bias, each with a clinical example and how to prevent it. Look up "confounding by indication" and "recall bias".
<i>Statistics at Square One</i> , chapter on study design and choosing a statistical test (online book)	T.D.V. Swinscow and M.J. Campbell, <i>BMJ</i>	Links study design to the analysis that follows.
CEBM study design pages — cebml.org	Centre for Evidence-Based Medicine, University of Oxford	Brief descriptions and a flow chart for identifying study designs (useful for "Name that design").
"The environment and disease: association or causation?", <i>Proceedings of the Royal Society of Medicine</i> 1965;58:295–300 (free online)	A. Bradford Hill	The original nine "viewpoints" for judging causation — short and very readable.
CONSORT extensions for cluster, non-inferiority and crossover trials (via equator-network.org)	CONSORT group	What is different about reporting these trial variants.
Risk-of-bias tools: RoB 2 and ROBINS-I — riskofbias.info	Developers of the Cochrane risk-of-bias tools	The official tools, guidance documents and templates for randomised (RoB 2) and non-randomised (ROBINS-I) studies of interventions.
QUADAS-2 (diagnostic accuracy studies) — University of Bristol, <i>Annals of Internal Medicine</i> 2011;155:529–536	P. Whiting and colleagues	The standard tool for appraising diagnostic accuracy studies.
AMSTAR 2 (systematic reviews) — amstar.ca ; <i>BMJ</i> 2017;358:j4008	B. Shea and colleagues	A 16-item tool for judging how far a systematic review can be trusted.
Newcastle–Ottawa Scale (cohort and case-control studies) — Ottawa Hospital Research Institute website	G. Wells and colleagues	A short star-based checklist widely used in systematic reviews of observational studies.
PRISMA 2020 statement, checklist and flow diagram — prisma-statement.org	PRISMA group	How systematic reviews are reported; use the flow-diagram template when reading a review.
STARD 2015 (diagnostic accuracy studies) — via equator-network.org	STARD group	What a diagnostic accuracy paper must report (sensitivity, specificity, flow of participants).
Cochrane Handbook , chapter on undertaking meta-analyses (forest plots, heterogeneity, I^2) — training.cochrane.org/handbook	Cochrane	Plain explanation of pooled estimates and heterogeneity; read the key-points box.

Weekend 3 (Sessions 5–6) — Data and analysis (with the FLUID-ICU course dataset)

Resource	Who publishes it	Why read it
* Statistics at Square One (online book), chapters on data display and summary, confidence intervals, t-tests and the chi-squared test	T.D.V. Swinscow and M.J. Campbell, <i>BMJ</i>	The clearest free introduction to the statistics in this course, with medical examples.
Statistics Notes series, <i>BMJ</i> (1994 onwards)	J.M. Bland and D.G. Altman	One-page notes on single ideas; start with "Absence of evidence is not evidence of absence" (<i>BMJ</i> 1995).
"Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations", <i>European Journal of Epidemiology</i> 2016;31:337–350 (open access)	S. Greenland, S.J. Senn, K.J. Rothman and colleagues	A list of common misreadings of p-values and CIs — the source behind "5 common misinterpretations". Read the numbered list; skip the technical parts.
"The ASA statement on p-values: context, process, and purpose", <i>The American Statistician</i> 2016;70:129–133 (open access)	R.L. Wasserstein and N.A. Lazar, American Statistical Association	Six short principles on what a p-value can and cannot tell you.
OpenEpi — openepi.com	Open-source project (Emory University and partners)	Free online calculators for 2×2 tables and sample size — the kind of tool used in the live demo.
REDCap — project-redcap.org; KoBoToolbox — kobotoolbox.org	Vanderbilt University (REDCap consortium); KoBoToolbox (non-profit)	Free electronic data capture tools with validation and audit trails; REDCap is free for non-profit institutions that join the consortium.
"Beyond bar and line graphs: time for a new data presentation paradigm", <i>PLoS Biology</i> 2015;13(4):e1002128 (open access)	T.L. Weissgerber and colleagues	Why "dynamite plots" hide data and what to use instead.
Statistics Notes : "Multiple significance tests: the Bonferroni method" (<i>BMJ</i> 1995;310:170) and "Survival probabilities (the Kaplan-Meier method)" (<i>BMJ</i> 1998;317:1572)	J.M. Bland and D.G. Altman	One-page explanations of multiple comparisons and Kaplan–Meier curves.
"Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls", <i>BMJ</i> 2009;338:b2393	J.A.C. Sterne and colleagues	MCAR, MAR and MNAR, complete-case analysis and multiple imputation, with reporting advice.
"Guidelines for the content of statistical analysis plans in clinical trials", <i>JAMA</i> 2017;318(23):2337–2343	C. Gamble and colleagues	What a statistical analysis plan (SAP) should contain — a checklist to use with the SAP template.
TRIPOD+AI statement, <i>BMJ</i> 2024;385:e078378 — tripod-statement.org	G.S. Collins and colleagues	How prediction models, including machine-learning models, should be reported; a good checklist for judging them.
"Calculating the sample size required for developing a clinical prediction model", <i>BMJ</i> 2020;368:m441	R.D. Riley and colleagues	Why overfitting happens and why "10 events per variable" is only a rough guide.

Resource	Who publishes it	Why read it
Clinical Data Analysis in R (free online course)	Neudata	Learn to clean, describe and analyse a clinical dataset yourself — the prerequisite for reproducing the FLUID-ICU analyses with the course R scripts, and for the advanced modules.

Weekend 4 (Sessions 7–8) — Protect, write and publish (with the group manuscript and peer review)

Ethics and good practice

Resource	Who publishes it	Why read it
* Declaration of Helsinki (2024 revision) — wma.net	World Medical Association	The core ethical principles for research involving people; short enough to read in 20 minutes.
ICH E6(R3) Guideline for Good Clinical Practice (2025) — ich.org	International Council for Harmonisation	The international GCP standard. Read the "Principles of ICH GCP" section first.
International Ethical Guidelines for Health-related Research Involving Humans (2016) — cioms.ch	Council for International Organizations of Medical Sciences (CIOMS) with WHO	Detailed guidance, including research with people who cannot consent and research in emergencies.
Free GCP e-learning — globalhealthtrainingcentre.tghn.org	The Global Health Network	Free online Good Clinical Practice courses. Check which version of ICH E6 a course covers.
ClinicalTrials.gov — clinicaltrials.gov ; WHO International Clinical Trials Registry Platform (ICTRP) — who.int/clinical-trials-registry-platform	US National Library of Medicine; World Health Organization	Where trials are registered and searched; ICTRP links the WHO-recognised primary registries.
PROSPERO — crd.york.ac.uk/prospéro	Centre for Reviews and Dissemination, University of York	International register for systematic review protocols.
SPIRIT 2025 statement (trial protocols) — equator-network.org	SPIRIT group	Checklist of what a trial protocol must contain; use it with the protocol template.

Integrity, authorship and AI

Resource	Who publishes it	Why read it
* Recommendations for the Conduct, Reporting, Editing, and Publication of Scholarly Work in Medical Journals — icmje.org	International Committee of Medical Journal Editors (ICMJE)	The four authorship criteria, conflicts of interest and the current position on AI-assisted work. Read "Defining the Role of Authors and Contributors".
COPE guidance, flowcharts and position statement "Authorship and AI tools" — publicationethics.org	Committee on Publication Ethics (COPE)	What to do about plagiarism, duplicate publication or authorship disputes, step by step.
CRedit — Contributor Role Taxonomy — credit.niso.org	NISO (National Information Standards Organization)	The 14 contributor roles to use in your publication plan and author statement.
ORCID — orcid.org	ORCID (non-profit)	Get a free researcher identifier before your first submission.
ICMJE disclosure form (conflicts of interest) — icmje.org	ICMJE	The standard form most journals use for declaring competing interests.

Reporting, publishing and appraisal

Resource	Who publishes it	Why read it
* EQUATOR Network — equator-network.org	EQUATOR Network	The library of reporting guidelines; use its "Toolkits" and the guideline finder to pick the right checklist.
CONSORT 2025 statement and checklist (randomised trials)	CONSORT group (via EQUATOR)	What a trial report must contain — used to appraise Paper 1.
STROBE statement and checklists (cohort, case-control, cross-sectional) — strobe-statement.org	STROBE initiative	What an observational study report must contain — used to appraise Paper 2.
SPIRIT 2025 (trial protocols), STARD 2015 (diagnostic accuracy), PRISMA 2020 (systematic reviews) — prisma-statement.org	Respective groups (via EQUATOR)	The other guidelines named in the course; open only the one that fits your design.
Think. Check. Submit. — thinkchecksubmit.org	Cross-industry initiative of publishers and library organisations	A short checklist to decide whether a journal is trustworthy before you submit.
Directory of Open Access Journals (DOAJ) — doaj.org	DOAJ	Check whether an open-access journal is listed and what it charges.
CASP checklists — casp-uk.net	Critical Appraisal Skills Programme	Free one-page appraisal checklists for each design; a natural next step after the course's 10-question checklist.
ICMJE Recommendations , section "Manuscript Preparation and Submission" — icmje.org	ICMJE	What goes in each part of a manuscript (title page, abstract, IMRaD sections, references) and in the submission package.
EQUATOR Network toolkits ("Writing research", "Using reporting guidelines") — equator-network.org	EQUATOR Network	Practical guidance on writing each section and on using checklists at submission.

Resource	Who publishes it	Why read it
AuthorAID — authoraid.info	INASP	Free online courses and mentoring in research writing and publishing for researchers in low- and middle-income countries.
medRxiv — medrxiv.org	Cold Spring Harbor Laboratory, BMJ and Yale University	The main preprint server for health research; check your target journal's preprint policy first.
COPE "Ethical guidelines for peer reviewers" — publicationethics.org	Committee on Publication Ethics	What reviewers are expected to do — useful when writing a response to reviewers and for the "Be the reviewer" exercise.

Going further (advanced modules — [Advanced_Modules/advanced_modules.md](#))

- **Cochrane Handbook**, chapters on risk of bias and on meta-analysis — for the meta-analysis module.
- **Statistics at Square One**, chapters on regression and survival analysis — preparation for the hands-on R and survival modules.
- **TRIPOD+AI** and **STARD 2015** — for the prediction-model and diagnostic-accuracy modules.

All resources were selected for being free, stable and from recognised organisations. Versions change (e.g. CONSORT, ICMJE): always use the current version on the publisher's site.

APPENDIX D

Planning pages

Use these pages to plan your group's two products. They follow the parts of the proposal template and the manuscript template, so that you can copy your notes straight into the Word files.

D.1 Proposal planner

Working title: _____

Part	Section	Our notes	Done
1 (S1–S2)	Problem statement — the local number		☐
1	PICO: P / I or E / C / O / T		☐
1	Research question (one sentence)		☐
1	FINER: what we changed		☐
1	Aim; 2–3 objectives; H0 / H1		☐
1	Five key references (in the reference manager)		☐
2 (S3–S4)	Design and why		☐
2	Setting; inclusion / exclusion criteria		☐
2	Exposure / intervention and comparator		☐
2	Primary outcome (what, how, when); secondary outcomes		☐
2	Two main biases and how we reduce them		☐
3 (S5–S6)	Variables table; mini-CRF; data dictionary		☐
3	Data management (capture, checks, storage, audit trail)		☐

Part	Section	Our notes	Done
3	Sample size (assumptions and source)		☐
3	Analysis plan; confounders; missing-data plan		☐
4 (S7)	Ethics committee; consent route; registration		☐
4	Confidentiality (codes, key file, storage)		☐
4	Feasibility: timeline, budget headings, recruitment per month		☐
4	Target journal (+ back-up); reporting guideline		☐
4	Authors and CRediT roles; abstract skeleton		☐

D.1.1 Timeline (Gantt chart)

Task	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Protocol and ethics approval												
Registration												
Training data collectors												
Recruitment and data collection												
Data cleaning												
Analysis												
Writing and submission												

D.1.2 Who does what in our group

Task	Who	By when	Done
Proposal Part 1			☐
Proposal Part 2			☐
Proposal Part 3			☐
Proposal Part 4			☐
Slides 1–6			☐
Slides sent to the course lead			☐
Final written proposal (2 weeks after S8)			☐

D.2 Manuscript planner

Our question (Q): _____

Key numbers (copied from our results pack — no other numbers):

Result	Estimate	95% CI	n	Where in the pack
Crude main effect				
Adjusted main effect				
Outcome per group (n/N or median)				
Further analysis (sensitivity / imputation / interaction)				
Missing data (how many, which variable)				

Section	Target words	Writer(s)	Draft done	Checked
Title page (simulated-data statement, CRediT)	—		☐	☐
Abstract	≤ 250		☐	☐
Introduction	~350		☐	☐
Methods	~550		☐	☐
Results	~450		☐	☐
Discussion	~650		☐	☐
Tables and figures	Table 1 + up to 2		☐	☐
References (10–20, Vancouver)	—		☐	☐
STROBE checklist with page numbers	—		☐	☐
AI-use statement	—		☐	☐

Discussion — six moves (one line each):

1. Answer: _____
2. Compare: _____
3. Explain: _____
4. Limitations (with direction): _____

6. Conclusion (same verb as the title): _____

Notes

Notes